From Roots to Branches*

The Role of Families in Structural Change

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Abstract

Family ties are often viewed as barriers to moving into expanding sectors and growing places, as relatives can keep workers rooted in their existing industries and locations. Yet family members who have already entered these sectors or settled in growing destinations can help others branch out by connecting them to new opportunities. This paper studies how these family branches can gradually overcome the barriers created by existing family roots during U.S. structural transformation from 1880 to 1970. I reconstruct family ties for 32 million individuals and show that relatives working in manufacturing help others enter and remain in the sector, nearby relatives discourage migration and migrants disproportionately move toward destinations where relatives already live. I embed these forces in a dynamic spatial model in which workers' choices affect other relatives and later generations. Initially, families' roots in agriculture and less industrialized places slow structural transformation. As relatives branch out into manufacturing and move to more industrialized places, however, the connections they create offset about 90% of this drag by 1970, raising the manufacturing share by 2.05 percentage points and advancing the transition by about seven years. Moving vouchers can generate lasting gains beyond their direct recipients by creating connections to new places for family members left behind.

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1 Introduction

Structural transformation is a central driver of economic growth, with the reallocation of workers from agriculture toward manufacturing and services among its most prominent characteristics (Kuznets, 1973). Because productivity differs across sectors and locations, barriers to these movements can slow structural change and reduce aggregate output. Family ties are often viewed as one such barrier because relatives can keep workers rooted in their current sectors and locations (Alesina & Giuliano, 2014; Enke, 2019; Ghosh, Hwang, & Squires, 2023; Schulz, 2022; Schulz, Bahrami-Rad, Beauchamp, & Henrich, 2019). This view overlooks how family ties can also facilitate reallocation, as relatives in expanding sectors can help workers find jobs while those at destinations can facilitate migration. As workers themselves branch out into new sectors and places, they create similar connections for other family members and later generations. As development proceeds, these new branches can increasingly offset the barriers created by existing family roots. This raises three questions. How much do family roots in existing sectors and places slow sector switching and migration? How much do branches into expanding sectors and growing destinations facilitate these transitions? How does the importance of these forces change across generations, and what do they imply for structural transformation and spatial policies?

I study these questions during U.S. structural transformation from 1880 to 1970 and make four contributions. First, I construct new data that trace extended family members, relatives outside the direct line of descent, across sectors, locations and generations, documenting how the places where they live and the sectors in which they work change as the economy develops. Second, I provide empirical evidence on both sides of the mechanism: relatives in current sectors and origins reinforce persistence, while those in alternative sectors and at destinations facilitate sector switching and migration. Third, I develop a dynamic spatial model to quantify how these forces affect the pace and level of structural transformation as family members move across sectors and places. Fourth, I use a one-generation moving voucher to study how temporary relocation policies affect not only their direct recipients but also other family members and later generations.

The first challenge is to measure these connections across generations. Doing so requires large-scale data that link individuals to extended relatives and trace where those relatives live and work over time. Family ties are particularly well suited to this analysis because relationships are largely given by birth rather than formed through common workplaces or locations and can remain important even after relatives move apart (McPherson, Smith-Lovin, & Cook, 2001; Roberts & Dunbar, 2011). I construct such data by combining the full-count U.S. censuses with Census Tree links and recursively exploiting household relationships to recover extended relatives across generations (Buckles, Haws, Price, & Wilbert, 2025; Ruggles et al., 2024). This construction identifies relatives who need not themselves appear in the ancestry links while

retaining their census information, including sector and location. By 1940, the resulting data contain approximately 32 million individuals whose family histories can be traced back to 1850. With these data, I establish two sets of empirical findings. First, the sectors in which relatives work and the places where they live change substantially during structural transformation. Among aunts and uncles, the manufacturing share rises from about 10% in 1880 to roughly 40% by 1930. Over the same period, the mean distance among relatives in different commuting zones rises from 542 to 849 kilometers and the share of connections extending more than 1,000 kilometers nearly doubles. Second, I find evidence of both roots and branches in sector and migration choices. Deaths of manufacturing-employed relatives reduce manufacturing employment among workers already in the sector and those initially outside it, consistent with relatives supporting both persistence and entry. Similarly, deaths of local non-employed relatives increase migration, while migrants disproportionately choose destinations where relatives already live, conditional on origin and common origin–destination attractiveness.

To understand how these individual effects shape structural transformation, I develop a dynamic spatial model in which workers choose sectors and locations. The role of family ties depends on where relatives live and work: those in the current sector and location can discourage switching and migration, while those in alternative sectors and at destinations can facilitate them. Crucially, workers' choices also change the connections available to other family members. Entering manufacturing establishes a family connection to that sector, while migration creates one at a new destination. Workers account for how their decisions affect their own descendants, but not relatives in other branches of the family, creating externalities across family members and generations. These effects can reinforce existing patterns of employment and settlement, because one worker's choice makes the same choice more attractive to other relatives. Whether this feedback slows or accelerates structural transformation depends on where family members live and work and how strongly these connections influence workers' choices.

To measure the strength of these mechanisms, I quantify the model for the United States from 1880 to 1970 using two sources of variation. First, deaths of parents, aunts and uncles identify how strongly family members affect sector and location choices. Second, using indirect inference, I exploit an agricultural productivity shock to estimate how workers respond to changing economic opportunities, informing the key choice elasticities in the model.

With these estimates, I examine how family ties affect structural transformation and how their influence changes over time. Two main results emerge. First, family ties account for a sizable component of the barriers workers face when changing sectors and places: in 1910, their contribution is equivalent to roughly 10–15 percent of sectoral and migration barriers. Second, their aggregate effect changes as relatives enter new sectors and places. In 1880, families are rooted in agriculture and less industrialized places, slowing structural transformation. By 1950, family ties lower the manufacturing share by 1.31 percentage points, or 1.8 percent, relative

to an economy in which they do not affect sector and location choices. Almost all of this drag comes from where relatives are located. Over time, however, relatives branch out into manufacturing and move to new destinations, creating connections that make these choices easier for other family members. These changes offset 55 percent of the initial drag by 1950 and about 90 percent by 1970, raising the manufacturing share in 1970 by 2.05 percentage points and advancing the transition by about seven years. Because these gains are concentrated in initially less industrialized regions, the connections created through sector switching and migration also help spread industrialization to places that begin the transition further behind (Eckert & Peters, 2022).

The same mechanism has implications for relocation policies that help workers leave lagging regions and reach new opportunities. Such assistance has been used in large-scale programs in China and Indonesia (Bazzi, Gaduh, Rothenberg, & Wong, 2016; Zhang, Xie, & Zheng, 2023). I study a one-generation moving voucher targeted at workers in low-manufacturing places. Family ties matter for this policy in two ways. First, subsidized moves create connections at destinations while weakening ties at origins, affecting relatives who never receive a voucher and later generations. By 1970, these spillovers account for 33 percent of the voucher's manufacturing gains and 22 percent of its earnings gains, while reducing the welfare gain of targeted families by 0.7 percentage points. Family members left behind lose welfare even as their earnings increase. Second, existing family ties matter for targeting: among similar destinations, the same budget induces about three times as many moves when more relatives already live there. Adjusting grants to account for the effects of relocation on other relatives modestly improves aggregate welfare and income without additional spending. Evaluating relocation policy only through its direct recipients therefore misses both the long-run benefits and the welfare costs of the connections their moves create for other family members.

Related literature This paper contributes to three strands of research. First, it relates to work on barriers to structural transformation and labor reallocation. Beyond productivity and demand (Boppart, 2014; Comin, Lashkari, & Mestieri, 2021; Ngai & Pissarides, 2007), the pace of structural change depends on workers' ability to move across sectors and places. Existing work emphasizes sector-specific human capital and experience, worker heterogeneity and costly sector switching (Artuç, Chaudhuri, & McLaren, 2010; Dix-Carneiro, 2014; Herrendorf & Schoellman, 2018; Lee & Wolpin, 2006; Porzio, Rossi, & Santangelo, 2022), while spatial models study how sectoral reallocation interacts with migration frictions, local productivity and trade (Adamopoulos, Brandt, Chen, Restuccia, & Wei, 2024; Budí-Ors & Pijoan-Mas, 2025; Cao, Chen, Xi, & Zuo, 2024; Eckert & Peters, 2022).[1] Kinship and local social ties are one source of these barriers: origin-based insurance and attachment can inhibit rural–urban, off-farm and

[1] See Donovan and Schoellman (2023) for a review of labor-market frictions in structural transformation.

geographic mobility (Ghosh et al., 2023; Munshi & Rosenzweig, 2016; Zabek, 2024). I show how the importance of these barriers changes with development. As workers change sectors and locations, they also change where relatives live and work, and hence which opportunities become accessible to other family members. This provides a microfoundation for why barriers to structural change can weaken, strengthen or shift over time.

Second, the paper is most directly related to work on social ties in migration and labor-market allocation. In migration, local connections can discourage leaving while connections at destinations facilitate moves and shape where migrants go (Blumenstock, Chi, & Tan, 2025; Büchel, Ehrlich, Puga, & Viladecans-Marsal, 2020; Ilyés, Boza, Lőrincz, & Eriksson, 2023; Munshi, 2003; Stuart & Taylor, 2021).[2] Quantitative work embeds social connections in spatial allocation (Koenen & Johnston, 2024; Sahai & Bailey, 2023; Zabek, 2024; Zerecero, 2021). A parallel literature shows that family and social connections shape access to employers, occupations and intergenerational specialization (Cassan, Keniston, & Kleineberg, 2021; Kramarz & Nordström Skans, 2014; Lo Bello & Morchio, 2022; Munshi, 2011; Patel & Vella, 2013), while stronger kinship can generate both geographic and occupational persistence (Ghosh et al., 2023).[3] This work highlights a common distinction: ties to current places and activities can reinforce persistence, while ties to alternatives can provide access. I examine both forces within the same extended family structure, across sectors and places, and quantify their roles in structural transformation in general equilibrium.

A further contribution is to study how workers' choices change these connections for other relatives and later generations. This relates to models in which social interactions shape occupational or spatial allocation (Cassan et al., 2021; Eckert & Kleineberg, 2024; Koenen & Johnston, 2024; Sahai & Bailey, 2023) and respond endogenously to economic choices (Fan & Zou, 2021; Kim, Patacchini, Picard, & Zenou, 2023). Baseler (2025) identifies migration spillovers from older to younger relatives using temporary labor-demand shocks from an infrastructure program in Thailand. Dai, Mookherjee, Munshi, and Zhang (2020) show that hometown business connections generate sectoral and spatial concentration, while Dai, Mookherjee, Munshi, and Zhang (2023) show that connections accumulated during an earlier transition can facilitate entry into new activities but impede subsequent reallocation. Munshi (2014) emphasizes that community ties can constrain individual mobility while facilitating collective moves into new occupations and places. I build on these insights by tracing how individual sector and location choices change the connections available to extended relatives across generations and quantifying their general-equilibrium effects on structural transformation, providing a family-based microfoundation for persistence and path dependence (Allen & Donaldson, 2020).

[2] Origin-based insurance and temporary migration are studied by Munshi and Rosenzweig (2016) and Morten (2019); see Munshi (2020) for a review.

[3] Related evidence on labor-market connections and social-ties support includes Glitz (2017), Compton and Pollak (2014) and Bau and Fernández (2021).

Finally, the paper contributes to work using linked historical and genealogical data. Population-scale genealogies reconstruct extended family relationships over long horizons (Kaplanis et al., 2018; Koylu, Guo, Huang, Kasakoff, & Grieve, 2021), while census-linking projects follow individuals and families across generations (Althoff, Brookes Gray, & Reichardt, 2026; Buckles et al., 2025; Helgertz et al., 2022). Existing work primarily uses these records to study intergenerational mobility and persistence (Feigenbaum, 2018; Long & Ferrie, 2013; Ward, 2023).[4] I extend these approaches by recursively exploiting full-count census household relationships to recover additional extended relatives who need not themselves appear in the ancestry tree data (Espín-Sánchez, Ferrie, & Vickers, 2023). Because these relatives remain linked to their census records, I observe where they live and which sectors they work in. This makes it possible to reconstruct family ties beyond the parental line across generations, together with their economic and geographic characteristics, and trace how these connections change during U.S. structural transformation.

The remainder of the paper proceeds as follows. Section 2 describes the construction of extended-family data. Section 3 presents the empirical evidence. Section 4 develops the quantitative model, and Section 5 describes its estimation. Section 6 reports the counterfactual exercises. Section 7 concludes.

2 Constructing family trees

I construct a new large-scale panel of extended family connections that follows relatives across households, generations and locations from 1850 to 1940. The contribution is to expand and classify extended kin connections across households using census household information and existing cross-census individual links. Existing population-scale genealogies assemble family relationships from user-contributed family trees (Kaplanis et al., 2018; Koylu et al., 2021), while census-based work reconstructs multigenerational links to study intergenerational mobility (Espín-Sánchez et al., 2023). I combine full-count U.S. census microdata with cross-census individual links from the Census Tree Project (Buckles et al., 2025) and recursively use within-household relationships to recover extended kin.[5][6] A relative can therefore enter the family structure through a linked household member even without being linked across census waves directly, broadening coverage. As households split, new generations appear and relatives move, the procedure carries family identifiers forward while retaining the economic and demographic

[4] For multigenerational and extended-family persistence, see Lindahl, Palme, Sandgren Massih, and Sjögren (2015) and Adermon, Lindahl, and Palme (2021).

[5] The Census Tree Project draws partly on genealogical information to construct its individual links. Because Census Tree Project crosswalks are unavailable for 1950, extending the construction to 1950 substantially reduces coverage.

[6] Alternative linking procedures were explored; see Althoff et al. (2026) for a systematic comparison. Their crosswalks improve accuracy along some dimensions but yield a smaller linked sample.

information recorded in each census.

Constructing family trees and relationships. The construction combines cross-census links, which follow individuals over time, with IPUMS relationship-to-head codes, imputed for 1850–1870 and recorded from 1880. I start in 1850 by assigning household heads and their inferred blood and half-blood relatives to common family lineages.[7][8] I then follow these individuals into later censuses using the Census Tree Project crosswalks. When a linked person appears in a new household, newly observed relatives are added to the existing lineage and can carry it forward in subsequent waves. Repeating this process through 1940 produces extended family trees spanning households, locations and generations.

Because an incorrect person link could otherwise spread to many relatives, I extend a lineage only when the implied relationships are consistent with the available relationship information in the new census. I also restrict propagation through spousal, descendant and collateral links when they would generate ambiguous assignments across lineages. Appendix A1.1 describes these checks and propagation rules as well as the overall algorithm in detail. The resulting family identifiers can therefore follow the same lineage even after its members no longer live together.

I next recover how individuals within each lineage are related. I assign each person a genealogical position in the family tree and compare these positions to identify parents and children, siblings, grandparents, aunts and uncles, nieces and nephews and cousins. Relationship-to-head codes alone do not uniquely identify every pair. Unresolved relationships are treated as ambiguous and are not used to extend blood lineages. Spouses do not enter one another's blood lineage, although pairwise kin measures can include relatives by marriage. Children inherit identifiers from both parental branches and can therefore belong to multiple lineages. When the same pair is connected through more than one lineage, I retain the relationship implied by the closest common ancestor so that each pair enters a given kin measure only once.[9] Figure A1.1.4 illustrates an example of the resulting family tree.

Coverage and representativeness. With this approach, roughly 30–40% of the U.S.-born adult population observed in 1940 can be linked to familial ancestors over the preceding 90 years.[10] The reconstruction includes approximately 32 million individuals in 1940 for whom at least one reconstructed family lineage can be traced back to 1850 (Table 1).

[7] This excludes unrelated co-residents such as lodgers and domestic workers.

[8] Initial household definitions and inferred ties can affect subsequent lineage assignments; census codes do not always distinguish biological, adoptive and step relationships.

[9] Remaining ties are resolved using the reconstruction order and, if necessary, the lower family identifier.

[10] These rates are broadly consistent with chaining decade-by-decade linkage success documented in Althoff et al. (2026).

Two potential sources of error arise in the reconstruction. First, within-census household relationships may be incorrectly inferred, miscoded or internally inconsistent. I therefore apply consistency checks before using them to extend family connections. Second, cross-census links may match the wrong individual. I reduce this risk by requiring support from at least two Census Tree linkage sources. These restrictions favor accuracy over coverage but cannot eliminate error totally.

A separate limitation is that the kin measures include only relatives observed in the census data at a given date, rather than every relationship in the reconstructed genealogy. This differs from genealogy-based data constructed from pre-existing ancestry trees, such as Koylu et al. (2021), where family links can be retained independently of whether both relatives are observed in a particular census. The advantage of the census-based approach is that relatives can be linked directly to their contemporaneous location and economic outcomes, but changes in measured kin geography can reflect census coverage and linkage as well as migration.

The construction also generates selection. Linkage success varies systematically with observable characteristics, so the linked sample need not reproduce the contemporaneous full census. In addition, tracing a lineage over several decades depends on survival, census observability and successful links across generations, which can further change the composition of the longitudinal sample (Antonie, Inwood, Minns, & Summerfield, 2020; Ruggles, Fitch, & Roberts, 2018). Anchoring lineages in 1850 also limits coverage of later immigrant branches and descendants of enslaved ancestors absent from the named population census. Relative to the full census, the unweighted linked sample is more likely to be white, rural, farm-resident and employed in agriculture, whereas differences in age and sex are considerably smaller (Table A1.4.2).

To reduce these observable compositional differences, I reweight the linked sample toward the corresponding full-count census using inverse-propensity weights following Ward (2023). Separately by census wave, I estimate the probability of appearing in the linked sample as a function of demographic, socioeconomic and geographic characteristics and weight linked individuals by the inverse of this probability.[11] The weights substantially reduce differences between the linked and full-count samples along the observed characteristics included in the propensity model (Bailey, Cole, & Massey, 2020). They cannot, however, correct selection on unobservables, restore family connections that are not recovered or eliminate false links.

Table A1.4.1 summarizes family size and generational depth. The median reconstructed family contains 22 linked members, with an interquartile range of 9 to 51 and spans roughly three generations from the initial anchor.[12] Figure A1.4.1 shows a fat-tailed family-size distribution, with a small number of large reconstructed lineages arising as distant lateral ties accumulate across waves.[13] Lineages with only 1–5 members can reflect linkage attrition and the conser-

[11] See Appendix A1.1.2 for details.

[12] Family size includes distant relatives such as second and third cousins.

[13] The empirical analysis focuses on ties no more distant than nieces and cousins.

vative exclusion of non-relatives and marriage-based upstream connections. Table 1 reports the average number of relatives by relationship type from 1880 to 1940. Extended kin account for a large share of the connections. Appendices A1.3 and A1.2 compare kin geography with independent measures of social and migration connectedness and report internal validation.

Table 1: Coverage and composition of reconstructed kin networks, 1880–1940

	1880	**1900**	**1910**	**1920**	**1930**	**1940**
Panel A. Composition of the kin network						
Vertical kin						
Parent	1.80	1.70	1.44	1.79	1.71	1.74
Child	1.74	1.69	1.42	1.79	1.71	1.74
Grandparent	0.42	0.55	0.32	0.39	0.38	0.33
Grandchild	0.44	0.55	0.33	0.41	0.38	0.35
Horizontal kin						
Sibling	3.48	3.06	2.46	2.99	2.64	2.45
Uncle/Aunt	1.59	1.70	1.42	1.54	1.45	1.38
Niece/Nephew	1.64	1.71	1.43	1.55	1.47	1.40
First cousin	2.39	2.97	2.29	2.57	2.27	2.11
All relatives	**13.50**	**13.93**	**11.11**	**13.03**	**12.01**	**11.50**
Panel B. Sample coverage						
Individuals in network (millions)	17.46	21.96	24.72	27.82	31.41	32.21
Share of full population (%)	34.9	29.0	26.9	26.3	25.6	24.4
Share of U.S.-born population (%)	51.6	43.4	40.3	38.2	35.4	31.2

Notes: Panel A reports the mean number of relatives of each type per individual in the reconstructed kin network, pooling relatives across commuting zones and employment sectors. Relationships are classified from reconstructed family paths. "All relatives" is the sum of the eight reported blood-relationship categories. Panel B reports the number of individuals in the network and its coverage relative to the full and U.S.-born census populations in each year.

Inferring deaths. To obtain family-specific shocks, I infer deaths from census records. Since population-wide individual vital statistics are unavailable before 1950, I combine repeated observations of marital status with longitudinal spouse links to infer mortality. IPUMS records whether individuals are married, single, divorced or widowed from 1880 onward. I classify a spousal death between census years t and $t+1$ when an individual is married at t, widowed at $t+1$ and the spouse observed at t cannot be found in the $t+1$ linkage universe. I then propagate the implied death through the reconstructed family network so that relatives can be assigned the shock even when they live in different households or locations.

The inferred-death measure has three main limitations. First, marital-status reporting error can generate misclassification, while deaths followed by remarriage before the next census may not be detected. Second, the procedure identifies deaths only for married individuals whose

surviving spouse remains observable and successfully linked across waves, so the resulting death sample is selected. Third, failures to link a surviving spouse can generate false inferred deaths. Table A1.6.1 reports summary statistics for the inferred-death sample. Appendix A1.5 documents correlations with state-level mortality rates. These shocks form the basis of the empirical analysis in the next section.

3 Family ties and labor reallocation

Family ties can both slow and facilitate the movement of workers from agriculture to manufacturing and from less industrialized to growing places. Relatives concentrated in agriculture and at origins may reinforce existing choices. Yet relatives who enter manufacturing or move elsewhere establish connections that can help other family members pursue the same opportunities. The role of family ties therefore depends not only on how strongly relatives influence one another, but also on where they work and live.

As structural transformation proceeds, the roots that initially anchor families can give rise to branches reaching new sectors and places. Understanding how these changing connections influence workers' choices requires tracing family relationships across sectors, locations and generations. Family ties are particularly useful for studying this mechanism for four reasons. First, they are largely determined by birth, rather than formed through common employment or residence, reducing selection into the relationships themselves. Second, they can remain important after relatives move apart or interact less frequently (Roberts & Dunbar, 2011). Third, migration by one family member may make the same destination more accessible to others (Baseler, 2025). Fourth, family connections retain substantial variation beyond ties associated with shared hometowns or birthplaces and continue to predict sector and destination choices after accounting for these common origins (Appendix A2.2). In this section, I first document how the sectoral and geographic distribution of relatives changes, then examine whether family connections influence sector and location choices. Since relatives' employment and residence remain endogenous, I complement descriptive evidence with event studies exploiting deaths of family members.

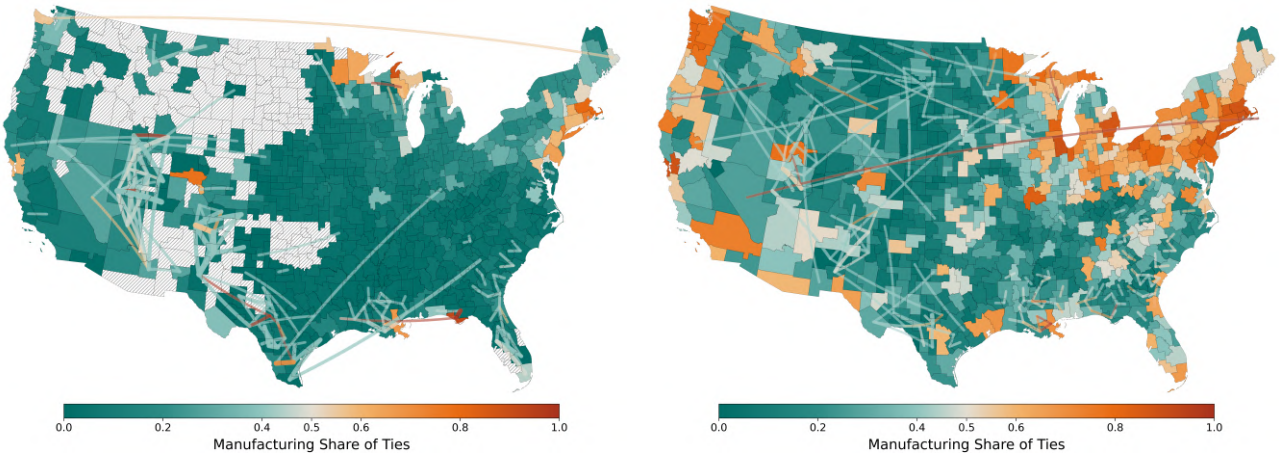
3.1 The changing reach of family ties

The relatives to whom workers are connected change markedly over the course of development. In 1880, family connections are concentrated in agriculture, but as manufacturing expands, more relatives enter the growing sector. Among aunts and uncles, for example, the manufacturing share rises from about 10% in 1880 to roughly 40% in 1930, both within and outside the individual's CZ. Cousins exhibit a similar increase. Thus, later generations are increasingly connected to family members who have already entered manufacturing.

Family ties also become less geographically concentrated. Between 1880 and 1930, the share of relatives living in the same CZ declines, while connections to relatives elsewhere span increasingly greater distances. Among relatives living in different CZs, mean distance rises from 542 to 849 km and median distance from 306 to 475 km. The share of these ties spanning more than 1,000 km nearly doubles, from 16.5% to 30.1%. Appendix A2.1 documents the patterns by relationship and the full distance distribution.[14] Family connections thus increasingly extend beyond workers' local labor markets, reaching relatives in more distant places.

Figure 1 shows how the changing geography of family ties accompanies the spread of manufacturing. CZ shading measures manufacturing's share of employment in agriculture and manufacturing, while lines identify strongly connected CZ pairs. To compare the concentration of bilateral ties across locations of different sizes, I follow Bailey, Cao, Kuchler, Stroebel, and Wong (2018a) and define relative connectedness as $RC_{od} = S_{od}/(S_o \times S_d)$, where S_{od} counts family ties between CZs o and d, and S_o and S_d are their respective totals of cross-CZ ties.

Figure 1: The changing geography and sectoral composition of family ties



(a) 1880 (b) 1930

Notes: Shading gives each CZ's manufacturing share among agriculture and manufacturing, with teal denoting lower and orange higher manufacturing shares. Lines show selected strong cross-CZ family connections. Line width increases with relative connectedness, as defined in the text, and line color gives the manufacturing share among relatives connected across the pair. Striped CZs have no sector-composition data.

By 1930, prominent family connections extend from the Midwest and Great Plains toward California and the Mountain West and from western locations toward the Northeast. Connections within the South and from Texas to other regions also become more visible. The sectoral composition of relatives connected across these places changes alongside manufacturing's geographic expansion. These patterns illustrate the formation of new family branches across

[14]Most of the increase in cross-CZ connections does not reflect how ties are constructed: within a census year, families reconstructed from the tree are only 1–3 percentage points more likely to span different CZs than families reconstructed 20 years earlier and followed forward.

sectors and locations but they do not establish whether these connections influence workers' decisions. That is the question I turn to next.

3.2 Family ties and sectoral reallocation

The changing distribution of relatives across sectors matters for structural transformation if their employment influences the choices of other family members. In particular, the same family influence could sustain employment in agriculture when most relatives work there, but facilitate manufacturing entry once some relatives have switched sectors. I begin by examining whether workers are more likely to work in sectors where their relatives are employed.

Workers with more local manufacturing relatives are more likely to work in manufacturing, while those with more agricultural relatives are more likely to work in agriculture. Following Eckert and Peters (2022), I focus on these two sectors, although services account for 33.9% of employment in 1880 and 52.6% in 1950.[15] Workers with 5–14 manufacturing relatives in their CZ are 7–8 percentage points more likely to work in manufacturing than workers with none. Among those with 15–19 agricultural relatives, the corresponding difference approaches 6 percentage points.[16] The relationships remain positive with CZ fixed effects and controls for individual characteristics and family size (Appendix Figure A2.3.1).[17] Raw employment rates show the same pattern (Appendix Figure A2.3.2).

These correlations are suggestive but need not reflect the influence of relatives. Families may share skills, preferences or other determinants of sector choice. To assess whether employment connections themselves matter, I study how workers respond when a relative employed in the same sector dies. Death-based designs have been used to examine connections among collaborators, co-inventors and coworkers (Azoulay, Graff Zivin, & Wang, 2010; Jäger & Heining, 2022; Jaravel, Petkova, & Bell, 2018; Jones & Olken, 2005; Oettl, 2012); Lin and Mo (2025) use a related approach to coworkers' sectoral connections.

Deaths of sector-employed relatives. The analysis uses a balanced panel of approximately 2.2 million individuals observed in the decennial censuses from 1900 to 1940. Treatment is the first death of a relative who was employed in the relevant sector and lived in the worker's CZ before death. I restrict eligible relatives to ages 18–40 to reduce concerns about predictable old-age mortality and anticipatory responses.[18] When two individuals are connected through multiple reconstructed family identifiers, their relationship is de-duplicated as in Section 2, so

[15] Industry employment is defined using IPUMS IND1950 codes. Manufacturing includes durable (306–399) and non-durable (406–499) manufacturing, while agriculture includes farming (105).

[16] Pairwise family relationships are de-duplicated as described in Section 2, so each relative is counted only once when reconstructed family trees overlap.

[17] Using the share of employed relatives in a sector yields similar results.

[18] Age and employment are measured in the last census in which the relative is observed alive, as the precise date of death is unavailable. Results are robust to alternative age restrictions.

each pair contributes at most once. Table A2.3.1 reports summary statistics for the estimation sample.

I estimate a stacked event study following Wing, Freedman, and Hollingsworth (2024).[19] Each stack compares workers experiencing their first eligible death in a particular census interval with workers who have not yet experienced one. For example, in the 1930 stack, treated workers lose a relative between 1920 and 1930. The 1920 census is the omitted period ($\tau = -1$) and 1930 is the first post-death observation ($\tau = 0$). Controls comprise never-treated and later-treated individuals while untreated while earlier-treated workers are excluded. A worker first treated in 1940 thus contributes to the 1930 stack only through 1930.[20] Identification requires that, conditional on the controls and fixed effects, treated and comparison workers would have followed parallel employment trends in the absence of a death.

Let i index individuals, f reconstructed families, s treatment-cohort stacks and t census years. Each worker is assigned a family identifier f following the de-duplication procedure described below. Let T_{if} be the first post-death census for worker i, e_s the treatment census for stack s and $D_{ifs} = \mathbf{1}\{T_{if} = e_s\}$. Define event time as $r_{ifst} = (t - e_s)/10$, in census decades, and let c^0_{ifs} denote the worker's CZ in the census preceding treatment in stack s. I estimate

$$\text{ManufEmp}_{ifst} = \alpha_{ifs} + \sum_{\tau \neq -1} \beta_\tau D_{ifs}\, \mathbf{1}\{r_{ifst} = \tau\} + \lambda_{c^0_{ifs} ts} + \gamma' A_{it} + \Gamma_t' X_f^{1900} + \varepsilon_{ifst}, \qquad (1)$$

where α_{ifs} are individual-by-stack fixed effects and $\lambda_{c^0_{ifs} ts}$ are pre-event-CZ-by-year-by-stack fixed effects. The latter assign workers to their pre-treatment CZ throughout the stack, allowing for local labor-market trends without conditioning on subsequent migration.[21] The vector A_{it} contains age controls, X_f^{1900} contains family size, the number of local manufacturing-employed relatives and other household characteristics measured in 1900. Interacting these initial characteristics with census-year effects allows employment trends to differ with initial family composition. Standard errors are clustered at the reconstructed family level f.[22]

Figure 2 shows that the death of a manufacturing-employed relative is followed by a persistent reduction in the worker's probability of employment in manufacturing. Pre-event estimates are small, while the post-death decline reaches approximately 0.5 percentage point three decades after the death, or 5% of the average manufacturing share (Panel a). The effect is not limited to immediate relatives. Twenty years after death, manufacturing employment falls by 0.76–0.83

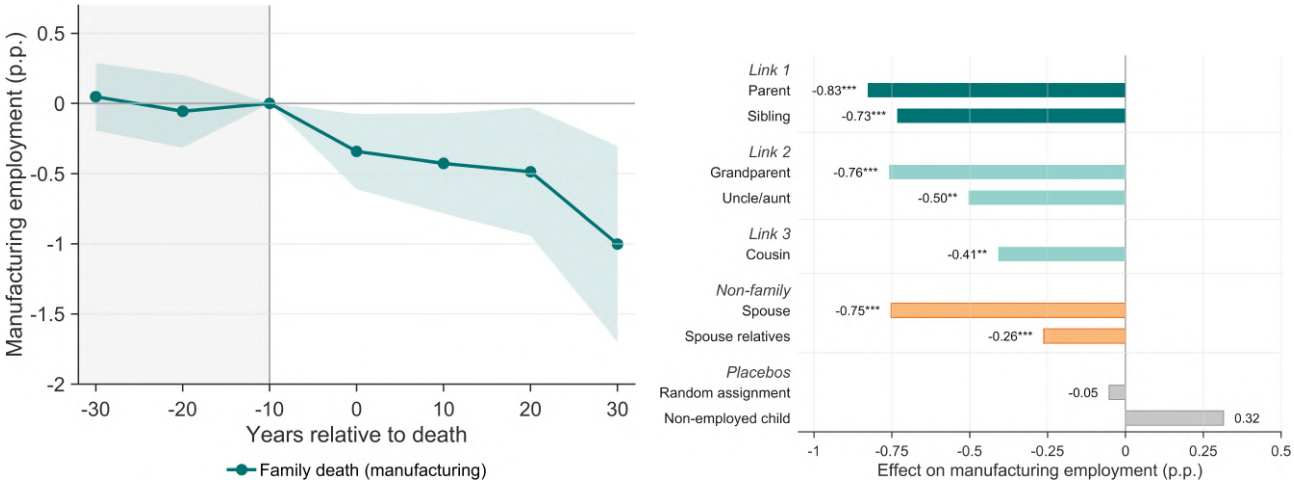
[19] Similar dynamic effects can be estimated using Borusyak, Jaravel, and Spiess (2024).

[20] Each stack includes up to three census waves before and after treatment. Observations are censored from the second eligible death onward. Approximately 11,000 individuals experience multiple eligible deaths. I apply the corrective weights of Wing et al. (2024), multiplied by population weights that reweight the linked sample toward the full census.

[21] Similar results are found with CZ-by-year-by-stack fixed effects from the location of the individual in 1900.

[22] If two individuals share multiple family identifiers, the de-duplication rule in Section 2 assigns their relationship to one family identifier. Never-treated individuals are assigned the earliest reconstructed family identifier available to them and remaining ties are resolved using the lower identifier.

Figure 2: Manufacturing employment after the loss of a sector-employed relative



(a) Event-study response (b) Response by relationship

Notes: Panel (a) reports estimates from equation (1); $\tau = -1$ is the final census before the death. Panel (b) reports the manufacturing response twenty years after the first eligible death separately by the focal individual's relationship to the deceased, together with placebo estimates. Shaded regions and vertical bars denote 95% confidence intervals. Standard errors are clustered at the de-duplicated family-identifier level.

percentage points following the loss of a parent, child, sibling or spouse (Panel b). The effect of losing an aunt or uncle is about 60% of the parent effect while the response to a cousin is approximately 0.41 percentage points. Thus, sectoral connections appear to matter beyond the household, although their influence declines with genealogical distance.

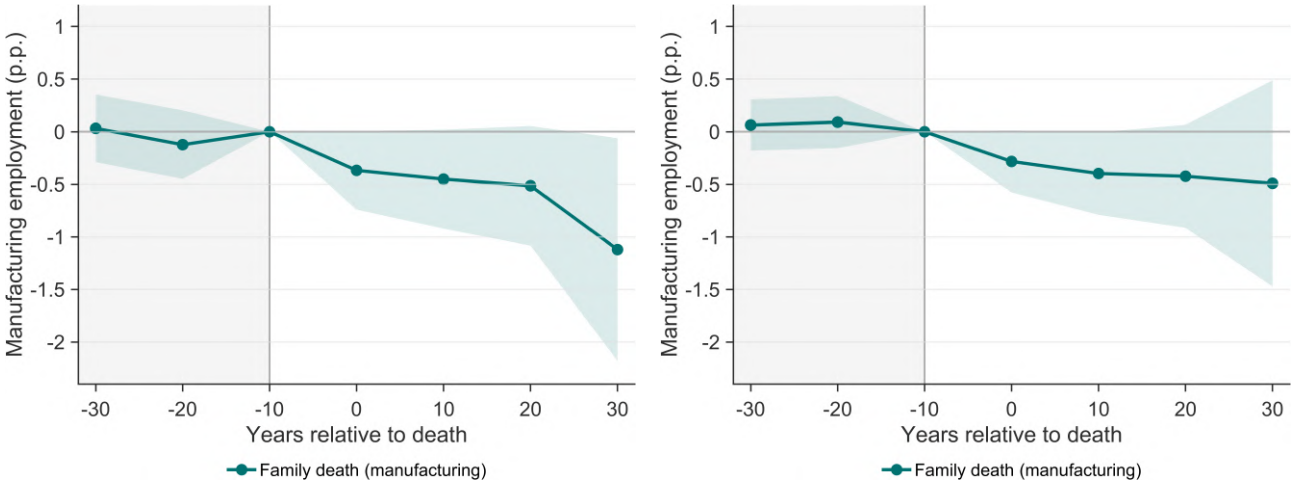
Sectoral persistence and entry. The decline in manufacturing employment following the death of a manufacturing relative can reflect lower persistence among workers already in the sector or reduced entry among those initially outside it. The former reinforces existing employment, while the latter allows manufacturing connections to spread within families. I distinguish these margins by estimating equation (1) separately by pre-death manufacturing status. Within each stack, treated and control workers are classified in the same pre-treatment census, such as 1920 for the 1930 stack.

Figure 3 shows negative post-death estimates in both samples, consistent with relatives supporting continued manufacturing employment among incumbents and facilitating entry among workers initially outside the sector.[23]

These responses could reflect bereavement, asset transfers or local sectoral shocks rather than the loss of sector-specific support. However, deaths of non-employed relatives generate no comparable manufacturing decline, while manufacturing-relative deaths are followed by little change in household composition or fixed assets. An exception is the decline in farm holdings

[23]Deaths of relatives employed in agriculture and services are similarly followed by declines in employment in the deceased relative's sector (Appendix Figure A2.3.3).

Figure 3: Deaths of manufacturing-employed relatives by workers' initial sector



(a) Initially in manufacturing (b) Initially outside manufacturing

Notes: The figure reports estimates from the stacked event-study specification in equation (1), separately for workers employed in manufacturing (Panel a) and outside manufacturing (Panel b) in the census preceding the death of a manufacturing-employed relative. Shaded regions and vertical bars denote 95% confidence intervals. Standard errors are clustered at the de-duplicated family-identifier level.

following agricultural-relative deaths. Occupational fatalities account for only a small share of deaths, and randomly reassigning manufacturing deaths across families within the same CZ generates no employment response, preserving local exposure while removing the family connection. Appendix A2.3.2 reports these checks and robustness to alternative age restrictions, weights, comparison groups and event windows.

The estimates do not identify precisely how relatives provide support. They may facilitate entry through job information, employer contacts and occupational knowledge (Calvó-Armengol & Jackson, 2004; Granovetter, 1973; Rees, 1966), while also helping workers maintain employment and earnings once established in a sector. Consistent with these mechanisms, occupational persistence is stronger where entry relies on workplace experience rather than schooling (Appendix A2.3.3). Employed relatives are also associated with lower unemployment and job search and faster subsequent wage growth in manufacturing, especially outside urban labor markets (Appendix A2.3.4). Additional evidence in Appendix A2.3 shows that manufacturing-relative deaths reduce earnings even among workers remaining in the same CZ and detailed industry, while deaths of supervisory relatives are associated with lower manufacturing employment and occupational standing. In agriculture, deaths reduce farm operation and residence, particularly when few other agricultural relatives remain, consistent with family labor and other productive support sustaining farming.

3.3 Family ties and spatial reallocation

Family ties can shape location choices through two opposing forces. Relatives nearby may make an origin more attractive, discouraging workers from leaving. Relatives who have settled elsewhere, by contrast, may make those destinations easier to reach. To distinguish these roles, I first examine migration following the death of a local relative and then study where migrants move in relation to existing family connections.

Family connections at origins. To isolate the value of family presence at the origin from contemporaneous employment connections, I study deaths of non-employed relatives living in the worker's CZ. The deceased are subject to the same age restriction as in the sectoral analysis. Because these relatives were not employed, their deaths reduce local family presence without directly removing an employment connection to a particular sector.

I apply the stacked event-study design to an indicator for changing CZ between consecutive censuses. Unlike employment, migration is measured over a census interval. Since the death is also observed between censuses, $\tau = 0$ includes the interval in which the death occurs; the order of the death and move within that interval is unknown.[24] The specification is

$$\text{Mig}_{ifst} = \alpha_{ifs} + \sum_{\tau \neq -1} \beta_\tau D_{ifs} \, \mathbf{1}\{r_{ifst} = \tau\} + \lambda_{c^0_{ifs} ts} + \gamma' A_{it} + \Gamma_t' X_f^{1900} + \varepsilon_{ifst}, \tag{2}$$

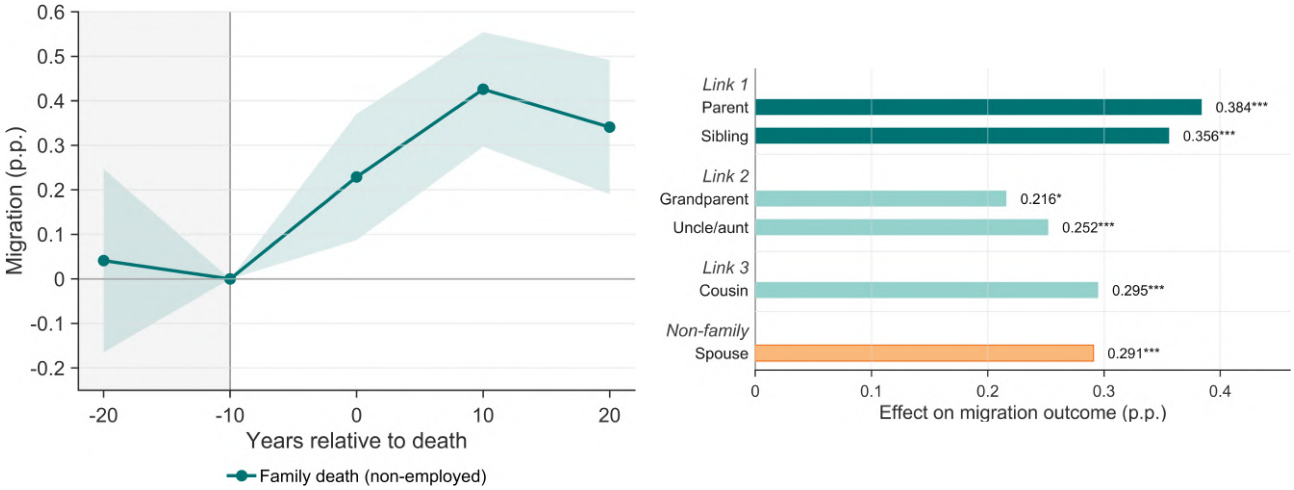
using the same fixed effects, age controls, 1900 family and household characteristics, weights and family-level clustering as equation (1).

Figure 4 shows that the loss of a local non-employed relative is followed by an increase in migration. The effect is 0.43 percentage points in the first unambiguously post-death interval and 0.34 percentage points in the next (Panel a). Relative to the baseline decadal migration rate of 19%, the first response amounts to an increase of about 2.3%. The magnitude is comparable to Koenen and Johnston (2024), who estimate that having a friend in the same CZ increases the probability of residing there by 0.3 percentage points. The response occurs following the deaths of parents, siblings, grandparents, aunts and uncles and cousins, as well as spouses (Panel b). Local family ties therefore appear to anchor workers beyond the immediate household and beyond blood relationships.

Nearby relatives may provide childcare and enable labor supply (Compton & Pollak, 2014), strengthen attachment to place (Zabek, 2024) or offer informal insurance (Morten, 2019; Munshi & Rosenzweig, 2016). Consistent with the importance of such support, the migration response to deaths is larger for households with young children (Appendix Table A2.4.1), and local non-employed relatives are associated with higher female employment and less housework

[24] The omitted period, $\tau = -1$, precedes the death. With location data from 1900 onward, treated migration cohorts are those of 1920 and 1930.

Figure 4: Migration after the loss of a local family tie



(a) Event-study response

(b) Response by relationship

Notes: Panel (a) reports estimates from equation (2); $\tau = -1$ is omitted and $\tau = 0$ is the migration interval in which the death occurs. Panel (b) reports estimated effects on migration separately by relationship, grouped by genealogical distance. The spouse estimate provides a comparison with a non-blood relationship. Shaded regions in Panel (a) denote 95% confidence intervals.

(Appendix Figure A2.4.1). Deaths of relatives outside the CZ do not generate a comparable migration response and the estimates are similar among non-homeowners. Together, these findings support the interpretation that nearby family presence increases the value of remaining at an origin.

Family connections at destinations. The same migration decisions that separate workers from relatives at origins can establish family connections in new places. Once a relative has settled elsewhere, that destination may become more accessible to other family members through information and support (Carrington, Detragiache, & Vishwanath, 1996; McKenzie & Rapoport, 2007; Munshi, 2003; Stuart & Taylor, 2021). Historical evidence also links family structure to geographic mobility (Ghosh et al., 2023). I therefore examine whether migrants disproportionately move toward destinations where relatives already live.

For reconstructed family identifier f, origin o, destination d and census t, let n_{fdt} count family members already living in $d \neq o$, and let y_{fodt} indicate whether a family member moves from o to d before the following census.[25] I estimate

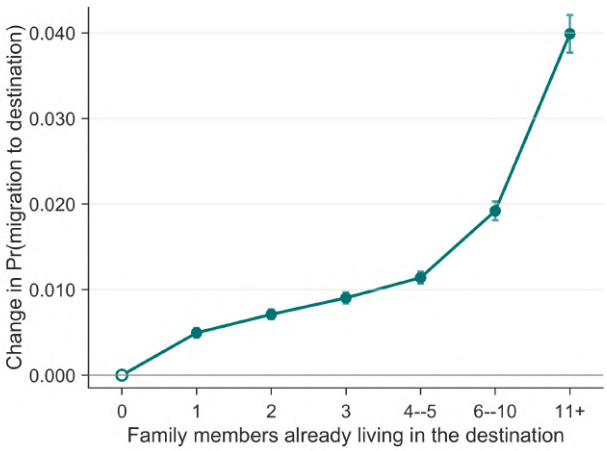
$$y_{fodt} = \sum_{j \geq 1} \beta_j \mathbf{1}\{n_{fdt} = j\} + \mu_{fot} + \delta_{odt} + \varepsilon_{fodt}, \tag{3}$$

where destinations without relatives form the omitted category. Family-by-origin-by-census

[25]Individuals may belong to multiple reconstructed lineages, so one move can contribute to more than one lineage-specific outcome, but is counted only once within each family identifier.

fixed effects μ_{fot} account for differences in migration propensities across families at a given origin, while origin-by-destination-by-census fixed effects δ_{odt} absorb common migration conditions for each corridor.[26] The comparison therefore holds fixed both the family-origin-period and conditions common to workers choosing a particular corridor. Appendix A2.2 further shows that this relationship persists after controlling for shared origins, including state of birth.

Figure 5: Migration toward family at destination



Notes: The figure reports estimates from equation (3), with destinations containing no relatives as the omitted category. Standard errors are two-way clustered by reconstructed family identifier and origin–destination corridor. Vertical bars denote 95% confidence intervals.

Figure 5 shows that migrants disproportionately choose destinations where relatives already live. The presence of even one relative is associated with a substantially higher probability that another family member moves to that destination, and the association grows with the number already there. By comparison, the probability of moving to a sampled destination without relatives is 0.035%. This evidence is descriptive: family members may share preferences for particular places or respond to opportunities not fully captured by the fixed effects. Nevertheless, the pattern is consistent with earlier migrants establishing connections that guide subsequent family migration. The branches created by one move can become places of settlement for those who follow.

Taken together, the evidence identifies a potential dynamic tension. Family ties support employment in the sectors where relatives work and increase the value of remaining near relatives at origins. Yet sector switching and migration also change the sectors and places to which subsequent family members are connected. The initial concentration of relatives in agriculture and less industrialized places may therefore slow reallocation, while the connections established by earlier movers may increasingly facilitate it. The empirical evidence documents these margins but cannot determine their aggregate balance or how it changes across generations. I turn

[26] I include destinations containing relatives, realized destinations and twelve randomly sampled non-chosen destinations without relatives, weighted by inverse inclusion probabilities.

to a dynamic spatial model to quantify how much new family branches offset the influence of existing roots on the pace and geography of structural transformation.

4 Quantitative model

The previous section highlights the role of family ties in reduced form, but its aggregate consequences depend on the joint response of wages, sector choices, migration and family connections. I therefore embed these mechanisms in a general-equilibrium model that evolves across space and generations.

Building on Eckert and Kleineberg (2024) and Eckert and Peters (2022),[27] the model combines non-homothetic demand à la Boppart (2014), trade and local congestion with an endogenous distribution of family connections across sectors and space. Family ties affect choices through two channels. Sector-specific connections lower the utility cost of entering a sector, capturing information, referrals and know-how documented in Section 3, while nearby relatives raise location utility through forms of home support such as childcare (Compton & Pollak, 2014) and place attachment as in Zabek (2024). These choices then reshape the sectoral and spatial distribution of relatives inherited by later cohorts. Lastly, parents also value their children's future opportunities, introducing an additional intergenerational link (Barro & Becker, 1989; Eckert & Kleineberg, 2024; Fogli & Guerrieri, 2019).

Environment. The economy has a finite set of regions $n \in N$, with $|N| = 455$ corresponding to commuting zones and two sectors $k \in \{A, M\}$. Time is discrete and one period is a 20-year interval, so that each model period corresponds approximately to one biological generation, as in Eckert and Peters (2022) and Eckert and Kleineberg (2024).[28] The population is organized in overlapping generations. Adults work, consume and choose where to live, while their children choose a sector before adulthood that remains fixed over the adult life cycle. This approximation separates sectoral from spatial adjustment within a generation. It is motivated by Eckert and Peters (2022), who show that much historical sectoral reallocation occurred within locations, and by Appendix A2.4.3, where migration induced by family deaths produces no detectable sectoral switching.

Generations are linked through a stylized family structure. Each adult belongs to a parental branch that gives rise to a mass ϕ of children in the next generation. These children form a sibling group. When they become adults, their sector and location choices determine where the

[27] These papers are part of a broader quantitative spatial literature studying the dynamics of development, trade and labor reallocation (Caliendo, Dvorkin, & Parro, 2019; Desmet, Nagy, & Rossi-Hansberg, 2018); see Redding and Rossi-Hansberg (2017) for a review.

[28] Model periods are spaced 20 years apart except between 1880 and 1910, where the destruction of the 1890 census creates a 30-year interval.

members of that sibling group live and work. From the perspective of their children, the other members of the group are aunts and uncles. I refer to these aunts and uncles as horizontal relatives. Each worker therefore inherits two types of family connections from the previous generation: one parental connection and a distribution of horizontal relatives across sectors and locations.

Population grows across generations at the common gross rate ϕ, which I impose equally across ancestral-origin groups.[29] Tracking the complete family tree behind this structure would nevertheless require carrying forward the histories of many relatives and potentially their simultaneous choices. I impose four restrictions that reduce this dimensionality while preserving the intergenerational family-tie mechanism.

Assumption 1. *Families are grouped by a permanent ancestral-origin type g, defined by the commuting zone of their linked 1850 ancestor.*

Rather than following each family separately, the model tracks the average distribution of relatives across sectors and locations among families with the same g. This preserves persistent differences in family geography across ancestral origins while abstracting from differences across individual families within the same origin group. Appendix A4.7 shows that the approximation preserves the aggregate transition reasonably well despite heterogeneity within g.

Assumption 2. *Workers' sector and location choices depend only on relatives from the previous generation: their parent and the parent's siblings.*

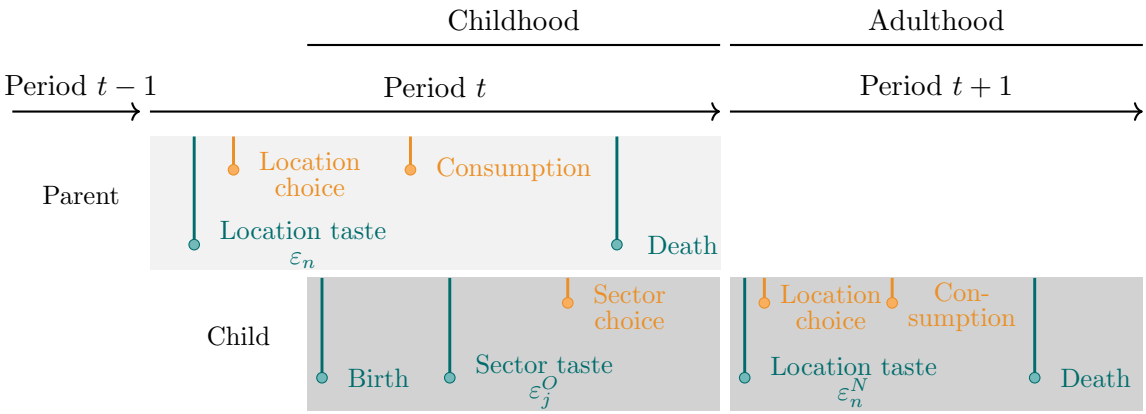
This makes inherited family connections predetermined when a worker chooses sector and location and limits the family history that must be carried forward to one generation. The cost is that the model does not capture direct effects from siblings, cousins or other relatives choosing in the same generation, but it substantially reduces history dependence and the state space of the model and reduces contemporaneous family complementarities that could generate additional multiplicity.

Assumption 3. *For each ancestral-origin type g, total mass of horizontal relatives r_g is fixed over time.*

This assumption isolates changes in the distribution of relatives from changes in their number. The reconstructed data do not cleanly identify changes in the number of relatives over time (Table 1), although U.S. fertility declined substantially over this period: the total fertility rate fell from about 4.2 births per woman in 1880 to about 3.0 in 1950. I therefore allow the sectoral and spatial distribution of relatives to evolve while holding the total mass of relatives r_g fixed. Population itself is not fixed and continues to grow at rate ϕ. Allowing the mass of relatives to vary could strengthen or weaken persistence depending on where those relatives live and work.

[29] The common growth rate ensures that differences in the mass of relatives across ancestral-origin groups do not mechanically generate differences in population growth or fertility.

Figure 6: Model timeline



Notes: A parent enters period t, chooses a location and consumes. Children are born in t, draw sector taste shocks and choose sectors before adulthood. As adults in $t+1$, they draw location shocks and choose where to live, becoming the parents of the next generation. Teal markers denote shocks and orange markers denote endogenous choices.

Appendix A5.3 shows that the main qualitative results are robust to alternative levels of the mass of horizontal relatives.

Assumption 4. *Conditional on their inherited family connections, workers make independent choices under Type-I extreme-value sector and location shocks.*

The shocks are independent across workers, alternatives and periods, with mean zero and dispersion parameters σ_O for sectors and σ_N for locations. Independence makes individual choice probabilities sufficient to determine how a sibling group is distributed across sectors and locations, rather than requiring its joint choices, while the extreme-value assumption delivers closed-form choice probabilities and expected values. The cost is that the model abstracts from direct coordination and correlated idiosyncratic choices among relatives.

Together, these restrictions allow the model to carry forward only the parental connection and the expected distribution of horizontal relatives for each ancestral-origin type. As one generation becomes the next, workers' sector and location choices reshape that distribution. Entry into manufacturing increases the mass of relatives in manufacturing inherited by the next generation, while migration increases the presence of relatives at the destination. The model can therefore capture how current choices change the family connections faced by later generations without tracking complete family trees. Figure 6 summarizes the timing.

Distribution of relatives. I summarize horizontal relatives by their overall mass and their distribution across sectors and locations. Consider an adult of ancestral type g who lives in location m and works in sector j. Let $\Phi_{g,t}(m, j; n, k)$ denote the average mass of horizontal

relatives in location n and sector k, conditional on the adult's state (m, j) and ancestral type g.

I write this mass as

$$\Phi_{g,t}(m, j; n, k) = r_g \lambda_{g,t}(n, k \mid m, j),$$

where $\lambda_{g,t}(n, k \mid m, j)$ is the share of horizontal relatives in location n and sector k, conditional on the adult's current state (m, j), and these shares sum to one across all locations and sectors. The scalar r_g is the total mass of horizontal relatives associated with one adult mass of ancestral type g, disciplined by the average number of reconstructed sibling links per adult in that ancestral-origin group.[30] The term $\lambda_{g,t}(n, k \mid m, j)$ is the share of that total mass located in (n, k), conditional on the adult being in (m, j).

Given Assumptions 3 and 4, the total mass of horizontal relatives remains fixed while its distribution can evolve. Conditional on their inherited family connections, siblings face the same systematic choice problem but draw independent idiosyncratic shocks. Their joint choice distribution therefore factorizes into individual choice probabilities, so the model updates the expected shares of relatives $\lambda_{g,t}$ using the sector and location choice probabilities derived below.

Consumption and structural transformation. Consumption follows PIGL preferences of Boppart (2014), as used by Eckert and Peters (2022). For nominal wages w and prices $p_{nt} = (p_{nAt}, p_{nMt})$, indirect utility and the agricultural expenditure share are

$$v(w, p_{nt}) = \frac{1}{\epsilon} \left(\frac{w}{p_{nAt}^{\varphi} p_{nMt}^{1-\varphi}} \right)^{\epsilon} - \nu \log \left(\frac{p_{nAt}}{p_{nMt}} \right), \tag{4}$$

$$\delta_A(w, p_{nt}) = \varphi + \nu \left(\frac{w}{p_{nAt}^{\varphi} p_{nMt}^{1-\varphi}} \right)^{-\epsilon}. \tag{5}$$

with

$$\delta_M(w, p_{nt}) = 1 - \delta_A(w, p_{nt}). \tag{6}$$

Because $\epsilon > 0$, the agricultural expenditure share falls with income, generating the non-homothetic Engel effects that drive structural change: as a region grows richer, demand tilts toward the manufacturing good, raising its relative wage and the return to entering it.

Children's sector choice. Before adulthood, a child of ancestral type g whose parent is in (n, j) chooses a sector, trading off a non-family entry cost, sector-specific family connections

[30] For example, a focal individual has around 1.4 aunts/uncles in 1880, so $r_g = 1.4$.

and the lifetime value associated with that sector. The systematic value of choosing sector k is

$$\mathcal{U}^k_{gnjt} = \theta_k F^k_{gnjt} - \tau^k_{gnjt} + V_{t+1}(n, k, g), \tag{7}$$

where τ^k_{gnjt} is a non-family entry cost, F^k_{gnjt} is the stock of sector-specific family connections (defined below), θ_k governs how strongly sector choice responds to these connections and $V_{t+1}(n, k, g)$ is the continuation value from the adult location problem defined below.

The term F^k_{gnjt} captures the sector-specific connections available to the child. Entering sector k may require information about jobs and employers as well as sector-specific skills or practical knowledge. Relatives already working in the sector can reduce these barriers through referrals, employer contacts and workplace experience (Calvó-Armengol & Jackson, 2004; Ioannides & Loury, 2004; Montgomery, 1991). I therefore model family ties as reducing non-wage barriers to sector entry, as in Takeda (2022). In the baseline, family connections do not directly affect wages conditional on sector and location, although they can raise earnings by facilitating access to better opportunities. This distinction is consistent with evidence that family connections facilitate job entry and access to higher-paying opportunities, while the wage effects of referrals vary across settings (Dustmann, Glitz, Schönberg, & Brücker, 2016; Glitz & Vejlin, 2021; Kramarz & Skans, 2014; Staiger, 2026).

Let $\Phi^k_{gnjt}(m) \equiv \Phi_{g,t}(n, j; m, k)$ denote the mass of horizontal relatives living in location m and working in sector k. I combine connections to parents and horizontal relatives into an effective stock of sector-k connections:

$$F^k_{gnjt} = \mathbf{1}\{j = k\} + \psi^k_R \Phi^k_{gnjt}(n), \tag{8}$$

where the parent's contribution is normalized to one and ψ^k_R measures the contribution of one unit of the local mass of horizontal relatives in sector k relative to the parent.

The child within its mass chooses the sector that maximizes $\mathcal{U}^k_{gnjt} + \sigma_O \varepsilon^O_k$, where ε^O_k is the sector taste shock from Assumption 4. Assumption 4 then implies the sector-choice probabilities and expected value

$$P^O_t(k \mid n, j, g) = \frac{\exp\left(\mathcal{U}^k_{gnjt}/\sigma_O\right)}{\sum_{\ell \in \{A,M\}} \exp\left(\mathcal{U}^\ell_{gnjt}/\sigma_O\right)}, \tag{9}$$

$$O_t(n, j, g) = \sigma_O \log \sum_{\ell \in \{A,M\}} \exp\left(\mathcal{U}^\ell_{gnjt}/\sigma_O\right). \tag{10}$$

The opportunity value $O_t(n, j, g)$ is the child's expected value before the sector taste shocks are realized and enters the parent's location choice below.

Parents' location choice. Adults choose where to live as a dynamic discrete choice, internalizing both their own utility and, through altruism, their children's future opportunities. This forward-looking migration problem follows the dynamic discrete-choice model of Kennan and Walker (2011) and is embedded here in a quantitative spatial environment related to Fogli and Guerrieri (2019) and Eckert and Kleineberg (2024).

Consider an adult of ancestral type g, whose origin is n and whose own fixed sector is j, who chooses destination m. The worker earns wage w^j_{mt}. Let $\mathcal{A}^j_{gmt}$ denote the pre-existing non-family amenity of destination m for this worker. The systematic value of destination m is

$$B^j_{gnt}(m) = \mathcal{A}^j_{gmt} - \rho \log L_{mt} + \log\left(v\left(w^j_{mt}, p_{mt}\right)\right) + \theta_H H_{gnjt}(m) - M^j_{gnmt} + \beta O_t(m, j, g), \quad (11)$$

where L_{mt} is total employment at the destination, M^j_{gnmt} is the migration cost, $H_{gnjt}(m)$ is the family amenity defined below and $\beta < 1$ is parental altruism. The final term, $O_t(m, j, g)$, is the value of the sectoral opportunities that destination m confers on the worker's children.

Proximity to relatives can facilitate childcare and female labor supply (Compton & Pollak, 2014), while related evidence shows that family peers affect mothers' labor-supply choices (Nicoletti, Salvanes, & Tominey, 2018). I capture these forms of family support as a location amenity.

Under Assumption 2, an adult's location choice depends on family connections inherited from the previous generation: the parent and the parent's siblings. The parent provides a connection to the location where the adult grew up, while the parent's siblings—the adult's aunts and uncles—provide connections to the places where they live. The relevant geographic distribution of relatives is therefore predetermined when the adult chooses where to live.

Consider an adult of ancestral type g who grew up in location n and works in sector j in period t. The adult's aunts and uncles belong to the parent's generation $t-1$, so their geographic distribution is described by $\Phi_{g,t-1}$. This distribution depends on the parent's sector q, which need not coincide with the adult's own sector j. Since the model no longer tracks q in the adult's state (n, j, g), I average the corresponding distributions of relatives using the probability of each parental sector conditional on the adult's childhood location and sector choice:

$$\Pr_{t-1}(q \mid n, j, g) = \frac{L^q_{gn,t-1} P^O_{t-1}(j \mid n, q, g)}{\sum_{\tilde{q} \in \{A,M\}} L^{\tilde{q}}_{gn,t-1} P^O_{t-1}(j \mid n, \tilde{q}, g)}.$$

The family amenity associated with destination m is then

$$H_{gnjt}(m) = \mathbf{1}\{m = n\} + \psi^H_R \sum_{q \in \{A,M\}} \Pr_{t-1}(q \mid n, j, g) \sum_{\ell \in \{A,M\}} \Phi^{\ell}_{gnq,t-1}(m). \quad (12)$$

The first term captures the parental connection to the childhood location n. For each possible parental sector q, the inner sum gives the mass of horizontal relatives at destination m, summing across the sectors in which these relatives work. The posterior weights average these distributions across parental sectors. The parameter ψ_R^H measures the contribution of horizontal relatives relative to the parent. This approximation simplifies the model by assigning the same expected distribution of relatives across locations to adults who share a childhood location, sector and ancestral type to reduce the history dependence and the state space.
The worker chooses the destination that maximizes $B_{gnt}^j(m) + \sigma_N \varepsilon_m^N$, where ε_m^N is the location taste shock from Assumption 4. Assumption 4 then implies the location-choice probabilities and expected value

$$P_t^N(m \mid n, j, g) = \frac{\exp\big(B_{gnt}^j(m)/\sigma_N\big)}{\sum_{r \in N} \exp\big(B_{gnt}^j(r)/\sigma_N\big)}, \tag{13}$$

$$V_t(n, j, g) = \sigma_N \log \sum_{m \in N} \exp\big(B_{gnt}^j(m)/\sigma_N\big)\,. \tag{14}$$

The parameter σ_N governs how sharply households sort on differences in location value. A higher σ_N makes location choices less sensitive to changes in wages, amenities and migration costs. Equations (13)–(14) use j for an adult's own sector. Thus, for a child who selected sector k in equation (9), the relevant location probability is $P_{t+1}^N(m \mid n, k, g)$: the parent's sector affects the child's sector choice, after which the child's own sector becomes the state for migration.

Evolution of family connections. Workers' sector and location choices determine where their relatives live and work in the next generation. To update these connections, I need to know where workers' parents lived and worked, since this determines which family connections they inherit. Workers in the same sector, location and dynasty g may come from different family backgrounds. Because the model tracks representative dynasties and not individual family trees, I first recover the share of workers coming from each parental origin and sector and then use these shares to determine the distribution of their relatives.
Consider parents of ancestral type g living in m and working in sector j in period t. Their children enter sector k with probability $P_t^O(k \mid m, j, g)$ and subsequently choose location n with probability $P_{t+1}^N(n \mid m, k, g)$. Using Bayes' rule, the share of workers in (n, k) whose parents lived in m and worked in j is

$$\Pr_t(m, j \mid n, k, g) = \frac{L_{gmt}^j P_t^O(k \mid m, j, g) P_{t+1}^N(n \mid m, k, g)}{\sum_{\tilde{m} \in N} \sum_{\tilde{j} \in \{A, M\}} L_{g\tilde{m}t}^{\tilde{j}} P_t^O(k \mid \tilde{m}, \tilde{j}, g) P_{t+1}^N(n \mid \tilde{m}, k, g)}. \tag{15}$$

The numerator gives the mass of workers coming from parents in (m, j), while the denominator sums over all possible parental origins and sectors.[31]

I then use these shares to determine where their horizontal relatives live and work. Conditional on where the parents lived and worked, horizontal relatives follow the same sector and location choice probabilities, reaching (n', k') with probability $P_t^O(k' \mid m, j, g)P_{t+1}^N(n' \mid m, k', g)$. Averaging across parental origins and sectors gives the expected distribution of relatives for workers in (n, k):

$$\lambda_{g,t+1}(n', k' \mid n, k) = \sum_{m \in N} P_{t+1}^N(n' \mid m, k', g) \sum_{j \in \{A,M\}} P_t^O(k' \mid m, j, g) \Pr_t(m, j \mid n, k, g). \tag{16}$$

Multiplying this distribution by the total mass of horizontal relatives r_g gives

$$\Phi_{g,t+1}(n, k; n', k') = r_g \lambda_{g,t+1}(n', k' \mid n, k). \tag{17}$$

Since the probabilities sum to one, the law of motion reallocates the fixed mass of horizontal relatives r_g across sectors and locations. This allows the model to track how family connections change across generations without following individual family trees.

Production and trade. Sectoral wages and prices are determined in general equilibrium through a standard Eaton–Kortum trade structure (Eaton & Kortum, 2002). Embedding this trade block in the dynamic spatial economy follows the broader approach of Caliendo et al. (2019). Each good is produced by competitive firms using labor,

$$q_{nt}^k = z_{nt}^k L_{nt}^k,$$

with Fréchet productivity draws of scale Z_{nt}^k and shape μ_k. Trade is subject to iceberg costs T_{nmt}^k, where n denotes the importing location and m the producing location. Because labor is the only input, unit production cost is the local wage. The expenditure share and sectoral price index are therefore

$$\pi_{nmt}^k = \frac{\left(w_{mt}^k T_{nmt}^k\right)^{-\mu_k} \left(Z_{mt}^k\right)^{\mu_k}}{\sum_{r \in N} \left(w_{rt}^k T_{nrt}^k\right)^{-\mu_k} \left(Z_{rt}^k\right)^{\mu_k}}, \tag{18}$$

$$p_{nt}^k = \Gamma^k \left[\sum_{m \in N} \left(w_{mt}^k T_{nmt}^k\right)^{-\mu_k} \left(Z_{mt}^k\right)^{\mu_k} \right]^{-1/\mu_k}. \tag{19}$$

[31] The common cohort-growth factor ϕ scales descendants from every parental origin and sector equally and therefore cancels from equation (15).

Aggregation and market clearing. The labor distribution evolves in parallel with the distribution of relatives. Let L^j_{gnt} denote adults of ancestral type g in sector j and location n at time t. Applying the sector and location choices to the entire adult population and scaling the next cohort by the common cohort-growth factor ϕ gives

$$L^k_{gm,t+1} = \phi \sum_{n \in N} P^N_{t+1}(m \mid n, k, g) \sum_{j \in \{A,M\}} P^O_t(k \mid n, j, g) L^j_{gnt}. \tag{20}$$

Thus, ϕ scales aggregate adult cohort growth, while the choice probabilities determine how each ancestral group is reallocated across sectors and locations.
Aggregate sectoral and local employment are

$$L^k_{nt} = \sum_g L^k_{gnt}, \qquad L_{nt} = \sum_{k \in \{A,M\}} L^k_{nt}.$$

Labor markets clear location by location and goods markets clear using the PIGL expenditure shares:

$$w^k_{nt} L^k_{nt} = \sum_{m \in N} \pi^k_{mnt} X^k_{mt}, \tag{21}$$

$$X^k_{nt} = \sum_g \sum_{j \in \{A,M\}} \delta_k\big(w^j_{nt}, p_{nt}\big)\, w^j_{nt} L^j_{gnt}. \tag{22}$$

The first condition equates the value of production in sector k and location n to expenditures on goods produced there across all importing locations. The second aggregates sector-k expenditure across households residing in location n.

Equilibrium. The endogenous state of the economy at date t is given by the distribution of workers across ancestral types, sectors and locations, L_t, together with the distribution of relatives, Φ_t. Let $S_t \equiv (L_t, \Phi_t)$ collect these endogenous state variables and let Ω_t denote the exogenous fundamentals of the economy, including productivity, trade costs, non-family amenities, sector-entry costs and migration costs. Following Caliendo et al. (2019) and Eckert and Kleineberg (2024), I define the equilibrium in two steps.

Temporary equilibrium. Conditional on the state S_t and current fundamentals Ω_t, a temporary equilibrium is a vector of wages, prices, trade shares and expenditures $\{w_t, p_t, \pi_t, X_t\}$ that satisfies the production and trade conditions in equations (18)–(19) and the market-clearing conditions in equations (21)–(22). Thus, conditional on the current labor distribution, the temporary equilibrium determines the within-period allocation and factor prices. The distribution of relatives Φ_t does not enter the static market-clearing conditions directly, but it is part of the dynamic choices because it affects sector and location choices.

Sequential competitive equilibrium. Given an initial state $S_0 = (L_0, \Phi_0)$, a path of fundamentals $\{\Omega_t\}_{t=0}^{\infty}$ and the structural parameters, a sequential competitive equilibrium is a sequence

$$\left\{V_t, O_t, P_t^N, P_t^O, L_t, \Phi_t, w_t, p_t, \pi_t, X_t\right\}_{t=0}^{\infty}$$

such that, at every date t: (i) sector choices satisfy equations (7)–(10); (ii) location choices satisfy equations (11)–(14); (iii) wages, prices, trade shares and expenditures constitute a temporary equilibrium given S_t and Ω_t; (iv) the labor distribution evolves according to equation (20); and (v) the distribution of relatives evolves according to equations (16)–(17).

Implications. The model allows family ties to both hinder and facilitate structural transformation. Initially, relatives concentrated in agriculture can favor continued agricultural employment. As workers enter manufacturing, however, they create connections that facilitate manufacturing entry for later generations. To illustrate this mechanism, consider a two-sector version of the model for one ancestral group g, in which all families remain in the same location n. Let Φ_t^M and Φ_t^A denote the expected masses of horizontal relatives working in manufacturing and agriculture that generation t inherits from the previous generation, with $\Phi_t^M + \Phi_t^A = r_g$. At the beginning of the transition, the initial distribution (Φ_0^M, Φ_0^A) is given, with most relatives initially working in agriculture. In subsequent generations, the distribution changes as workers enter different sectors. As in the quantitative model, these connections enter sector-choice utility through $\theta_k \psi_R^k$. To isolate the role of horizontal relatives, I assume that children face the same sector-choice incentives regardless of their parents' sector, abstracting from differences in parental support and family composition. Manufacturing entry and the changing distribution of relatives then satisfy

$$\begin{aligned} P_t^{O,M,F} &= \Lambda\left(a_t^O + \frac{\theta_M \psi_R^M \Phi_t^M - \theta_A \psi_R^A \Phi_t^A}{\sigma_O}\right), \\ \Phi_{t+1}^M = r_g P_t^{O,M,F}, &\qquad \Phi_{t+1}^A = r_g(1 - P_t^{O,M,F}), \end{aligned} \tag{23}$$

where $\Lambda(z) = (1 + e^{-z})^{-1}$, $P_t^{O,M,F}$ is the probability of entering manufacturing and a_t^O collects sector-entry costs, expected adult values and the direct parental-sector contribution. The law of motion follows because the sector choices of generation t determine the distribution of horizontal relatives inherited by generation $t+1$. Thus, while Φ_0 is predetermined, the subsequent distributions Φ_t are determined endogenously by earlier manufacturing choices. Appendix A3.1 derives these equations and states the simplifying assumptions.

Roots and branches. Compare three paths under the same opportunities apart from horizontal-relative support: $P_t^{O,M,0}$ without effects from horizontal relatives, $P_t^{O,M,\text{fix}}$ with the initial distribution of horizontal relatives (Φ_0^M, Φ_0^A) held fixed in every generation, and $P_t^{O,M,F}$ with family connections changing endogenously according to equation (23). The total contribution

of horizontal relatives to manufacturing decomposes as

$$\underbrace{P_t^{O,M,F} - P_t^{O,M,0}}_{\text{total effect of relatives}} = \underbrace{P_t^{O,M,\text{fix}} - P_t^{O,M,0}}_{\text{roots}} + \underbrace{P_t^{O,M,F} - P_t^{O,M,\text{fix}}}_{\text{branches}}. \tag{24}$$

Taking manufacturing log odds, the two contributions can be expressed directly in terms of the model parameters:

$$\text{logit}(P_t^{O,M,F}) - \text{logit}(P_t^{O,M,0}) = \underbrace{\frac{\theta_M \psi_R^M \Phi_0^M - \theta_A \psi_R^A \Phi_0^A}{\sigma_O}}_{\text{roots}} + \underbrace{\frac{\theta_M \psi_R^M + \theta_A \psi_R^A}{\sigma_O} \left(\Phi_t^M - \Phi_0^M\right)}_{\text{branches}}. \tag{25}$$

Roots capture the initial balance between agricultural and manufacturing connections, weighted by their importance for sector choice. Branches measure how changes in that distribution affect manufacturing entry: new manufacturing connections replace agricultural ones.

The relative contribution of horizontal relatives becomes positive when the manufacturing share of inherited horizontal relatives exceeds

$$\frac{\Phi_t^M}{r_g} > \frac{\theta_A \psi_R^A}{\theta_M \psi_R^M + \theta_A \psi_R^A} \quad \Longleftrightarrow \quad P_t^{O,M,F} > P_t^{O,M,0}. \tag{26}$$

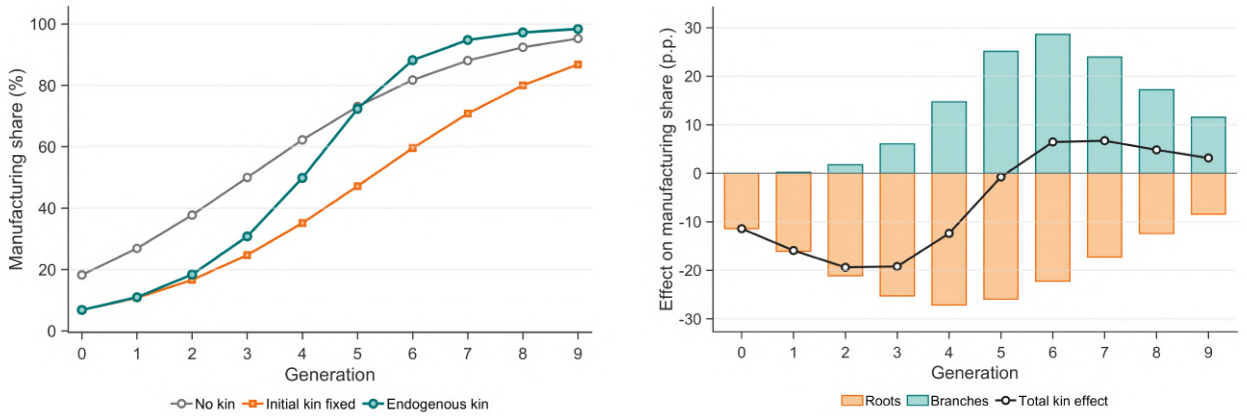
The threshold is one half when family connections provide equally strong support in both sectors. It is lower when manufacturing connections are more valuable relative to agricultural connections, so fewer manufacturing relatives are needed to overcome the initial drag. With the illustrative parameters in Figure 7, the threshold is about 51%, and branches gradually offset the negative roots effect before eventually more than compensating for it.

This example isolates horizontal sector connections. In the quantitative model, parental-sector differences, migration and the distribution of relatives across locations also affect choices. The aggregate contribution of family ties additionally incorporates changes in equilibrium wages and prices.

Family ties create spillovers across relatives. These effects are not fully internalized by the worker making the initial choice. Through continuation values and parental altruism, workers take into account the consequences of their choices for their own descendants. They do not take into account how the same choices change the connections inherited by their siblings' children. A worker who enters manufacturing becomes a manufacturing connection for nieces and nephews, while a worker who migrates becomes a destination connection for them.

This creates an externality across family members. Its strength is governed by the same parameters above: $\theta_k \psi_R^k / \sigma_O$ determines the effect of a sector-specific connection to horizontal relatives, $\theta_H \psi_R^H / \sigma_N$ the effect of a location connection to horizontal relatives and r_g the total mass of horizontal relatives through which these effects operate. Because the total mass of horizontal

Figure 7: Roots and branches along the manufacturing transition



(a) Manufacturing transition

(b) Roots and branches

Notes: Panel (a) compares manufacturing shares without effects from horizontal relatives, with the initial distribution of horizontal relatives held fixed and with their distribution changing endogenously. Panel (b) decomposes the difference between the endogenous and no-horizontal-relative-support paths into roots ($P_t^{O,M,\text{fix}} - P_t^{O,M,0}$) and branches ($P_t^{O,M,F} - P_t^{O,M,\text{fix}}$), in percentage points.

relatives is fixed, moving one horizontal relative from the origin to the destination both removes a connection at the origin and creates one at the destination, changing destination-versus-origin log odds by $2\theta_H \psi_R^H / \sigma_N$.

Extensions. The baseline isolates the effects of family ties through sector entry and location amenities. Two additional patterns motivate extensions: workers with same-sector family members experience faster wage growth and schooling partly substitutes for family connections in sector entry. The appendix captures these patterns by allowing family connections to affect individual wages (Appendix A3.2.1) and schooling to serve as a formal substitute for family-based sector entry. These additional channels are quantitatively secondary to the main results and are therefore excluded from the baseline model.

5 Model quantification

I quantify the model for the U.S. economy at four dates between 1880 and 1950: 1880, 1910, 1930 and 1950, with 1970 as the terminal period. I proceed sequentially. First, I take demand, trade, congestion and altruism parameters from external sources and initialize the labor distribution and family connections from the data. Second, deaths of parents and aunts or uncles identify the strength of family connections relative to the dispersion in idiosyncratic sector and location preferences. Third, a migration-gravity equation identifies migration costs, while the static trade block recovers productivity. Fourth, I estimate the dispersion parameters governing sector

and location choice by indirect inference using quasi-experimental variation from the Dust Bowl, an agricultural productivity shock. Finally, I invert the model to recover the remaining non-family amenities and sector-entry costs. Table 2 summarizes the resulting parameterization.

5.1 External parameters and initialization

External parameters. I take the externally disciplined parameters from the literature in three steps. First, the parameters governing non-homothetic demand are taken from Eckert and Peters (2022), while parental altruism is disciplined using Eckert and Kleineberg (2024). Second, trade costs are constructed from the historical county-to-county transportation-cost data of Hornbeck and Rotemberg (2024) and the trade elasticities for agriculture and manufacturing are taken from Yang (2024). Appendix A4 describes the aggregation of transportation costs to commuting zones. Third, population congestion is set to the value used by Eckert and Peters (2022), following Allen and Donaldson (2020).

Initial distribution of relatives. I initialize the labor distribution from the Census and the distribution of horizontal relatives from the linked genealogical data. The latter requires two objects: the total mass of horizontal relatives associated with each ancestral type and the distribution of those relatives across sectors and locations.

For an adult i of ancestral-origin type g, let $R_{i0}(n,k)$ denote the number of reconstructed and de-duplicated siblings of i who live in region n and work in sector k in 1860.[32] These siblings become aunts and uncles from the perspective of the adult's children. I measure the average mass of horizontal relatives for ancestral type g as

$$r_g = \mathbb{E}\left[\sum_{n\in N}\sum_{k\in\{A,M\}} R_{i0}(n,k) \,\middle|\, g(i)=g\right]. \tag{27}$$

This equation averages the total number of reconstructed horizontal relatives across all individuals with ancestral origin g, regardless of where those relatives live or in which sector they work.

I then use the data to determine where these relatives live and in which sectors they work. For adults of type g observed in state (m,j), define

$$\lambda_{g,0}(n,k \mid m,j) = \frac{\mathbb{E}[R_{i0}(n,k) \mid g(i)=g,\ (m_i,j_i)=(m,j)]}{\sum_{n'\in N}\sum_{k'\in\{A,M\}} \mathbb{E}[R_{i0}(n',k') \mid g(i)=g,\ (m_i,j_i)=(m,j)]}. \tag{28}$$

This equation gives the share of horizontal relatives located in each region-sector cell (n,k)

[32] The same is done for the uncles/aunts for parents in 1880.

among adults with ancestral origin g who themselves live in m and work in j. Thus, r_g determines the total mass of horizontal relatives, while $\lambda_{g,0}(n,k \mid m,j)$ determines how that mass is distributed across sectors and locations. By construction, these shares sum to one.
Consistent with Assumption 3 in Section 4, the model assigns the same total mass of horizontal relatives r_g to all current states within ancestral type g. The initial model-implied stock of relatives is therefore

$$\Phi_{g,0}(m,j;n,k) = r_g \lambda_{g,0}(n,k \mid m,j).$$

I hold r_g fixed thereafter and use equation (16) to evolve the sectoral and geographic distribution of horizontal relatives endogenously.

Cohort growth. I use the initial average horizontal degree to discipline common cohort growth, separately from changes in the distribution of relatives. Let $\bar{r}$ denote the initial-population-weighted average of r_g across ancestral types. I set the common gross cohort-growth factor to

$$\phi = \frac{1+\bar{r}}{2}. \tag{29}$$

The term $1+\bar{r}$ is the implied average surviving sibling-group size, including the individual themself. Dividing by two converts the sibling-group size generated by a parental pair into gross cohort growth per adult. In the data, $\bar{r} = 1.4$, implying $\phi = 1.2$. This fits the overall population growth rate well during this time period. The same ϕ applies to all ancestral types, so differences in r_g do not themselves generate differential population growth. Finally, I initialize $P^O_{1860}(\cdot)$ using observed sector choices between 1860 and 1880.

5.2 Estimating the family-connection parameters

The family-connection block contains two sets of parameters: the value of geographically proximate relatives in location choice and the value of local sector-specific family connections in sector choice. The parameters are

$$\Theta^F = \left\{\psi^H_R, \Theta_H, \psi^A_R, \psi^M_R, \Lambda_A, \Lambda_M\right\}, \qquad \Theta_H \equiv \frac{\theta_H}{\sigma_N}, \qquad \Lambda_k \equiv \frac{\theta_k}{\sigma_O}.$$

The parent is the reference connection and is normalized to contribute one unit. The parameters ψ^H_R and ψ^k_R measure the contribution of an aunt-or-uncle connection relative to a parent. The parameters Θ_H and Λ_k measure the value of a parental connection relative to the dispersion in idiosyncratic location and sector preferences. Once σ_N and σ_O are estimated, I recover the corresponding utility parameters θ_H and θ_k.
To identify these parameters, I exploit deaths of relatives that remove particular connections. I use three linked twenty-year panels: 1900–1920, 1910–1930 and 1920–1940. For a panel

Table 2: Parameterization of the quantitative model

Symbol	Description	Value	Discipline
A. External and calibrated parameters			
μ_A, μ_M	Trade elasticities	$\{12, 6.5\}$	Yang (2024)
ϵ	Engel elasticity	0.8	Eckert and Peters (2022)
ν	PIGL parameter	1	Eckert and Peters (2022)
φ	Asymptotic agricultural share	0.01	Eckert and Peters (2022)
ρ	Congestion	0.15	Allen and Donaldson (2020)
β	Parental altruism	0.231	Eckert and Kleineberg (2024)
ϕ	Cohort growth	1.2	Eq. (29)
B. Idiosyncratic-choice dispersion			
σ_N	Location taste dispersion	0.51	Indirect inference
σ_O	Sector taste dispersion	0.60	Indirect inference
C. Family-connection parameters			
ψ_R^H	Aunt/uncle location weight	0.874	Death ratio, Eq. (33)
$\Theta_H \equiv \theta_H/\sigma_N$	Parent location value / dispersion	0.024	Parent-death effect, Eq. (33)
ψ_R^A, ψ_R^M	Aunt/uncle sector weights	$\{0.311, 0.138\}$	Death ratios, Eq. (35)
Λ_A, Λ_M	Parent sector values / dispersion	$\{0.0208, 0.0487\}$	Parent-death effects, Eq. (35)
D. Recovered fundamentals			
$\mathcal{A}_{gmt}^j$	Location-sector amenities	—	Model inversion
Z_{nt}^k	Productivity	—	Trade inversion
τ_{gnjt}^k	Sector-entry costs	—	Model inversion
M_{gnmt}^j	Migration costs	—	Gravity and σ_N

Notes: Panel A reports calibrated and externally disciplined parameters. Panel B reports the dispersion of idiosyncratic tastes in the location and sector choices; larger values imply weaker responses to systematic payoff differences. Panel C reports family-connection parameters estimated from the death variation described in Section 5.2. Panel D lists fundamentals recovered by model inversion or auxiliary gravity equations. The parental connection is normalized to one in both the location and sector margins. Horizontal relationship weights refer to the parent's siblings.

beginning in t, eligible deaths occur between t and $t + 10$ and choices are measured in $t + 20$. Let $a = (g, m, j)$ index an ancestral-type-location-sector cell. Within each cell, I construct the pre-death population-weighted mean number of relevant connections lost and the corresponding choice log odds.

Step 1: geographic family attachment. I identify the value of geographically proximate relatives using deaths of co-located *non-employed* parents and aunts or uncles. Restricting the deceased to non-employed relatives isolates the loss of local family support from any contemporaneous change in sector-specific family employment connections. In the model, a parent contributes one unit to the family amenity at the origin. An aunt or uncle contributes ψ_R^H, weighted by the probability that the horizontal connection is inherited through parental sector q. Let $\pi_{gnjt}(q) \equiv \Pr_{t-1}(q \mid n, j, g)$ denote this probability.

To see how these deaths identify the location parameters, define $\Delta H_{gnjt}(m) \equiv H_{gnjt}(m) - H_{gnjt}(n)$ and $\Delta O_{gnjt}(m) \equiv O_t(m,j,g) - O_t(n,j,g)$. The log odds of leaving the baseline region n rather than remaining there are then

$$\ell^N_{gnjt} \equiv \log \frac{\sum_{m \neq n} P^N_{t+1}(m \mid n,j,g)}{P^N_{t+1}(n \mid n,j,g)} = \log \sum_{m \neq n} \exp\left\{ \kappa^j_{gnt}(m) + \Theta_H \Delta H_{gnjt}(m) + \frac{\beta}{\sigma_N} \Delta O_{gnjt}(m) \right\}. \tag{30}$$

where $\Theta_H = \theta_H/\sigma_N$ and $\kappa^j_{gnt}(m)$ collects current non-family differences in the payoff from living in destination m rather than remaining in n, including wages, amenities, congestion and migration costs. The final term captures differences in children's continuation opportunities across locations.

A co-located parent death lowers $H_{gnjt}(n)$ by one while leaving family support at alternative destinations unchanged. An aunt-or-uncle death inherited through parental sector q lowers origin support by $\psi^H_R \pi_{gnjt}(q)$. Importantly, neither death mechanically changes O_t: the deceased parent or aunt or uncle of the current adult is a grandparent or more distant relative from the perspective of the next generation and therefore does not enter its family connections by Assumption 2. Restricting the deceased to non-employed relatives further ensures that the death does not simultaneously remove a sector-specific employment connection.

Let P denote a parent death and R an aunt-or-uncle death. The model therefore implies

$$\ell^{N,\text{post}}_P - \ell^{N,\text{pre}}_P = \Theta_H + \xi^N_P, \qquad \ell^{N,\text{post}}_R - \ell^{N,\text{pre}}_R = \Theta_H \psi^H_R \pi_{gnjt}(q) + \xi^N_R, \tag{31}$$

where ξ^N_P and ξ^N_R collect changes in non-family location payoffs and continuation values not mechanically generated by the loss of family support. Conditional on controls and fixed effects, the identifying assumption is that these residual changes are unrelated to the timing of the death. The parent coefficient therefore identifies Θ_H, while the aunt-or-uncle coefficient relative to the parent coefficient identifies ψ^H_R.

Guided by this mapping, I estimate

$$\bar{\ell}^N_{a,t+20} = \alpha_a + \delta_{njqt} + \beta^H_P \overline{\text{Dth}}^{P,H}_{at} + \beta^H_R \overline{\text{Dth}}^{R,H,\pi}_{at} + X'_{at}\Gamma_N + \varepsilon^H_{at}, \tag{32}$$

where $\overline{\text{Dth}}^{P,H}_{at}$ is the mean number of co-located non-employed parental connections lost and $\overline{\text{Dth}}^{R,H,\pi}_{at} \equiv \overline{\pi_{gnjt}(q)\,\text{Dth}^{R,H}_{at}}$ is the corresponding aunt-or-uncle mass weighted by the parental-sector probability implied by the model. Parent deaths require no weighting because the parental connection enters the location amenity with unit weight. The fixed effects δ_{njqt} compare individuals with the same origin, own sector, parental sector and period, while X_{at} contains predetermined distribution of relatives and local characteristics.

The remaining identification requirement is an exclusion restriction on the timing of deaths.

Conditional on the fixed effects and controls, residual variation in parent and aunt-or-uncle deaths must be unrelated to changes in non-family determinants of subsequent migration:

$$\mathbb{E}\left[\widetilde{\mathrm{Dth}}_{at}^{P,H}\,\varepsilon_{at}^{H}\right]=0, \qquad \mathbb{E}\left[\widetilde{\mathrm{Dth}}_{at}^{R,H,\pi}\,\varepsilon_{at}^{H}\right]=0.$$

The assumption requires that, after conditioning on the initial state and predetermined characteristics, the timing of a death is unrelated to shocks that would independently change the individual's incentive to migrate. Origin-by-own-sector-by-parental-sector-by-year fixed effects absorb shocks common to workers in the same initial local state, while ancestry fixed effects and predetermined controls absorb persistent differences in the distribution of relatives and mortality exposure. Restricting the deceased to non-employed relatives additionally rules out the direct sector-employment channel described above. Under this exclusion restriction,

$$\widehat{\Theta}_H=\widehat{\beta}_P^H, \qquad \widehat{\psi}_R^H=\frac{\widehat{\beta}_R^H}{\widehat{\beta}_P^H}. \tag{33}$$

Table 3: Effects of family ties in location and sector choice

	Location choice	Sector choice	
	Home support	Agriculture	Manufacturing
	(1)	(2)	(3)
Panel A. Death coefficients			
Parent	0.024***	0.0206***	0.0489***
	(0.002)	(0.003)	(0.012)
Aunt/uncle	0.021***	0.006***	0.007***
	(0.002)	(0.001)	(0.003)
Panel B. Implied aunt/uncle weights relative to parents			
Aunt/uncle relative to parent, $\widehat{\psi}_R$	0.874	0.305	0.141
Observations	962,098	611,273	611,273

Notes: Column (1) reports the parent and aunt-or-uncle coefficients from equation (32). Columns (2)–(3) report the corresponding sector-specific estimates from equation (34). Panel B reports the implied weight on one aunt-or-uncle connection relative to one parental connection. Ratios are calculated using the unrounded coefficient estimates. Standard errors are reported in parentheses. $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$.

The estimates imply $\widehat{\Theta}_H=0.024$ and $\widehat{\psi}_R^H=0.874$. Thus, horizontal relatives receive substantial weight in location choice relative to the normalized parental connection.

Step 2: local family connections and sector choice. I next use deaths of co-located sector-employed parents and aunts or uncles to identify the sector-specific family parameters.

This restriction follows equation (8): sector access depends on relatives who live in the same region and work in the relevant sector. In sector k, an employed parent contributes one unit to F^k, while each employed aunt or uncle contributes ψ_R^k.

Let $\ell \neq k$ denote the alternative sector. The log odds of choosing sector k rather than ℓ are

$$\ell_{gnjt}^k = \Lambda_k F_{gnjt}^k - \Lambda_\ell F_{gnjt}^\ell - \frac{\tau_{gnjt}^k - \tau_{gnjt}^\ell}{\sigma_O} + \frac{V_{t+1}(n,k,g) - V_{t+1}(n,\ell,g)}{\sigma_O},$$

where $\Lambda_h = \theta_h/\sigma_O$. The first two terms capture sector-specific family support, the third other differences in sector-entry costs and the final term the value of the subsequent location choice.

Consider the death of a co-located relative employed in sector k. A parent death reduces F^k by one, while an aunt-or-uncle death reduces it by ψ_R^k. The direct effect therefore lowers the log odds of choosing k by Λ_k and $\Lambda_k\psi_R^k$, respectively. There is one additional mechanical effect. Because sector choice precedes the individual's subsequent location choice, a co-located sector-employed relative also provides geographic support at the origin. Its death can therefore change the continuation value of choosing sector k relative to ℓ.

Let Ω_r^k denote this continuation-value response for a death of relationship type $r \in \{P, R\}$. The model-implied change in sector-k log odds is

$$\ell^{k,\text{post}(r)} - \ell^{k,\text{pre}} = -\Lambda_k d_{F,r}^k + \Omega_r^k, \qquad d_{F,P}^k = 1, \quad d_{F,R}^k = \psi_R^k.$$

The first term captures the loss of sector-specific family support and the second the induced change in the value of the subsequent location choice. Without the latter, the parent response identifies Λ_k, while the aunt-or-uncle response relative to the parent identifies ψ_R^k. Appendix A4.4 derives Ω_r^k from the location problem. The correction depends on the difference in the probability of remaining at the origin after choosing sectors k and ℓ. If location behavior is identical across sectors, it cancels exactly.

I estimate

$$\overline{\ell}_{a,t+20}^k = \alpha_a^k + \delta_{njt}^k - \beta_P^k \overline{\text{Dth}}_{at}^{P,k} - \beta_R^k \overline{\text{Dth}}_{at}^{R,k} + X_{at}'\Gamma_k + \varepsilon_{at}^k, \tag{34}$$

where the death variables measure the mean numbers of co-located sector-k parental and aunt-or-uncle connections lost during the first decade of the panel. The fixed effects absorb common shocks within the initial local and sectoral state, while X_{at} contains the predetermined distribution of relatives, connections to the other sector, local characteristics and lagged sector-specific migration rates.

Identification requires that, conditional on these fixed effects and controls, death timing be unrelated to shocks that independently change the relative attractiveness of sectors k and ℓ. The age restrictions focus on premature deaths, while the pre-event evidence in the empirical section and the comparison with deaths of non-employed relatives address potential alternative

explanations based on pre-existing trends or the consequences of losing a relative unrelated to sector-specific employment connections.
Under this exclusion restriction, the estimated coefficients satisfy

$$\beta_P^k = \Lambda_k - \overline{\Omega}_P^k, \qquad \beta_R^k = \Lambda_k \psi_R^k - \overline{\Omega}_R^k,$$

where the bars denote continuation-value corrections averaged over the identifying variation. Appendix A4.4 shows that these corrections are quantitatively small, at most 2.8%. Using the corrected values, I compute:

$$\widehat{\Lambda}_k = \widehat{\beta}_P^k - \bar{\Omega}_P^k, \qquad \widehat{\psi}_R^k = \frac{\widehat{\beta}_R^k + \overline{\Omega}_R^k}{\widehat{\Lambda}_k}, \qquad k \in \{A, M\}. \tag{35}$$

The resulting parameters are

$$\widehat{\Lambda}_A = 0.0208, \qquad \widehat{\Lambda}_M = 0.0487, \qquad \widehat{\psi}_R^A = 0.311, \qquad \widehat{\psi}_R^M = 0.138.$$

Thus, one aunt-or-uncle connection receives about 31 percent of the weight of a parent in agriculture and 14 percent in manufacturing.

5.3 Migration costs and location values

I construct migration distance using the least-cost path between CZ centroids on the historical railroad network from Atack (2016), allowing travel off the rail network to be more costly. The construction adapts the least-cost network approach of Donaldson and Hornbeck (2016) to obtain a reduced-form proxy for passenger migration costs.[33]
Let $\widetilde{d}_{nmt}$ denote the log of rail-adjusted least-cost distance between origin n and destination m, with the within-CZ value normalized to zero. Let F_{gnmt}^j denote the migration flow of ancestral type g and sector j from n to m. The location-choice equation implies

$$\mathbb{E}\left[F_{gnmt}^j\right] = \exp\left(\text{orig}_{gnt}^j + \text{dest}_{gmt}^j + \widehat{\Theta}_H H_{gnjt}(m) - \lambda_t^j \widetilde{d}_{nmt}\right), \tag{36}$$

where, within each year-sector regression, the origin fixed effect absorbs origin mass and the location-choice denominator, while the destination fixed effect absorbs components of destination utility common across origins. Because $H_{gnjt}(m)$ varies jointly with origin and destination, it is not absorbed by these fixed effects. I therefore estimate equation (36) by PPML (Silva & Tenreyro, 2006), treating the previously estimated $\widehat{\Theta}_H H_{gnjt}(m)$ as known. The coefficient λ_t^j then identifies the distance-related component of migration costs.

[33] Appendix A4.3 provides the details and sensitivity to the off-rail cost multiplier.

The distance coefficient identifies migration costs relative to the location taste dispersion:

$$\frac{M^{j}_{gnmt}}{\sigma_N} = \lambda^{j}_{t}\widetilde{d}_{nmt}. \tag{37}$$

After estimating σ_N, migration costs in utility units are $M^{j}_{gnmt} = \sigma_N \lambda^{j}_{t}\widetilde{d}_{nmt}$.
The estimated destination values, together with the family-proximity and distance terms, imply the location inclusive value

$$\frac{V_t(n,j,g)}{\sigma_N} = \log \sum_{m \in N} \exp\left(\text{dest}^{j}_{gmt} + \widehat{\Theta}_H H_{gnjt}(m) - \widehat{\lambda}^{j}_{t}\widetilde{d}_{nmt}\right). \tag{38}$$

Railroad placement is not random. Lines may have been built along economically important corridors that also attracted migration and shaped where families settled. Origin and destination fixed effects absorb differences associated with the two endpoints, but not unobserved bilateral attractiveness correlated with rail connectivity. I therefore interpret $\widehat{\lambda}^{j}_{t}\widetilde{d}_{nmt}$ as a reduced-form measure of bilateral migration costs rather than the causal effect of railroad access. This concern does not identify the family-proximity coefficient Θ_H, which is estimated separately from the death variation above, but it can affect the interpretation of the recovered distance component.

5.4 Indirect inference

I next estimate the dispersion of idiosyncratic location and sector tastes, (σ_N, σ_O). These parameters determine how strongly choices respond to differences in systematic utility. A larger value implies more dispersed idiosyncratic tastes and therefore a weaker response of choices to a given change in wages, costs or amenities. They cannot, however, be recovered directly from observed migration or sectoral responses to wages. In particular, with non-homothetic preferences, a given change in wages or productivity does not translate into a constant change in utility, so there is no simple elasticity mapping from observed migration responses to σ_N (Eckert & Peters, 2022). I therefore use the Dust Bowl as an external productivity shock and choose (σ_N, σ_O) so that the full model reproduces its empirical migration and sectoral responses. The estimation follows the indirect-inference approach of Gouriéroux, Monfort, and Renault (1993), Bilal and Rossi-Hansberg (2023) and Rodríguez-Clare, Ulate, and Vásquez (2026).
The Dust Bowl provides a large and spatially heterogeneous agricultural productivity disturbance (Hornbeck, 2012, 2023; Liu, 2026). I construct two sets of twenty-year response moments. For sector choice, I follow children younger than 15 at baseline and measure manufacturing employment twenty years later, separately by parental sector. For migration, I follow adults older than 25 and measure whether they leave their baseline CZ, separately by initial sector.

In both cases, I estimate the differential responses of medium- and high-erosion CZs relative to low-erosion CZs using CZ and state-by-year fixed effects and predetermined controls. Appendix A4.5 reports the specifications and samples.

I separately recover the size of the underlying agricultural productivity disturbance from the static goods-market inversion, which does not depend on (σ_N, σ_O). By 1950, recovered agricultural productivity is 0.179 log points lower in medium-erosion CZs and 0.226 log points lower in high-erosion CZs, with no comparable pre-Dust-Bowl differences. As an external check, farm products per acre fall by 0.176 and 0.241 log points in the same two groups. The similarity of these magnitudes is not imposed by the inversion and supports the size of the productivity disturbance used for identification.

For each candidate $\sigma = (\sigma_N, \sigma_O)$, I first combine the family-connection parameters identified relative to taste dispersion with each candidate (σ_N, σ_O) to recover their utility-unit counterparts and re-invert the model so that the candidate parameterization rationalizes the observed pre-counterfactual choices. I then solve two equilibrium transitions. The first follows the observed economy, including the estimated Dust Bowl productivity disturbance. The second removes only that component of agricultural productivity while holding the remaining recovered fundamentals fixed. Sector choices, migration, wages, prices and family connections are allowed to adjust endogenously in both economies.

Applying the same auxiliary regressions used in the data to the simulated outcomes gives a vector of model-implied responses $b^{\text{model}}(\sigma)$. I choose

$$\widehat{\sigma} = \arg\min_{\sigma \in \Sigma} \left[b^{\text{model}}(\sigma) - \widehat{b}^{\text{data}} \right]' W \left[b^{\text{model}}(\sigma) - \widehat{b}^{\text{data}} \right], \qquad W = I. \tag{39}$$

Thus, the dispersions are identified by the response of migration and sector choice to an independently measured productivity shock, while the family-connection parameters are disciplined separately by the death variation above. Appendix A4.5 provides the full construction, including the candidate-specific inversion, the simulated auxiliary regressions and the maintained identifying assumptions.

Estimation results and model fit. The estimation yields $(\widehat{\sigma}_N, \widehat{\sigma}_O) = (0.51, 0.60)$. These estimates are of a similar order of magnitude to dispersion parameters used in related quantitative spatial and sectoral-choice models (Eckert & Kleineberg, 2024; Eckert & Peters, 2022; Rodríguez-Clare et al., 2026). At the selected values, I perform the final inversion and use the resulting fundamentals and family-connection parameters in the quantitative exercises that follow.

Appendix A4.6.3 reports several untargeted checks on the resulting economy. Recovered agricultural and manufacturing productivity covary with independent measures of sectoral value per worker that are not used in the inversion. The model also reproduces the spatial pattern of

employment growth from 1880 to 1950, including the shift from agriculture toward manufacturing in initially agricultural places. Finally, the family-connection block matches the untargeted cross-CZ distribution of aunts and uncles.

6 Quantitative importance of family ties

The quantitative analysis uses the model to assess the economic importance of the family connections identified in the empirical analysis. I first quantify how much family ties contribute to the barriers workers face when changing sectors and locations. I then decompose their aggregate effect on structural transformation into the role of the initial distribution of relatives and the changes in connections created by earlier workers, before examining each force in turn and its implications for spatial policies.

6.1 Family ties and aggregate structural change

One explanation for persistent wage gaps across sectors is that workers face barriers to moving across occupations and places (Donovan & Schoellman, 2023). Since these barriers are not directly observed, quantitative models recover them as residual sectoral and spatial wedges that rationalize observed choices (Eckert & Kleineberg, 2024; Takeda, 2022). Family ties have long been viewed as one source of such barriers because they can keep workers attached to their current sector and place. The linked family data and model structure allow me to quantify this channel directly. I first measure how much sector- and location-specific family connections contribute to the recovered barriers. I then ask how these connections affect structural transformation.

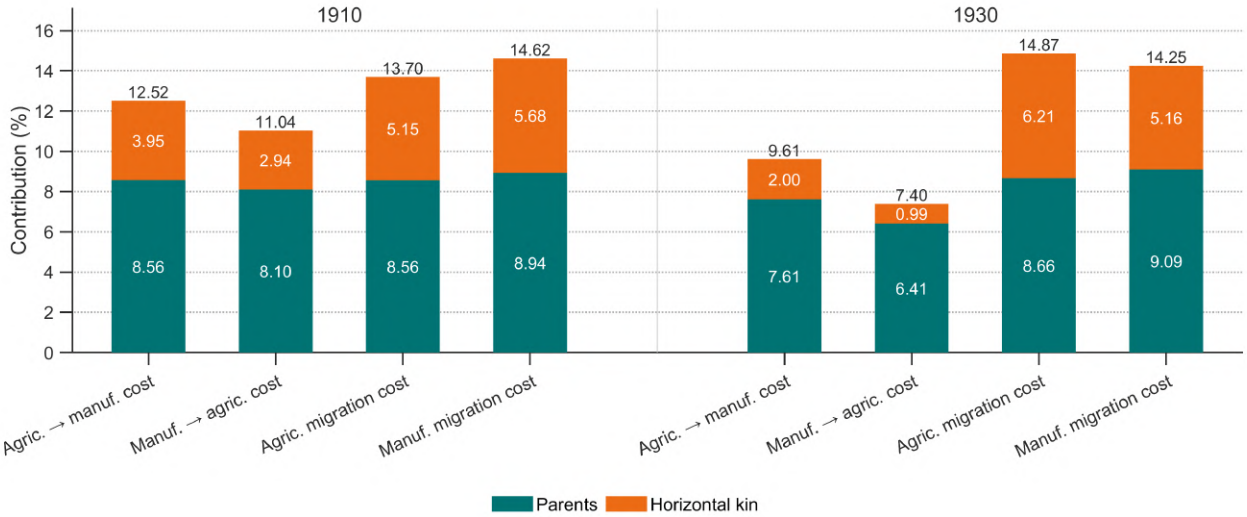
To measure this contribution, I separately remove parental and horizontal-relative support from each choice equation and re-invert the model, holding observed choices fixed. For migration, I remove the benefit of proximity to each type of relative. For sector switching, I remove the advantage of entering the parent's sector or the access provided by horizontal relatives. In each exercise, the other choice equation remains unchanged. Let X_{it} denote the baseline recovered cost and $X_{it}^{(-c)}$ its value after removing family component c. I calculate

$$\mathcal{L}_{ct}^{X} = 100 \frac{\sum_i \ell_{it} \left| X_{it}^{(-c)} - X_{it} \right|}{\sum_i \ell_{it} |X_{it}|}, \tag{40}$$

where ℓ_{it} is employment in ancestral-origin-sector group i. This statistic measures the employment-weighted adjustment in recovered costs needed to reproduce observed choices without the family component.

Figure 8 shows that family ties contribute substantially to both sectoral and spatial barriers.

Figure 8: Family ties and barriers to reallocation



Notes: The figure reports the contribution of family ties to recovered sector-entry and migration barriers in 1910 and 1930. Bars show the employment-weighted re-inversion adjustment relative to the recovered object, $\mathcal{L}$, defined in equation (40). Each bar decomposes the adjustment into parental and horizontal-relative components.

In 1910, removing family support requires adjustments equal to 12.52 percent of the recovered agriculture-to-manufacturing entry cost and 11.04 percent of the reverse cost. For migration, the corresponding adjustments are 13.70 percent in agriculture and 14.62 percent in manufacturing. Horizontal relatives also matter, accounting for 32 percent of the family-tie component of the agriculture-to-manufacturing barrier.

These contributions change over development. Between 1910 and 1930, the family-tie component falls to 9.61 percent for agriculture-to-manufacturing entry and 7.40 percent for the reverse transition. Its contribution to migration barriers instead rises to 14.87 percent in agriculture and remains broadly stable at 14.25 percent in manufacturing. Family ties thus remain important as structural transformation proceeds, but their relative contribution shifts from sectoral entry toward spatial adjustment.

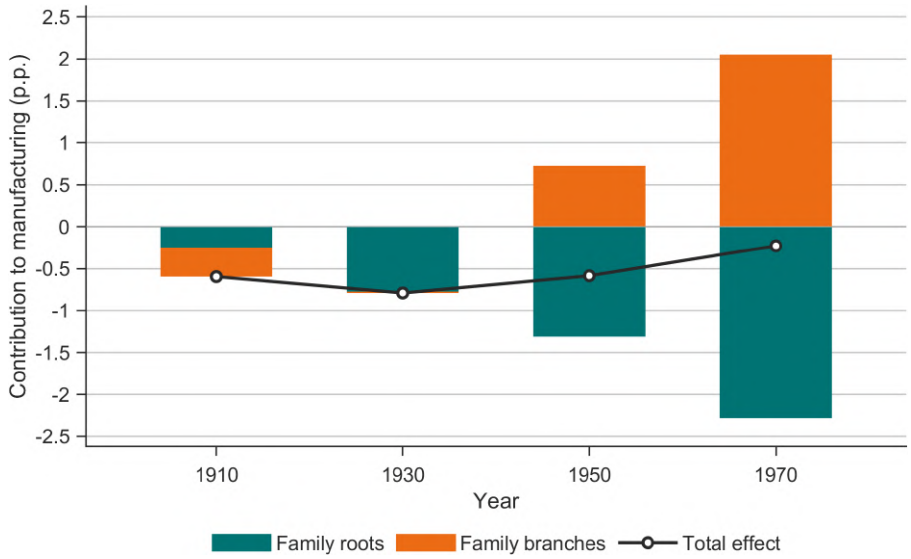
The aggregate contribution of roots and branches. The contribution of family ties to recovered barriers does not, by itself, establish how these connections affect structural transformation. Their aggregate effect also depends on where relatives are initially established and how this distribution changes with workers' choices. I separate these forces by comparing three economies with the same non-family fundamentals. In the first, relatives do not affect sector or location choices. In the second, their sectoral and spatial distribution remains fixed at its observed 1880 pattern. In the third, connections to relatives evolve endogenously as workers enter different sectors and move across locations. Let M_t^0, M_t^{fix} and M_t^F denote their respective

national manufacturing shares. The overall contribution of relatives can be written as

$$\underbrace{M_t^F - M_t^0}_{\text{overall effect}} = \underbrace{M_t^{\text{fix}} - M_t^0}_{\text{roots}} + \underbrace{M_t^F - M_t^{\text{fix}}}_{\text{branches}}. \tag{41}$$

Roots capture the effect of the initial distribution of relatives, while branches capture the additional effect of changes in that distribution across generations. The latter can initially reinforce the drag from roots, but as relatives enter manufacturing and move to growing places, their new connections increasingly facilitate reallocation.

Figure 9: Family roots and branches along structural transformation



Notes: The figure decomposes the effect of relatives on the national manufacturing share, in percentage points, using equation (41). Roots compare the economy with relative connections held at their 1880 distribution with an economy without relative support. Branches compare the endogenous economy with the fixed-1880 economy. The line reports their sum, imposing a steady state in 1970 and constant fundamentals after 1950. Appendix Figure A5.4.1 reports the extended series until 2010.

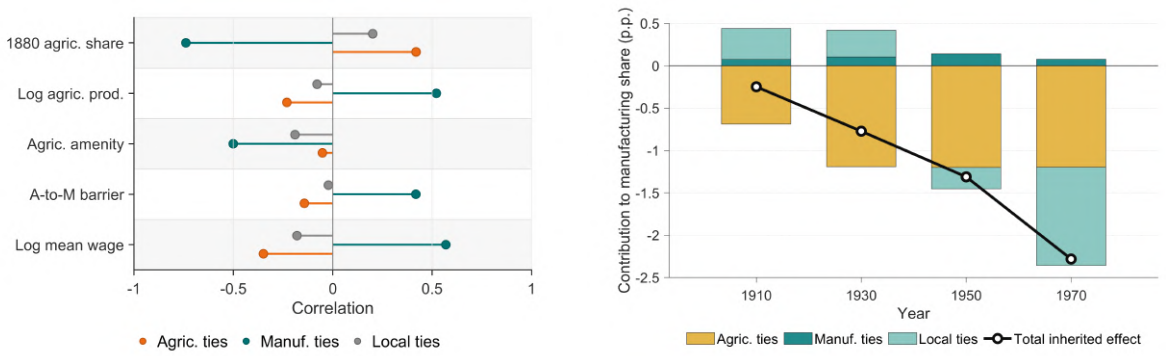
Figure 9 shows that the initial distribution of family connections slows structural transformation, while the connections created by earlier movers become increasingly important. Initially, these changes reinforce rather than offset the drag from roots. As families branch out, however, their contribution becomes positive and offsets about 90 percent of the negative roots effect by 1970, leaving a small negative overall effect. Beyond 1970, the model predicts that branches eventually more than offset roots, turning the overall contribution of family ties positive (Appendix A5.3). I next examine why inherited connections slow reallocation, before turning to the branches that help later generations reach expanding sectors and places.

6.2 The role of family roots in structural change

The initial distribution of family roots. The effect of family ties on development depends not only on the size of the barrier but also on where relatives live and which sectors they work in. Panel 10a shows that these connections are systematically related to the economic conditions families face in 1880. Agricultural and local connections are more common among families originating in relatively agricultural and lower-opportunity places, while manufacturing connections are more common where manufacturing is already more attractive. The initial distribution of family members therefore may reinforce the sectoral and spatial allocation at the start of the transition.

The aggregate effects of roots. Given this uneven distribution of family members across sectors and places, I next quantify how these roots affect aggregate structural change. To do so, I compare an economy without relative support with an otherwise identical economy in terms of fundamentals in which relative connections are present but their sectoral and spatial distribution is held constant at its observed 1880 pattern. This isolates the effect of roots on the movement out of agriculture, before allowing migration and sector switching to change where relatives live and work.

Figure 10: Family roots and structural transformation



(a) Family connections in 1880

(b) Manufacturing effects of family roots

Notes: Panel (a) reports CZ-level correlations in 1880 between agricultural, manufacturing and local family connections and selected local characteristics inverted from the model. Panel (b) reports contributions to the national manufacturing share in percentage points from holding the observed 1880 family distribution constant relative to an economy without relative support. Bars report the exact Shapley decomposition into agricultural, manufacturing and local connections. The black line reports the total effect of roots.

Panel 10b shows that roots become a larger drag on structural transformation as the economy moves away from its initial allocation. By 1950, holding the distribution of family members fixed at its 1880 pattern lowers the manufacturing share by 1.311 percentage points relative to an economy without relative support. This amounts to around 1.8 percent of the 1950

manufacturing share. Agricultural connections account for most of the decline, reducing the manufacturing share by 1.20 points. Manufacturing connections offset 0.14 points, while local connections reduce the share by another 0.25 points. By 1970, the total effect reaches −2.28 percentage points, equivalent to 4.1 percent of the increase in manufacturing between 1880 and 1970.

The drag from roots grows over time. As manufacturing expands, connections to agriculture increasingly reinforce a declining sector, while local connections continue to favor the places in which families were concentrated in 1880. The same initial distribution of family connections therefore becomes more restrictive as economic activity shifts across sectors and space.

Where those relatives are located is central to this result. Appendix A5.2 keeps the same total mass of relatives observed in 1880 but redistributes relatives across CZs in proportion to population. Under this population-proportional distribution, family ties lower the 1950 manufacturing share by only 0.03 percentage points. The observed geography accounts for the remaining 1.28 points of the 1.31-point effect. The initial drag therefore comes primarily from the spatial distribution of relatives rather than from the number of family members alone.

The effect operates through both sectoral change within places and the spatial distribution of employment. In 1950, 0.688 percentage points of the 1.311-point decline come from lower manufacturing shares within CZs and 0.603 points from changes in where employment is located, with only a small interaction term. These roots therefore slow structural transformation both by affecting sector choice within places and by affecting movement across places.

These results show why family roots can act as a barrier to reallocation. Workers inherit connections to particular sectors and places, and those connections become more or less valuable as economic opportunities change. The analysis so far has isolated roots by holding the 1880 distribution of relatives constant. In practice, migration and sector switching create new connections to sectors and places, branches, for other family members and later generations. I next quantify how these branches affect the pace of structural transformation.

6.3 The role of family branches in structural change

The previous subsection quantified the effect of family roots by holding the sectoral and spatial distribution of relatives fixed at its 1880 pattern. I now allow migration and sector switching to change where relatives live and work, creating new connections, branches, for later generations. I compare the full model with an otherwise identical economy in which the distribution of relatives remains fixed at its observed 1880 pattern, while other economic fundamentals evolve identically in both economies. The difference measures how these new family branches affect subsequent structural transformation.

The previous results show that family roots in agriculture and less industrialized places slow reallocation. This need not remain true as structural transformation proceeds, however, because

the same migration and sector switching that drive the transition create new connections for later generations. I therefore ask how much these branches offset the initial drag.

Allowing these connections to evolve raises the national manufacturing share by 0.727 percentage points in 1950 and by 2.053 points in 1970 relative to keeping the distribution of relatives fixed at its 1880 pattern. This accounts for 3.7 percent of the entire increase in the manufacturing share between 1880 and 1970. These branches advance structural transformation by approximately seven years: the full-model economy reaches the manufacturing share of the fixed-1880 economy in 1970 already by 1963. The new connections offset 55 percent of the initial drag from family roots by 1950 and approximately 90 percent by 1970. By the end of the period, the net effect of family ties on the national manufacturing share is therefore only -0.228 percentage points.

The additional manufacturing is concentrated in places that begin the transition further behind. Table 4 shows that, by 1970, allowing family connections to change raises the manufacturing share in the South by 3.1 percentage points and in CZs that were in the bottom third of the 1880 manufacturing distribution by 3.8 points. The share of the population living in a manufacturing-majority CZ rises by 1.9 percentage points. Relative to the overall transition, these effects are especially important for spatial convergence. Family branches explain about 4 percent of the national rise in manufacturing, but 17 percent of the narrowing of the manufacturing gap between the most and least industrialized thirds of CZs and 19 percent of the decline in the dispersion of manufacturing shares across CZs. They also account for 8 percent of the across-CZ component of structural change, compared with 3 percent of the within-CZ component. The new connections therefore contribute disproportionately to the geographic diffusion of industrialization and the catch-up of places that begin the transition behind (Eckert & Peters, 2022).

Figure 11 examines how the effects of family branches change across sectors and places. Panel 11a separates the manufacturing effect into four channels. Sector persistence captures how relatives working in the parent's sector encourage children to enter the same sector, while cross-sector pull captures how relatives in other sectors facilitate switching. Similarly, origin persistence captures how relatives at the origin encourage workers to remain, while destination pull captures how relatives elsewhere facilitate migration.

Panel 11a shows that sectoral changes matter earlier than spatial changes. In 1950, sector persistence and cross-sector pull account for 0.718 of the 0.727-percentage-point manufacturing effect, while origin persistence and destination pull nearly offset one another. By 1970, the sectoral channels contribute 1.1 percentage points and the spatial channels 0.9, so changes in the geography of family connections account for 44 percent of the long-run effect. The spatial channel takes longer to emerge because migration must first change where relatives live before those new locations affect the choices of later generations.

Table 4: Effects of family branches on structural transformation

Panel A. Aggregate structural transformation

	Full	Fixed 1880	Effect
Manufacturing share in 1970 (%)	79.7	77.6	+2.1 pp
Year reaching fixed-1880 manufacturing share in 1970	1963	1970	7 years earlier

Panel B. Spatial incidence

	Full	Fixed 1880	Effect
Population in manufacturing-majority CZs (%)	90.8	88.9	+1.9 pp
Manufacturing share in the South (%)	71.3	68.2	+3.1 pp
Manufacturing share in bottom-third CZs (%)	62.9	59.1	+3.8 pp

Panel C. Contribution to structural change and convergence

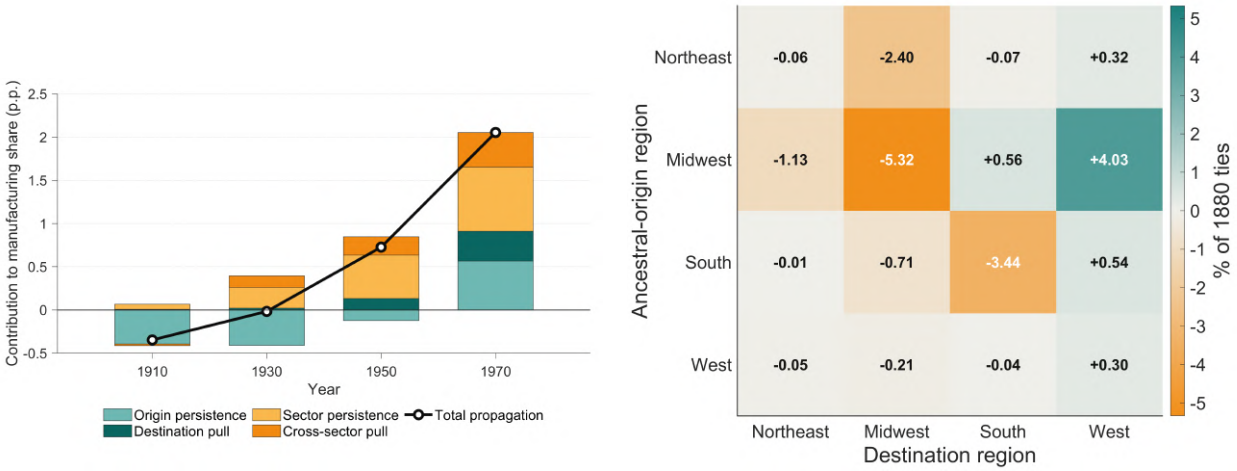
	Full-model change	Share explained by branches
Manufacturing share (pp)	55.1	4%
Within CZs (pp)	48.7	3%
Across CZs (pp)	6.4	8%
Manufacturing share, bottom-third CZs (pp)	60.1	6%
Top–bottom third manufacturing-share gap (pp)	−16.8	17%
SD of CZ manufacturing shares (pp)	−6.5	19%

Notes: The full model allows family connections to change endogenously with sector and location choices. The fixed-1880 counterfactual holds the sectoral and spatial distribution of relatives at its observed 1880 pattern, while parental connections and other economic fundamentals evolve as in the full model. Panels A and B report outcomes in the two economies and their difference. Panel C reports the share of the 1880–1970 change in each outcome explained by branches, calculated as the difference between the full and fixed-1880 economies relative to the change in the full model. Thirds of CZs are defined by their manufacturing share in 1880. For the timing measure, the full-model path is linearly interpolated between model periods.

The decomposition also separates access to new opportunities from the persistence of earlier reallocation. By 1970, destination pull and cross-sector pull contribute a combined 0.707 percentage points, about one third of the total effect. Origin persistence and sector persistence contribute 1.346 points, about two thirds. Most of the long-run effect therefore comes not from repeatedly drawing workers away from the same original sector and place, but from reinforcing the sectors and places reached by earlier generations.

The spatial channel makes this mechanism particularly clear. Origin persistence is negative in 1950 but positive by 1970 because the relevant origin changes across generations. Once a family branch migrates, a destination for one generation becomes the origin of the next. The same mechanism operates across sectors: once a branch enters manufacturing, later relatives inherit more connections to manufacturing. As families branch out, the sectors and places they reach become new roots that reinforce the allocation created by earlier migration and sector

Figure 11: Family branches and structural transformation



(a) Manufacturing effects of branches

(b) Changing geography of family branches

Notes: Panel (a) reports the Shapley decomposition of the manufacturing-share effect of evolving family connections into origin persistence, destination pull, sector persistence and cross-sector pull. Origin and destination refer to successive generations. The black line reports the total effect. Panel (b) reports the net change between 1880 and 1970 in population-weighted connections to relatives for each ordered pair of Census regions, excluding connections at the ancestral CZ itself, as a percentage of the 1880 stock. Rows denote ancestral-origin regions and columns destination regions.

switching rather than the one observed in 1880. Panel 11b shows how migration reshapes family connections across regions. Between 1880 and 1970, connections from families with Midwestern roots to relatives in the West increase by 4 percent of the 1880 stock, while connections to other Midwestern locations decline by 5.3 percent. Connections within the South also decline, while connections to the West increase for families from every ancestral region. These changes illustrate how families gradually establish connections in new destinations, making them more accessible to later generations.

Two additional exercises in Appendix A5.3 clarify how the contribution of branches accumulates. About three quarters of the 1970 manufacturing effect comes from higher manufacturing shares within CZs, while the remainder comes from changes in the spatial distribution of employment. The effect also requires repeated rounds of adjustment. Allowing family connections to change only through 1910 generates essentially none of the eventual manufacturing gain, while allowing them to change through 1930 already produces 1.5 of the eventual 2.1 percentage points.

6.4 Spatial policies and family ties

The preceding results suggest that migration can gradually overcome the constraints imposed by the family roots and induce structural transformation. This raises a policy question: can

relocation assistance accelerate this process and who benefits when family connections change? A subsidized move can establish new destination branch for relatives and later generations, but it also removes support from those left behind. To study this trade-off, I examine a one-generation relocation voucher in 1880.

The voucher targets the 137 commuting zones in the bottom 30 percent of the manufacturing-share distribution, concentrated in the South and lower Midwest (Appendix Figure A5.5.1). These CZs contain 20.9 percent of the population but only 2.25 percent of national manufacturing employment. Workers receive a grant if they move to a CZ outside the targeted set. A budget of $119.4 million, equal to one percent of the 1880 national wage bill and financed by a uniform lump-sum tax, provides $723 per recipient, or 20.8 percent of the annual wage. The scale is comparable in spirit to major relocation programs in Indonesia and China (Bazzi et al., 2016; Zhang et al., 2023).[34]

The cost of leaving family roots. Relocation requires workers to leave behind valuable family support. For each origin–destination pair, I calculate the transfer compensating a worker for the difference in family support between the two places. Figure 12 shows a median of $157, approximately 12 percent of annual earnings and 22 percent of the voucher; the 75th and 90th percentiles are $249 and $353. Only 6 percent of pairs offer greater support at the destination. Existing family roots thus constitute a substantial private cost of moving.

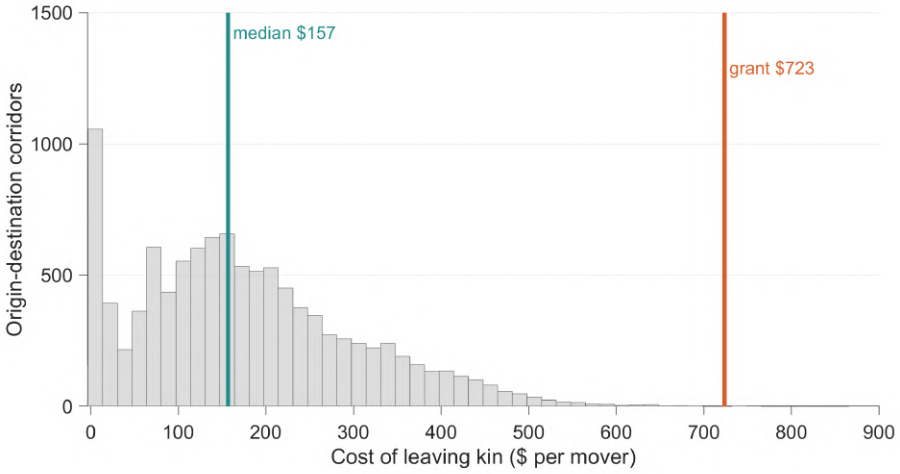


Figure 12: The value of family connections lost through relocation

Notes: Distribution of compensating transfers for differences in family support across targeted origin–destination pairs. Teal: median ($157). Orange: benchmark voucher ($723).

[34]Indonesia's Transmigration Program relocated about two million people between 1979 and 1988, while China's Anti-Poverty Relocation Program relocated more than 9.6 million people between 2016 and 2020. Results are qualitatively similar for budgets between 0.25% and 1.25% of the 1880 wage bill.

Family branches and intergenerational spillovers. Leaving family roots imposes a private cost, but also changes the connections available to other relatives: fewer remain at the origin and new branches arise at the destination. To isolate these effects, I compare the full model with an economy that reproduces the same initial sector and location choices but gives later generations the connections to aunts and uncles they would have had without the voucher. Parental connections continue to change as in the full model. The difference thus measures the contribution of policy-induced changes in connections to aunts and uncles.

The voucher raises the share leaving targeted CZs in 1880 from 20.5 to 23.0 percent, equivalent to additional moves by about 0.5 percent of the 1880 labor force. Each 100 induced initial moves generate 80.2 further moves through 1970, including 18.5 attributable to changing connections to aunts and uncles. The effects thus extend beyond the subsidized generation.

Following Yang (2024), I measure welfare for movers and those left behind, including their descendants. Let $\bar{u}^{q}_{s|t,a}$ denote average log consumption-and-amenity utility for group a and its descendants in generation s, traced from generation t, in economy $q \in \{0, \text{voucher}\}$. The discounted welfare difference is

$$\Delta \mathcal{W}_{t,a} = (1-\beta) \sum_{\ell=0}^{\infty} \beta^{\ell} \left(\bar{u}^{\text{voucher}}_{s_\ell|t,a} - \bar{u}^{0}_{s_\ell|t,a} \right),$$

where $s_0 = t$, s_ℓ indexes subsequent generations and β is parental altruism. The percentage consumption equivalent is

$$\lambda_{t,a} = 100 \left[\exp(\Delta \mathcal{W}_{t,a}) - 1 \right]. \tag{42}$$

A value of 0.5 represents a permanent 0.5 percent gain in the group's consumption-and-amenity index. Appendix A5.5 provides the full derivation.

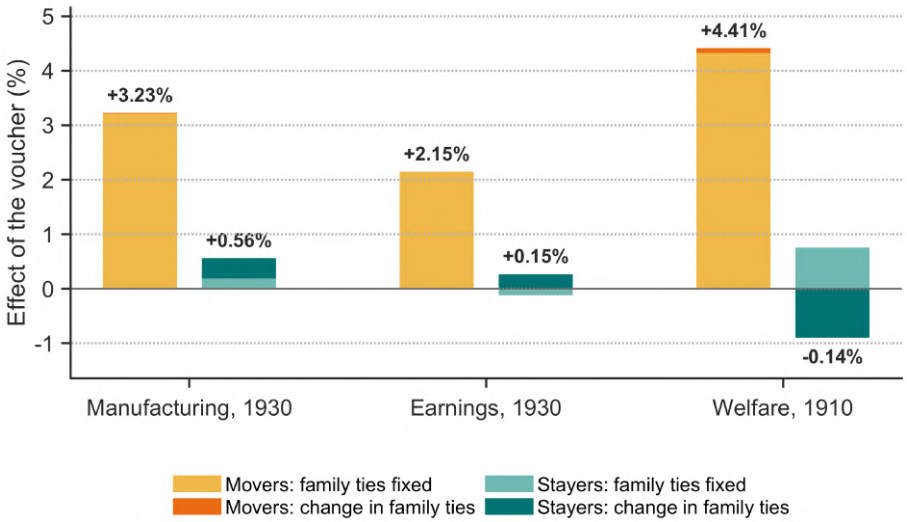
Figure 13 compares workers who leave targeted CZs under the voucher with those left behind ("stayers"), including their descendants. It reports manufacturing employment and earnings in 1930 and welfare in 1910; darker segments isolate the effects of changing connections to aunts and uncles.

In 1930, the voucher increases manufacturing employment and earnings among targeted families by 1.5 and 0.9 percent. The respective gains are 3.2 and 2.1 percent for movers, compared with 0.6 and 0.2 percent for those left behind. For the latter, changes in connections to aunts and uncles explain about two thirds of the manufacturing gain and turn an earnings decline into a small increase. By 1970, these changes account for 33 percent of the voucher's manufacturing effect and 22 percent of its earnings effect.[35]

Welfare effects differ. In 1910, targeted families gain 0.9 percent, but changes in connections to

[35] Allowing only origin or only destination connections to aunts and uncles to change, I find that origin connections account for 63 percent of their contribution to manufacturing in 1970 and 68 percent of that to earnings, destination connections for 16 and 24 percent and the remainder is their interaction.

Figure 13: Family connections and the effects of moving vouchers



Notes: Moving voucher with a budget equal to one percent of the 1880 wage bill. Effects are relative to no policy for movers (workers leaving targeted CZs under the voucher) and stayers (those staying in the period of the voucher), including their descendants. Manufacturing employment and average earnings are measured in 1930 (percent) and welfare in 1910 (consumption equivalent in percent, equation (42)). Darker segments show the contribution of policy-induced changes in connections to aunts and uncles, comparing the full model with one in which later generations retain the corresponding connections from the no-policy economy. Parental connections evolve as in the full model in both economies.

aunts and uncles reduce the gain by 0.7 percentage points. Movers gain 4.4 percent, with little contribution from these changes. Those left behind lose 0.1 percent overall: a 0.9 percent loss from changing family connections outweighs other benefits. Thus, new branches may improve future employment while the loss of local support reduces welfare at origins, consistent with evidence on informal insurance (Morten, 2019; Munshi & Rosenzweig, 2016).

Accounting for family spillovers in relocation policy. Existing relatives at destinations make moving easier: with the same budget, vouchers directed toward destinations with more family connections induce about three times as many initial moves as those directed toward places with fewer connections (Appendix Figure A5.5.2). But moves that are easier to induce need not benefit other relatives most. Leaving an origin with many relatives can weaken valuable support, while reaching a destination with few relatives can establish a new branch.

I therefore classify moves into six types based on family connections at origins and destinations. For each type c, I restrict the \$723 voucher to those moves and compare economies with and without the resulting changes in connections to aunts and uncles. I convert the welfare difference into 1880 dollars per additional move, $\hat{e}_c$. These spillovers range from $-$\$199 for moves from origins with many relatives to $+$\$212 for moves from origins with fewer relatives to destinations

with fewer existing connections (Appendix Table A5.5.5). I set the family-adjusted grant to

$$s_c = \max\{0, \$723 + \hat{e}_c - \lambda\},$$

where λ holds total spending at \$119.4 million, favoring moves with more beneficial family spillovers.

Table A5.5.7 shows that family-adjusted vouchers redirect 8.8 percent of spending and induce slightly fewer initial moves, but improve longer-run outcomes. By 1970, the manufacturing effect on targeted families rises from 0.44 to 0.52 percent and the earnings effect from 0.19 to 0.23 percent. For those left behind, the manufacturing gain rises from 0.26 to 0.36 percent and the 1910 welfare loss falls from 0.14 to 0.06 percent. The aggregate welfare loss decreases from 0.259 to 0.248 percent, recovering about half the welfare cost from changes in connections to aunts and uncles.

These results show that the benefits of relocation depend not only on how many workers move, but also on which family connections their moves weaken or create. By accounting for these effects on other relatives, relocation policy can generate larger long-run gains in manufacturing and earnings while reducing welfare losses, even with fewer initial moves.

7 Conclusion

This paper shows that family ties can both slow and accelerate structural transformation. Relatives in current sectors and origins reinforce persistence, while relatives in expanding sectors and at destinations facilitate sector switching and migration. Because workers who switch sectors and move change where their relatives work and live, these effects build across generations. In the quantitative model, families' roots in agriculture and less industrialized places initially slow structural transformation, but the new branches created as relatives enter manufacturing and move to new destinations offset 55% of this initial drag by 1950 and about 90% by 1970.

These dynamics have direct implications for relocation policy. A subsidized move creates opportunities for relatives at new destinations but also weakens family support at the origin. In a moving-voucher experiment, these changes in family connections account for one third of the long-run manufacturing effect but reduce welfare for those left behind. Existing family members at destinations also make moving vouchers more effective: the same budget induces about three times as many moves when more relatives already live there. Adjusting grants to account for these effects modestly improves aggregate welfare. Relocation policies that consider only direct recipients or economic opportunities therefore overlook both the benefits and costs that extend across families and generations.

Two extensions are particularly natural. First, marriage and household formation are taken as given, even though marriage combines family geographies and may itself respond to migration

and occupational opportunities. Endogenizing family formation would allow structural change to reshape family connections through an additional margin. Second, the analysis stops at the agriculture-to-manufacturing transition. The rise of services may place different weight on education, credentials and access to particular occupations, potentially changing both how family ties matter and how quickly these connections adapt. Studying these margins would help establish how the role of family ties evolves across later stages of development.

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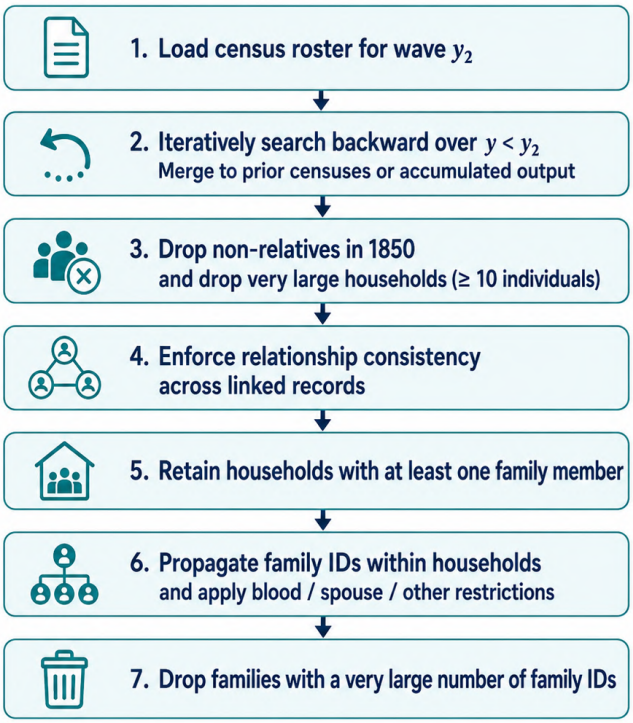
A1 Family ties construction

A1.1 Detailed procedure

This appendix describes in detail how I construct multigenerational family networks for the United States using decennial census waves from 1850 to 1940. The objective is to assign each individual observed in a given census wave a set of extended-family identifiers that can cross household and region boundaries, while maintaining internal consistency with observed household rosters and relationship codes. The key challenge is that households can contain multiple lineages (e.g., married children co-residing with parents, widowed parents joining adult children, siblings living together) and a given individual may simultaneously belong to multiple family groups (e.g., through marriage). The procedure therefore iteratively links individuals across census waves, anchors households to previously observed family identifiers and propagates those identifiers within households only when the implied relationship structure is consistent. The family seed is initialized in 1850 on the household and the household head is defined as the original ancestor. The algorithm then proceeds sequentially by census wave. For each target wave from 1860–1940, it searches backward across earlier waves using crosswalk files that map individuals to their prior-wave individual identifiers. Whenever a match is found, the algorithm retrieves the individual's last observed household identifier and relationship code in the earlier wave. These inherited identifiers become anchors for constructing extended family memberships in the current census wave. This is until until the end of the sample, in 1940. Because a single erroneous link can mechanically contaminate an entire household if family identifiers are propagated without checks, the procedure is deliberately conservative. It implements (i) initial restrictions in 1850, (ii) a relationship-consistency check that compares inferred kinship structure across the old and new household rosters and (iii) a set of propagation rules that restrict how family identifiers can spread through the household when the link connecting households is via a spouse, via a descendant ("linking from below") or via collateral relatives (e.g., cousins or nephews). Figure A1.1.1 summarizes the overall construction pipeline described below.

Data inputs. The construction combines three inputs. The first is the full-count census in each wave taken from IPUMS. For every person, the census identifies the historical identifier (`histid`), household identifier (`serial`), the individual's relationship to the household head (`relate`) and, when available, links to co-resident parents (`momloc` for mothers and `poploc` for fathers). The second input is the set of Census Tree Project crosswalks, which link individuals across census waves (Buckles et al., 2025). The third input is a relationship-pair crosswalk that I construct, as described below. This crosswalk maps the relationships of two household members to the head into their relationship to each other. For example, if one person is coded

Figure A1.1.1: Steps for constructing multigenerational family ties



Notes: This figure describes the construction of the family ties. Individuals are linked backward across censuses, inconsistent or implausible relationships are removed and family identifiers are propagated to construct multigenerational networks.

as a child of the head and another as the parent of the head, the crosswalk classifies the first person as the grandchild of the second. When a pair of relationship codes is not uniquely informative, I flag them to be ambiguous.

Iterative backward linkage and anchoring. For each target census wave, I search backward through all earlier waves for linked relatives. The set of possible source waves expands over time: for the 1860 census I search only 1850, while for the 1940 census I search 1930, 1920, 1910, 1900, 1880, 1870, 1860 and 1850. I start from the full roster in the target wave and create fields that record the first successful historical anchor: the linked person in an earlier census, the year of that link, the household in which the person was previously observed and the person's relationship to the head in that earlier household.

The backward search proceeds year by year. When a target-wave individual is linked to an earlier, more recent census through the crosswalk, I retrieve the family information already attached to that earlier observation. For links to 1850, this information comes directly from the raw 1850 household structure. For links to later source years, it comes from the family-ties

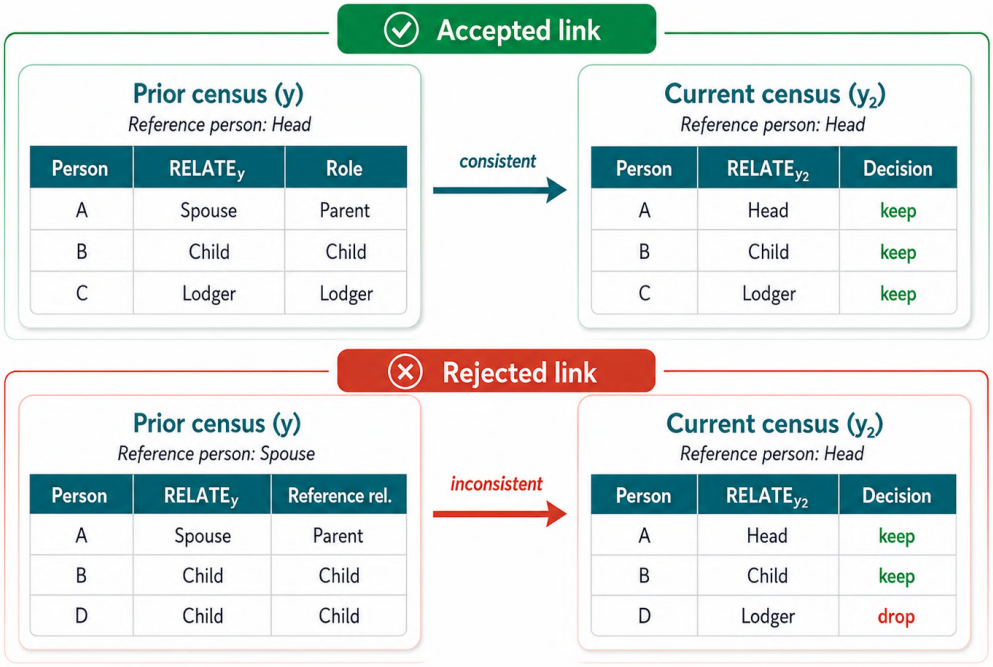
file that has already been constructed for that year. Thus, family labels accumulate recursively: a person linked to 1910 can inherit family groups that were themselves built from links to 1900, 1880 and earlier waves.
Each individual is assigned at most one primary historical anchor within a target wave. Once a valid backward match is found, that person is removed from later searches over older waves. This rule avoids conflicts when an individual has multiple possible cross-census links. Additional family connections are still added through within-household relationships and marriage links, but the historical anchor used to bring forward inherited family labels is unique.

Restrictions in 1850. The 1850 census provides limited relationship detail. In particular, several categories of co-residents are not consistently classifiable as blood relatives in a way that is stable under forward propagation. To avoid introducing spurious kin links at the baseline of the panel, I drop from the set of anchored individuals those whose last observed household is in 1850 and who are coded as non-relatives (`RELATE` codes ≥ 1061) or as "other relatives" (`RELATE` code 1001). This restriction is intentionally conservative: it reduces the risk that ambiguous household ties in 1850 generate false extended family connections in later waves.

Relationship consistency check. A central concern in this linking procedure is that a single incorrect person-level link can cause an entire household to inherit incorrect family identifiers. The algorithm therefore implements a relationship-consistency check designed to detect mismatches between the implied kinship structure of the old household and the observed structure of the new household.
The check proceeds as follows. Within each new household in a given census wave, consider the subset of individuals who were successfully linked to some prior household. Among those linked individuals, the algorithm selects a reference person based on a deterministic "seniority" ordering of relationship codes that prioritizes older generations and household heads or spouses. Table A1.1.2 presents the ordering. Intuitively, the most senior linked person is the most informative anchor because their relationships to other household members are typically measured with less ambiguity.
Once the reference person is determined, the algorithm constructs two mappings for every person in the household. The first mapping uses the linked individuals' prior-wave relationship codes to infer each person's relationship to the reference person in the old household. The second mapping uses the current-wave relationship codes to infer each person's relationship to the reference person in the new household. Both mappings rely on the constructed relationship-pair crosswalk, which converts a pair of relationships to the household head into an inferred relationship between the two individuals, together with an ambiguity flag when the relationship is not uniquely determined (e.g., "other relatives").

A merge is retained if the inferred relationship of each individual to the reference person agrees across the old and new mappings *and* the individuals originate from the same prior household, or if the individual is the linking person itself, allowing household composition to change. The procedure does not reject cases in which either side of the relationship mapping is missing, since missing mappings frequently correspond to boarders or ambiguous codes for which the crosswalk cannot uniquely determine kinship or to cases in which the relationship cannot be measured because the individuals were present in different households earlier.
Figure A1.1.2 illustrates how the relationship-consistency check works in a stylized example.

Figure A1.1.2: Relationship-consistency check: accepted vs. rejected links



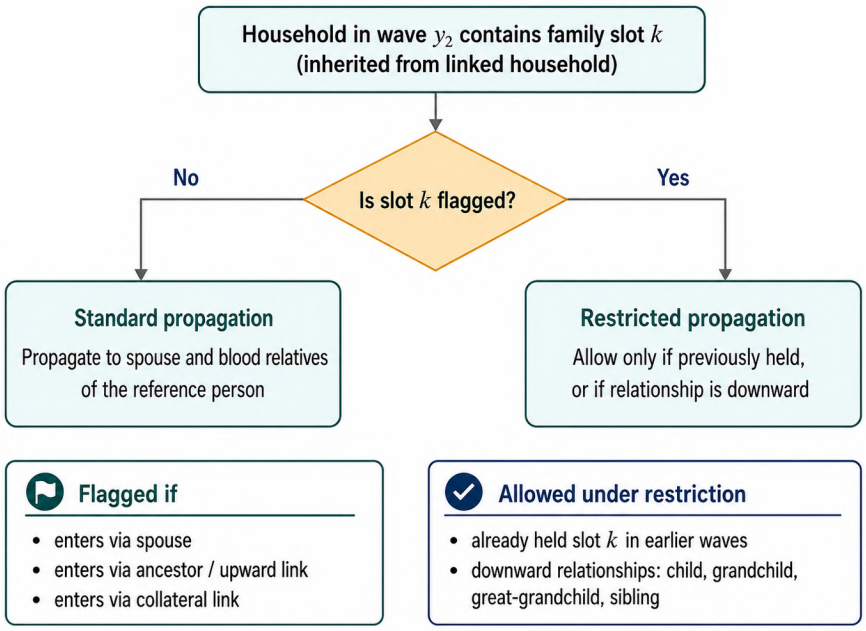
Notes: The algorithm enforces stability of inferred kin roles across linked census waves. Links are retained when role changes are uninformative (e.g., lodgers) but rejected when a previously identified family member appears with an incompatible role.

The procedure applies stricter checks when an individual's relationship in the old household is clearly identified (e.g., as a child, parent or sibling) but is inconsistent with the structure of the new household. When the relationship cannot be inferred in one or both households, often because the individual is a lodger or has an ambiguous relationship code, the procedure does not automatically reject the match. Such records are not used to establish kin connections in subsequent steps.

Propagation of family identifiers within households. After the consistency check, the algorithm retains only households in which at least one member carries an inherited family identifier. This household-level propagation allows family trees to expand over time while ensuring that every family label assigned in a new wave is anchored to at least one credible backward link. The reference anchor is chosen first by the presence of an inherited family identifier and then by seniority. Within the remaining households, the procedure enumerates the number of distinct inherited prior-household identifiers present. In practice, this number can exceed one: a household can contain multiple linked lineages.

Propagation is then carried out family-slot by family-slot. For each slot k, the algorithm defines a reference person within the new household and uses the relationship-pair crosswalk to infer whether each member of the household is a blood relative or spouse of that reference person. Individuals inherit slot k if they are classified as a blood relative or spouse or if they already carried that family identifier from prior waves. Individuals who are neither blood relatives nor spouses and who did not previously carry that identifier do not inherit it. This ensures that family identifiers do not mechanically spill over to boarders, servants, or unrelated co-residents.

Figure A1.1.3: Within household propagation



Notes: This figure shows how the family ties are linked within households. Family identifiers propagate to spouses and blood relatives; flagged links are retained only when previously observed or permitted by the relationship direction.

Cleaning propagation. Even when a household contains a valid historical anchor, not all anchors are equally informative for forward propagation across generations. The code tracks three cases in which the connecting relationship is likely to create spurious upward or cross-lineage assignments if used without restriction.

First, I flag cases where the family slot enters the household through a spousal tie. Such links are informative for marriage connections but, without additional information, can blur lineage boundaries if used to infer broader blood relationships of co-residents, for example parents-in-law or cousins of the household head. Second, I flag cases where the connection is through descendants (grandchildren or great-grandchildren). These links can generate ambiguous upward propagation because descendants may carry a family identifier inherited through a parent who is not connected to the lineage of the grandparent used as the current anchor. Third, I flag collateral links such as cousins, nephews or nieces and grand-nieces or grand-nephews, where extending the identifier further can similarly blur lineage boundaries.

If a family slot is flagged by any of these indicators, propagation in the next iteration is restricted: individuals who would inherit the slot solely through an upward or collateral mapping are prevented from doing so unless they are themselves in downward relationships (child, grandchild, great-grandchild, or sibling) or already carried the identifier historically.

These restrictions are designed to preserve lineage membership rather than co-residence. Spousal, descendant and collateral links can identify valid kin connections, but they are also the cases in which a small linkage error can generate implausible assignments across several generations. Restricting their subsequent propagation therefore trades some coverage for greater precision. I additionally require cross-census links to be supported by at least two of the linking specifications incorporated in the Census Tree Project. The resulting files record each individual's inherited family identifiers and position within those lineages, which form the basis for the sectoral, spatial and relationship-specific kin measures used in the analysis.

Table A1.1.1 provides a summary of the number of linkages after the cleaning.

A1.1.1 Inferring pairwise relationships and tree positions

The construction above assigns each individual to one or more lineages, but does not yet establish how household members are related to one another or their positions within a family tree. I recover this information using two complementary tools. First, a relationship-pair crosswalk infers relationships between any two household members based on their relationships to the household head. Second, a compact tree-position encoding records each individual's genealogical path from the head and carries it forward across census waves. Together, these tools establish relationships among non-head household members and organize them into a coherent, multigenerational family tree.

Table A1.1.1: Linkage and cleaning summary by census wave

Wave	Raw roster	Drop non-rel. in 1850 & large HH	Drop incons. rel. check	Link within HH	Within HH rel. check	Final ties
1860	27,451,053	24,922,901	24,623,640	13,837,017	12,458,274	12,458,274
1870	38,515,225	35,470,487	35,161,290	15,055,802	13,175,659	13,175,659
1880	50,140,482	46,686,457	46,293,189	18,991,034	17,464,960	17,464,295
1900	75,824,712	70,986,557	70,654,648	24,863,913	21,971,465	21,961,359
1910	92,043,618	85,822,285	85,197,264	28,129,945	24,802,154	24,728,487
1920	105,730,985	100,614,636	99,883,063	31,319,830	28,077,928	27,815,220
1930	122,777,512	117,731,763	116,879,915	36,204,604	32,287,648	31,411,504
1940	131,902,398	127,927,495	127,070,751	37,344,793	33,562,530	32,213,482

Notes: "Raw roster" corresponds to the initial roster size in each census wave. "Drop non-rel. in 1850 & large HH" applies the early-wave filtering rule for 1850 (non-relatives/other relatives) and large-household exclusions. "Drop incons. rel. check" applies the crosswalk-based relationship consistency filter. "Link within HH" reports the number of individuals remaining in households with at least one family member (post household-presence filter). "Within HH rel. check" corresponds to the post across-time linkage count and "Final ties" is the final sample after all cleaning steps.

The relationship-pair crosswalk. The IPUMS `related` variable records each person's relationship to the household head, but not how other household members are related to one another. To recover these relationships, I construct a crosswalk for every ordered pair of `related` codes. Given the relationships of individuals i and r to the household head, the crosswalk determines how i is related to r and flags cases where this relationship is ambiguous.
Some relationships can be determined directly. If r is the household head, i's relationship to r is simply given by i's own `related` code. If i is the head, the relationship is reversed: the head is the parent of someone recorded as the head's child, the child of someone recorded as the head's parent and so on. When neither individual is the head, I infer their relationship by combining their respective relationships to the head. For example, if r is a child of the head, another child of the head is r's sibling, a parent of the head is r's grandparent and a sibling of the head is r's aunt or uncle. Some combinations do not identify a unique relationship: a grandchild of the head, for instance, could be r's own child or a niece or nephew. The crosswalk applies analogous rules to other relationships, including spouses, grandparents, cousins and in-laws, and distinguishes blood, step, half and adopted relationships wherever the codes permit.

Ambiguity. Because the crosswalk relies only on each person's relationship to the household head, some relationships cannot be determined uniquely. For example, when the head has several children, the codes do not reveal which child a child-in-law is married to. Similarly, two

Table A1.1.2: Seniority ordering for linking

Seniority	RELATE code	IPUMS label
Panel A. Core household roles		
1	1011	Grandparent
2	0501	Parent
3	1021	Aunt or uncle
4	0101	Head/householder
5	0201	Spouse
6	0701	Sibling
7	0301	Child
8	1041	Cousin
9	1031	Nephew, niece
10	0901	Grandchild
Panel B. Step relations		
11	1012	Step grandparent
12	0502	Step parent
13	0702	Step/half/adopted sibling
14	0303	Stepchild
15	0903	Step grandchild
16	1033	Step/adopted nephew/niece
Panel C. In-laws (affinal ties)		
17	1013	Grandparent-in-law
18	1022	Aunt-/uncle-in-law
19	0601	Parent-in-law
20	0602	Step parent-in-law
21	0801	Sibling-in-law
22	0802	Step/half sibling-in-law
23	0401	Child-in-law
24	0402	Step child-in-law
25	0904	Grandchild-in-law
26	1034	Grand niece/nephew

Notes: This table documents the deterministic mapping used to construct the variable seniority. Seniority is assigned from the IPUMS RELATE code (here shown with leading zeros) and is used to prioritize individuals for linking and for tie-propagation ordering. Lower values denote higher priority.

individuals recorded as parents of the head are likely to be spouses, but this cannot be established from their codes alone. Step and adopted relationships introduce further uncertainty: two individuals who would otherwise be classified as siblings may instead be half-siblings, step-siblings or adoptive siblings. The crosswalk flags such cases as ambiguous and assigns pairs whose relationship cannot be determined to the "other relative" category. I treat ambiguous and non-blood relationships conservatively. They may be retained as observed ties, but are not used to extend blood lineages to earlier generations. This prevents uncertain relationships from generating additional inferred connections throughout the family tree. I also use the crosswalk

to check consistency across household matches. For each individual, I infer their relationship to the senior reference person in both the old and new household and compare the two.

Tree positions as genealogical paths. To make the full kin structure explicit, I assign every individual a position in the family tree, written as a path from the household head. A position is a sequence of elementary steps: p moves up to a parent, c moves down to a child and s moves across to a spouse, with the head as the root ("`0`"). Direct relatives use a single step (spouse `s`, child `c`, parent `p`); relatives reached only by traversing an intermediate node that is not itself observed carry a "`?`" for that node. Table A1.1.3 lists the encoding: a sibling is "`p?.c`" (up to an unobserved shared parent, then down to a child), a grandparent "`p?.p`", an aunt or uncle "`p?.p?.c`", a niece or nephew "`p?.c?.c`", a cousin "`p?.p?.c?.c`" and a great-grandchild "`c?.c?.c`"; in-laws append a spousal step, so a child-in-law is "`c?.s`" and a parent-in-law "`s?.p`".

Carrying positions forward. Positions are seeded in 1850, where each person is placed relative to the head of their 1850 household. When a household is linked to a later wave, members who are already in the tree keep their existing position and newly observed co-residents are assigned a position by appending the step implied by the crosswalk to the position of the linked reference person. A single chain of links can thus build up multi-step paths—a grandchild first observed in a later wave inherits the reference's position as a prefix and adds the steps down to its own generation—so paths lengthen and the tree deepens as waves accumulate from 1850 to 1940.

Pairwise relationships and final cleaning. The tree positions and generation indices allow me to recover the relationship between two individuals who share a lineage. I classify pairs using their relative position in the tree and the number of generations separating them. This identifies spouses, parents and children, siblings, grandparents and grandchildren, aunts and uncles, nieces and nephews and cousins. Because a spousal step and a blood step both count as a single link in the tree, the resulting categories include the corresponding in-law relationships; for example, parents-in-law are grouped with parents and siblings-in-law with siblings.

I apply three additional cleaning rules to these pairwise relationships. First, the same pair can be connected through more than one lineage. When the same pair is connected through more than one lineage, I retain the relationship implied by the closest common ancestor so that each pair enters a given kin measure only once. Remaining ties are resolved using the reconstruction order and, if necessary, the lower family identifier. Second, more distant relationships are retained only when both sides of the connection can be traced through direct lineage links rather than through previously flagged spousal, descendant or collateral propagation. These are the distributions in which small linkage errors are most likely to accumulate across generations.

Third, I remove distant relationships that imply implausible age differences given the inferred kinship category. Pairs with missing generation information or unresolved tree positions are left unclassified and do not enter the relationship-specific measures. Appendix A1.2 validates these restrictions.

Table A1.1.3: Tree-position path encoding

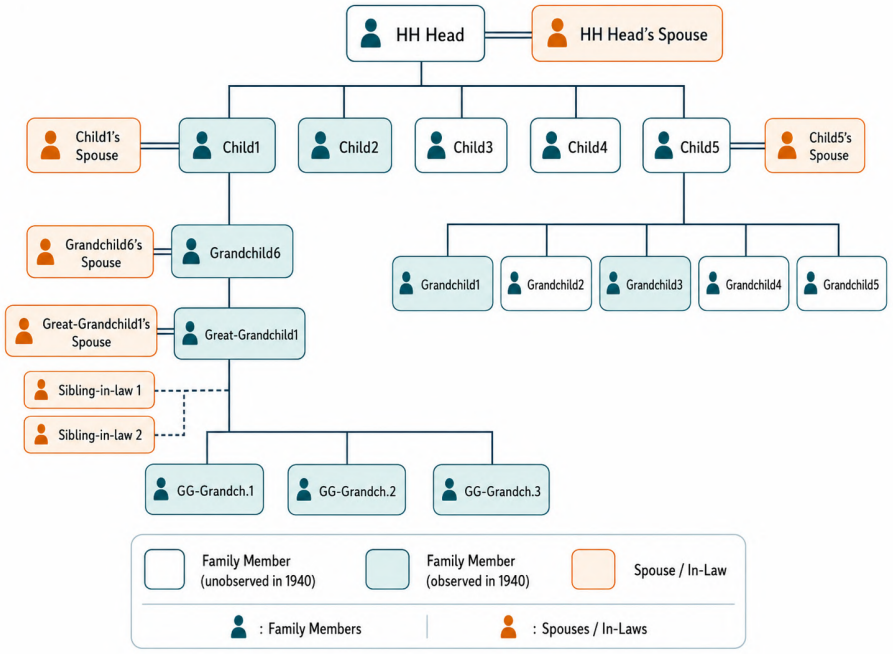
Relationship to reference	**Tree path**
Head (root)	`0`
Spouse	`s`
Child	`c`
Parent	`p`
Sibling	`p?.c`
Grandchild	`c?.c`
Grandparent	`p?.p`
Aunt or uncle	`p?.p?.c`
Niece or nephew	`p?.c?.c`
Cousin	`p?.p?.c?.c`
Great-grandchild	`c?.c?.c`
Grand-niece/nephew	`p?.c?.c?.c`
Child-in-law	`c?.s`
Parent-in-law	`s?.p`
Sibling-in-law	`p?.c?.s`

Notes: Each individual's tree position is a path from the household head built from three elementary steps: `p` (up to a parent), `c` (down to a child) and `s` (across to a spouse), with the head as the root (`0`). A "`?`" marks an intermediate node that is inferred from the relationship-pair crosswalk rather than directly observed in the household roster; a sibling (`p?.c`) is thus reached by going up to a shared parent and back down to a child. A terminal index equal to the person number distinguishes, e.g., one sibling from another. Step, half, adopted and in-law variants share the same path skeleton as their blood counterparts.

A1.1.2 Reweighting linked individuals

This appendix describes how I reweight the linked family data to reduce observable differences associated with selective linkage. Because family ties can only be constructed for individuals who are successfully linked across censuses, the resulting sample is not a random draw from the population. Linkage success varies systematically with demographic characteristics, location and socioeconomic status. Ignoring this selection would bias any estimates that use family structure or network composition. To address this issue, I reweight the linked sample using an inverse propensity-score approach that balances it toward the full census along observed

Figure A1.1.4: Family tree example



Notes: This figure is an illustrative example of a reconstructed family tree descending from an ancestor head. Children and spouses are organized hierarchically. Teal boxes denote individuals present in the 1940 census. Orange boxes denote individuals who are not members of the focal household head's blood lineage; they may nevertheless belong to other reconstructed family lineages.

characteristics. This proceudre follows Ward (2023) and Althoff et al. (2026).

The procedure begins by stacking, for each census year, the linked sample (individuals for whom family ties are observed) together with the full underlying census population in that year. Individuals are classified according to whether they are observed in the linked sample or not. I then estimate, separately by census year, a reduced-form model for the probability that an individual is successfully linked. The specification conditions on a rich set of pre-determined individual characteristics that are observed for both linked and unlinked individuals. These include demographic attributes (age group, sex, race, marital status), birthplace and state of residence, urban versus rural status, sector and skill group, industry sector, farming status, literacy (when available) and indicators for internal migration and immigration.

Using this model, I compute for each individual the predicted probability of being linked. The key identifying assumption is selection on observables: conditional on the included characteristics, linkage is assumed to be as good as random. Under this assumption, inverse-probability weighting recovers the population distribution of the characteristics included in the propensity model in expectation. Intuitively, individuals who are unlikely to be linked but nevertheless appear in the linked sample receive higher weight, while individuals who are very likely to be linked receive lower weight.

Linked individuals are assigned a weight equal to the ratio of the unconditional linkage rate in the census to their predicted linkage probability. This normalization ensures that the weighted linked sample reproduces the size of the linked population while matching the covariate distribution of the full census. Individuals who are not linked are dropped after the weights are constructed.

This reweighting step is performed separately for each census year to allow the linkage process and its correlates to vary over time. After reweighting, all family-level and individual-level moments used in estimation and counterfactual analysis are computed using these weights. The weighted linked sample therefore matches the full census more closely along the observed dimensions included in the propensity model.

The main strength of this approach is transparency and flexibility. It directly addresses selection into linkage using a large set of observables that capture demographic composition, geography and economic structure. It also nests simpler reweighting schemes—such as weighting by age or region alone—as special cases. It does not correct for selection on unobserved characteristics and only partially corrects for survival bias.

A1.2 Internal validation

This appendix assesses the internal reliability of the reconstructed kin network using the census records, the Census Tree crosswalks and the reconstructed lineage files. The exercises validate three distinct stages of the construction. First, I test whether alternative cross-census linking paths identify the same individual. Second, I test whether relationships inferred from the reconstructed network agree with relationships independently observed in a later census. Third, I examine how reliability changes with genealogical distance and the route through which lineage identifiers propagate. The last exercise disciplines the cleaning rules described in Appendix A1.1.

Cross-census identity links. I first test the identity links underlying the network. Whenever both a chained and a direct longer-horizon Census Tree link exist, the two routes should identify the same individual. I compare the chain 1900 $\rightarrow$ 1910 $\rightarrow$ 1920 with the direct 1900 $\rightarrow$ 1920 link on a deterministic one-in-sixteen sample of 1900 identifiers. Among individuals for whom both routes are observed, the two links identify the same 1920 individual in 99.67 percent of cases.

Table A1.2.1 shows similarly high agreement across observable characteristics. Differences by sex, race, nativity, urban status and region are small. The identity links therefore exhibit high internal consistency both overall and across the main demographic groups used in the analysis.

Table A1.2.1: Transitive consistency of cross-census identity links

Dimension	Group	Pairs	Agreement
All		1,475,252	99.67%
Sex	Male	783,728	99.50%
	Female	691,524	99.86%
Race	White	1,429,813	99.68%
	Non-white	45,439	99.38%
Residence	Rural	983,545	99.69%
	Urban	491,707	99.63%
Nativity	U.S.-born	1,345,120	99.67%
	Foreign-born	130,132	99.63%
Region	Midwest	620,399	99.70%
	Northeast	386,843	99.65%
	South	388,883	99.65%
	West	79,127	99.65%

Notes: The chained $1900 - 1910 - 1920$ link is compared with the direct $1900 - 1920$ Census Tree link on a deterministic one-in-sixteen sample of 1900 identifiers. The sample contains individuals for whom both routes exist; the absence of a direct long-horizon link is not treated as a disagreement. Characteristics are measured in 1900.

Relationship holdout. I next test relationships inferred by the network against information from a later census that is not used to form the ties. For a target census t_2, I take the reconstructed network as of the preceding wave t_1 and carry it forward using only the $t_1 \rightarrow t_2$ Census Tree link. The household structure observed in t_2 is then used to evaluate the predicted relationship. Where available, I use the census mother, father and spouse pointers; otherwise I use the relationship-pair crosswalk described in Appendix A1.1.

I conduct the exercise for 1900, 1910, 1920 and 1940 using a one-in-twenty sample of households in each target census. The validation unit is a pair of co-residents who already shared a lineage identifier before the target wave.

Because two individuals can share several lineage identifiers, I de-duplicate the validation sample in the same way as the network measures used in the analysis: each unordered pair enters once through the lineage implying the closest genealogical relationship. This yields 3.46 million distinct pairs from 5.66 million individual-by-lineage records.

I predict how two individuals are related using their reconstructed positions in the family tree, without looking at their relationships in the target census household. The prediction is based on how many generations separate them and how closely they are connected through the tree. Because the reconstructed paths cannot always distinguish blood relatives from relatives by marriage, I group these categories when comparing predicted relationships with those recorded in the census. I calculate agreement only when both the census and the reconstructed family

tree identify a specific relationship. Otherwise, I exclude the pair rather than count it as an error.

Agreement across waves. Table A1.2.2 reports the holdout results. Across the four target waves, the reconstructed relationship agrees exactly with the later census for 96.95 percent of verifiable pairs and at the broader close- versus extended-kin level for 98.83 percent. Exact agreement exceeds 96.7 percent in 1910, 1920 and 1940 and is somewhat lower in 1900, when the reconstructed network is based on fewer preceding census waves.

De-duplicating to the closest lineage improves agreement in every target wave. The gain ranges from about 0.3 percentage points in 1910 to 1.3 points in 1900, consistent with redundant, more distant lineage paths being less informative than the closest genealogical connection.

Table A1.2.2: Out-of-sample relationship agreement

Target wave	Pairs	Verifiable	Agreement	Broad class
Closest lineage				
1900 (from 1880)	255,010	239,605	95.50%	98.56%
1910 (from 1900)	958,204	925,524	97.38%	99.00%
1920 (from 1910)	1,033,095	989,467	97.11%	98.97%
1940 (from 1930)	1,210,862	1,159,247	96.76%	98.84%
Pooled	3,457,171	3,313,843	96.95%	98.83%
Individual-by-lineage records				
1900	338,685	318,703	94.25%	98.34%
1910	1,482,188	1,439,117	97.05%	98.92%
1920	1,638,711	1,578,776	96.69%	98.78%
1940	2,195,678	2,119,840	95.93%	98.53%
Pooled	5,655,262	5,456,436	96.35%	98.69%

Notes: Relationships are predicted from the reconstructed network as of the preceding wave and carried forward using the Census Tree link only. No target-wave household information enters the prediction. Verifiable pairs are those for which the target census identifies a relationship and the network assigns a kin category. Broad class groups parent–child, sibling and spouse ties as close kin and grandparent, aunt/uncle and cousin ties as extended kin. The sample is a one-in-twenty household sample in each target census.

Precision and recall by relationship. Overall agreement can conceal errors in rare relationship categories. Table A1.2.3 therefore reports precision, the fraction of predicted relationships confirmed by the census and recall, the fraction of relationships observed in the census recovered by the network.

Close relationships are measured accurately in both dimensions. Before the final distant-kin cleaning, precision is 98.73 percent for parent–child links, 98.25 percent for siblings and 95.06

percent for spouses, with recall above 94 percent for all three. The cleaning rules leave these relationships essentially unchanged.

Extended relationships are more difficult to validate because few extended kin co-reside and can therefore be observed directly in the census. Their validation moments are consequently based on much smaller and selected samples and errors are more concentrated in these categories because they require longer reconstructed paths. The final cleaning substantially raises precision for grandparents and aunts or uncles, while reducing recall by discarding the least reliable distant paths.

Table A1.2.3: Precision and recall by relationship

		Before distant-kin cleaning		After distant-kin cleaning	
Relationship	Observed	Precision	Recall	Precision	Recall
Parent–child	1,872,634	98.73%	95.53%	98.73%	95.53%
Sibling	905,400	98.25%	95.54%	98.47%	95.25%
Spouse	576,503	95.06%	94.87%	95.06%	94.87%
Grandparent	11,778	39.42%	79.39%	68.49%	74.14%
Aunt/uncle	528,763	30.38%	26.59%	66.31%	15.51%
Cousin	764,401	32.99%	33.67%	54.32%	21.95%

Notes: Closest-lineage pairs pooled across the 1900, 1910, 1920 and 1940 target waves. "Observed" is the number of relationships recorded by the target census and is the denominator of recall. All columns apply closest-lineage de-duplication. The final two columns additionally restrict distant relationships to direct lineage paths and impose the age-plausibility rule described in Appendix A1.1. Ambiguous or unmappable census relationships are excluded from the validation.

Validation using census parent identifiers. The census identifies parents living in the same household, providing a direct way to validate reconstructed parent–child and sibling relationships. Among reconstructed parent–child pairs living together, the census confirms the reconstructed father in 97.23 percent of cases and the reconstructed mother in 97.85 percent. In fewer than one percent of cases, the census identifies a different parent. Cases where the census does not identify a parent are reported separately and are not counted as disagreements. Sibling relationships show the same pattern. Most reconstructed sibling pairs either share both observed parents or share one parent, as expected for full siblings, half-siblings and households in which only one parent is observed. Direct contradictions are rare. Table A1.2.4 shows similar results across waves; the weaker information in 1900 is driven primarily by missing parent pointers rather than contradictory links.

Propagation depth and entry route. The principal weakness of the reconstructed network arises when lineage identifiers are propagated through long genealogical paths. Table A1.2.5

Table A1.2.4: Validation against census parent identifiers

	1900	1910	1920	1940
Parent–child pairs: fathers				
Confirmed	95.57%	97.28%	97.50%	97.31%
Contradicted	0.39%	0.46%	0.64%	0.65%
Pointer missing	4.05%	2.26%	1.85%	2.04%
Parent–child pairs: mothers				
Confirmed	96.93%	97.87%	97.98%	97.93%
Contradicted	0.46%	0.55%	0.62%	0.62%
Pointer missing	2.61%	1.59%	1.40%	1.45%
Sibling pairs				
Both parents shared	56.56%	82.30%	82.05%	82.28%
One parent shared	25.28%	13.34%	12.96%	12.33%
Contradicted	0.12%	0.13%	0.11%	0.14%
No usable pointer	17.93%	4.23%	4.83%	5.20%

Notes: Closest-lineage pairs. For parent–child pairs, the older individual is checked against the younger individual's `momloc` or `poploc` according to the parent's sex. For sibling pairs, the two individuals are compared using both parent identifier. Missing pointers are reported separately and are not treated as contradictions. A small residual category of sibling pairs is identified from the household relationship codes rather than usable parent pointers and is omitted from the displayed rows.

reports relationship agreement by the number of tree steps separating a pair and by the route through which the lineage identifier was inherited.

Direct-blood paths remain highly reliable through two steps and retain 87.90 percent agreement at three steps. Spousal, descendant and collateral routes also perform well at two steps, but their accuracy falls substantially at three steps, to between 48 and 58 percent. At four steps, agreement is below one third for every route and at five or more steps it is below 15 percent. The loss of reliability is therefore concentrated in longer propagation paths rather than in the short connections that constitute most of the reconstructed network.

This pattern motivates restricting distant propagation rather than imposing a general restriction on extended kin. The cleaning rules retain short paths, preserve longer direct-blood connections where they remain informative and remove the distant propagation routes for which the validation indicates substantially greater classification error.

Choice of cleaning rules. Table A1.2.6 compares alternative restrictions for distant relationships. Starting from closest-lineage de-duplication, requiring paths of three or more steps to follow the direct-blood route raises overall agreement from 96.95 to 97.56 percent and extended-kin precision from 45.75 to 50.16 percent. Adding the age-plausibility restriction raises agreement further to 97.71 percent and extended-kin precision to 60.05 percent, while retaining

Table A1.2.5: Agreement by propagation depth and entry route

Tree steps	Direct blood	Spouse entry	Descendant	Collateral
1	97.33%	96.62%	87.02%	25.95%
2	97.90%	97.41%	95.85%	95.15%
3	87.90%	58.22%	48.24%	51.23%
4	13.15%	30.85%	31.07%	27.76%
≥ 5	14.42%	8.70%	1.40%	3.30%
Pair records	3,167,672	2,353,333	111,092	23,165

Notes: Individual-by-lineage pair records pooled across the four target waves. The route refers to the way the lineage identifier reached the pair. Direct blood means that neither side inherited the identifier through a spousal, descendant or collateral route. Records at zero tree steps cannot be assigned a relationship and are omitted from the agreement rows; the bottom row reports all records by entry route.

substantially more extended-kin recall than a blanket two-step depth cap. This combination is used in the baseline construction.

The alternatives clarify the trade-off. Restricting all relationships to two steps produces slightly higher overall agreement, 97.79 percent, but lowers extended-kin recall to 38.29 percent and yields slightly lower extended-kin precision than the adopted rule. A generation-only classifier discards information contained in tree position, while requiring confirmation through two separate lineages removes a large share of otherwise well-measured kin links. The adopted rule therefore targets the part of the network for which the holdout exercise identifies a clear loss of reliability without discarding informative extended relationships more generally.

Table A1.2.6: Alternative cleaning rules for distant kin

		All pairs		Extended kin	
Rule	Pairs	Agreement	Broad	Precision	Recall
Closest lineage only	3,457,171	96.95%	98.83%	45.75%	56.42%
Direct blood for $d \geq 3$	3,420,401	97.56%	99.37%	50.16%	49.18%
Depth cap at $d \leq 2$	3,356,049	97.79%	99.56%	59.30%	38.29%
Generation-only classifier	3,457,171	97.44%	99.41%	59.42%	44.65%
Require two lineages	1,434,824	96.77%	99.32%	51.04%	16.37%
Drop unknown-branch paths	3,361,722	96.95%	98.83%	45.76%	54.83%
Age plausibility for $d \geq 3$	3,457,171	97.14%	99.01%	49.76%	55.73%
Generation-only and direct blood	3,420,401	97.62%	99.48%	60.65%	42.07%
Direct blood and age plausibility	3,420,401	97.71%	99.52%	60.05%	48.61%

Notes: Closest-lineage pairs pooled across the four target waves. Extended kin comprise grandparent, aunt/uncle and cousin relationships. Recall uses the fixed set of extended relationships recorded by the target census in the closest-lineage sample, so dropping pairs cannot mechanically improve recall. The final row combines the two distant-kin restrictions used in the baseline construction. Because the same validation sample is used to compare rules, differences across rows are used for rule selection rather than as an independent out-of-sample test.

Genealogical consistency and de-duplication. I conclude with two mechanical checks on the reconstructed network. First, I search for distributions inconsistent with a valid genealogy. Such cases are rare. A small fraction of individual-by-lineage records place two individuals on the same reconstructed tree node, imply a generation difference larger than the number of tree steps, or lack a generation index. Age-based contradictions among predicted parent–child and sibling pairs are similarly uncommon. Records for which the relationship cannot be resolved are not used in relationship-specific measures.

Second, I quantify the importance of de-duplicating overlapping lineages. Multiple lineage membership becomes more common as the reconstructed panel deepens because the same pair can become connected through several ancestors. In the pooled validation sample, 64.2 percent of individual-by-lineage pair records belong to a pair reachable through more than one lineage. Treating these lineage records as distinct kin pairs would therefore mechanically inflate kin counts and increasingly do so in later waves. The baseline measures instead retain each unordered pair once through the lineage that implies the closest genealogical connection.

A1.3 External benchmarks for the family network measures

In this section I assess the family network measures using external data against three external sources: the Social Connectedness Index of Bailey, Cao, Kuchler, Stroebel, and Wong (2018b), historical migration networks from Stuart and Taylor (2021) and measures of close-knit commuities by Raz (2025). For the first comparison, I construct a comparable Family Connectedness Index (FCI) for two locations i, j:

$$FCI_{i,j} = \frac{Fam_{i,j}}{Fam_i \times Fam_j}$$

where $Fam_{i,j}$ is the total number of family members from region i living in j and Fam_c is the population of commuting zone c. Table A1.3.1 shows a positive correlation between the historical family network and the modern Social Connectedness Index.

For the historical comparison, I use the network connectedness index of Stuart and Taylor (2021), who construct birth-town networks using census and Duke data for 1916–1936. The network index measures the excess correlation in location choices among migrants from the same birth town relative to nearby towns. Formally, it equals the expected increase in flows from j to k when a randomly chosen migrant makes the same move. I investigate their correlation with a measure of connectedness, controlling for migration distance:

$$\text{Share FM}_{i,j} = \frac{Fam_i, j}{Fam_i}$$

Their measure uses origin states and county destinations, so I aggregate county destinations to commuting zones. Table A1.3.1 again shows a positive correlation.[36]

Table A1.3.1: Correlation of Family Network Measures with Existing Data Sources

VARIABLES	(1) SCI (Bailey et al. 2018)	(2) Network Index (Stuart & Taylor 2021)
FCI (standar.)	0.337***	
	(0.124)	
Share FM		5.467**
		(2.269)
Observations	261,002	2,818
R-squared	0.256	0.135

Notes: The unit of observation is a commuting-zone pair. *** denotes significance at the 1 percent level and ** at the 5 percent level. Regressions include log distance between CZs as a control.

I further assess my data with measures of close-knit communities compiled by Raz (2025). He compiles measures constructed from full-count U.S. censuses using first names of individuals. The Local Name Index (LNI), adapted from Fryer and Levitt (2004) is given first name n, locality l, gender g and census year t

$$LNI_{nlgt} = 100 \times \frac{\Pr(n \mid l, g, t)}{\Pr(n \mid l, g, t) + \Pr(n \mid -l, g, t)}.$$

Names that are disproportionately common in a locality receive high scores, while names that are common elsewhere receive low scores. The county-level index is the average score of local children.

I construct an analogous statistic using traced family lineages rather than children's first names. For each traced lineage f and commuting zone l, I define

$$FAM_LNI_{fl} = 100 \times \frac{\Pr(f \mid l)}{\Pr(f \mid l) + \Pr(f \mid -l)}.$$

Here $\Pr(f \mid l)$ is lineage f's population-weighted share of the traced population in commuting zone l and $\Pr(f \mid -l)$ is its share among traced individuals elsewhere in the country. Each individual is assigned the score of their own lineage–commuting-zone cell and the CZ-level measure is the population-weighted average across residents. A high value therefore identifies places whose residents disproportionately belong to lineages that remain geographically concentrated

[36] I will validate these measures more with recent social connectedness variables from Raz (2025).

there. Unlike names, lineage membership is not chosen anew each generation, so the measure is constructed using all traced family members rather than only children.

I then compare the family-network LNI the LNI index computed by Raz (2025) as well as other measures of close-knit commuities such as intra-community marriage, tightness of family norms, religious homogeneity, kinship propinquity and a summary strength-of-family-ties index. All measures are originally defined at the county level. I aggregate them to commuting zones using population weights the crosswalk files by Eckert, Gvirtz, Liang, and Peters (2020).

Table A1.3.2: Family-network concentration and historical measures of close-knit communities

	Family-network LNI	R^2	Observations
Panel A. Community structure and norms			
Local Name Index	2.375***	0.314	722
	(0.364)		
Intra-Community Marriage	0.016	0.007	722
	(0.013)		
Tight Norms Index	0.165***	0.050	722
	(0.040)		
Religious Homogeneity Index	0.204***	0.056	722
	(0.062)		
Panel B. Kinship and family ties			
Median Kinship Propinquity	0.038***	0.164	722
	(0.008)		
Kinship Propinquity Rate	0.016***	0.049	722
	(0.003)		
Strength of Family Ties Index	0.164***	0.031	722
	(0.047)		

Notes: The table relates family-network concentration to historical measures of close-knit communities from Raz (2025). Each row reports a separate commuting-zone-level regression. The dependent variable is the measure listed in the first column, aggregated from counties to commuting zones using population weights. The explanatory variable is the family-network Local Name Index (LNI). Standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

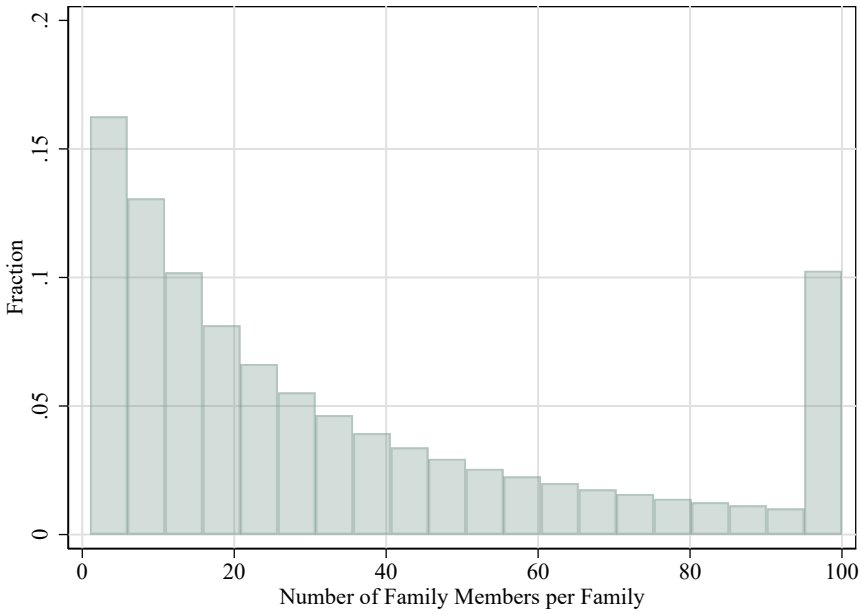
A1.4 Summary statistics of families

Table A1.4.1: Summary statistics of family characteristics in 1940

Characteristic	25th Percentile	Median	75th Percentile
Family Size (No. of Members)	9	22	51
Generational Depth (Avg.)	2.8	3.2	3.5
Number of families per Indiv.	1	2	3

Notes: This table reports summary statistics at the family level. Family size refers to the total number of individuals linked to a focal person. Generational depth is the average number of generations represented in a family calculated from the initial household head in 1850.

Figure A1.4.1: Distribution of family members



Notes: This figure shows the distribution of family sizes in 1940 from the final dataset.

Table A1.4.2: Summary statistics: full census vs. linked family-ties samples

	Full census	Ties (unweighted)	Ties (weighted)	Unw. − Census	Wtd. − Census
Observations (Millions)	132	32.2	32.2		
Panel A: Demographics and residence					
Age (years)	31.05	30.92	31.37	-0.13	0.31
Male (%)	50.14	50.89	49.60	0.75	-0.54
White (%)	89.80	99.18	88.42	9.38	-1.38
Married (%)	45.85	48.75	45.97	2.90	0.12
Urban (%)	55.58	42.03	56.86	-13.56	1.27
Homeownership (%)	44.14	50.29	47.41	6.15	3.27
Panel B: Employment and sector					
Agriculture employment (%)	18.39	24.01	16.47	5.62	-1.92
Manufacturing employment (%)	34.11	32.27	33.25	-1.84	-0.86
Self-employed (%)	19.27	25.25	18.94	5.98	-0.33
Farm resident (%)	23.08	31.19	21.97	8.11	-1.11
Panel C: Mobility					
Moved across states (%)	5.40	5.71	5.61	0.30	0.21
Moved within state (%)	53.24	51.16	52.53	-2.08	-0.71
Panel D: Monetary outcomes					
Income (real $)	44199	42295	45271	-1904	1072
Rent value (real $)	66.25	68.34	63.26	2.09	-2.99

Notes: Entries are sample means (pooled across the census waves used in the analysis). "Ties" samples consist of individuals for whom extended-family links can be constructed. The weighted ties column applies inverse-propensity weights for linkage so that the linked sample matches the full census on observed characteristics. Percentage variables are reported in percent. Income and rent values are reported in the same real units as used in the main analysis.

Table A1.4.3: Summary statistics by individual characteristics

Variable	Subgroup	FM in Origin	FM not in Origin	Avg. Distance (km)	Avg. Mig. Rate	Direct Mig. Rate	Direct Mig. Share (%)
Race	Non-White	44.44	19.67	543.95	20.52	6.62	32.29
	White	18.77	13.74	542.92	15.58	6.28	40.32
Sex	Female	16.49	10.91	642.76	10.19	4.19	41.09
	Male	19.01	13.87	538.91	15.93	6.41	40.25
Age	> 30	18.56	13.35	549.85	13.65	5.41	39.61
	16–30	20.13	14.67	526.29	23.62	9.89	41.89
Marital Status	Not Married	16.31	11.53	654.16	11.60	4.59	39.57
	Married	19.11	13.91	535.23	15.97	6.44	40.33
College	No	19.80	14.17	519.66	14.60	6.22	42.61
	Yes	11.80	11.01	665.30	23.00	6.68	29.05
Sector	Agriculture	21.88	15.34	423.06	13.05	6.44	49.33
	Manufacturing	20.25	14.68	548.73	16.80	6.82	40.60
	Services	16.21	12.48	571.36	17.35	6.01	34.66
	Not Employed	16.39	12.42	712.12	11.81	4.72	40.01
Born In-State	No	10.92	13.02	679.77	22.85	8.36	36.57
	Yes	22.93	14.35	409.75	11.89	5.22	43.94
Parent	No	15.36	12.61	619.59	16.04	6.24	38.91
	Yes	20.03	14.14	516.46	15.45	6.30	40.76
Homeownership	No	18.35	13.43	533.54	24.45	9.39	38.39
	Yes	19.29	14.42	548.10	9.21	4.15	45.08
Income	Below Median	20.89	14.79	549.45	17.98	7.94	44.18
	Above Median	15.60	11.65	571.88	18.26	5.23	28.65

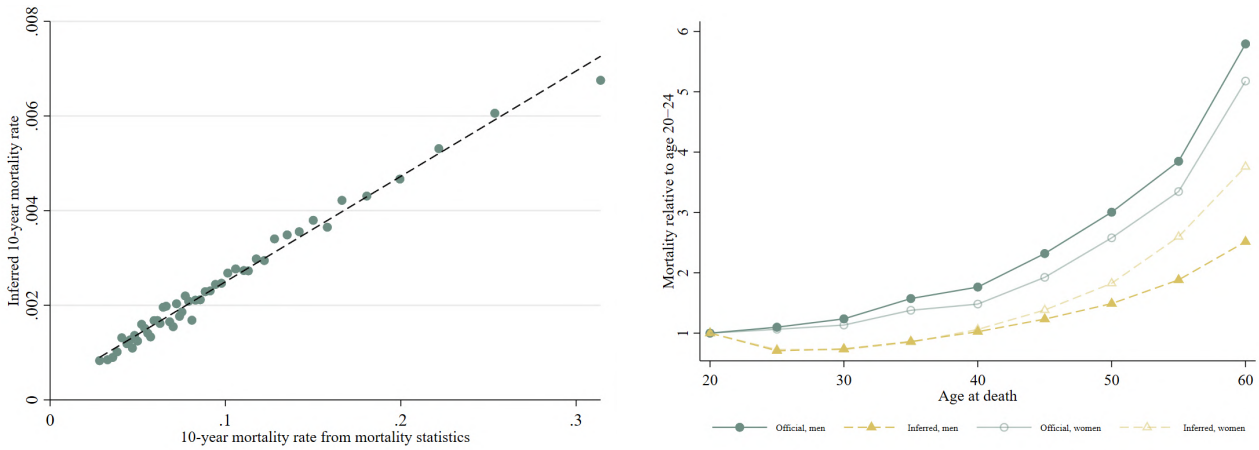
Notes: The table reports subgroup-level averages of family members (FM), migration behavior, and distances. Distances are measured in kilometers. Direct migration share refers to the share of migrants who move directly rather than sequentially. All values are rounded to two decimal places.

A1.5 External benchmark for inferred deaths

Deaths are inferred when an individual who is married in census t reports being widowed in census $t + 10$. I validate this measure against official mortality statistics from the U.S. death-registration area, digitized from the annual *Mortality Statistics* volumes of the U.S. Bureau of the Census by Miller (2006). Because the inferred measure requires marriage, widowhood and successful linkage across censuses, its level need not coincide with population mortality. I therefore compare variation across state, period, sex and age cells and the age profile of mortality.

Figure A1.5.1 shows that the inferred measure closely tracks these independent mortality patterns. Across 2,169 state-by-period-by-sex-by-age cells, the population-weighted correlation between inferred and official mortality is 0.838. Panel (b) shows that the inferred deaths also reproduce the increase in mortality with age for both men and women, although the gradient is flatter at older ages.

Figure A1.5.1: Validation of inferred deaths



(a) Mortality across cells

(b) Mortality by age and sex

Notes: Panel (a) compares inferred ten-year mortality with official mortality across state-by-period-by-sex-by-age cells. Points are weighted by the population eligible to generate an inferred death and the dashed line shows the weighted linear fit. Panel (b) compares age profiles separately by sex, normalizing mortality to one at ages 20–24. Official mortality comes from the historical death-registration statistics. Inferred deaths are identified from transitions from married to widowed status across consecutive censuses.

A1.6 Summary statistics of death variables

Table A1.6.1: Summary Statistics For Death Variables

	Dead individuals	**Dead < 40**	**Married (census)**
	(1)	(2)	(3)
Observations	73,330	21,629	49,730,378
Panel A: Demographics			
Age (years)	46.50	32.39	41.07
Male (share)	0.70	0.68	0.50
White (share)	0.89	0.84	0.91
Panel B: Sector and residence			
Agriculture (share)	0.22	0.15	.25
Manufacturing (share)	0.35	0.37	0.34
Services (share)	0.44	0.47	0.4
Farm resident (share)	0.19	0.16	0.24
Homeowner (share)	0.54	0.38	0.48
In labor force (share)	0.70	0.72	0.53
Panel C: Outcomes			
Occupation score	25.36	25.33	24.29
House value	5,488.79	4,399.91	6,366.03
Rent value	40.65	31.78	47.50
Literate (share)	0.97	0.98	0.95

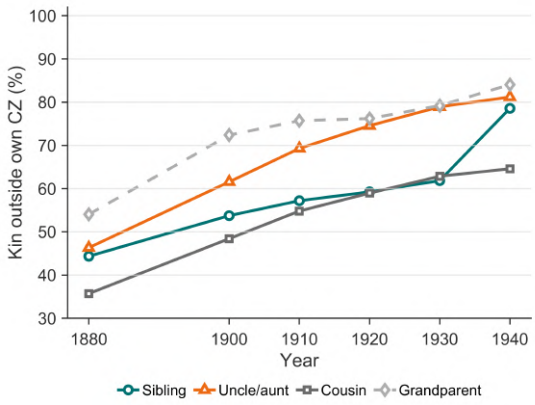
Notes: Entries report sample means. "Dead individuals" are defined by as people whose spouse became widowed between 1930-1940; "Dead < 40" "Dead individuals" that are below 40 years old. Married individuals are defined by having having a spouse present in the household (`marst=1`). Additional sample counts: number of married individuals linkable to the ties data = 10,533,803; number of constructed ties = 31,411,504. Variables measured as shares take values in $[0, 1]$.

A2 Empirics

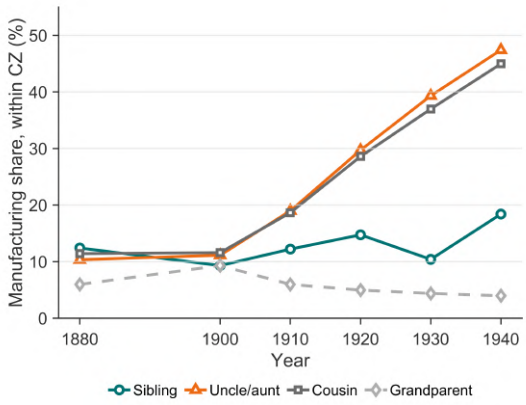
A2.1 The evolution of family networks across sectors and space

This appendix provides additional descriptive evidence on how family networks changed during structural transformation. The main text shows that cross-CZ family connections became both more geographically dispersed and more oriented toward manufacturing. Here I document that these patterns are visible across relationship types and throughout the distance distribution.

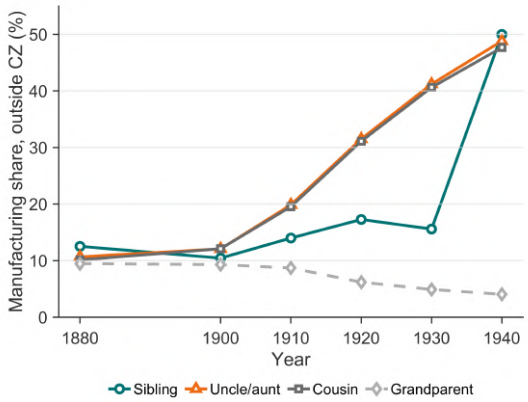
Figure A2.1.1: Family networks across sectors and space



(a) Relatives outside own CZ



(b) Manufacturing, local relatives



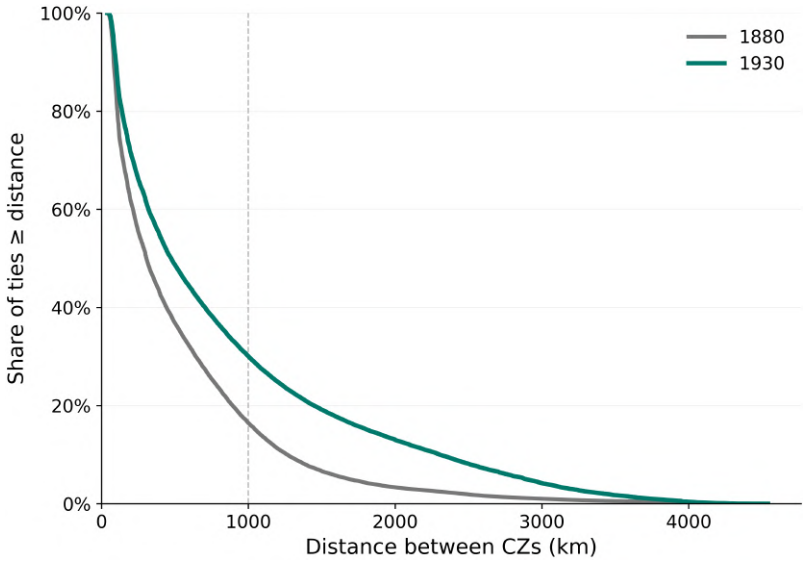
(c) Manufacturing, non-local relatives

Notes: Panel (a) reports the share of relatives living outside the individual's CZ by relationship and census year. Panels (b) and (c) report the manufacturing share among relatives employed in agriculture or manufacturing, separately for relatives living within and outside the individual's CZ.

Panel (a) shows a broad increase in the geographic dispersion of family networks. By 1940, roughly 80% of aunts and uncles and grandparents and about two-thirds of cousins live outside the individual's CZ. Panels (b) and (c) show a simultaneous shift in sectoral composition. Manufacturing becomes increasingly prevalent among aunts and uncles and cousins both within

and outside the individual's CZ, while the pattern is weaker for siblings and grandparents.

Figure A2.1.2: Spatial dispersion of cross-CZ family ties



Notes: Share of cross-CZ family ties spanning at least a given distance in 1880 and 1930. Distances are measured between CZ centroids.

The increase in geographic dispersion is not driven only by a few very long-distance connections. Among cross-CZ ties, mean distance rises from 542 km in 1880 to 849 km in 1930, the median from 306 to 475 km and the 90th percentile from 1,266 to 2,303 km. The share extending more than 1,000 km increases from 16.5% to 30.1%. Together, these patterns show that structural transformation coincided with a broad reorientation of family networks across both sectors and space.

A2.2 Kinship and place-based social connections

The main analysis focuses on family connections, but kinship is not the only social relationship that can shape economic choices. Workers from the same hometown may exchange information, help one another find jobs or follow similar migration routes even when they are not related. Migrants from the same state or country of birth may likewise concentrate in particular destinations. A natural concern is therefore that the predictive power of kinship simply reflects these broader place-based connections.

I examine this possibility by asking whether kinship predicts sector and migration choices among workers who share the same hometown and birthplace. The exercise is complementary to the quasi-experimental evidence in the main text. It is not intended to identify the causal effect of kinship relative to other forms of social connection. Instead, it asks whether family ties contain information about workers' choices beyond the broader origin-based connections that they share with other people from the same place.

Migration. I use linked individuals in the full-count censuses from 1900 to 1940. For each census year $t \in \{1910, 1920, 1930\}$, I select linked individuals aged 18–45 and observe their location ten years later. For computational tractability, I take a 2% random sample of individuals and pair each person with each of the 40 largest commuting zones (CZs) other than the person's current CZ. A person's hometown is the county in which they live at t. Birthplace is the state of birth for the U.S.-born and country of birth for the foreign-born, since the census does not report town of birth.

For person i, origin county $c(i)$, birthplace $b(i)$ and potential destination d, I estimate

$$\text{Move}_{idt} = \beta \, \text{Kin}_{idt} + \alpha_{c(i)dt} + \gamma_{b(i)dt} + \varepsilon_{idt}, \tag{A2.2.1}$$

where $\text{Move}_{idt} = 100$ if person i lives in destination d at $t + 10$. The variable Kin_{idt} equals one if a sibling, grandparent, aunt or uncle, or cousin lives in d at time t.

The fixed effects absorb two broad sources of place-based social connection. Origin-county $\times$ destination $\times$ year fixed effects absorb anything shared by people currently living in the same county that makes a particular destination attractive. This includes established migration routes from the hometown: for example, the share of residents of county c at $t - 10$ who live in d at t. Birthplace $\times$ destination $\times$ year fixed effects analogously absorb destination-specific connections shared by people born in the same state or foreign country. The coefficient β is therefore identified from differences in whether individuals have relatives at a destination after absorbing the migration patterns associated with both their current hometown and their birthplace.

Family connections remain strongly related to destination choice. With only origin-county and destination fixed effects, having kin in a destination raises the probability of moving there by 1.74 percentage points. Absorbing the full set of hometown-to-destination and birthplace-to-destination fixed effects reduces the coefficient to 0.95 percentage points, compared with a migration rate of 0.11% for person–destination pairs without kin. Thus, more than half of the original association remains and workers are about ten times as likely to move to a destination where relatives already live.

The result is not driven by a small set of established hometown migration routes. Among destinations to which nobody from the individual's hometown had moved during the previous decade, kin at the destination still raise migration by 0.52 percentage points, relative to a baseline migration probability of 0.014%. The association is also present for destinations more than 500 kilometers from the origin and among foreign-born workers, for whom the birthplace fixed effects absorb country-of-origin-specific destination patterns. Estimating the effects of hometown and birthplace migration shares directly leads to the same conclusion: both predict destination choice, but adding family ties leaves their coefficients largely unchanged. Kinship therefore captures variation in destination access that is distinct from these broader origin-based

migration patterns.

Sector choice. I use a similar exercise to distinguish sector-specific family connections from broader characteristics shared by workers from the same places. The sample contains a 30% draw of agricultural workers aged 18–45 who are linked to the following census. I estimate

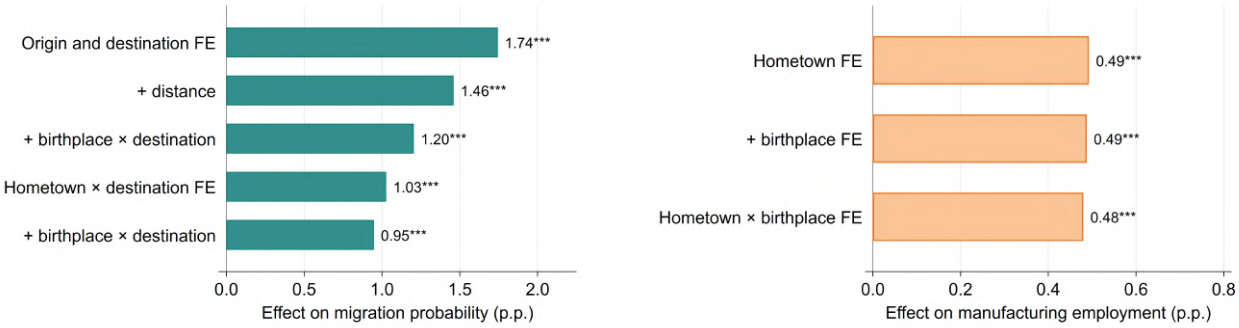
$$\text{Manuf}_{i,t+10} = \theta\,\text{KinM}_{it} + \lambda\,\text{KinE}_{it} + \kappa\,n_{it} + \delta_{c(i)b(i)t} + u_{it}, \tag{A2.2.2}$$

where KinM_{it} indicates that a sibling, grandparent, aunt or uncle works in manufacturing, KinE_{it} indicates that at least one such relative is employed and n_{it} is the number of observed relatives. The specification includes origin-county × birthplace × year fixed effects. It therefore compares workers from the same hometown and birthplace while conditioning on whether they have employed relatives and on the number of relatives observed. Standard errors are clustered by origin county.

Having a relative in manufacturing raises the probability that an agricultural worker enters manufacturing over the next decade by 0.48 percentage points, relative to a baseline manufacturing-entry rate of 6.75%. The estimate remains 0.43 percentage points after adding age × year fixed effects. What matters is not simply having relatives who are economically active. Conditional on manufacturing kin, the indicator for having any employed relative enters negatively. Manufacturing relatives also reduce the probability of remaining in agriculture by 1.10 percentage points, while the corresponding increase in services is smaller and imprecisely estimated. The association is therefore specific to the sector in which relatives work rather than to employment among relatives more generally.

Interpretation. These exercises show that the role of kinship in the main analysis is not simply a proxy for broad migration patterns shared by people from the same hometown or birthplace. Family connections predict both where workers move and which sector they enter after flexibly absorbing these place-based similarities. The distinction is particularly clear for migration: relatives predict moves even toward destinations with no recent migration from the worker's hometown.

The exercise does not imply that family ties are the only relevant social connections. The historical censuses do not observe friendships, neighbors or other non-kin relationships directly, and unobserved preferences may also be correlated within families. The results instead establish a narrower point that is important for the quantitative analysis: kinship contains substantial destination- and sector-specific information that is not captured by broad origin-based social connections.



(a) Destination choice

(b) Manufacturing entry

Figure A2.2.1: Kinship beyond hometown and birthplace connections

Notes: Panel (a) reports estimates of β from equation (A2.2.1). Panel (b) reports estimates of θ from equation (A2.2.2). Outcomes are measured in percentage points. Each specification successively adds the fixed effects or controls indicated in the figure. In panel (a), hometown denotes the county of residence at time t and birthplace denotes the state of birth for the U.S.-born and country of birth for the foreign-born. All fixed effects are interacted with census year. Standard errors are clustered by origin county. Stars denote significance at the 10%, 5% and 1% levels.

Table A2.2.1: Kinship beyond place-based connections

	Coefficient	s.e.	Mean, no kin	Observations
Panel A. Destination choice: kin at destination				
All	0.948	(0.024)	0.105	19,231,125
By census year				
1910	1.239	(0.047)	0.107	5,469,044
1920	1.071	(0.041)	0.127	6,370,686
1930	0.706	(0.028)	0.085	7,391,395
By birthplace				
Born in the United States	0.948	(0.024)	0.105	19,004,821
Foreign-born	0.803	(0.174)	0.133	180,489
By destination				
No recent hometown migration	0.519	(0.037)	0.014	7,887,628
Positive recent hometown migration	1.003	(0.027)	0.170	11,342,216
At least 500 km from origin	0.556	(0.023)	0.051	14,846,361
Panel B. Sector choice: kin in manufacturing				
Manufacturing	0.480	(0.106)	6.747	1,325,569
Manufacturing, age × year FE	0.425	(0.106)	6.747	1,325,569
Services	0.235	(0.135)	12.471	1,325,569
Agriculture	−1.104	(0.193)	66.536	1,325,569

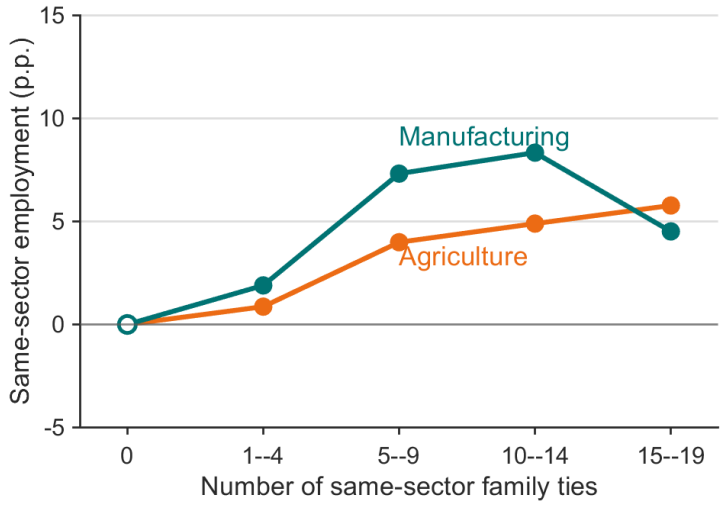
Notes: Outcomes are measured in percentage points. Panel A estimates equation (A2.2.1) with origin county × destination × year and birthplace × destination × year fixed effects on the indicated subsample. "No recent hometown migration" denotes destinations at which no resident of the individual's origin county at $t-10$ lives at t. Panel B estimates equation (A2.2.2) with origin county × birthplace × year fixed effects for the indicated sector at $t+10$. Standard errors are clustered by origin county.

A2.3 Family connections and sectoral choice

A2.3.1 Sectoral concentration in family networks

Figure A2.3.1 reports the conditional relationship between sector employment and the number of same-sector relatives residing in the individual's CZ. The regressions absorb CZ fixed effects and control for individual characteristics and total family size, with individuals who have no same-sector relatives as the omitted category. Sector employment rises strongly with same-sector family presence in both manufacturing and agriculture.

Figure A2.3.1: Sectoral ties and sector choice



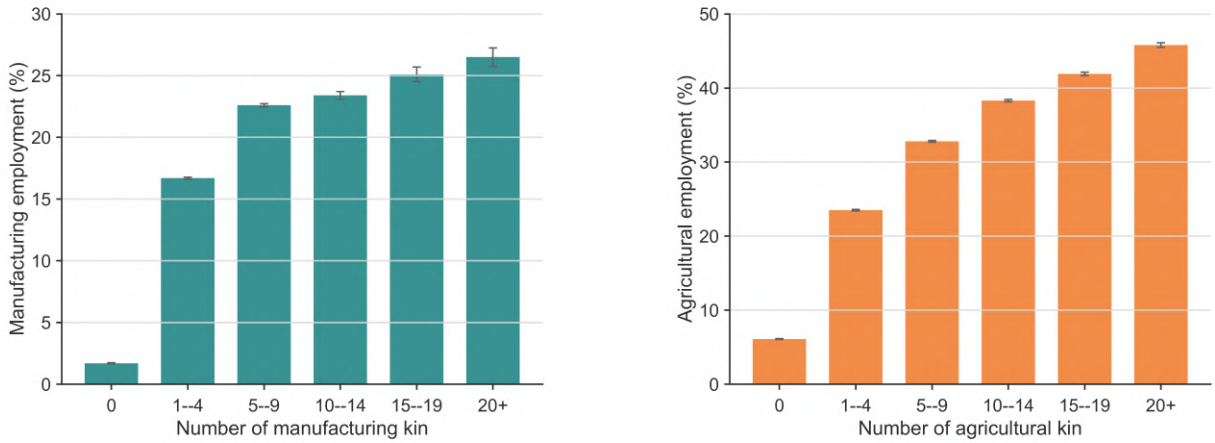
Notes: Coefficients from regressions of sector employment on bins of same-sector relatives residing in the individual's CZ; zero same-sector relatives is the omitted category. Regressions include CZ fixed effects, individual controls and total family size. Vertical bars denote 95% confidence intervals. $N = 9{,}485{,}355$.

Figure A2.3.2 reports the corresponding unconditional sectoral employment rates in 1900. The raw gradients are large in both sectors. Manufacturing employment is below 2% among individuals with no local manufacturing relatives and exceeds 20% among individuals with at least five. Agricultural employment rises from roughly 6% among individuals with no local agricultural relatives to more than 30% among those with at least five and exceeds 40% at higher levels of agricultural family presence.

A2.3.2 Deaths of sector-employed relatives: robustness

The main text focuses on the loss of manufacturing-employed relatives because manufacturing is the central destination sector in the structural-transformation analysis. This section reports

Figure A2.3.2: Sectoral employment and same-sector family ties



(a) Manufacturing (b) Agriculture

Notes: Mean sectoral employment by the number of same-sector relatives residing in the individual's CZ in 1900. Pairwise family relationships are de-duplicated as described in Section 2. Vertical bars denote 95% confidence intervals.

complementary evidence across sectors and examines whether the manufacturing response can be explained by more general consequences of a family death.

Summary statistics This section documents the empirical sample and its characteristics relative to the census.

Alternative sectors. Figure A2.3.3 reports analogous event studies around the deaths of relatives employed in agriculture and services. In each panel, the outcome is employment in the deceased relative's sector. The direction of the response is similar to the manufacturing result: following the loss of a sector-employed relative, subsequent employment in that sector declines. The relationship between sector-specific family connections and sectoral attachment is therefore not unique to manufacturing.

Placebo deaths and specification robustness. A concern with the death design is that a family death may affect subsequent outcomes even when the deceased relative does not provide a sector-specific labor-market connection. I therefore repeat the analysis using deaths of non-employed relatives. Such deaths may remove emotional, financial or household support, but they do not remove a contemporaneous employment connection to manufacturing. The corresponding estimate is close to zero.

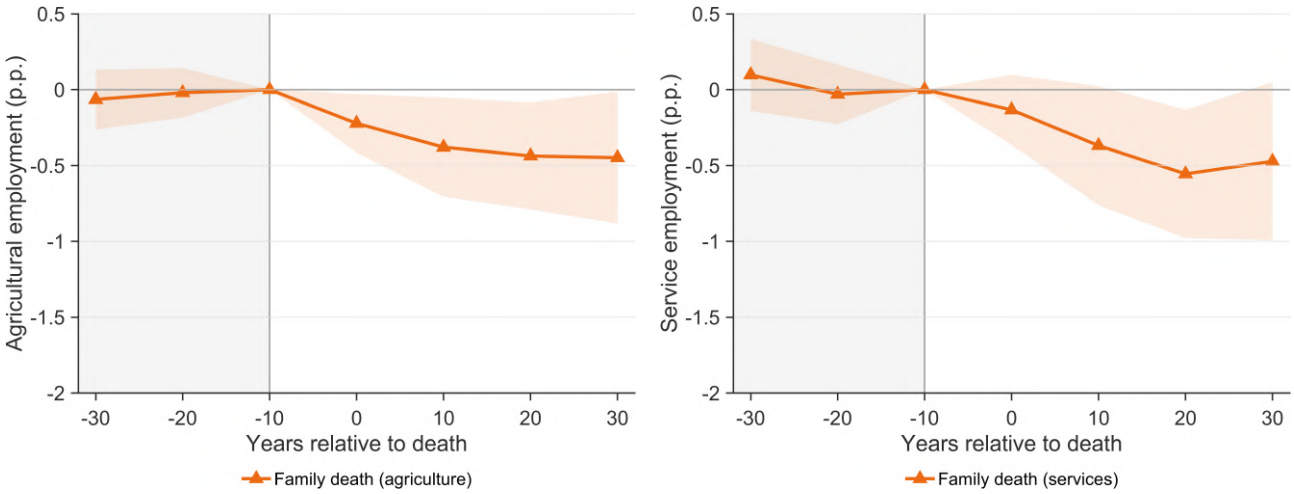
Table A2.3.2 reports this placebo together with alternative specifications of the manufacturing-death design. The estimates are similar when individual and family controls are omitted and under alternative functional forms. The stability of the effect across specifications, together

Table A2.3.1: Empirical sample summary statistics

	Mean				Normalized difference	
	Full sample	Treated	Non-treated	Census	Full vs. census	Treated vs. non-treated
Panel A. Individual characteristics						
Age	32.198	31.498	32.204	33.281	-0.084	-0.045
Female	0.487	0.465	0.488	0.490	-0.005	-0.044
White	0.912	0.963	0.912	0.894	0.061	0.215
Married	0.573	0.554	0.573	0.562	0.023	-0.038
Number of children	1.171	1.059	1.172	1.147	0.015	-0.069
Owns home	0.547	0.514	0.548	0.475	0.145	-0.068
Urban	0.511	0.635	0.510	0.524	-0.027	0.255
Farm residence	0.282	0.170	0.283	0.267	0.034	-0.274
In school	0.134	0.136	0.134	0.113	0.072	0.006
In manufacturing	0.102	0.175	0.102	0.105	-0.009	0.215
In agriculture	0.141	0.087	0.142	0.130	0.034	-0.171
Panel B. Family network						
Relatives in family tree	10.181	12.059	10.164	—	—	0.077
Relatives in manufacturing	1.418	2.303	1.409	—	—	0.413
Relatives in agriculture	3.460	2.973	3.465	—	—	-0.127
Relatives in services	2.801	3.126	2.798	—	—	0.106
Relatives not employed	2.502	3.657	2.492	—	—	0.062
Has a relative in manufacturing	0.770	0.897	0.769	—	—	0.350
Has a relative in agriculture	0.863	0.829	0.863	—	—	-0.096
Manuf. share of employed relatives	0.202	0.302	0.201	—	—	0.471
Person-years	22,115,970	177,768	21,938,202			

Notes: The table reports means of the linked panel at the person-year level, pooling census waves from 1900 to 1940, with each person-year counted once. All sample columns use post-stratification weights, the ratio of full-count census to linked-sample observations within cells defined by age band, sex, race, state, urban status and farm residence. Treated individuals have a manufacturing-relative death between 1900 and 1940. Panel B counts relatives in the individual's family tree at the census, excluding the individual. Panels B has no census counterpart. Normalized differences are $(\bar{x}_1 - \bar{x}_2)/\sqrt{(s_1^2 + s_2^2)/2}$.

Figure A2.3.3: Deaths of sector-employed relatives in agriculture and services



(a) Agriculture-employed relative (b) Services-employed relative

Notes: The figure reports estimates from the stacked event-study specification in equation (1). Panel (a) reports event-time coefficients around the first eligible death of an agriculture-employed relative and Panel (b) around the first eligible death of a services-employed relative. In each panel, the outcome is employment in the deceased relative's sector. Event time zero is the first census following the death and the omitted period is the preceding census. The specification includes individual-by-stack fixed effects and pre-event-CZ-by-year-by-stack fixed effects, together with age controls and predetermined family and household characteristics. Observations are censored from the second eligible death onward. Regressions use the corrective stack weights together with the population reweighting weights. Standard errors are clustered at the de-duplicated family-identifier level. Vertical bars and shaded regions denote 95% confidence intervals.

with the absence of a comparable response following deaths of non-employed relatives, supports the interpretation that the manufacturing result is linked to the loss of a sector-specific family connection rather than to family bereavement alone.

Other outcomes around the death. I next examine whether the manufacturing response coincides with changes in other household outcomes that could indicate a broader disruption following the death. Table A2.3.3 reports responses in employment, household-head status, number of children, marital status and farm residence. For manufacturing deaths, there is little evidence of corresponding discontinuities in these outcomes. This makes a broad change in household composition or labor-force attachment less consistent with the manufacturing result. The agricultural case warrants somewhat more caution. Deaths of agriculture-employed relatives are accompanied by a decline in farm ownership or farm residence in some specifications. Because the farm is both a productive asset and a place of employment, agricultural deaths may affect sectoral outcomes through succession, inheritance or asset reallocation in addition to the loss of a sector-specific labor-market connection.

Additional heterogeneity. Table A2.3.4 reports the manufacturing response separately by gender and race. The estimates are broadly similar across groups.

Table A2.3.2: Sectoral choice and family deaths: robustness

	Dependent variable: manufacturing employment				
	Baseline (1)	No controls (2)	PPMLE (3)	Non-employed death (4)	Agricultural death (5)
Years relative to death					
	0.0478	0.0470	−0.0118	0.0811	−0.148
−30	(0.123)	(0.123)	(0.0901)	(0.259)	(0.179)
	−0.0554	−0.0617	−0.0486	0.168	−0.083
−20	(0.132)	(0.133)	(0.0546)	(0.251)	(0.058)
−10					
	−0.342**	−0.340**	0.0558	0.0798	−0.010
0	(0.137)	(0.139)	(0.0352)	(0.241)	(0.037)
	−0.427**	−0.428**	−0.1807***	−0.0129	0.196***
+10	(0.182)	(0.183)	(0.0541)	(0.223)	(0.035)
	−0.486**	−0.404*	−0.258***	0.160	0.208***
+20	(0.233)	(0.225)	(0.0670)	(0.259)	(0.044)
	−1.002***	−0.973***	−0.451***	0.0924	0.207***
+30	(0.356)	(0.356)	(0.0940)	(0.212)	(0.058)
Observations	22,115,970	22,115,970	3,727,444	22,115,970	22,115,970
R-squared	0.358	0.358	–	0.358	0.356
Mean outcome	36.85	36.85	36.85	36.85	36.85
Mean manuf./agric. death	0.02	0.02	0.02	0	0.02
Individual FE	✓	✓	✓	✓	✓
CZ-year FE	✓	✓	✓	✓	✓
Individual controls	✓	×	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Estimation	OLS	OLS	PPMLE	OLS	OLS

Notes: The dependent variable is manufacturing employment. Columns (1)–(3) report the effect of losing a manufacturing-employed relative. Column (2) omits individual and family controls, while column (3) uses PPMLE. Column (4) replaces manufacturing deaths with deaths of non-employed relatives as a placebo. Column (5) considers deaths of agricultural-employed relatives, including CZ-year fixed effects. Blank event-time cells indicate coefficients not reported for that specification. The baseline stacked specification includes individual-by-stack fixed effects and pre-event-CZ-by-year-by-stack fixed effects, together with age controls and predetermined family and household characteristics. Observations are censored from the second eligible death onward. The baseline and other specifications in columns (1)–(5) use corrective stack weights and population reweighting weights. Standard errors, reported in parentheses, are clustered at the de-duplicated family-identifier level. The sample for columns (1)–(5) covers census observations from 1900 to 1940. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Robustness of the stacked event-study design This appendix examines the sensitivity of the manufacturing death event study to treatment-cohort composition, event-time support, the choice of comparison group and population weighting. I estimate the event study around the first eligible death of a manufacturing-employed relative. Because deaths are observed only between censuses, a relative observed alive in census year y and inferred to have died before census $y+10$ is treated as dying during the interval $(y, y+10]$. Event time 0 is the first census after the death and the final census before the death, $t = -10$, is the omitted period.

The estimation panel is decadal and spans 1900–1940, covering five census waves. The resulting treatment cohorts are 1910, 1920, 1930 and 1940. Within each stack, treated individuals are compared with never-treated individuals and with later-treated individuals whose first eligible death occurs after the end of that stack's event window. Individuals who have already been

Table A2.3.3: Demographic outcomes and family deaths

	Dependent variable				
	Employed (1)	House (2)	Number of children (3)	Married (4)	Farm (5)
Death (agric.) −30 years					-0.001
					(0.036)
Death (agric.) −20 years					0.003
					(0.026)
Death (agric.)					0.008
					(0.011)
Death (agric.) +10 years					-0.097***
					(0.014)
Death (agric.) +20 years					-0.125***
					(0.014)
Death (agric.) +30 years					-0.142***
					(0.016)
Death (manuf.) −30 years	-0.002	-0.010	-0.184***	-0.041*	
	(0.036)	(0.036)	(0.059)	(0.024)	
Death (manuf.) −20 years	0.013	-0.020	-0.053	-0.018	
	(0.026)	(0.032)	(0.034)	(0.017)	
Death (manuf.)	-0.009	-0.005	-0.040	-0.032**	
	(0.024)	(0.025)	(0.040)	(0.013)	
Death (manuf.) +10 years	0.003	0.012	-0.161**	-0.033	
	(0.029)	(0.022)	(0.060)	(0.021)	
Death (manuf.) +20 years	-0.023	0.041	-0.090	-0.001	
	(0.033)	(0.033)	(0.093)	(0.020)	
Death (manuf.) +30 years	-0.033	0.046	-0.186	-0.024	
	(0.043)	(0.050)	(0.148)	(0.037)	
Observations	22,115,970	21,789,776	21,845,180	21,845,180	21,845,180
R-squared	0.560	0.457	0.638	0.694	0.580
Mean outcome	0.63	0.52	0.89	0.45	0.36
Mean manuf./agric. death	0.02	0.02	0.02	0.02	0.02
Individual FE	✓	✓	✓	✓	✓
CZ-year FE	✓	✓	✓	✓	✓
Individual controls	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Estimation	OLS	OLS	OLS	OLS	OLS

Notes: The table reports responses of demographic and household outcomes to the first eligible death of a manufacturing-employed relative using the stacked event-study specification in equation (1). Columns consider employment, household-head status, number of children, marital status and farm residence. All specifications include individual-by-stack fixed effects and pre-event-CZ-by-year-by-stack fixed effects, together with age controls and predetermined family and household characteristics. Observations are censored from the second eligible death onward. Regressions use the corrective stack weights together with the population reweighting weights. Standard errors, reported in parentheses, are clustered at the deduplicated family-identifier level. The sample covers census observations from 1900 to 1940. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

treated, or whose first eligible death occurs within the event window, are excluded from the control group for that stack. Observations are censored from the second eligible death onward. The baseline specification includes individual-by-stack fixed effects and pre-event-CZ-by-year-by-stack fixed effects. The latter hold location fixed at its pre-event value, thereby absorbing changes in the initial local labor market without conditioning on subsequent migration. Controls include age and predetermined family and household characteristics interacted with census-year effects. Corrective stacked-event-study weights account for the repeated use of controls

Table A2.3.4: Sectoral choice and family deaths: heterogeneity

	Dependent variable: manufacturing employment				
	All	Men	Women	White	Non-white
	(1)	(2)	(3)	(4)	(5)
Death (manuf.) −30 years	0.0478	0.0612	0.388	0.0809	-1.610
	(0.123)	(0.127)	(0.370)	(0.119)	(3.247)
Death (manuf.) −20 years	-0.0554	-0.0284	0.0222	-0.0181	-5.131
	(0.132)	(0.147)	(0.296)	(0.127)	(3.793)
Death (manuf.)	-0.342**	-0.343**	-0.140	-0.341***	2.482
	(0.137)	(0.150)	(0.313)	(0.124)	(3.036)
Death (manuf.) +10 years	-0.427**	-0.413**	-0.670*	-0.423**	-4.159
	(0.182)	(0.205)	(0.377)	(0.165)	(3.680)
Death (manuf.) +20 years	-0.486**	-0.446*	-0.533	-0.415*	1.011
	(0.233)	(0.250)	(0.507)	(0.225)	(3.624)
Death (manuf.) +30 years	-1.002***	-0.945**	-0.692	-0.933***	
	(0.356)	(0.413)	(0.612)	(0.353)	
Observations	22,115,970	18,922,530	3,193,440	22,078,300	37,670
R-squared	0.358	0.394	0.215	0.359	0.578
Mean outcome	36.85	36.85	36.85	36.85	36.85
Mean manuf./agric. death	.02	.02	.02	.02	.02
Individual FE	✓	✓	✓	✓	✓
CZ-year FE	✓	✓	✓	✓	✓
Individual controls	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Estimation	OLS	OLS	OLS	OLS	OLS

Notes: The table reports heterogeneity in the effect of losing a manufacturing-employed relative on subsequent manufacturing employment using the stacked event-study specification in equation (1). Columns split the sample by gender and race. All specifications include individual-by-stack fixed effects and pre-event-CZ-by-year-by-stack fixed effects, together with age controls and predetermined family and household characteristics. Observations are censored from the second eligible death onward. Regressions use the corrective stack weights together with the population reweighting weights. Standard errors, reported in parentheses, are clustered at the de-duplicated family-identifier level. The sample covers census observations from 1900 to 1940. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

across stacks and are combined with the population weights used to make the linked sample representative of the full census.

Table A2.3.5 varies features of this design one at a time. The *unweighted* specification removes the population weights while retaining the corrective stacked weights. The *balanced post window* keeps only cohorts observed through +30; because the panel ends in 1940, this leaves the 1910 treatment cohort alone. The *balanced pre-trend window* fixes cohort composition over the pre-treatment window through $t = -20$. The *never-treated controls* specification removes later-treated individuals from the comparison group altogether. Finally, the *fully balanced individuals* specification requires complete observation over every census needed for the corresponding stack and event window. All other features of the baseline specification are held fixed within each exercise.

Event-time support is necessarily asymmetric in a finite panel. Observations close to treatment draw on all four cohorts, whereas longer leads and lags are identified by progressively fewer cohorts. In particular, +10 is supported by the 1910–1930 cohorts, +20 by 1910 and 1920 and +30 by the 1910 cohort alone; Table A2.3.6 reports the corresponding sample support.

The balanced-window exercises therefore trade comparability of cohort composition against precision and, at the longest horizons, identify effects from substantially narrower treatment populations.

Table A2.3.5: Stacked event study: robustness to cohort composition, controls and panel support

	−30	−20	0	+10	+20	+30
Baseline	0.048	-0.055	-0.342**	-0.427**	-0.486**	-1.002***
	(0.123)	(0.132)	(0.137)	(0.182)	(0.233)	(0.356)
Unweighted	0.047	-0.055	-0.327**	-0.426**	-0.484*	-0.982***
	(0.123)	(0.134)	(0.135)	(0.183)	(0.284)	(0.374)
Balanced post window	—	—	-0.830***	-0.636	-0.705**	-0.721**
			(0.298)	(0.404)	(0.331)	(0.317)
Balanced pre-trend window	—	0.189	-0.252	-0.316	-0.424*	—
		(0.202)	(0.199)	(0.217)	(0.253)	
Never-treated controls only	0.047	-0.055	-0.327**	-0.426**	-0.483*	-0.538*
	(0.123)	(0.134)	(0.135)	(0.183)	(0.284)	(0.320)
Fully balanced individuals	-0.015	0.043	-0.497*	-0.259	-0.242	-0.361
	(0.458)	(0.327)	(0.301)	(0.441)	(0.311)	(0.311)

Notes: The dependent variable is manufacturing employment; $t = -10$ is omitted. The baseline uses corrective stacked-event-study and population weights. The unweighted row drops the population weights; the remaining rows modify cohort, event-window or comparison-group support as described in the text. Dashes indicate event times outside the corresponding estimation window. Standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The estimates are stable across the main alternatives. Removing population weights produces almost identical coefficients throughout the event window. Restricting the comparison group to never-treated individuals also leaves the estimates through +20 essentially unchanged, indicating that the baseline results are not driven by the use of not-yet-treated observations as controls. Estimates from balanced windows are less precise because they rely on fewer treatment cohorts, but continue to show a decline in manufacturing employment following the death. In particular, the balanced post specification, which is identified entirely from the 1910 cohort, yields negative estimates at every available post-treatment horizon.

The exercises also clarify how the longest-horizon estimate should be interpreted. The baseline +30 coefficient is identified only from the 1910 treatment cohort and is therefore not directly comparable to shorter-horizon coefficients that average across several cohorts. Its larger magnitude may reflect either the evolution of the response with time since treatment or differences specific to the earliest cohort. The balanced-post estimate confirms a persistent negative response within that cohort, but is smaller than the baseline +30 estimate. I therefore place

greater weight on the common negative post-treatment pattern than on comparisons of coefficient magnitudes across horizons with changing cohort support.

Table A2.3.6: Treated and control support by event time

Event time	Cohorts included	No. cohorts	Treated individuals	Control ind.
−30	1930, 1940	2	13,668	2,324,726
−20	1920, 1930, 1940	3	16,914	3,492,041
−10	1910, 1920, 1930, 1940	4	18,807	4,660,978
0	1910, 1920, 1930, 1940	4	18,807	4,660,978
+10	1910, 1920, 1930	3	11,203	3,492,041
+20	1910, 1920	2	5,056	2,324,726
+30	1910	1	1,853	1,160,481
Total	—	—	**86,308**	**22,115,970**

Notes: The table reports event-time support in the stacked event-study sample. Cohorts are defined by treatment year. The omitted event period is −10.

A2.3.3 Occupational entry routes and family persistence

The death design establishes that losing a sector-employed relative lowers subsequent attachment to that sector. I next ask whether the strength of family persistence varies with the availability of alternative routes into an occupation. This provides complementary evidence on the interpretation of sector-specific family ties as a labor-market input.

Classification of occupational entry routes. I classify occupations using the 1949 *Occupational Outlook Handbook* (U.S. Bureau of Labor Statistics, 1949). The Handbook contains separate reports for 209 occupations outside agriculture, including explicit discussions of training and entry for 181 occupations. I classify an occupation as having a *workplace route* when the Handbook identifies apprenticeship, helper or trainee positions, on-the-job training, learning from experienced workers, promotion from related employment or prior occupational experience as a route into the occupation. I classify an occupation as having a *school route* when the Handbook identifies vocational, trade or technical school, classroom instruction or correspondence instruction as a route of preparation.

I map the Handbook reports to detailed 1950 occupation codes. Because several Handbook reports can map into a single census occupation, I assign a route only when the underlying reports agree; ambiguous codes are excluded from the route-specific coefficients. This procedure yields 33 workplace-only occupations and 10 occupations with a school route. Airplane mechanics remain in the occupational choice set but load on neither route coefficient, producing 44 occupation clusters in the estimation.

Table A2.3.7: Detailed occupations by route of entry

Panel A. Workplace route only (33 occupations)			
500	Bakers	565	Paperhangers
501	Blacksmiths	575	Pressmen and plate printers
502	Bookbinders	582	Shoemakers and repairers
503	Boilermakers	585	Structural metal workers
505	Cabinetmakers	592	Tool makers and die makers
513	Cranemen, derrickmen, hoistmen	593	Upholsterers
515	Electricians	594	Craftsmen, n.e.c.
520	Electrotypers and stereotypers	621	Attendants, auto service
522	Excavating and road machinery op.	622	Blasters and powdermen
524	Forgemen and hammermen	624	Brakemen, railroad
530	Glaziers	635	Filers, grinders, polishers
533	Inspectors, n.e.c.	641	Furnacemen and smeltermen
535	Job setters, metal	642	Heaters, metal
541	Locomotive engineers	643	Laundry and dry cleaning op.
542	Locomotive firemen	670	Painters, except construction
560	Millwrights	681	Switchmen, railroad
563	Opticians and lens grinders		
Panel B. School route also available (10 occupations)			
504	Brickmasons and stonemasons	573	Plasterers
510	Carpenters	574	Plumbers and pipe fitters
525	Furriers	591	Tinsmiths and sheet metal workers
550	Automobile mechanics	644	Meat cutters
553	Railroad and car shop mechanics		
564	Painters, construction		

Notes: Detailed OCC1950 occupations for which the underlying Handbook reports agree on the route of entry. Codes with conflicting classifications are excluded.

Empirical specification. The estimation sample uses occupational outcomes from 1920, 1930 and 1940, with 1900 and 1910 supplying predetermined occupational histories. I retain family-by-generation-by-year cells with at least one predetermined occupational tie somewhere in the choice set, since cells without such a tie contain no within-cell variation in the family-experience measure.

For family f, generation g, year t and occupation o, I estimate

$$\mathrm{E}[n_{fgto} \mid \cdot] = \exp\{\beta_W D_{fgto}\mathbf{1}\{\text{workplace only}\}_o + \beta_S D_{fgto}\mathbf{1}\{\text{school route}\}_o + \alpha_{fgt} + \delta_{ot}\}\,, \tag{A2.3.1}$$

where n_{fgto} is the number of employed family members in occupation o and D_{fgto} indicates that a relative outside the current family-generation-year cell held occupation o in an earlier census. Family-by-generation-by-year fixed effects α_{fgt} absorb the family's overall occupational allocation, while occupation-by-year fixed effects δ_{ot} absorb occupation size and aggregate changes in

Table A2.3.8: Examples of occupational entry routes in the 1949 Handbook

Occupation (OCC1950)	p.	Passage
Workplace route only		
Melters, foundry (641)	262	"As a rule, there are no apprenticeships or other organized training programs provided for melters. ... The usual way to get one of the more skilled jobs is to begin as a furnace helper or less skilled melter and gradually to pick up the trade."
Drop-hammer operators (524)	266	"Drop-hammer operation is learned on the job, but a regular apprenticeship is not ordinarily provided. ... Workers usually begin as helpers and after sufficient experience advance to the job of heater."
Inspectors, machinery parts (533)	276	"Skilled inspectors are obtained from the ranks of machine-tool operators and other machine-shop workers, or by upgrading less skilled inspectors."
Millwrights (560)	286	"Entry into this occupation is usually through a millwright apprenticeship or equivalent on-the-job training. ... Inexperienced workers may be hired as helpers or trainees and pick up the occupation while working."
School route also available		
Painters, construction (564)	177	"Sometimes the painter learns his job by taking courses in a public trade school."
Automobile mechanics (550)	198	"Young people under 18 should, if possible, complete at least 2 years of high school before beginning on-the-job training or entering a vocational school."
Meat cutters (644)	355	"Vocational schools in some cities offer instruction in meat cutting."
Carmen, railroads (553)	344	"On some roads, a few young men 18 to 25 years of age who have completed a technical high-school course are taken on as special apprentices."

Notes: Selected passages from the 1949 *Occupational Outlook Handbook*. Page numbers refer to printed pages. Ellipses omit intervening text. The passages illustrate how workplace and school routes are coded.

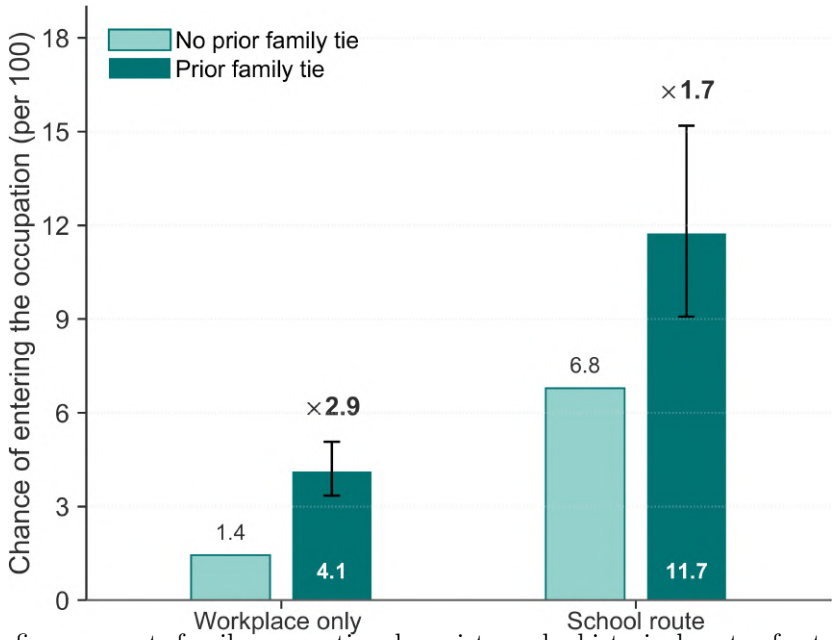
occupational demand. Identification therefore comes from whether predetermined family experience disproportionately predicts entry into particular occupations within the same family-generation-year choice set.

The final estimation sample contains approximately 16.3 million occupation-choice observations from 225,942 reconstructed family identifiers. Because the route classification varies at the occupation level, standard errors are two-way clustered by family identifier and detailed occupation.

Results. Figure A2.3.4 shows a pronounced difference across entry routes. The estimated family-persistence coefficient is $\widehat{\beta}_W = 1.051$ for workplace-only occupations and $\widehat{\beta}_S = 0.548$

for occupations in which a school route is also available. The difference, $\widehat{\beta}_W - \widehat{\beta}_S = 0.503$, is statistically significant ($p = 0.005$). In proportional terms, predetermined family experience is therefore substantially more predictive of occupational entry when preparation occurs primarily through workplace-based routes.

Figure A2.3.4: Family occupational persistence by route of entry



Notes: The figure reports family occupational persistence by historical route of entry, estimated from equation (A2.3.1). Workplace-only occupations have no school-based entry route identified by the 1949 *Occupational Outlook Handbook*. Standard errors are two-way clustered by reconstructed family identifier and occupation. Vertical bars denote 95% confidence intervals.

Robustness. Table A2.3.9 reports alternative measures of predetermined family occupational experience and several robustness exercises. The difference between workplace and school-route occupations remains similar when family experience is measured by the number of prior family ties, when the sample is restricted to high-confidence occupation crosswalks, among craftsmen and when very small occupations are excluded. Reassigning occupational ties across families eliminates the relationship, providing a falsification check that the result is tied to actual family occupational histories rather than occupation-level differences alone.

A2.3.4 Family ties and labor-market outcomes

The occupational-entry evidence suggests that family connections can facilitate access to information, contacts and workplace knowledge. I examine two complementary implications of this interpretation. First, employed relatives may facilitate job finding and reduce unemployment. Second, same-sector relatives may improve workers' access to sector-specific opportunities after

Table A2.3.9: Family occupational persistence by route of entry

	β_W	β_S	$\beta_W - \beta_S$
Main specification	1.051	0.548	0.503
	(0.105)	(0.131)	(0.178)
log(1 + number of ties)	1.468	0.730	0.738
	(0.144)	(0.174)	(0.238)
Craftsmen only	1.010	0.537	0.473
	(0.116)	(0.121)	(0.167)
Minimum occupation size: 1,000	1.012	0.549	0.462
	(0.099)	(0.132)	(0.175)
Minimum occupation size: 5,000	0.852	0.541	0.311
	(0.086)	(0.128)	(0.151)
Ties reassigned across families	-0.004	-0.005	0.001
	(0.009)	(0.004)	(0.009)

Notes: PPML estimates based on equation (A2.3.1). All specifications include family-by-generation-by-year and occupation-by-year fixed effects. Standard errors are two-way clustered by reconstructed family identifier and occupation. The main specification uses an indicator for any predetermined family occupational tie. Subsequent rows vary the family-tie measure or estimation sample. The final row randomly reassigns predetermined occupational ties across families.

entry, resulting in faster subsequent wage growth. I also ask whether these relationships differ between urban and non-urban labor markets, where alternative sources of information and matching may be more readily available.

For the job-finding outcomes, I estimate

$$Y_{i,t+1} = \alpha + \beta_F \,\mathrm{FamTie}_{i,t} + \beta_U \,\mathrm{Urban}_{i,t} + \beta_{FU} \,(\mathrm{FamTie}_{i,t} \times \mathrm{Urban}_{i,t}) + \lambda' X_{it} + \varepsilon_{i,t+1}, \quad \text{(A2.3.2)}$$

where $Y_{i,t+1}$ is either an indicator for looking for work or an indicator for unemployment, $\mathrm{FamTie}_{i,t}$ measures local employed family presence and X_{it} contains the controls and fixed effects reported in the table. The looking-for-work specification is estimated among nonworkers, while the unemployment specification uses the working-age population.

For workers, I estimate an analogous specification for subsequent wage growth,

$$\Delta \log w_{i,t+1} = \alpha + \beta_F^s \,\mathrm{FamTie}_{i,t}^s + \beta_U^s \,\mathrm{Urban}_{i,t} + \beta_{FU}^s \left(\mathrm{FamTie}_{i,t}^s \times \mathrm{Urban}_{i,t}\right) + \lambda_s' X_{it} + \varepsilon_{i,t+1}, \quad \text{(A2.3.3)}$$

where $\mathrm{FamTie}_{i,t}^s$ measures family presence in the worker's sector and the regressions are estimated separately for agricultural and manufacturing workers.

Table A2.3.10 reports the results. Among nonworkers outside urban areas, family presence is associated with a 2.87-percentage-point lower probability of looking for work. The interaction

with urban status is positive, implying a smaller association of approximately 1.46 percentage points in urban areas. Family ties are also associated with a 0.60-percentage-point lower unemployment probability outside urban areas, with the association again substantially attenuated in urban labor markets. These outcomes do not measure search duration directly, but they are consistent with employed relatives facilitating job finding or reducing the incidence of unsuccessful search.

The wage results provide complementary evidence on outcomes after sector entry. Among manufacturing workers outside urban areas, a family tie is associated with approximately 2.2 log points faster subsequent hourly-wage growth. The corresponding estimate for agricultural workers is smaller and statistically indistinguishable from zero. In manufacturing, the family-tie association is substantially weaker in urban areas, as indicated by the negative interaction between urban status and family presence.

Table A2.3.10: Family ties and labor-market outcomes

VARIABLES	(1) Looking for work	(2) Unemployed Ages 15–60	(3) Δ Log hourly wage Agriculture	(4) Δ Log hourly wage Manufacturing
Urban	-0.0144***	-0.00905***	0.214***	0.0608***
	(0.00207)	(0.00144)	(0.0603)	(0.00691)
Family tie	-0.02872***	-0.00602***	0.00916	0.0222***
	(0.00145)	(0.00135)	(0.0207)	(0.00594)
Urban × family tie	0.0141***	0.00368***	-0.0443	-0.0510***
	(0.00210)	(0.00135)	(0.0643)	(0.00742)
Observations	4,958,082	20,918,419	45,832	465,224
R-squared	0.090	0.041	0.737	0.643
Mean outcome	0.0596	0.0446		
CZ × year FE	✓	✓		
Individual controls	✓	✓	✓	✓
Cluster	CZ	CZ	CZ	CZ
Sample	Nonworkers	All	Agricultural workers	Manufacturing workers

Notes: Columns (1) and (2) relate local employed family presence to indicators for looking for work and unemployment, respectively. Column (1) is restricted to nonworkers; column (2) includes individuals ages 15–60. Columns (3) and (4) relate same-sector family presence to subsequent changes in log hourly wages among agricultural and manufacturing workers, respectively. Each specification includes urban status and its interaction with the family-tie measure. The additional individual controls and fixed effects are reported in the table. Standard errors are clustered by CZ. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Family support and sectoral persistence. Family connections may help workers remain employed in a sector after entering it. To examine the potential channels, I follow workers employed in manufacturing or agriculture in 1930 and study their subsequent outcomes following the death of a relative employed in the same sector. For each initial sector $k \in \{A, M\}$, I

estimate

$$Y_{it} = \beta_t^k D_i^k + \alpha_{c_{i,1930} \times j_{i,1920}, t} + \gamma_{1t} \text{Age}_{i,1930} + \gamma_{2t} \text{Age}_{i,1930}^2 + X_{i,1930}' \delta_t + \varepsilon_{it}, \tag{A2.3.4}$$

where D_i^k indicates that a relative employed in sector k died between 1930 and 1940. The fixed effects interact the worker's 1930 commuting zone with their sector in 1930. The vector $X_{i,1930}$ includes urban residence, farm residence and the number of children. I estimate the specification separately by outcome and initial sector, using population weights and clustering standard errors by individual. To examine heterogeneity, I interact the death indicator with predetermined characteristics of the worker or distinguish deaths by characteristics of the deceased.

Table A2.3.11 provides additional evidence on how relatives may sustain manufacturing employment beyond initial entry. First, earnings decline even among workers who remain in the same commuting zone and detailed industry, consistent with relatives helping workers maintain valuable employment opportunities. Second, relatives in supervisory occupations may provide workplace support. Their deaths are associated with a 0.8 percentage point decline in manufacturing employment and a 1.5-point reduction in occupational score, suggesting that workplace sponsorship may contribute to sectoral persistence.

Table A2.3.11: Channels of manufacturing persistence

	Effect		
Panel A. Employment and earnings			
In manufacturing, 1940 (pp)	-1.12^{**} (0.5)		
Log earnings, same CZ and industry (×100)	-1.8^{***} (0.46)		
Panel B. Supervisory relatives			
	Supervisor	Other	p-value
In manufacturing, 1940 (pp)	-1.88^{**} (0.88)	-0.93^{*} (0.51)	0.071
Occupational score, 1940	-1.52^{***} (0.27)	-0.27^{***} (0.06)	0.022
Lower-scoring occupation (pp)	6.9^{**} (2.9)	1.4 (1.5)	0.025

Notes: The sample comprises workers employed in manufacturing in 1930. Panel A reports estimates from equation (A2.3.4). The earnings regression restricts the sample to workers remaining in the same commuting zone and detailed industry. Panel B distinguishes deaths of relatives in supervisory occupations, including foremen and managers, from other manufacturing deaths. The last column reports p-values from tests of equality between the two coefficients. Standard errors, reported in parentheses, are clustered at the family level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A2.3.12 examines whether family labor and other productive support help sustain agricultural employment. First, the death of an agricultural relative is followed by declines in agricultural employment, farm operation and farm residence. These effects are concentrated among workers with fewer other agricultural relatives, consistent with surviving family members substituting for the deceased. Second, the deceased's role in farming matters. Deaths of relatives who were not farm operators are associated with particularly large declines in agri-

cultural employment and farm operation, suggesting that family labor and other productive inputs help sustain farming.

Table A2.3.12: Channels of agricultural persistence

	Effect		
Panel A. Agricultural employment and farm activity			
In agriculture, 1940 (pp)	-0.8^{***} (0.21)		
Farm operator, 1940 (pp)	-1.0^{***} (0.35)		
Farm residence, 1940 (pp)	-3.2^{***} (0.81)		
Panel B. Other agricultural relatives			
	Few	Many	p-value
In agriculture, 1940 (pp)	-2.6^{***} (0.75)	-0.2 (0.7)	0.002
Farm operator, 1940 (pp)	-2.4^{***} (0.55)	0.2 (0.7)	< 0.001
Panel C. Deceased farm operator			
	No	Yes	p-value
In agriculture, 1940 (pp)	-2.41^{**} (1.22)	-0.67^{***} (0.15)	0.075
Farm operator, 1940 (pp)	-2.50^{***} (0.74)	-0.91 (0.67)	0.018

Notes: The sample comprises workers employed in agriculture in 1930. Panel A reports estimates from equation (A2.3.4). Panel B distinguishes workers with fewer than 5 versus at least 5 other agricultural relatives in 1930, excluding the deceased. Panel C distinguishes deaths of farm operators from deaths of other agricultural relatives. Farm operation is proxied by agricultural self-employment. The last column reports p-values from tests of equality between the two coefficients. Binary outcomes are in percentage points. Standard errors, reported in parentheses, are clustered by family. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A2.3.5 Persistence of sectoral differences across generations

The main text studies changes in sectoral employment following the loss of an existing sector-specific family connection. This subsection asks a complementary descriptive question: whether sectoral differences associated with an ancestor's occupational transition remain visible among later descendants.

I begin with individuals employed in agriculture in 1860 and distinguish ancestors who entered manufacturing by 1870 from those who remained in agriculture. Let

$$T_a = \mathbf{1}\{\text{ancestor } a \text{ enters manufacturing by } 1870\},$$

and let $G_{ia} \in \{1, \ldots, 5\}$ denote descendant i's genealogical distance from ancestor a. I estimate

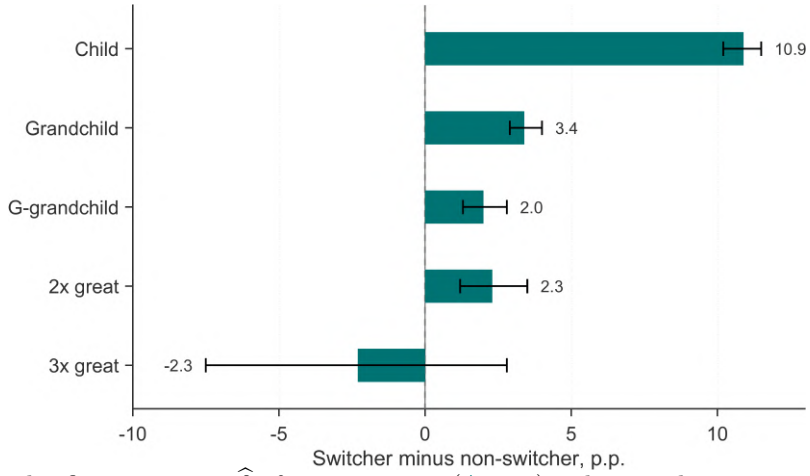
$$100 \times \mathbf{1}\{\text{sector}_{iat} = M\} = \alpha_{c_a t} + \lambda_{G_{ia}} + \sum_{g=1}^{5} \beta_g \left[T_a \times \mathbf{1}\{G_{ia} = g\}\right] + Z_a' \Gamma_{G_{ia}} + \varepsilon_{iat}, \quad \text{(A2.3.5)}$$

where $\alpha_{c_a t}$ are ancestor-origin-CZ-by-year fixed effects, $\lambda_{G_{ia}}$ are generation fixed effects and Z_a contains predetermined characteristics of ancestor a measured in 1860, including literacy, farm

ownership, age, urban status and sex. The coefficients β_g therefore compare manufacturing employment among descendants of observationally similar agricultural ancestors from the same origin CZ according to whether the ancestor subsequently entered manufacturing.

Figure A2.3.5 reports the estimates. Descendants of ancestors who entered manufacturing are approximately 9–11 percentage points more likely to work in manufacturing in the first descendant generation. The difference declines with genealogical distance but remains positive among later descendants. Sectoral differences associated with the ancestor's transition therefore remain visible beyond the immediately following generation.

Figure A2.3.5: Manufacturing persistence across generations



Notes: The figure reports $\widehat{\beta}_g$ from equation (A2.3.5), where g denotes genealogical distance from the agricultural ancestor observed in 1860. The comparison is between descendants of ancestors who entered manufacturing by 1870 and descendants of ancestors who remained in agriculture, conditional on ancestor-origin-CZ-by-year fixed effects and predetermined ancestor characteristics. Standard errors are clustered at the root-lineage level. Vertical bars denote 95% confidence intervals.

A2.4 Families and migration choice

The main text establishes two spatial margins of family connections. First, losing a non-employed relative at origin raises subsequent out-migration. Second, migration is disproportionately directed toward destinations where members of the individual's family network already live. This appendix reports additional evidence on the origin mechanism.

A2.4.1 Heterogeneity in the response to losing an origin tie

Table A2.4.1 examines heterogeneity in the out-migration response following the death of a local non-employed relative. The specifications follow equation (2) and differ only in the indicated subsample. Because the death removes a geographically proximate family connection without removing a contemporaneous employment tie, these splits help characterize the types of households for which local family support appears most valuable.

The point estimates are larger among households with young children and in rural areas. I interpret this heterogeneity cautiously: the subsamples differ along several dimensions and the estimates do not separately identify childcare, household production or informal insurance. The pattern is nevertheless consistent with locally provided family inputs being particularly valuable where household needs are greater or formal substitutes are less readily available.

Table A2.4.1: Out-migration after a family death: heterogeneity

	Dependent variable: migration						
	All	W.o. 5yrs old	With 5yrs old	Poor	Rich	Urban	Rural
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Death (non-emp.) −20 years	0.0413	0.0851	-0.216	-0.00706	0.0417	-0.331	-0.149
	(0.105)	(0.183)	(0.186)	(0.151)	(0.149)	(0.553)	(0.131)
Death (non-emp.)	0.229***	0.159	0.327***	0.327***	0.145*	0.149	0.191**
	(0.0724)	(0.139)	(0.0848)	(0.123)	(0.0862)	(0.260)	(0.0943)
Death (non-emp.) +10 years	0.426***	0.375***	0.681***	0.432***	0.385***	0.167	0.309***
	(0.0656)	(0.121)	(0.0956)	(0.0750)	(0.0904)	(0.282)	(0.117)
Death (non-emp.) +20 years	0.341***	0.250*	0.546***	0.433***	0.265**	-0.578	0.249
	(0.0770)	(0.133)	(0.143)	(0.102)	(0.104)	(0.424)	(0.161)
Observations	18,701,551	14,506,810	4,194,741	9,434,322	9,267,229	2,793,325	15,908,226
Mean outcome	19.0	19.0	19.0	19.0	19.0	19.0	19.0
Individual FE	✓	✓	✓	✓	✓	✓	✓
Individual controls	✓	✓	✓	✓	✓	✓	✓
Estimation	OLS	OLS	OLS	OLS	OLS	OLS	OLS

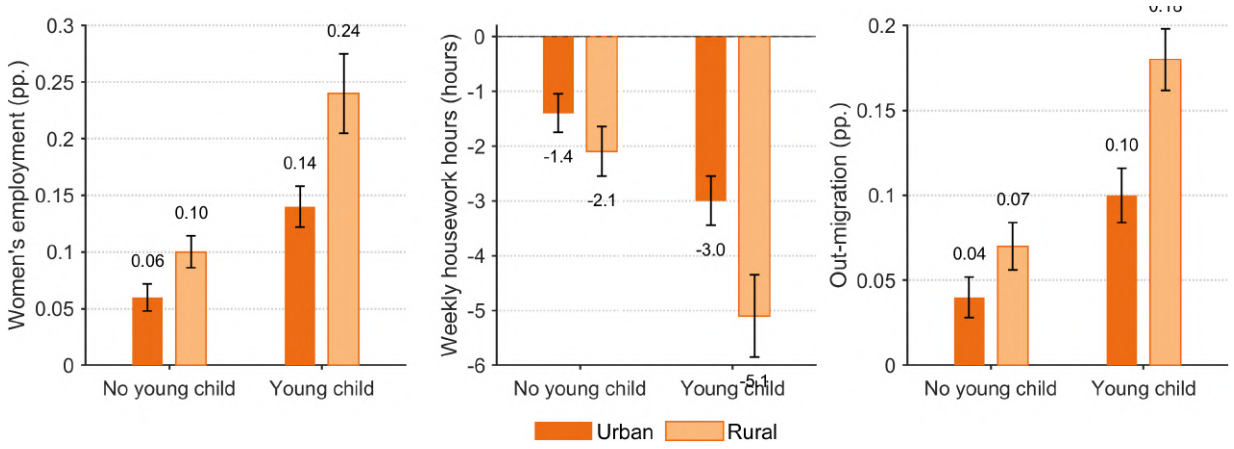
Notes: The table reports heterogeneity in the response of out-migration to the first eligible death of a local non-employed relative using the stacked event-study specification in equation (2). The outcome is an indicator for moving across commuting zones between consecutive censuses. Columns report estimates for the indicated subsamples. Event time zero is the migration interval containing the death; subsequent intervals are post-death. All specifications include individual-by-stack fixed effects and pre-event-origin-CZ-by-year-by-stack fixed effects, together with age controls and predetermined family and household characteristics. Observations are censored from the second eligible death onward. Regressions use the corrective stack weights together with the population-reweighting weights. Standard errors, reported in parentheses, are clustered at the de-duplicated family-identifier level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A2.4.2 Evidence on locally provided family support

The origin-death design is consistent with geographically proximate relatives providing services that raise the value of remaining nearby. I examine this interpretation using household outcomes for which local family support may be particularly relevant.

Figure A2.4.1 combines two pieces of evidence. First, the presence of a local non-employed relative is associated with higher female employment and fewer hours devoted to housework. These relationships are particularly pronounced among households with young children and outside urban areas. Second, the out-migration response following the death of such a relative is larger for similar households. The first set of estimates is descriptive, while the migration estimates come from the death design in equation (2).

Figure A2.4.1: Family ties, household constraints and place attachment



Notes: The figure reports relationships between local non-employed family presence and women's employment and weekly housework hours, together with the out-migration response following the death of a local non-employed relative. Estimates are shown by the presence of a child below age five and by urban status. The household-outcome estimates are descriptive associations; the migration estimates are based on equation (2). Whiskers denote 95% confidence intervals. The migration sample contains 1,119,284 observations.

Two additional exercises help distinguish geographically provided support from a general response to family bereavement. I find no comparable migration response following the death of a relative living outside the individual's CZ. The migration response is also similar among individuals who do not own their home, making a housing-asset channel less likely to account for the result. I therefore interpret the origin-death evidence as consistent with the loss of locally provided family support rather than as identifying any single underlying service.

A2.4.3 Migration and sectoral switching

The quantitative model specifies sector choice before location choice. I provide additional empirical evidence on this sequencing by asking whether migration induced by the loss of a

local non-employed relative is accompanied by movement out of the worker's initial sector.
The identifying shock is the same death of a local non-employed relative used in the origin-migration design. These deaths increase out-migration but, as shown in the sectoral placebo exercises in Appendix A2.3.2, do not generate a detectable change in sectoral employment on average. This is consistent with the shock removing locally provided family support without directly removing a sector-specific employment connection.
I focus on workers initially employed in agriculture and use the death shock as an instrument for out-migration. The first stage is

$$\text{OutMig}_{it} = \pi \ \text{Death}_{it} + \alpha_i + \lambda_{ct} + \gamma' X_{it} + u_{it}, \tag{A2.4.1}$$

where Death_{it} captures exposure to the death of a local non-employed relative and the remaining controls follow the migration specification. I then estimate

$$\text{NonAgric}_{i,t+1} = \beta \, \widehat{\text{OutMig}}_{it} + \alpha_i + \lambda_{ct} + \gamma' X_{it} + \varepsilon_{it}. \tag{A2.4.2}$$

Table A2.4.2 shows that the migration response generated by the family shock is not accompanied by a detectable increase in non-agricultural employment. The point estimate is small and statistically indistinguishable from zero. This result is consistent with migration operating primarily as a location-adjustment margin within a worker's existing sectoral attachment in this historical setting.
I interpret this exercise as supporting evidence for the model's sequencing rather than as a general estimate of the causal effect of migration on sector choice. The exclusion restriction requires the death of a non-employed relative to affect subsequent sector choice only through migration. The absence of a sectoral reduced-form response to these deaths supports this restriction, but does not rule out all other direct effects of losing local family support.

Table A2.4.2: Sectoral returns to migration

VARIABLES	(1) Agric.	(2) Manuf.
Migration	-0.0224	0.0224
	(0.0746)	(0.0746)
Observations	2,357,596	2,357,596
R-squared	0.003	0.003
Mean outcome	0.768	0.232
Kleibergen-Paap F-statistic	16.88	16.88
First-stage R-squared	0.612	0.612
Individual FE	✓	✓
State-Year FE	✓	✓
Individual controls	✓	✓
Cluster	Individual	Individual
Pop. weighted	✓	✓
Sample	Agric./Manuf. workers	Agric./Manuf. workers

Notes: The unit of observation is an individual; standard errors are clustered at the individual level. Agric./Manuf. is an indicator for employment in agriculture or manufacturing scaled to 1 or 0. Individual controls include age, the number of family members in agriculture and manufacturing and marital status. The sample consists of children of parents working in agriculture or manufacturing and spans 1900–1940 in 10-year intervals. *** denotes significance at the 1 percent level; ** denotes significance at the 5 percent level; * denotes significance at the 10 percent level.

A3 Theory

A3.1 Stylized model

This appendix develops a simple model to show why family ties can initially slow structural transformation but later facilitate it. When relatives work mainly in agriculture, their sector-specific support favors agricultural employment. As workers enter manufacturing, they become manufacturing connections for later generations. I refer to the effect of preserving the initial distribution of relatives as *roots*, and to the additional effect of connections created as that distribution evolves as *branches*. The objective is to derive these forces and the condition under which branches overcome roots. I first derive the sector-choice problem from the quantitative model, then hold location fixed and impose common sector-choice incentives to obtain a closed law of motion. I finally explain what changes when migration is reintroduced. Wages, prices and other payoffs unrelated to horizontal relatives are treated as exogenous.

Environment and assumptions. Consider one ancestral-origin group g and two sectors, agriculture A and manufacturing M. Generations are indexed by t. As in the quantitative model, children choose a sector in period t and work in that sector as adults in $t+1$. Each child inherits connections to horizontal relatives: the siblings of the child's parent, who are the child's aunts and uncles. These relatives belong to the previous generation and have already chosen their sectors and locations when the child makes a choice. Their expected total mass is $r_g > 0$, which remains constant across generations, as in Assumption 3 of the quantitative model.

The sector-choice value in the quantitative model, for a child whose parent lives in n and works in sector j, is

$$\mathcal{U}^k_{gnjt} = -\tau^k_{gnjt} + V_{t+1}(n,k,g) + \theta_k \left[\mathbf{1}\{j=k\} + \psi^k_R \Phi_{g,t}(n,j;n,k)\right], \qquad k \in \{A,M\}.$$

The first two terms are the sector-entry cost and the expected adult value, the indicator captures the parental-sector connection, and $\Phi_{g,t}(n,j;n,k)$ is the mass of horizontal relatives working in sector k at the child's location. To isolate how the choices of one generation change these connections for the next, I impose three simplifying assumptions.

First, all families remain in the same location n, so that $P^N_{t+1}(n \mid n,k,g) = 1$ and there is no migration decision. *Second*, all children of type g have the same inherited distribution of horizontal relatives and the same non-horizontal-relative sector values in each period, regardless of their parent's sector. I therefore replace the parent-specific baseline $-\tau^k_{gnjt} + V_{t+1}(n,k,g) + \theta_k \mathbf{1}\{j=k\}$ with a common value $\bar{\mathcal{U}}^k_t$. This holds the direct parental-sector contribution fixed within the common baseline rather than letting it change with the distribution of parents'

sectors. The values $\bar{\mathcal{U}}_t^k$ may change over time but are held fixed across the counterfactuals. *Third*, children draw independent Type-I extreme-value sector shocks of dispersion σ_O, as in Assumption 4 of the quantitative model. With common systematic payoffs, each child's sector-choice probability is also the expected fraction of the cohort, including a sibling group, choosing that sector. These assumptions remove the dependence of relatives' distributions and choices on individual parental histories and close the stylized model.

Let Φ_t^{nM} and Φ_t^{nA} denote the expected masses of horizontal relatives working in manufacturing and agriculture, respectively, available to children choosing at t. Since $\Phi_t^{nM} + \Phi_t^{nA} = r_g$, the inherited composition can be summarized by one number,

$$x_t \equiv \frac{\Phi_t^{nM}}{r_g} \in [0,1], \qquad \Phi_t^{nM} = r_g x_t, \qquad \Phi_t^{nA} = r_g(1-x_t). \tag{A3.1.1}$$

Thus x_t is the share of a child's horizontal relatives who work in manufacturing. For example, $x_t = 0.06$ means that 6% of the child's expected horizontal-relative mass works in manufacturing and 94% in agriculture. The initial composition x_0 is predetermined. The model below determines how that composition changes from one generation to the next.

Sector choice. Under these assumptions, $\Phi_{g,t}(n, j; n, k) = \Phi_t^{nk}$ for every parental sector j. The sector-choice value above therefore reduces to

$$\mathcal{U}_t^k = \bar{\mathcal{U}}_t^k + \theta_k \psi_R^k \Phi_t^{nk}, \qquad k \in \{A, M\}. \tag{A3.1.2}$$

The coefficient $\theta_k \psi_R^k > 0$ measures the benefit of one unit of local horizontal-relative mass working in sector k. With independent Type-I extreme-value shocks of scale σ_O, the manufacturing-choice probability is

$$p_t^F \equiv P_t^O(M) = \frac{\exp(\mathcal{U}_t^M/\sigma_O)}{\exp(\mathcal{U}_t^M/\sigma_O) + \exp(\mathcal{U}_t^A/\sigma_O)} = \Lambda\left(\frac{\mathcal{U}_t^M - \mathcal{U}_t^A}{\sigma_O}\right), \tag{A3.1.3}$$

where $\Lambda(z) \equiv (1 + e^{-z})^{-1}$ and the superscript F denotes the economy with endogenous horizontal-relative support. Taking log odds shows how the two types of support enter with opposite signs:

$$\operatorname{logit}(p_t^F) \equiv \log \frac{p_t^F}{1 - p_t^F} = \frac{\bar{\mathcal{U}}_t^M - \bar{\mathcal{U}}_t^A}{\sigma_O} + \frac{\theta_M \psi_R^M \Phi_t^{nM} - \theta_A \psi_R^A \Phi_t^{nA}}{\sigma_O}.$$

Define $a_t \equiv (\bar{\mathcal{U}}_t^M - \bar{\mathcal{U}}_t^A)/\sigma_O$ and $b_k \equiv \theta_k \psi_R^k / \sigma_O > 0$. Substituting the inherited relative masses from equation (A3.1.1) and inverting the log odds yields

$$p_t^F = \Lambda\left[a_t + b_M \Phi_t^{nM} - b_A \Phi_t^{nA}\right] = \Lambda[a_t + r_g((b_M + b_A)x_t - b_A)]\,. \tag{A3.1.4}$$

Agricultural connections reduce manufacturing entry relative to an economy without horizontal-relative support, while manufacturing connections increase it. Because the total mass of horizontal relatives is fixed, an increase in x_t creates manufacturing connections while replacing agricultural ones. The index a_t captures changes in economic opportunities that would affect manufacturing without horizontal-relative support. For example, increasing relative manufacturing wages or falling entry costs can raise a_t over time, although these changes are not determined within this stylized exercise. In the main-text illustration, $p_t^F = P_t^{O,M,F}$ and $a_t = a_t^O$.

Evolution of horizontal relatives. The sector choices made by children at t determine the sectoral distribution of their sibling group when its members become adults at $t+1$. A fraction p_t^F of that group enters manufacturing and a fraction $1 - p_t^F$ enters agriculture in expectation. These adults become the aunts and uncles of the following generation. Their expected masses are therefore

$$\Phi_{t+1}^{nM} = r_g p_t^F, \qquad \Phi_{t+1}^{nA} = r_g(1 - p_t^F), \qquad x_{t+1} = p_t^F. \tag{A3.1.5}$$

This is also the simplified version of the quantitative model's law of motion. In the full model, workers in location n and sector k at $t+1$ may have parents who lived in different locations m and worked in different sectors j at t. The share with each parental background follows Bayes' rule:

$$\Pr_t(m, j \mid n, k, g) = \frac{L_{gmt}^j P_t^O(k \mid m, j, g) P_{t+1}^N(n \mid m, k, g)}{\sum_{\tilde{m} \in N} \sum_{\tilde{j} \in \{A,M\}} L_{g\tilde{m}t}^{\tilde{j}} P_t^O(k \mid \tilde{m}, \tilde{j}, g) P_{t+1}^N(n \mid \tilde{m}, k, g)}.$$

Here L_{gmt}^j is the mass of parents from ancestral group g living in m and working in j, and the two probabilities describe their children's sector and subsequent location choices. For each parental background, relatives follow the same sector and location choice distributions. Averaging across these backgrounds gives

$$\lambda_{g,t+1}(n', k' \mid n, k) = \sum_{m \in N} \sum_{j \in \{A,M\}} \Pr_t(m, j \mid n, k, g) P_t^O(k' \mid m, j, g) P_{t+1}^N(n' \mid m, k', g).$$

The full model multiplies this share by r_g to obtain the mass of relatives in each sector and location. In the stylized model, everyone remains in n, so the location-choice probability is one for n and zero elsewhere. Because children also have the same sector-choice probabilities regardless of their parent's sector, $P_t^O(k' \mid n, j, g) = p_t(k')$ for all j, where $p_t(M) = p_t^F$ and $p_t(A) = 1 - p_t^F$. The conditional distribution therefore simplifies to

$$\begin{aligned} \lambda_{g,t+1}(n, k' \mid n, k) &= \sum_{j \in \{A,M\}} P_t^O(k' \mid n, j, g) \Pr_t(n, j \mid n, k, g) \\ &= p_t(k') \sum_{j \in \{A,M\}} \Pr_t(n, j \mid n, k, g) = p_t(k'). \end{aligned}$$

Multiplying by the fixed mass r_g as in equation (17) gives $\Phi_{t+1}^{nk'} = r_g p_t(k')$, which is equation (A3.1.5). Thus the posterior over where parents lived and worked drops out because the choice probabilities are identical across these backgrounds; the full quantitative model needs the posterior precisely because this homogeneity does not generally hold.
Dividing the expected manufacturing-relative mass by r_g yields $x_{t+1} = p_t^F$. Given the predetermined x_0, the sequence of non-relative payoffs $\{a_t\}$ and parameters (r_g, b_A, b_M), equations (A3.1.4) and (A3.1.5) determine the entire transition without tracking further family histories. The common cohort-growth factor affects population levels but not these normalized shares.

Three paths: roots and branches. To distinguish the influence of inherited connections from the consequences of their evolution, consider three paths facing the same $\{a_t\}$. In the first, horizontal-relative utility effects are removed, while all other payoffs are unchanged. In the second, horizontal relatives matter for choices but the original share x_0 is held fixed in every generation. In the third, horizontal relatives matter and their composition evolves according to equation (A3.1.5). The corresponding manufacturing probabilities are

$$\begin{aligned} p_t^0 &= \Lambda(a_t) && \text{(no horizontal-relative effects)}, \\ p_t^{\text{fix}} &= \Lambda[a_t + r_g((b_M + b_A)x_0 - b_A)] && \text{(initial distribution fixed)}, \\ p_t^F &= \Lambda[a_t + r_g((b_M + b_A)x_t - b_A)], \quad x_{t+1} = p_t^F && \text{(endogenous connections)}. \end{aligned} \tag{A3.1.6}$$

The first path sets the effect of horizontal relatives on sector choice to zero, while keeping the baseline a_t unchanged. The fixed-distribution path keeps the initial composition unchanged even though workers enter manufacturing; it is not required to satisfy the endogenous law of motion. The fixed and endogenous paths have identical manufacturing probabilities at $t = 0$ because both inherit x_0. Adding and subtracting p_t^{fix} gives the exact decomposition

$$\underbrace{p_t^F - p_t^0}_{\text{total effect of relatives}} = \underbrace{p_t^{\text{fix}} - p_t^0}_{\text{roots}} + \underbrace{p_t^F - p_t^{\text{fix}}}_{\text{branches}}. \tag{A3.1.7}$$

Roots measure the effect of preserving the initial composition of horizontal relatives, while *branches* measure the additional effect of allowing that composition to change. The decomposition is exact in manufacturing-share levels even though choice probabilities are nonlinear, because the differences add up algebraically. Neither term must have a particular sign: roots are negative when initial relative connections favor agriculture sufficiently, and branches are positive when the endogenous manufacturing-relative share rises above x_0. These are partial-equilibrium comparisons because the common a_t is not recalculated across paths.

The transition from roots to branches. Comparing manufacturing choices in the endogenous and no-horizontal-relative paths in equation (A3.1.6), the two paths are equal when $b_M\Phi_t^{nM} = b_A\Phi_t^{nA}$. Substituting $\Phi_t^{nM} = r_g x_t$ and $\Phi_t^{nA} = r_g(1 - x_t)$ gives the threshold

$$x^* \equiv \frac{b_A}{b_M + b_A} = \frac{\theta_A \psi_R^A}{\theta_M \psi_R^M + \theta_A \psi_R^A}. \tag{A3.1.8}$$

The difference in manufacturing log odds can then be written as

$$\text{logit}(p_t^F) - \text{logit}(p_t^0) = r_g(b_M + b_A)(x_t - x^*), \tag{A3.1.9}$$

where $\text{logit}(p) \equiv \log[p/(1-p)]$. Since the logit function is strictly increasing and $r_g, b_M, b_A > 0$, horizontal relatives reduce manufacturing entry relative to the no-horizontal-relative benchmark when $x_t < x^*$ and increase it when $x_t > x^*$. If the initial distribution satisfies $x_0 < x^*$, roots remain negative in every generation, while branches more than offset them once $x_t > x^*$.
The threshold depends on the relative strength of manufacturing and agricultural connections, rather than the total mass of relatives r_g, which affects the magnitude of their influence and the evolution of x_t. Importantly, crossing the threshold means that horizontal relatives increase manufacturing employment relative to the no-horizontal-relative benchmark, not simply that manufacturing employment rises over time. Whether branches eventually overcome roots depends on whether x_t reaches this threshold.

Why branches can accelerate the transition. The changing composition of relatives also affects the pace of manufacturing entry. Taking log odds in equation (A3.1.4) gives $\text{logit}(p_t^F) = a_t + r_g[(b_M + b_A)x_t - b_A]$. Subtracting this expression between consecutive generations yields

$$\text{logit}(p_t^F) - \text{logit}(p_{t-1}^F) = (a_t - a_{t-1}) + r_g(b_M + b_A)(x_t - x_{t-1}), \qquad t \geq 1. \tag{A3.1.10}$$

The first term is the change in manufacturing log odds in the economy without horizontal-relative effects. The second is positive whenever the inherited share of manufacturing relatives rises. For $t \geq 2$, this second term equals $r_g(b_M + b_A)(p_{t-1}^F - p_{t-2}^F)$ because $x_t = p_{t-1}^F$.

A3.2 Model extensions

The baseline deliberately restricts the effects of family ties to the two margins identified most directly in the empirical analysis: local sector-specific family connections lower sector-entry barriers through Q_{gnjt}^k, while geographically proximate relatives raise location utility through $H_{gnjt}(m)$. This appendix considers two extensions. The first allows sector-specific family connections to influence wages directly. The second introduces schooling as a formal substitute

for inherited sector-specific connections. In each case, the extension is written to nest the corresponding baseline specification.

A3.2.1 Family connections and wages

In the baseline model, relatives help workers enter a sector through information, referrals and sector-specific knowledge, but workers in the same sector and location earn the same wage. Relatives may also help workers obtain better-paying jobs within a sector. I extend the model by allowing family connections to affect the wages individuals earn, in addition to their probability of entering a sector.

Consider a child of ancestral type g whose parent lives in location n and works in sector j. If the child enters sector k and works in location d, the wage earned is

$$\widetilde{w}^{k}_{gnjt+1}(d) = w^{k}_{d,t+1} + \gamma^{W}_{k}\left[\mathbf{1}\{j=k\}\, w^{k}_{nt} + \psi^{W,k}_{R}\sum_{m\in N}\mathcal{K}^{W}_{nm}\,\Phi^{k}_{gnjt}(m)\, w^{k}_{mt}\right], \tag{A3.2.1}$$

where

$$\mathcal{K}^{W}_{nm} = \left(1+\frac{\text{dist}_{nm}}{100\,\text{km}}\right)^{-\delta_k}. \tag{A3.2.2}$$

The first term is the market wage in sector k at destination d. The second captures the additional earnings associated with relatives working in that sector. The parent's contribution applies when the parent works in sector k, while aunts and uncles contribute according to their wages, locations and expected mass of relatives. The parameter γ^{W}_{k} measures the effect of the parental wage connection, while $\psi^{W,k}_{R}$ measures the contribution of horizontal relatives relative to the parent. The distance parameter δ_k determines how quickly the influence of relatives declines with distance.

I replace the market wage with $\widetilde{w}^{k}_{gnjt+1}(d)$ wherever individual earnings enter the worker's choices. Family connections can therefore affect both the sector a worker enters and the wage earned within that sector. The original sector-entry mechanism remains unchanged. Setting $\gamma^{W}_{A} = \gamma^{W}_{M} = 0$ recovers the baseline model.

Estimation. To estimate whether relatives affect individual wages, I exploit deaths between 1940 and 1950 that remove existing family connections. I focus on children aged 8–17 in 1940 who lived with a parent and can be linked to the 1950 census. In the model, the death of a relative removes a wage connection whose value depends on the relative's sector and earnings. I therefore interact deaths of parents and aunts or uncles with their wages in 1940, allowing the effects to differ between agriculture and manufacturing. Death indicators by relative and child sector account for the loss of family support independently of wages. I also control for the origin commuting zone, the child's destination and sector, parental sector and individual

characteristics. The outcome is the child's annual wage income in 1950. The ratio of the aunt-or-uncle coefficient to the parental coefficient provides an estimate of the relative weight of horizontal relatives, $\psi_R^{W,k}$.

Table A3.2.1 reports the results. All four death interactions are negative, as predicted by a positive wage contribution from relatives, but none is statistically significant. I calibrate the distance decay of wage connections to horizontal relatives using the model's estimated migration-distance elasticities, with $\delta_A = 1.120$ and $\delta_M = 1.054$. The implied weights on horizontal relatives are 0.62 in agriculture and 0.61 in manufacturing.

Table A3.2.1: Effects of relatives' deaths on individual wages

	Child's sector	
	Agriculture	Manufacturing
Death interacted with relative's wage		
Parent	−0.339	−0.231
	(0.270)	(0.428)
Aunt or uncle	−0.210	−0.141
	(0.190)	(0.110)
Implied weight on horizontal relatives		
Relative to parent, $\widehat{\psi}_R^{W,k}$	0.619	0.610
Observations	92,908	
Clusters (1940 CZ)	711	
R^2	0.116	

Notes: The dependent variable is the child's annual wage income in 1950. The sample includes children aged 8–17 in 1940 who work for wages in agriculture or manufacturing in 1950. All coefficients are estimated jointly. Regressions include destination-CZ-by-sector, origin-CZ, parent-sector-by-child-sector, age and weeks-worked fixed effects, death indicators by relative and child sector, relative wages by sector and child and parent characteristics. The weight on horizontal relatives is calculated as the ratio of the aunt-or-uncle coefficient to the parental coefficient within each sector. Standard errors in parentheses are clustered by 1940 CZ.

A3.2.2 Schooling as a formal substitute for family connections

The baseline sector-entry value is linear in the inherited family stock, $Q_{gnjt}^k = \theta_k F_{gnjt}^k$. I now introduce schooling as a formal alternative to this inherited route. The extension is constructed so that shutting down the formal route returns the baseline family contribution exactly.

Let $\mathcal{E}_{nt} > 0$ denote the stock of formal school resources available to a child whose parent lives in n. Because $F_{gnjt}^k = \log\big(\exp(F_{gnjt}^k)\big)$, I represent the family route by the positive input

$\mathcal{F}^k_{gnjt} \equiv \exp\left(F^k_{gnjt}\right)$. Family and formal inputs are combined through

$$I^k_{gnjt} = \left[\omega^k_F \exp\left(\chi F^k_{gnjt}\right) + (1-\omega^k_F)\mathcal{E}^{\chi}_{nt}\right]^{1/\chi}, \qquad \chi \equiv \frac{\eta-1}{\eta}, \tag{A3.2.3}$$

where $\eta > 1$ is the elasticity of substitution between inherited family access and formal school resources. The sector-entry contribution becomes

$$Q^{k,S}_{gnjt} = \theta_k \log I^k_{gnjt}. \tag{A3.2.4}$$

This normalization preserves the baseline. As $\omega^k_F \to 1$, equation (A3.2.3) gives $I^k_{gnjt} = \exp\left(F^k_{gnjt}\right)$, and therefore $Q^{k,S}_{gnjt} = \theta_k F^k_{gnjt} = Q^k_{gnjt}$. When $\eta > 1$, schooling and inherited family access are gross substitutes: an expansion of formal resources lowers the marginal value of an additional sector-specific family connection.

I allow school resources to increase with local development while retaining residual variation in local school supply:

$$\mathcal{E}_{nt} = \mathcal{S}_{nt}\bar{w}^{\xi_s}_{n,t-1}, \qquad \xi_s > 0, \tag{A3.2.5}$$

where $\bar{w}_{n,t-1}$ is the lagged employment-weighted average wage and $\mathcal{S}_{nt}$ captures school-supply variation not mechanically generated by local wages. Using the lagged wage prevents the child's contemporaneous sector choice from feeding directly back into the formal input entering that choice.

The extension changes only the sector-entry contribution Q^k_{gnjt}. The distribution of relatives $\Phi_{g,t}$, its law of motion, the adult location problem, production and trade remain unchanged. Schooling nevertheless affects the future distribution of relatives indirectly because changes in sector choices alter the distribution of descendants that enters the law of motion for $\lambda_{g,t+1}$.

Estimation. The extension adds ξ_s, the CES curvature χ, the sector-specific family weights ω^k_F and the corresponding sector-entry value scales. I estimate ξ_s from the relationship between local wage growth and school expenditure per enrolled child, instrumenting wage growth with baseline industry shares interacted with leave-one-CZ-out national industry wage growth. The preferred specification yields $\widehat{\xi}_s = 0.337$, with a first-stage F-statistic of 10.863.

To identify substitution, define $q^k_i = F^k_i$, $e_i = \log \mathcal{E}_i$ and $x^k_i = q^k_i - e_i$. The CES elasticity of the sector-entry index with respect to the family input is

$$s^k_{Fi} = \text{invlogit}\left(\alpha_k + \chi x^k_i\right), \qquad \alpha_k = \log \frac{\omega^k_F}{1-\omega^k_F}. \tag{A3.2.6}$$

Deaths of same-sector relatives identify the family margin at different values of x^k_i. I estimate the death-induced family response in twelve pre-determined relative-effectiveness groups and

fit

$$\widehat{m}^k_{F,r} = \Lambda_k \,\text{invlogit}\big(\alpha_k + \chi \bar{x}^k_r\big) + \varepsilon^k_r. \tag{A3.2.7}$$

Variation in the family-loss response across relative effectiveness identifies χ; the horizontal location of the schedule identifies ω^k_F; and its level identifies Λ_k. I impose a common curvature across sectors.

Table A3.2.2: Sector-entry parameters

	Agriculture	Manufacturing
Panel A. CES curvature		
Curvature, $\widehat{\chi}_k$	0.857**	0.936**
	(0.289)	(0.274)
Number of groups	12	12
First-stage observations	318,287	74,162
Panel B. Common curvature and elasticity		
Common curvature, $\widehat{\chi}$	0.899***	
	(0.199)	
Implied substitution elasticity, $\widehat{\eta} = 1/(1-\widehat{\chi})$	9.86	
Panel C. Sector-specific entry parameters		
Scale-free sector-entry value, $\widehat{\Lambda}_k$	0.0698	0.0724
CES family weight, $\widehat{\omega}^k_F$	0.584	0.385

Notes: Panel A reports the unrestricted sector-specific curvature estimates $\widehat{\chi}_k$ from equation (A3.2.7). Panel B reports the common curvature $\widehat{\chi}$. Panel C reports $\widehat{\Lambda}_k$ and $\widehat{\omega}^k_F$. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

School expansion and its interaction with family geography I use the extension to repeat the dynamic family accounting for the historical increase in school expenditure. The exercise compares the response without family connections, with the initial distribution of horizontal relatives held fixed and in the full model with endogenous changes in family connections.

A3.2.3 Model extension results

A4 Quantification

This section describes the data sources, the construction of the variables used in the empirical analysis and the procedures undertaken to estimate the structural parameters of the model for the U.S. between 1880 and 1950.

Table A3.2.3: Model extension

	1910	1930	1950	1970
Panel A. Baseline model				
Inherited kin connections	−0.25	−0.77	−1.30	−2.27
Propagation through extended kin	−0.34	−0.02	+0.68	+2.03
Panel B. Schooling extension				
Inherited kin connections	+0.1	−1.26	−1.66	−2.46
Propagation through extended kin	−0.28	−0.04	+0.72	+2.27
Panel C. Kin effects on wages				
Inherited kin connections	−0.36	−0.86	−1.18	−2.00
Propagation through extended kin	−0.18	+0.09	+0.53	+1.95

Notes: The table reports the effects of inherited kin connections and their propagation across generations. Panel A reports the baseline model. Panels B and C consider extensions allowing for schooling and kin effects on individual wages, respectively.

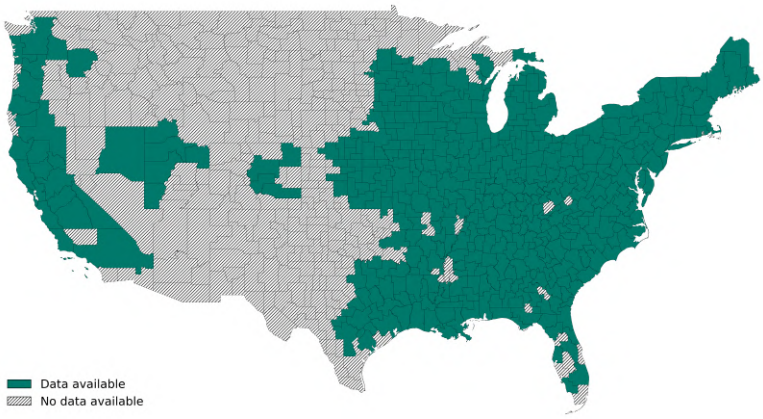
A4.1 Data

The primary data source is the complete-count U.S. Population Census from 1850 to 1950, accessed through IPUMS USA (Ruggles et al., 2024) in order to construct migration flows and parental-child industry transitions. In addition, information on agricultural wages is obtained from the Agricultural Census (1870–1940) and information on manufacturing salaries from the Manufacturing Census (1880–1940), both accessed through IPUMS NHGIS (Schroeder et al., 2025). To harmonize the geographic structure over time, consistent county-to-commuting-zone crosswalks developed by Eckert et al. (2020) are employed.

Regions. I follow the regional definition of Eckert and Peters (2022) and use commuting zones (CZs). County-to-CZ crosswalks are from Eckert et al. (2020) and respect historical county boundaries. A balanced sample from 1880 to 1950 yields 455 CZs. Table A4.1.1 compares the model sample with the full U.S. economy. The agricultural share among agricultural and manufacturing workers aged 20–40 is close to the national share early in the sample, although the fit deteriorates over time. The model sample contains roughly 23–25% of the U.S. population.

Sectoral and age definition. I focus on agriculture and manufacturing, the two sectors that account for most employment over the historical period and for which comparable wage measures are available (Eckert & Peters, 2022). Workers are classified using industry (`IND1950`) codes. Agricultural workers have industry code 105. Manufacturing workers have industry codes 306–499, covering durable and non-durable manufacturing. Construction and mining are

Figure A4.1.1: Region sample



Notes: This figure shows the 455 commuting zones that are used in the quantitative analysis. Source: Eckert et al. (2020).

Table A4.1.1: Sectoral employment and population in model vs. census, 1880–1950

Year	Agric. share among A+M %	Employment share Agric. %	Employment share Manuf. %	Population millions
Panel A. Model commuting zones, ages 20–40				
1880	68.3	36.8	17.1	12.54
1910	43.8	19.8	25.4	23.20
1930	28.6	11.7	29.3	29.48
1950	18.2	7.4	33.4	35.07
Panel B. United States, ages 20–40				
1880	75.8	44.2	14.1	16.29
1910	58.2	28.8	20.7	32.25
1930	42.9	18.6	24.8	41.13
1950	28.6	11.4	28.5	49.54
Panel C. United States, all ages				
1880	78.6	49.0	13.3	49.76
1910	65.2	35.3	18.8	92.02
1930	50.9	23.5	22.7	122.76
1950	35.6	14.3	25.9	152.26

Notes: The table reports sectoral employment shares and population totals for the model sample and the full United States. The first column reports agricultural employment as a share of agricultural plus manufacturing employment. The next two columns report agricultural and manufacturing employment as shares of the relevant population. Population is reported in millions.

excluded because wages for these groups are not consistently observed before 1940. Children are individuals aged 0–20. Parents are individuals aged 20–40 with at least one co-resident

child (`NCHILD > 0`).

Migration Flows. Migration flows are measured using individual-level links from the Census Linking Tree Project (Buckles et al., 2025) for 1880-1910, 1910-1930 from IPUMS Multigenerational Longitudinal Project for 1930-1950 (Helgertz et al., 2023). Parents are defined as individuals aged 20–40 with at least one co-resident child (`NCHILD > 0`). Gross migration flows and migration probabilities are constructed by following individuals 20/30 years in the future with a single crosswalk (no chained crosswalks over two consecutive 10-year periods).[37] Migration flows across regions are computed by aggregating them to the CZ level.

Wages. Manufacturing wages for 1880–1920 are obtained from the Manufacturing Census (NHGIS) as total wages divided by the number of workers, aggregated to the commuting-zone level. Since manufacturing wages are not available for 1930, I use the average wages from the 1920 Manufacturing Census and the 1940 individual census. For 1950, individual-level census data allow direct computation of average wages for manufacturing workers aged 20–40. Agricultural wages are measured as expenditures on farm labor (1870, 1900, 1920) per county, divided by the number of farm laborers aged 20–60. The model covers slightly different time periods so I use the wages in 1870 for the 1880 period, the wages in 1900 for 1910 period and the wages in 1920 for 1930. While self-employed farmers' incomes are excluded, validation against Iowa state census data and comparisons with 1950 commuting-zone income distributions suggest the measure is reasonable. In cases where individual wages are present, I do not residualize them using Mincer regressions so that they are consistent with the averages for 1880-1920.

Productivity. Figure A4.1.3 depicts the geographical distribution of soil erosion taken from Hornbeck (2012).

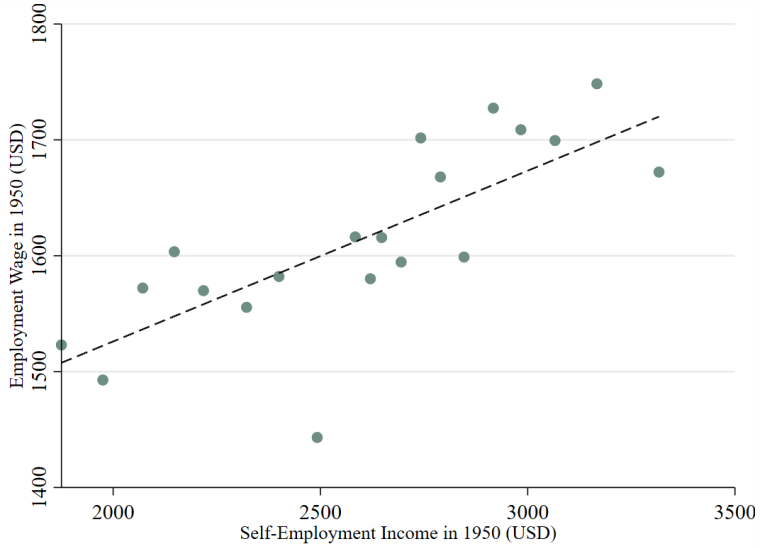
Trade Costs. I construct bilateral trade costs between regions using the county-to-county transportation cost data from Hornbeck and Rotemberg (2024). Their data digitize the U.S. transportation network by decade from 1830 to 1920 and report the cost of shipping one ton of goods between county centroids, holding counties fixed at 1890 boundaries. Following Hornbeck and Rotemberg (2024), I convert dollar-per-ton costs into iceberg trade costs by assuming an average value of $35 per ton of agricultural goods in 1890 following Eckert and Peters (2022):

$$T_{odt} = 1 + \frac{t_{odt}}{35}, \tag{A4.1.1}$$

where t_{odt} is the reported transportation cost from county o to county d in decade t. I then aggregate the resulting county-level trade cost matrix to the regional definition.

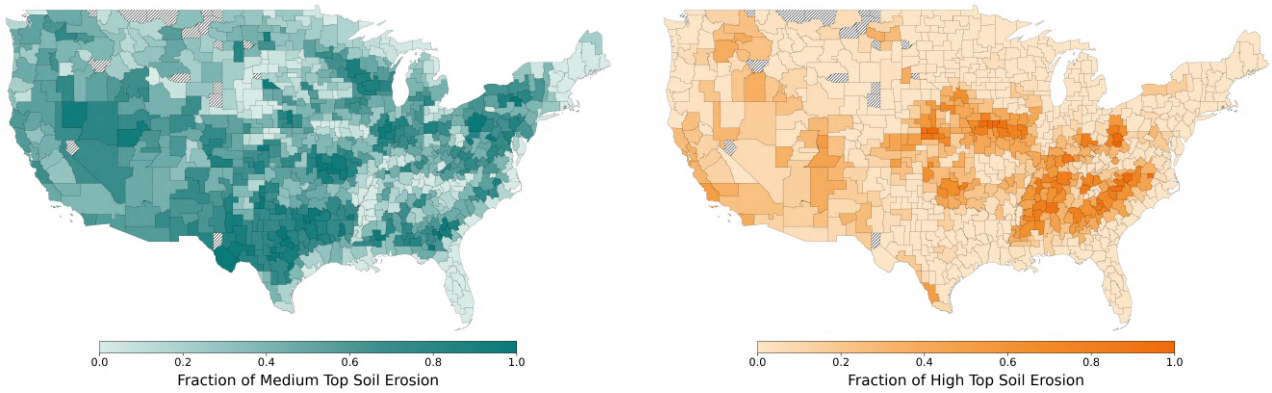
[37] This is done for consistency because there is no census data for 1890.

Figure A4.1.2: Agricultural income comparison: self-employed (total) vs wage income



Notes: This figure shows a binned scatter plot between self-employed income from farmer individuals and wage income from the same industry group within CZs in 1950. The dotted line represents the fitted line. Observations: 227 CZs. Source: Eckert et al. (2020); Helgertz et al. (2023).

Figure A4.1.3: Geographical Distribution of Soil Erosion



(a) Medium Erosion (b) High Erosion

Notes: Panel (a) shows the geographical distribution of medium soil erosion across CZs per agricultural area. Panel (b) shows the geographical distribution of high soil erosion across CZs per agricultural area. High degrees of soil erosion are defined as regions with more than 75 percent of topsoil lost and potential numerous or deep gullies, while medium erosion is characterized by areas with 25 percent to 75 percent of topsoil lost and potentially some gullies (Hornbeck, 2012).

A4.2 Construction of migration distance

Migration costs in the model are governed by an elasticity on the distance between CZ pairs. Rather than using straight-line distance between CZ centroids, I construct a rail-adjusted effective distance that accounts for the historical railroad network, following the least-cost-path

approach used in the market-access literature (Donaldson, 2018; Donaldson & Hornbeck, 2016). This approach is motivated by evidence that rail access shaped population movement and settlement directly, not only goods trade (Atack, Bateman, Haines, & Margo, 2010).

Railroad segments are taken from the Atack (2016) historical railroad shapefile, which records the year each segment entered operation (1826–1916) along with its geographic path. Commuting zones are the 1990 boundaries used throughout the paper (Eckert et al., 2020; Tolbert & Sizer, 1996), for which I take the polygon centroid as the CZ's location.

For each migration wave $t \in \{1880, 1910, 1930\}$ I build the cumulative rail network in place at the start of the wave, keeping all segments with an operational year at or before t. Segment vertices are snapped to a 1 km grid so that shared endpoints across adjacent segments are merged into common nodes, yielding a connected, routable graph in which edge weights equal true segment length. Each CZ centroid is linked by a straight-line "access edge" to its $K = 20$ nearest rail nodes, proxying for the unpaved wagon/road trip required to reach the railroad; no digitized historical road network is used for this leg. The grid resolution and K are computational choices made for tractability.[38]

For every ordered CZ pair (o, d) and wave t, I compute the least-cost route on the combined graph, where travel on the rail network costs 1 per unit distance and travel off the network costs κ per unit distance:

$$\text{distance}_t(o,d) \;=\; \min_{\text{routes}} \Big[\text{km}_{\text{on-rail}} \times 1 \;+\; \text{km}_{\text{off-rail}} \times \kappa \Big], \tag{A4.2.1}$$

solved by Dijkstra's shortest-path algorithm from every CZ centroid. By construction, $\text{eff_dist}_t(o,d)$ is bounded between the Euclidean distance (perfect rail connection) and $\kappa\times$ Euclidean distance (no useful rail access, pure wagon travel), and is floored at the latter to handle pairs with no nearby rail. This yields a distinct effective-distance matrix for each wave, so that the same CZ pair becomes relatively closer over time as the rail network expands.

The parameter κ is the ratio of the per-mile cost of off-rail (wagon) transport to on-rail transport. Historical freight-rate evidence puts wagon haulage at roughly an order of magnitude more expensive per ton-mile than rail (Donaldson & Hornbeck, 2016; Fogel, 1964; Taylor, 1951); complementary hedonic evidence confirms rail access was capitalized into antebellum land values (Craig, Palmquist, & Weiss, 1998). Motivated by this range, I set $\kappa = 10$.[39]

A4.3 Estimation of migration costs

I estimate migration costs from gross migration flows implied by the location-choice probabilities. Let $F^{j}_{gnn't}$ denote the mass of adults of ancestral type g and sector j who move from origin

[38] The size of K does not matter much for the effective distance with a median deviation of below 0.1%.

[39] The results are robust to $\kappa \in \{3, 5, 10\}$.

n to destination n' in period t. Given the location-choice probability in equation (13),

$$F^j_{gnn't} = L^j_{gnt} P^N_t(n' \mid n, j, g), \tag{A4.3.1}$$

where L^j_{gnt} is the mass of type-g, sector-j adults residing in origin n.
From equation (11), the location-choice probability can be written as

$$P^N_t(n' \mid n, j, g) = \exp\left\{ \frac{1}{\sigma_N}\left[\mathcal{A}^j_{gn't} - \rho \log L_{n't} + v\left(w^j_{n't}, p_{n't}\right) + \beta O_t(n', j, g)\right] + \Theta_H H_{gnjt}(n') - \frac{M^j_{gnn't}}{\sigma_N} - \log \sum_{n'' \in N} \exp\left(\frac{B^j_{gnt}(n'')}{\sigma_N}\right) \right\}, \tag{A4.3.2}$$

where

$$\Theta_H \equiv \frac{\theta_H}{\sigma_N}. \tag{A4.3.3}$$

Substituting equation (A4.3.2) into equation (A4.3.1) gives

$$\mathbb{E}\left[F^j_{gnn't}\right] = \exp\left\{ \log L^j_{gnt} - \log \sum_{n'' \in N} \exp\left(\frac{B^j_{gnt}(n'')}{\sigma_N}\right) + \frac{1}{\sigma_N}\left[\mathcal{A}^j_{gn't} - \rho \log L_{n't} + v\left(w^j_{n't}, p_{n't}\right) + \beta O_t(n', j, g)\right] + \Theta_H H_{gnjt}(n') - \frac{M^j_{gnn't}}{\sigma_N} \right\}. \tag{A4.3.4}$$

This expression separates naturally into origin- and destination-specific components. Define the destination component as

$$\text{destFE}^j_{gn't} \equiv \frac{1}{\sigma_N}\left[\mathcal{A}^j_{gn't} - \rho \log L_{n't} + v\left(w^j_{n't}, p_{n't}\right) + \beta O_t(n', j, g)\right], \tag{A4.3.5}$$

which contains all components of destination utility that vary with g, n', j and t, but not with the origin n.
Similarly, define the origin component as

$$\text{origFE}^j_{gnt} \equiv \log L^j_{gnt} - \log \sum_{n'' \in N} \exp\left[\text{destFE}^j_{gn''t} + \Theta_H H_{gnjt}(n'') - \frac{M^j_{gnn''t}}{\sigma_N}\right]. \tag{A4.3.6}$$

The first term is the mass of potential migrants at the origin, while the second is the location-choice inclusive value. Both are common across destinations for a given origin state (g, n, j, t) and can therefore be absorbed by an origin fixed effect.

Using equations (A4.3.5)–(A4.3.6), expected migration flows can be written as

$$\mathbb{E}\left[F^{j}_{gnn't}\right] = \exp\left(\text{origFE}^{j}_{gnt} + \text{destFE}^{j}_{gn't} + \Theta_H H_{gnjt}(n') - \frac{M^{j}_{gnn't}}{\sigma_N}\right). \tag{A4.3.7}$$

Importantly, the kin term is not part of either fixed effect. The object $H_{gnjt}(n')$ depends jointly on the origin state and the destination: two adults of the same ancestral type and sector considering the same destination can have different kin exposure there depending on their origin. It therefore cannot in general be absorbed by separate origin and destination fixed effects. If not origin or destination fixed effect can be estimated for a group, I assign them the weighted average for a given destination or origin region by sector.

I parameterize normalized bilateral migration costs as

$$\frac{M^{j}_{gnn't}}{\sigma_N} = \lambda^{j}_{t}\widetilde{d}_{nn't}, \tag{A4.3.8}$$

where $\widetilde{d}_{nn't}$ is the log of rail-adjusted least-cost distance between n and n', normalized to zero for within-CZ moves. Substituting equation (A4.3.8) into equation (A4.3.7) yields the estimating equation

$$\mathbb{E}\left[F^{j}_{gnn't}\right] = \exp\left(\text{origFE}^{j}_{gnt} + \text{destFE}^{j}_{gn't} + \widehat{\Theta}_H H_{gnjt}(n') - \lambda^{j}_{t}\widetilde{d}_{nn't}\right). \tag{A4.3.9}$$

I estimate equation (A4.3.9) by PPML (Silva & Tenreyro, 2006), using all origin-destination flows, including stayers. Because Θ_H is identified separately from the family-death design, I treat $\widehat{\Theta}_H H_{gnjt}(n')$ as a known offset rather than re-estimating its coefficient. The coefficient λ^{j}_{t} therefore identifies distance-related migration costs net of kin exposure.

The estimated distance coefficient identifies migration costs relative to the location-choice scale. After estimating σ_N, I recover migration costs in utility units as

$$\widehat{M}^{j}_{gnn't} = \widehat{\sigma}_N \widehat{\lambda}^{j}_{t}\widetilde{d}_{nn't}. \tag{A4.3.10}$$

Table A4.3.1 reports the estimated distance coefficients.

A4.4 Sector deaths and continuation-value adjustment

This appendix derives the mapping between the sector-death coefficients and the structural kin parameters and quantifies the error from ignoring the effect of a sector-employed relative on future location values. As in Section 5.2, $r = P$ denotes a parent death and $r = R$ an

Table A4.3.1: Distance Elasticity of Migration Flows by Sector and Year

	1880		1910		1930	
	Agric. (1)	Manuf. (2)	Agric. (3)	Manuf. (4)	Agric. (5)	Manuf. (6)
Log(distance)	-1.152***	-1.144***	-1.222***	-1.135***	-1.406***	-1.290***
	(0.013)	(0.018)	(0.015)	(0.022)	(0.019)	(0.022)
Observations	9,568,230	1,222,099	14,706,667	5,022,271	11,810,302	8,457,209
Pseudo R^2	0.559	0.542	0.591	0.529	0.675	0.626
Origin clusters	448	433	454	454	455	455
Destination clusters	448	444	454	454	455	455
Estimation	PPML	PPML	PPML	PPML	PPML	PPML

Notes: The dependent variable is the migration flow between origin and destination commuting zones for ancestry group g by origin sector. Each column reports a separate PPML regression by year and sector. All specifications include ancestry-group × origin and ancestry-group × destination fixed effects. Standard errors, reported in parentheses, are robust to heteroskedasticity and two-way clustered by origin and destination commuting zone. The constant is omitted from the table. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

aunt-or-uncle death. For any object x,

$$\Delta^{(r)}x \equiv x^{\text{post}(r)} - x^{\text{pre}}$$

denotes its post-death minus pre-death change following a death of type r.

Sector log odds. For sector k relative to the alternative sector $\ell \neq k$, the logit choice probabilities imply

$$\ell^k_{gnjt} \equiv \log \frac{P^O_t(k \mid n, j, g)}{P^O_t(\ell \mid n, j, g)} = \Lambda_k F^k_{gnjt} - \Lambda_\ell F^\ell_{gnjt} - \frac{\tau^k_{gnjt} - \tau^\ell_{gnjt}}{\sigma_O} + \frac{V_{t+1}(n, k, g) - V_{t+1}(n, \ell, g)}{\sigma_O}. \tag{A4.4.1}$$

A death of a sector-k relative reduces the sector-access stock by

$$d^k_{F,P} = 1, \qquad d^k_{F,R} = \psi^k_R.$$

The resulting change in sector-k log odds can therefore be written as

$$\Delta^{(r)}\ell^k = -\Lambda_k d^k_{F,r} + \Omega^k_r + \xi^k_r, \tag{A4.4.2}$$

where

$$\Omega^k_r \equiv \frac{\Delta^{(r)}V_H(k) - \Delta^{(r)}V_H(\ell)}{\sigma_O} \tag{A4.4.3}$$

captures the continuation-value response generated directly by the loss of geographic kin support. The residual term ξ_r^k collects changes in relative sector barriers and non-kin continuation values.

Continuation-value response. Let $d_{H,r}$ denote the reduction in location support caused by a death of type r. A parent contributes one unit, so $d_{H,P} = 1$; for an aunt or uncle, $d_{H,R}$ is the corresponding horizontal-location contribution implied by the kin block.
Consider a relative located at the origin n. Conditional on sector h, write the pre-death inclusive value as

$$V(h) = \sigma_N \log \left[\exp\left\{ \frac{B^h(n)}{\sigma_N} \right\} + \sum_{m \neq n} \exp\left\{ \frac{B^h(m)}{\sigma_N} \right\} \right]. \tag{A4.4.4}$$

The death lowers origin utility by $\theta_H d_{H,r}$ while leaving the utility of alternative destinations unchanged. The post-death inclusive value is therefore

$$V^{\text{post}(r)}(h) = \sigma_N \log \left[\exp\left\{ \frac{B^h(n)}{\sigma_N} \right\} e^{-\Theta_H d_{H,r}} + \sum_{m \neq n} \exp\left\{ \frac{B^h(m)}{\sigma_N} \right\} \right]. \tag{A4.4.5}$$

Let

$$s_h \equiv P_{t+1}^N(n \mid n, h, g)$$

denote the pre-death probability of remaining at the origin. Factoring out the pre-death denominator gives

$$\Delta^{(r)} V_H(h) = \sigma_N \log \left[1 - s_h \left(1 - e^{-\Theta_H d_{H,r}} \right) \right]. \tag{A4.4.6}$$

Substituting this expression into equation (A4.4.3) yields

$$\Omega_r^k = \frac{\sigma_N}{\sigma_O} \log \left[\frac{1 - s_k \left(1 - e^{-\Theta_H d_{H,r}} \right)}{1 - s_\ell \left(1 - e^{-\Theta_H d_{H,r}} \right)} \right]. \tag{A4.4.7}$$

The geographic component therefore cancels exactly when $s_k = s_\ell$. Otherwise,

$$\operatorname{sign}(\Omega_r^k) = -\operatorname{sign}(s_k - s_\ell). \tag{A4.4.8}$$

For small Θ_H,

$$\Omega_r^k \simeq -\frac{\sigma_N}{\sigma_O} \Theta_H d_{H,r} (s_k - s_\ell). \tag{A4.4.9}$$

Thus the location channel affects the coefficient mapping only when the two sector choices imply different subsequent location behavior.

Identification and mapping error. Let $\widetilde{D}_z^r$ denote death exposure residualized on the fixed effects and controls in equation (34). The identifying restriction is

$$\mathbb{E}\left[\widetilde{D}_z^r \xi_{r,z}^k\right] = 0. \tag{A4.4.10}$$

This requires residual death exposure to be unrelated to shocks that independently change relative sector barriers or non-kin continuation values. Origin-CZ-by-parental-sector-by-year fixed effects absorb sectoral shocks common to workers in the same initial state, while ancestry fixed effects and predetermined controls absorb persistent differences across ancestral groups. Lagged sector-specific migration rates are also included and strongly predict subsequent migration rates, helping absorb persistent heterogeneity in future location behavior.
Under equation (A4.4.10), the regression coefficient satisfies

$$\beta_r^k = \Lambda_k d_{F,r}^k - \overline{\Omega}_r^k, \tag{A4.4.11}$$

where $\overline{\Omega}_r^k$ is the continuation-value correction averaged over the identifying variation in the regression. Using the simpler mapping $\beta_r^k = \Lambda_k d_{F,r}^k$ therefore produces the error

$$\underbrace{\beta_r^k - \Lambda_k d_{F,r}^k}_{\text{mapping error}} = -\overline{\Omega}_r^k. \tag{A4.4.12}$$

The corresponding corrected parameters are

$$\Lambda_k^{\text{corr}} = \beta_P^k + \overline{\Omega}_P^k, \qquad \psi_R^{k,\text{corr}} = \frac{\beta_R^k + \overline{\Omega}_R^k}{\beta_P^k + \overline{\Omega}_P^k}. \tag{A4.4.13}$$

Magnitude. The continuation-value adjustment is quantitatively small. Average probabilities of remaining at the origin are 0.734 for agricultural workers and 0.744 for manufacturing workers. Using the model-implied sector-specific location probabilities together with

$$\widehat{\sigma}_N = 0.51, \qquad \widehat{\sigma}_O = 0.60, \qquad \widehat{\Theta}_H = 0.024, \qquad \widehat{\psi}_R^H = 0.874,$$

equation (A4.4.7) gives

$$\overline{\Omega}_P^M = -0.000205, \qquad \overline{\Omega}_P^A = 0.000205,$$

and

$$\overline{\Omega}_R^M = -0.000179, \qquad \overline{\Omega}_R^A = 0.000179.$$

Applying equation (A4.4.13) would change the parental sector parameters from

$$\Lambda_M = 0.0489 \quad \text{to} \quad \Lambda_M^{\text{corr}} = 0.0487,$$

and from

$$\Lambda_A = 0.0206 \quad \text{to} \quad \Lambda_A^{\text{corr}} = 0.0208.$$

These correspond to changes of approximately 0.4% and 1.0%, respectively. The implied horizontal relationship weights change from

$$\psi_R^M = 0.141 \quad \text{to} \quad \psi_R^{M,\text{corr}} = 0.138,$$

and from

$$\psi_R^A = 0.305 \quad \text{to} \quad \psi_R^{A,\text{corr}} = 0.311,$$

or by approximately 2.1% in either case. Equivalently, the absolute aunt-or-uncle continuation corrections are about 2.6% of the manufacturing aunt coefficient and 2.8% of the agricultural aunt coefficient.

Because the adjustment is small relative to the estimated sector-kin effects and leaves the substantive conclusions unchanged, the baseline quantification uses the direct coefficient mapping in equation (35). Equation (A4.4.13) shows that explicitly accounting for the location continuation-value channel has negligible quantitative effects on the recovered kin parameters.

A4.5 Indirect Inference

This appendix provides the details of the indirect-inference procedure used to estimate the location- and sector-choice dispersion parameters (σ_N, σ_O). I first describe the empirical Dust Bowl moments and the agricultural productivity disturbance, then explain how candidate dispersion parameters enter the model inversion, how I construct the corresponding Dust Bowl counterfactuals and how the empirical and simulated moments are matched.

A4.5.1 Empirical moments

The Dust Bowl generated large and persistent losses in agricultural land quality and economic activity across locations with different erosion severity (Hornbeck, 2012). These shocks induced substantial migration and longer-run adjustment (Hornbeck, 2012, 2023; Liu, 2026). I combine full-count Census microdata (Ruggles et al., 2024), longitudinal links from the IPUMS Multigenerational Longitudinal Panel (Helgertz et al., 2023) and the Census Tree (Buckles et al., 2025), soil-erosion measures from Hornbeck (2012) and local historical controls from Fishback (2017).

Let c index commuting zones and t the baseline census year, with outcomes measured twenty years later. Because erosion severity is time invariant, I interact medium- and high-erosion

indicators with baseline-year indicators and estimate

$$Y_{i,c,t+20} = \alpha + \lambda_c + \lambda_{s(c),t} + \gamma' X_{i,c,t} + \sum_{r \neq 1910} \mathbf{1}\{t = r\} \left[\beta_{M,r} \text{MErosion}_c + \beta_{H,r} \text{HErosion}_c\right] + \varepsilon_{i,c,t+20}, \tag{A4.5.1}$$

where 1910 is the baseline cohort, λ_c are CZ fixed effects and $\lambda_{s(c),t}$ are state-by-baseline-year fixed effects. Medium and high erosion are defined from cumulative topsoil loss, with low-erosion CZs omitted (Hornbeck, 2012). The controls $X_{i,c,t}$ contain predetermined individual and household characteristics and local Depression-era controls. Standard errors are clustered by CZ.

For sector choice, the sample consists of children younger than 15 at baseline and the outcome is manufacturing employment twenty years later, estimated separately by parental sector.[40] For migration, the sample consists of adults older than 25 and the outcome indicates whether the individual moves across commuting zones, estimated separately by initial sector. The resulting erosion-by-cohort coefficients correspond to changes in the sector-choice and out-migration probabilities generated by the model.

Let $\widehat{b}^{\text{data}}$ collect the empirical coefficients used as indirect-inference moments. The same samples, erosion definitions and auxiliary specifications are applied to simulated model outcomes below.

A4.5.2 Productivity estimates

The indirect-inference exercise also requires the magnitude of the agricultural productivity disturbance that generates these behavioral responses. Rather than mapping agricultural revenue per acre into productivity through an auxiliary production function, I estimate the erosion effect directly on the agricultural productivity shifter Z_{ct}^A recovered from the quantitative model.

Agricultural productivity is recovered for each CZ and model period from the goods-market inversion described in Appendix A4.6.1. This inversion combines observed wages and employment with trade costs, expenditure shares and the Eaton–Kortum trade structure to recover Z_{ct}^A. Importantly, this recovery does not use the location- and sector-choice dispersion parameters (σ_N, σ_O) subsequently estimated by indirect inference.

I estimate

$$\log Z_{ct}^A = \alpha_c + \alpha_{s(c),t} + \sum_{\tau \in \{1910,1950\}} \left[\beta_{M,\tau} \,\text{MErosion}_c + \beta_{H,\tau} \,\text{HErosion}_c\right] \mathbf{1}\{t = \tau\} + \gamma' X_{ct} + \varepsilon_{ct}, \tag{A4.5.2}$$

[40] Manufacturing contains durable industries (IND1950 306–399) and non-durable industries (IND1950 406–499). Agriculture is farming (IND1950 105).

Table A4.5.1: Erosion exposure, sector choice and migration

	Children			Household heads	
	Farmer children		Non-farmer children	Migration	
	Agriculture	Manufacturing	Agriculture	Farmer	Non-farmer
	(1)	(2)	(3)	(4)	(5)
Panel A. Children's sector choice					
Pre-trends					
1900 × Medium erosion	1.017	-0.910*	-0.314		
	(0.628)	(0.532)	(0.404)		
1900 × High erosion	0.617	-0.664	-0.342		
	(0.772)	(0.569)	(0.500)		
Post-treatment					
1930 × Medium erosion	-3.87***	2.27***	-0.71		
	(1.05)	(0.82)	(0.51)		
1930 × High erosion	-6.23***	3.26***	-0.97		
	(1.30)	(1.15)	(0.64)		
Panel B. Household-head migration					
Pre-trends					
1900 × Medium erosion				-0.755	-0.639
				(0.887)	(0.425)
1910 × Medium erosion				-0.354	0.241
				(0.669)	(0.343)
1900 × High erosion				-0.671	0.633
				(1.011)	(0.446)
1910 × High erosion				-0.945	-0.222
				(0.846)	(0.342)
Post-treatment					
1930 × Medium erosion				1.12***	1.00***
				(0.40)	(0.37)
1930 × High erosion				0.93**	0.17
				(0.42)	(0.46)
Observations	4,112,128	4,112,128	7,541,146	4,441,056	19,507,083
R^2	0.109	0.055	0.063	0.045	0.043
Mean outcome in 1910	51.00	23.00	8.00	5.00	8.00
Mean medium-erosion exposure	44.00	44.00	41.00	44.00	40.00
Mean high-erosion exposure	19.00	19.00	11.00	18.00	11.00
CZ fixed effects	Yes	Yes	Yes	Yes	Yes
State × year fixed effects	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes

Notes: The table reports difference-in-differences estimates relating medium and high erosion exposure to children's sector choices and household-head migration. Columns (1)–(2) restrict the sample to children of farmers; column (3) uses children of non-farmers. Columns (4)–(5) report migration outcomes for farmer and non-farmer household heads, respectively. Coefficients are multiplied by 100 and are therefore expressed in percentage points. All specifications include commuting-zone fixed effects, state-by-year fixed effects and individual controls. Standard errors, clustered by commuting zone, are reported in parentheses. The omitted erosion category is low exposure. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

where 1930 is the omitted year. CZ fixed effects absorb permanent differences across locations and state-by-year fixed effects absorb shocks common to CZs within a state. The preferred specification additionally includes the local controls reported in Table A4.5.2. Standard errors are clustered by CZ.

Columns (3)–(4) of Table A4.5.2 report the estimates. The pre-Dust-Bowl coefficients are small and statistically indistinguishable from zero: in the preferred specification, the 1910 coefficients are 0.008 for medium erosion and −0.019 for high erosion. By 1950, agricultural productivity is 0.179 log points lower in medium-erosion CZs and 0.226 log points lower in high-erosion CZs relative to low-erosion CZs.

I use these estimates to construct the Dust Bowl component of agricultural productivity,

$$\widehat{\Delta \log Z}^{DB}_{c,1950} = \widehat{\beta}_{M,1950}\,\mathrm{MErosion}_c + \widehat{\beta}_{H,1950}\,\mathrm{HErosion}_c. \tag{A4.5.3}$$

The observed transition retains this component. In the no-Dust-Bowl counterfactual used for indirect inference, I remove it from agricultural productivity:

$$\log Z^{A,\mathrm{noDB}}_{c,1950} = \log Z^{A,\mathrm{obs}}_{c,1950} - \widehat{\Delta \log Z}^{DB}_{c,1950}. \tag{A4.5.4}$$

Because the estimated Dust Bowl components are negative, removing them raises agricultural productivity in medium- and high-erosion locations relative to the observed economy.

Comparison with agricultural outcomes in the data. As an additional validation of the model-recovered productivity disturbance, I estimate the same event-study specification using log farm products per acre. Measures of agricultural production or revenue per acre are standard outcomes in empirical work on the Dust Bowl; related measures also enter the quantitative analysis in Yang (2024).

Columns (1)–(2) of Table A4.5.2 show that farm products per acre decline by 0.176 log points in medium-erosion CZs and 0.241 log points in high-erosion CZs by 1950. These magnitudes are close to the corresponding model-recovered productivity effects of 0.179 and 0.226 log points. The agreement is not imposed by the productivity inversion and provides an external check that the erosion-related decline in Z^A is of a magnitude consistent with the deterioration observed directly in agricultural production.

A4.5.3 Candidate-specific model inversion

The empirical responses identify how strongly workers react to the Dust Bowl, while the productivity estimates identify the size of the disturbance generating that response. For each candidate pair

$$\sigma = (\sigma_N, \sigma_O),$$

Table A4.5.2: Estimating the effect of soil erosion on agricultural productivity

	Data		Model	
	Log farm products per acre		Log agricultural productivity	
	(1)	(2)	(3)	(4)
Panel A. Medium versus low erosion				
1910 × Medium erosion	0.130	0.130	0.001	0.008
	(0.116)	(0.117)	(0.059)	(0.060)
1950 × Medium erosion	-0.176*	-0.176*	-0.186	-0.179
	(0.096)	(0.098)	(0.117)	(0.117)
1970 × Medium erosion	-0.053	-0.053		
	(0.128)	(0.129)		
1990 × Medium erosion	0.004	0.004		
	(0.148)	(0.150)		
Panel B. High versus low erosion				
1910 × High erosion	0.273**	0.273**	-0.036	-0.019
	(0.128)	(0.128)	(0.054)	(0.056)
1950 × High erosion	-0.241***	-0.241**	-0.231*	-0.226*
	(0.092)	(0.094)	(0.118)	(0.118)
1970 × High erosion	-0.268*	-0.268*		
	(0.140)	(0.142)		
1990 × High erosion	-0.446**	-0.446**		
	(0.187)	(0.190)		
Observations	1,857	1,857	1,665	1,647
R^2	0.849	0.849	0.939	0.939
CZ fixed effects	✓	✓	✓	✓
State × year fixed effects	✓	✓	✓	✓
Log population		✓		✓
Years	1910–1990	1910–1990	1910–1950	1910–1950
Cluster	CZ	CZ	CZ	CZ

Notes: The unit of observation is a CZ by year. Columns (1)–(2) use the log value of farm products per acre as the measure of observed agricultural productivity. Columns (3)–(4) use agricultural productivity recovered from the quantitative model. Medium- and high-erosion CZs are compared with low-erosion CZs, with 1930 as the omitted year. All specifications include CZ and state-by-year fixed effects. Columns (2) and (4) additionally control for log population. Standard errors, clustered at the CZ level, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

I recover the remaining structural costs and amenities using the model inversion in Appendix A4.6. The kin parameters identified earlier are expressed relative to the corresponding taste dispersion. For a candidate σ_N, the estimated location-attachment coefficient and migration-cost schedule are mapped into utility units according to

$$\theta_H = \sigma_N \Theta_H, \qquad M^j_{gnmt} = \sigma_N \lambda^j_t \widetilde{d}_{nmt}.$$

Given these objects and observed location-choice probabilities, the location inversion nets out

kin amenities and bilateral migration costs to recover the non-kin location amenities that rationalize observed migration choices.
Similarly, for sector choice, $\theta_k = \sigma_O \Lambda_k$. Given the candidate σ_O, observed sector-choice odds and the corresponding continuation values, I recover the non-kin sector-entry costs after netting out the kin contribution Q^k_{gnjt}. Because continuation values themselves depend on σ_N and the equilibrium path, the recovered sector-entry costs are candidate specific.
Thus, each candidate (σ_N, σ_O) implies a complete set of fundamentals that exactly rationalizes the observed baseline choices while preserving the kin effects identified relative to taste dispersion from the empirical kin variation. This inversion is repeated before evaluating the Dust Bowl response for that candidate.

A4.5.4 Model-based Dust Bowl effects

For each candidate σ, I solve two equilibrium transitions using the candidate-specific recovered fundamentals.

Observed transition (Dust Bowl). The first transition follows the observed economy and includes the estimated Dust Bowl component of agricultural productivity. Sector choices, migration, wages, prices and the kin distribution respond endogenously. Because observed fundamentals end in 1950, I hold them fixed thereafter for the final transition used to evaluate continuation values.

Counterfactual transition (no Dust Bowl). The second transition holds all non-Dust-Bowl fundamentals and recovered wedges fixed but removes the estimated erosion-induced component of agricultural productivity in equation (A4.5.4). I then resolve the model for the counterfactual equilibrium path.
Let $Y^{\text{DB}}_{i,c,t+20}(\sigma)$ and $Y^{\text{noDB}}_{i,c,t+20}(\sigma)$ denote the corresponding model-implied outcomes. Their difference,

$$\tau^{\text{model}}_{i,c,t+20}(\sigma) = Y^{\text{DB}}_{i,c,t+20}(\sigma) - Y^{\text{noDB}}_{i,c,t+20}(\sigma), \tag{A4.5.5}$$

isolates the model-implied Dust Bowl response while allowing sector choice, migration, wages, prices and kin connections to adjust in equilibrium. All non-Dust-Bowl fundamentals are held fixed across the two paths.

A4.5.5 Simulated regressions and indirect inference

To make the simulated moments directly comparable with the empirical estimands, I apply the same auxiliary design used in equation (A4.5.1) to the model-generated outcomes. For each

candidate $\sigma = (\sigma_N, \sigma_O)$, I aggregate model outcomes to the corresponding ancestry-group-by-CZ cells and estimate

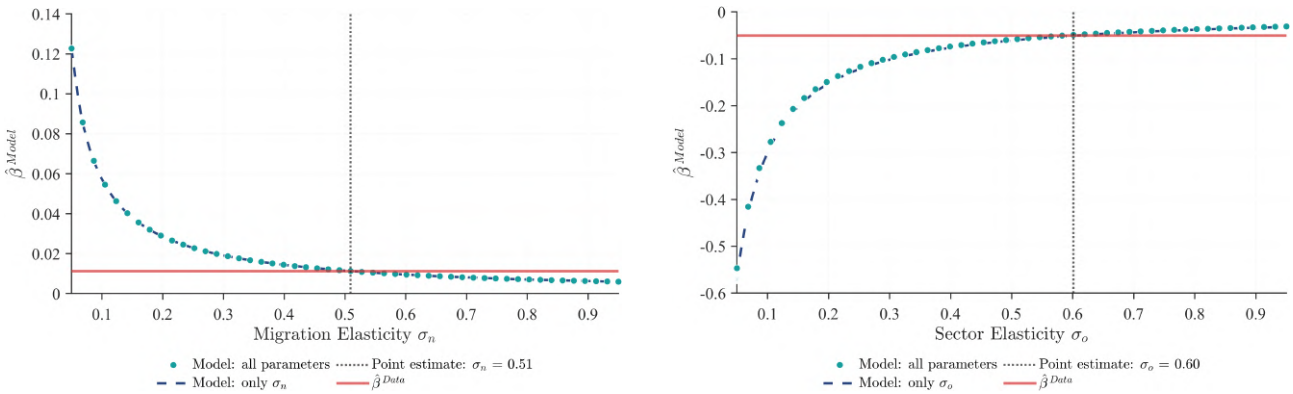
$$Y^{\text{model}}_{g,c,t+20}(\sigma) = \alpha^{\text{model}}(\sigma) + \lambda^{\text{model}}_{c}(\sigma) + \lambda^{\text{model}}_{s(c),t}(\sigma) + \sum_{r \neq 1910} \mathbf{1}\{t = r\} \left[\beta^{\text{model}}_{M,r}(\sigma)\text{MErosion}_c + \beta^{\text{model}}_{H,r}(\sigma)\text{HErosion}_c\right] + u^{\text{model}}_{g,c,t+20}(\sigma). \tag{A4.5.6}$$

I use the same omitted cohort and fixed-effect structure as in the empirical auxiliary regression.[41] Let $b^{\text{model}}(\sigma)$ collect the same erosion-by-cohort coefficients from equation (A4.5.6) that enter $\widehat{b}^{\text{data}}$. The indirect-inference estimator is

$$\widehat{\sigma} = \arg\min_{\sigma \in \Sigma} m(\sigma)' W m(\sigma), \qquad m(\sigma) = b^{\text{model}}(\sigma) - \widehat{b}^{\text{data}}, \qquad W = I. \tag{A4.5.7}$$

The identity weighting matrix gives equal weight to the targeted auxiliary coefficients. The current implementation evaluates the objective on a 50×50 grid over (σ_N, σ_O). Figure A4.5.1 reports the resulting fit.

Figure A4.5.1: Indirect inference results



(a) Location-choice dispersion

(b) Sector-choice dispersion

Notes: Panel (a) shows how the model-implied migration response to the Dust Bowl varies with the location-choice dispersion parameter σ_N. Panel (b) shows the corresponding sectoral response as a function of the sector-choice dispersion parameter σ_O. The selected pair minimizes equation (A4.5.7).

A4.5.6 Timing and identifying assumptions

The indirect-inference exercise relies on four maintained assumptions.

First, I treat the Dust Bowl as unanticipated in 1930. Once the shock occurs, agents understand the subsequent productivity path used in the transition. This prevents pre-1930 choices from

[41] The empirical regression is estimated at the individual level, whereas the simulated regression uses ancestry-group-by-CZ cells generated by the model.

responding to a shock that had not yet occurred.

Second, observed fundamentals are unavailable after 1950. I therefore hold the 1950 fundamentals fixed for the final model period used to evaluate continuation values. This terminal assumption affects future values but does not introduce an additional post-1950 shock.

Third, erosion affects workers through agricultural productivity rather than directly changing migration costs or non-kin sector-entry costs. Baseline farm-ownership controls, heterogeneity by ownership status and alternative specifications provide checks on this restriction (Liu, 2026; Yang, 2024).

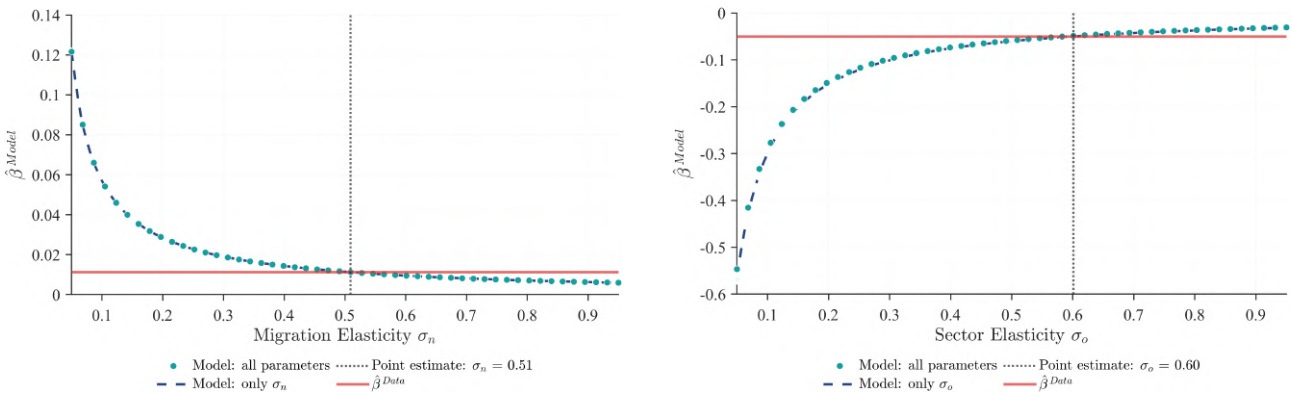
Fourth, the no-Dust-Bowl counterfactual removes the estimated productivity disturbance symmetrically while holding other recovered fundamentals fixed. The exercise therefore abstracts from additional state-dependent responses, such as borrowing constraints, that would themselves change because of the shock.

Under these assumptions, the variation identifying (σ_N, σ_O) comes from the response of migration and sector choice to an independently measured agricultural productivity disturbance rather than from cross-sectional differences in the levels of those choices.

A4.5.7 Indirect inference without social ties

For the counterfactual analysis without social ties, I re-estimate the dispersion parameters because their minimizing values could in principle change when the kin mechanism is removed. Figure A4.5.2 documents the results. In the current implementation, the minimizing values are unchanged for both σ_N and σ_O.

Figure A4.5.2: Indirect inference results without social ties



(a) Location-choice dispersion

(b) Sector-choice dispersion

Notes: Panel (a) shows the model-implied migration response as a function of σ_N in the model without social ties. Panel (b) shows the corresponding sectoral response as a function of σ_O.

A4.6 Model inversion

A4.6.1 Productivity

This subsection describes how sector–location productivity is recovered from observed wages, employment, trade costs and expenditure shares. The inversion follows the logic of Takeda (2022) and combines labor-market clearing with the Eaton–Kortum trade structure and the PIGL non-homothetic demand system used in the main text (Boppart, 2014; Eckert & Peters, 2022).

For each sector $k \in \{A, M\}$ and location n, labor-market clearing requires that total labor income equals total expenditure on varieties produced in that location,

$$w_{nt}^k L_{nt}^k = \sum_{i \in N} \pi_{int}^k X_{it}^k, \tag{A4.6.1}$$

where w_{nt}^k is the sector-k wage in location n, L_{nt}^k is employment, π_{int}^k is the share of expenditure in destination i on varieties sourced from origin n and X_{it}^k is total expenditure in location i on sector k.

Expenditure is determined by the PIGL demand system. An individual in location i and sector j with nominal labor income w_{it}^j spends a share $\delta_k(w_{it}^j, p_{it})$ on good k, where from (5) the agricultural share is

$$\delta_A(y, p_{it}) = \varphi + \nu \left(\frac{y}{(p_{it}^A)^{\varphi}(p_{it}^M)^{1-\varphi}} \right)^{-\epsilon}. \qquad \delta_M = 1 - \delta_A, \tag{A4.6.2}$$

Because $\epsilon > 0$, the agricultural share falls with real income, so richer locations spend relatively less on agriculture—the non-homothetic Engel force that drives structural change. Aggregating across the two sectors of workers in location i, total sector-k expenditure is

$$X_{it}^k = \delta_k(w_{it}^A, p_{it}) w_{it}^A L_{it}^A + \delta_k(w_{it}^M, p_{it}) w_{it}^M L_{it}^M. \tag{A4.6.3}$$

where each sector's workers spend out of their own labor income at their own PIGL shares.

Trade shares follow the Eaton–Kortum structure,

$$\pi_{int}^k = \frac{(w_{nt}^k T_{int}^k)^{-\mu_k} (Z_{nt}^k)^{\mu_k}}{\sum_{r \in N} (w_{rt}^k T_{irt}^k)^{-\mu_k} (Z_{rt}^k)^{\mu_k}} \tag{A4.6.4}$$

with iceberg trade costs T_{int}^k, trade elasticity μ_k and unit cost x_{it}^k; under the linear production technology unit costs equal wages, $x_{it}^k = w_{it}^k$. Substituting (A4.6.3) and (A4.6.4) into (A4.6.1)

gives

$$w_{nt}^k L_{nt}^k = \sum_{i\in N} \frac{(x_{nt}^k T_{int}^k)^{-\mu_k}(Z_{nt}^k)^{\mu_k}}{\sum_{j\in N}(x_{jt}^k T_{ijt}^k)^{-\mu_k}(Z_{jt}^k)^{\mu_k}} \left[\delta_k(w_{it}^A, p_{it})\, w_{it}^A L_{it}^A + \delta_k(w_{it}^M, p_{it})\, w_{it}^M L_{it}^M\right]. \tag{A4.6.5}$$

Factoring out the origin-specific productivity and unit-cost terms,

$$\begin{aligned} w_{nt}^k L_{nt}^k = (x_{nt}^k)^{-\mu_k}(Z_{nt}^k)^{\mu_k} \sum_{i\in N} \frac{(T_{int}^k)^{-\mu_k}}{\sum_{j\in N}(x_{jt}^k T_{ijt}^k)^{-\mu_k}(Z_{jt}^k)^{\mu_k}} \\ \times \left[\delta_k(w_{it}^A, p_{it})\, w_{it}^A L_{it}^A + \delta_k(w_{it}^M, p_{it})\, w_{it}^M L_{it}^M\right], \end{aligned} \tag{A4.6.6}$$

so productivity solves

$$\begin{aligned} (Z_{nt}^k)^{\mu_k} = w_{nt}^k L_{nt}^k (x_{nt}^k)^{\mu_k} \Bigg(\sum_{i\in N} \frac{(T_{int}^k)^{-\mu_k}}{\sum_{j\in N}(x_{jt}^k T_{ijt}^k)^{-\mu_k}(Z_{jt}^k)^{\mu_k}} \\ \times \left[\delta_k(w_{it}^A, p_{it})\, w_{it}^A L_{it}^A + \delta_k(w_{it}^M, p_{it})\, w_{it}^M L_{it}^M\right] \Bigg)^{-1}. \end{aligned} \tag{A4.6.7}$$

To express the inversion in observable prices, use the Eaton–Kortum sectoral price equation $p_{nt}^k = \Gamma^k \left(\sum_{j\in N}(x_{jt}^k T_{njt}^k)^{-\mu_k}(Z_{jt}^k)^{\mu_k}\right)^{-1/\mu_k}$, which implies

$$\left(\frac{p_{it}^k}{\Gamma^k}\right)^{-\mu_k} = \sum_{j\in N}(x_{jt}^k T_{ijt}^k)^{-\mu_k}(Z_{jt}^k)^{\mu_k}. \tag{A4.6.8}$$

Substituting (A4.6.8) into (A4.6.7) yields the closed-form productivity expression

$$\begin{aligned} Z_{nt}^k = (w_{nt}^k L_{nt}^k)^{1/\mu_k}\, x_{nt}^k \Bigg(\sum_{i\in N}(T_{int}^k)^{-\mu_k} \left(\frac{p_{it}^k}{\Gamma^k}\right)^{\mu_k} \\ \times \left[\delta_k(w_{it}^A, p_{it})\, w_{it}^A L_{it}^A + \delta_k(w_{it}^M, p_{it})\, w_{it}^M L_{it}^M\right] \Bigg)^{-1/\mu_k}. \end{aligned} \tag{A4.6.9}$$

The system (A4.6.9) is solved by fixed-point iteration: starting from a guess for $\{Z_{nt}^k\}$, compute prices from the Eaton–Kortum price equation, form the PIGL expenditure shares (A4.6.2), evaluate the right-hand side and update productivity until convergence.

A4.6.2 Sector-entry costs and location amenities

This subsection describes how I invert observed sector and location choices into the remaining residual objects of the model: time-varying sector-entry costs and location-sector amenities. The kin parameters are estimated before the inversion. Hence, conditional on a candidate pair of

choice scales (σ_N, σ_O), the value of sector-specific kin connections and geographically proximate kin support are known. The inversion therefore treats the kin terms as structural components of the same choice problems used in the quantitative model rather than as additional reduced-form controls.

Step 0: inputs, kin objects and normalizations. For each date t, ancestral-origin type g, current location n, candidate destination m, current sector j and child sector k, the inversion takes as inputs the observed sector-choice probabilities $P_t^O(k \mid n, j, g)$, the migration flows used in the gravity estimation, employment, wages, prices, the current kin state, normalized migration costs and the kinship parameters. The choice dispersion parameters are not fixed prior to indirect inference. Instead, for each candidate $\sigma = (\sigma_N, \sigma_O)$, I perform the inversion conditional on that candidate. The final reported fundamentals are evaluated at $\widehat{\sigma} = (\widehat{\sigma}_N, \widehat{\sigma}_O)$. For an adult of ancestral type g currently in location n and sector j, define

$$\Phi^k_{gnjt}(m) \equiv \Phi_{g,t}(n, j; m, k), \tag{A4.6.10}$$

so that $\Phi^k_{gnjt}(m)$ is the expected mass of horizontal kin located in m and employed in sector k. The sector-specific kin stock entering the child's sector choice is

$$F^k_{gnjt} = \mathbf{1}\{j = k\} + \psi^k_R \Phi^k_{gnjt}(n), \tag{A4.6.11}$$

because only same-region, same-sector horizontal kin enter the baseline sector-access stock. Conditional on the candidate sector-choice scale,

$$Q^k_{gnjt} = \theta_k F^k_{gnjt}, \qquad \theta_k = \sigma_O \Lambda_k. \tag{A4.6.12}$$

The geographically provided kin stock entering location choice is

$$H_{gnjt}(m) = \mathbf{1}\{m = n\} + \psi^H_R \sum_q \Pr_{t-1}(q \mid n, j, g) \sum_\ell \Phi^\ell_{gnq,t-1}(m). \tag{A4.6.13}$$

and its utility value is

$$\theta_H H_{gnjt}(m), \qquad \theta_H = \sigma_N \Theta_H. \tag{A4.6.14}$$

Thus, the parent is normalized to contribute one unit in both kin margins. Horizontal kin receive the estimated relationship-specific weights ψ^k_R and ψ^H_R. In the baseline model,

$$Q^k_{gnjt} = \theta_k F^k_{gnjt}.$$

The same inversion applies to the schooling extension in Appendix A3.2.2 after replacing Q^k_{gnjt}

by the corresponding schooling-augmented sector-entry value.
Sector-choice probabilities identify only relative sector utilities. For each parental sector j, I normalize the cost of remaining in the parent's sector to zero:

$$\tau^{j}_{gnjt} = 0 \qquad \text{for all } (g, n, j, t). \tag{A4.6.15}$$

The time-varying cost of entering the alternative sector $k \neq j$ is then identified from relative sector-choice probabilities. Thus, at each date the model recovers the two directional switching costs τ^{M}_{gnAt} and τ^{A}_{gnMt}.
Location-choice probabilities likewise identify destination values only up to an additive constant. I use the same destination-value normalization in every step of the inversion and preserve that normalization across counterfactuals. The normalization affects the levels assigned to residual amenities but not the location probabilities generated by the model.

Step 1: recover normalized destination values and adult continuation values. The migration-gravity equation in Appendix A4.3 is estimated with the kin term $\widehat{\Theta}_H H_{gnjt}(m)$ included as a known offset. It is therefore useful to define the non-kin destination value

$$\mathcal{D}^{j}_{gmt} \equiv \mathcal{A}^{j}_{gmt} - \rho \log L_{mt} + v\big(w^{j}_{mt}, p_{mt}\big) + \beta O_t(m, j, g). \tag{A4.6.16}$$

The structural migration-flow equation can consequently be written as

$$\mathbb{E}\big[F^{j}_{gnmt}\big] = \exp\left[\text{orig}^{j}_{gnt} + \frac{\mathcal{D}^{j}_{gmt}}{\sigma_N} + \Theta_H H_{gnjt}(m) - \lambda^{j}_{t}\widetilde{d}_{nmt}\right]. \tag{A4.6.17}$$

The destination fixed effect recovered from the PPML regression therefore satisfies

$$\text{dest}^{j}_{gmt} = \frac{\mathcal{D}^{j}_{gmt}}{\sigma_N}, \tag{A4.6.18}$$

up to the destination-value normalization imposed in the gravity estimation. Conditional on a candidate σ_N, the destination value in utility units is therefore

$$\mathcal{D}^{j}_{gmt} = \sigma_N \text{dest}^{j}_{gmt}. \tag{A4.6.19}$$

The same result can be seen directly from migration odds. For any two candidate destinations m and r,

$$\begin{aligned}\mathcal{D}^{j}_{gmt} - \mathcal{D}^{j}_{grt} = \sigma_N \log \frac{P^{N}_{t}(m \mid n, j, g)}{P^{N}_{t}(r \mid n, j, g)} \\ - \theta_H \left[H_{gnjt}(m) - H_{gnjt}(r)\right] + M^{j}_{gnmt} - M^{j}_{gnrt}.\end{aligned} \tag{A4.6.20}$$

Thus, once bilateral migration costs and kin support are removed, destination odds recover the relative non-kin destination value.

Given the normalized destination values, the adult continuation value follows directly from the location-choice log sum:

$$V_t(n,j,g) = \sigma_N \log \sum_{m \in N} \exp\left[\frac{\mathcal{D}^j_{gmt} + \theta_H H_{gnjt}(m) - M^j_{gnmt}}{\sigma_N}\right]. \tag{A4.6.21}$$

Using $\theta_H = \sigma_N \Theta_H$, $M^j_{gnmt} = \sigma_N \lambda^j_t \widetilde{d}_{nmt}$, the same expression can be written entirely in normalized units as

$$\frac{V_t(n,j,g)}{\sigma_N} = \log \sum_{m \in N} \exp\left[\text{dest}^j_{gmt} + \Theta_H H_{gnjt}(m) - \lambda^j_t \widetilde{d}_{nmt}\right]. \tag{A4.6.22}$$

The origin fixed effect provides an equivalent representation. Because the gravity equation is estimated on flows,

$$\text{orig}^j_{gnt} = \log L^j_{gnt} - \frac{V_t(n,j,g)}{\sigma_N}, \tag{A4.6.23}$$

under the same fixed-effect normalization. Hence,

$$V_t(n,j,g) = \sigma_N \left[\log L^j_{gnt} - \text{orig}^j_{gnt}\right]. \tag{A4.6.24}$$

Equations (A4.6.22) and (A4.6.24) therefore provide two equivalent ways of recovering the adult continuation value from the migration block. I use (A4.6.21).

Step 2: recover time-varying sector-entry costs. A child whose parent is in state (n, j) chooses sector k before making the next-period adult location decision. The systematic sector-choice value is

$$U^k_{gnjt} = Q^k_{gnjt} - \tau^k_{gnjt} + V_{t+1}(n,k,g). \tag{A4.6.25}$$

Importantly, once the child chooses k, that sector becomes the child's own sector state. The continuation value is therefore $V_{t+1}(n,k,g)$, not a continuation value indexed by the parent's sector j.

With $\tau^j_{gnjt} = 0$, the sector-choice log odds imply, for the alternative sector $k \neq j$,

$$\tau^k_{gnjt} = \left[Q^k_{gnjt} + V_{t+1}(n,k,g)\right] - \left[Q^j_{gnjt} + V_{t+1}(n,j,g)\right] - \sigma_O \log \frac{P^O_t(k \mid n,j,g)}{P^O_t(j \mid n,j,g)}. \tag{A4.6.26}$$

Observed sector-choice odds therefore identify the residual cost of entering the alternative sector at date t, after accounting for inherited sector-specific kin access and differences in future location values across sectors.

The inversion is performed separately at each date. For parents initially employed in agriculture, equation (A4.6.26) recovers τ^{M}_{gnAt}, the cost of entering manufacturing. For parents initially employed in manufacturing, it recovers τ^{A}_{gnMt}, the cost of entering agriculture. These residual costs are allowed to evolve over time and therefore absorb changes in sector-entry barriers not otherwise captured by the observed kin state, continuation values or other explicit model components.

Step 3: compute children's opportunity values. Given the recovered time-varying sector-entry costs, sector-specific kin values and adult continuation values, the opportunity value available to the next generation is

$$O_t(n,j,g) = \sigma_O \log \sum_{k\in\{A,M\}} \exp\left[\frac{Q^k_{gnjt} - \tau^k_{gnjt} + V_{t+1}(n,k,g)}{\sigma_O}\right]. \tag{A4.6.27}$$

This is the same sector-choice inclusive value that enters the parent's location problem through altruism. The object therefore links the sector and location blocks without introducing an additional reduced-form continuation term.

Step 4: recover location-sector amenities. Once $O_t(m,j,g)$ has been recovered, I can decompose the destination value in equation (A4.6.16). Rearranging gives

$$\mathcal{A}^j_{gmt} = \mathcal{D}^j_{gmt} + \rho \log L_{mt} - v\big(w^j_{mt}, p_{mt}\big) - \beta O_t(m,j,g). \tag{A4.6.28}$$

The recovered amenity inherits the additive normalization of the destination fixed effects. This normalization is held fixed throughout the quantitative exercises. Counterfactual outcomes depend on utility differences and are therefore invariant to a common additive renormalization applied consistently to the destination values and the associated residual objects.

Step 5: implementation and terminal values. The inversion is conditional on a candidate pair (σ_N, σ_O) inside the indirect-inference procedure. For each candidate, I first convert the objects estimated in the kin and gravity blocks into utility units:

$$\theta_H = \sigma_N \Theta_H, \qquad \theta_k = \sigma_O \Lambda_k, \qquad M^j_{gnmt} = \sigma_N \lambda^j_t \widetilde{d}_{nmt}, \qquad \mathcal{D}^j_{gmt} = \sigma_N \mathrm{dest}^j_{gmt}. \tag{A4.6.29}$$

I then compute $V_t(n,j,g)$ at each observed date directly from equation (A4.6.22). Thus, the observed location-choice block supplies the adult continuation values along the historical path. Sector choice at the final observed date T requires V_{T+1}, which is not directly recovered from an observed $T+1$ migration distribution. I obtain this terminal continuation value from the terminal extension used in the quantitative model during the counterfactuals. For the amenity

inversion, I assume that the amenities and sectoral barriers are the same in 1970 as they are in 1950.
Conditional on these continuation values, for each observed date t I perform the following steps:

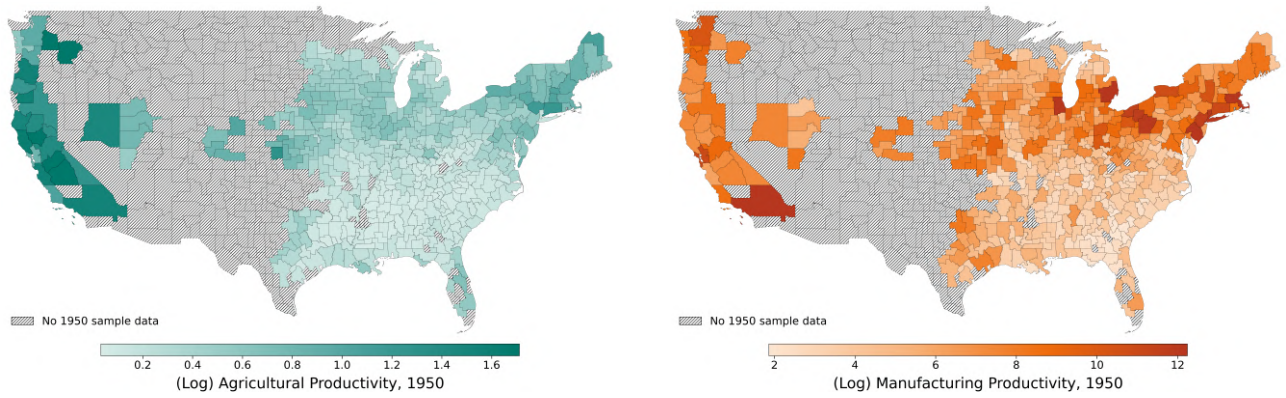
1. recover the two directional time-varying sector-entry costs, τ^{M}_{gnAt} and τ^{A}_{gnMt}, from equation (A4.6.26);
2. compute $O_t(n, j, g)$ from equation (A4.6.27);
3. recover $\mathcal{A}^{j}_{gmt}$ from equation (A4.6.28).

A4.6.3 Model inversion: additional results

This appendix reports additional inversion results that complement the main validation exercise. I report the geography of recovered fundamentals, additional untargeted validation moments, the evolution of sectoral costs and the transition in family input shares.

Geography of recovered fundamentals. The inversion recovers, for each commuting zone and year, sectoral productivities, sectoral amenities and market-support prices. These objects are not targeted on external spatial series. Figure A4.6.1 maps recovered productivity in 1950. Agricultural productivity is highest in the West and Great Plains and lowest in the industrial Northeast, while manufacturing productivity is concentrated in the Northeast and around the Great Lakes. Figure A4.6.2 maps recovered amenities in 1880. These objects vary systematically across space without being directly targeted in the inversion.

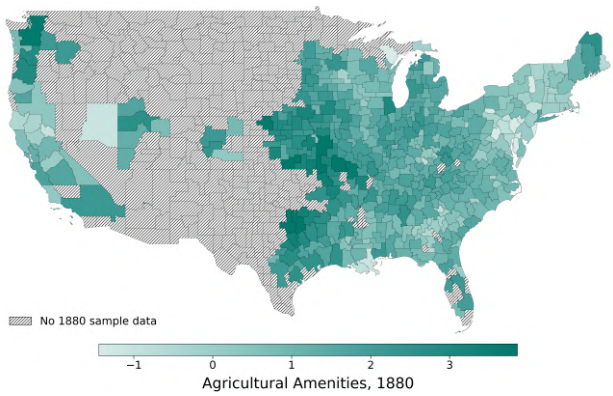
Figure A4.6.1: Geographical distribution of productivity, 1950



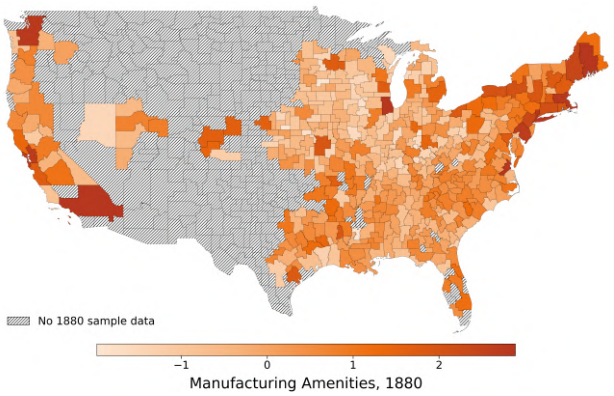
(a) Agricultural productivity (b) Manufacturing productivity

Notes: Recovered log sectoral productivity across commuting zones in 1950.

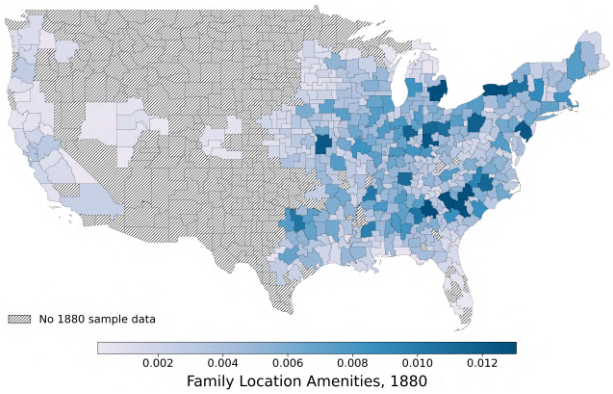
Figure A4.6.2: Geographical distribution of amenities, 1880



(a) Agricultural amenities



(b) Manufacturing amenities



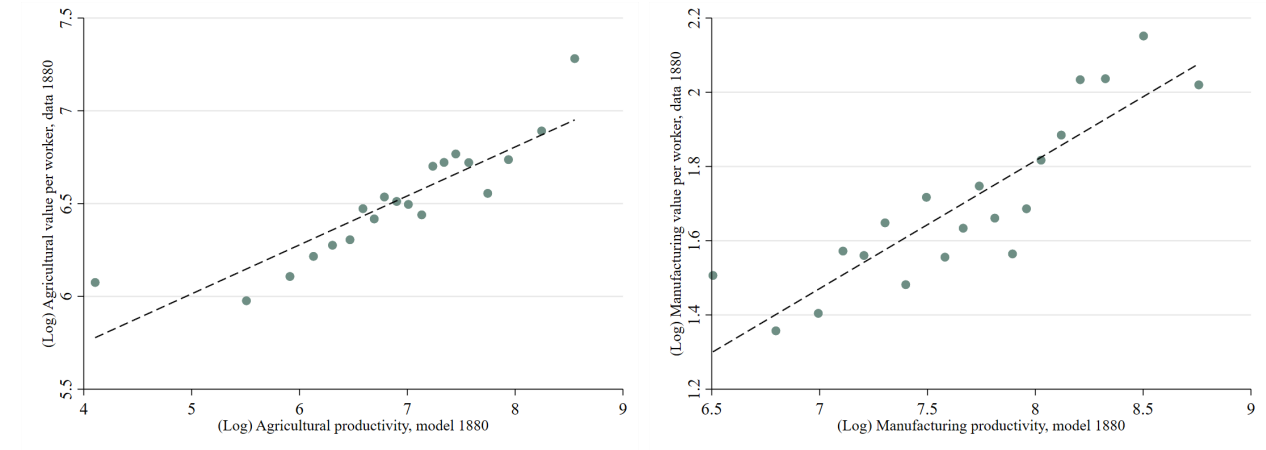
(c) Kin location amenities

Notes: Recovered sectoral amenities across commuting zones in 1880.

Additional untargeted validation. Figure A4.6.3 reports the full productivity validation. Recovered agricultural productivity aligns with agricultural output per worker in both 1880 and 1950, although the output-per-worker series is not used in the inversion. Figure A4.6.4 reports the corresponding employment-growth validation for both sectors. The model reproduces the spatial dynamics of structural change: agricultural employment grows slowly or declines in initially agricultural places, while manufacturing employment grows fastest in those same places.

Kin structure. Figure A4.6.5 provides additional validations of the kin block. The figure compares the model-implied and observed commuting-zone distribution of uncles and aunts in 1880, an untargeted kinship moment. The close fit indicates that the network law of motion generates a realistic spatial distribution of extended kin.

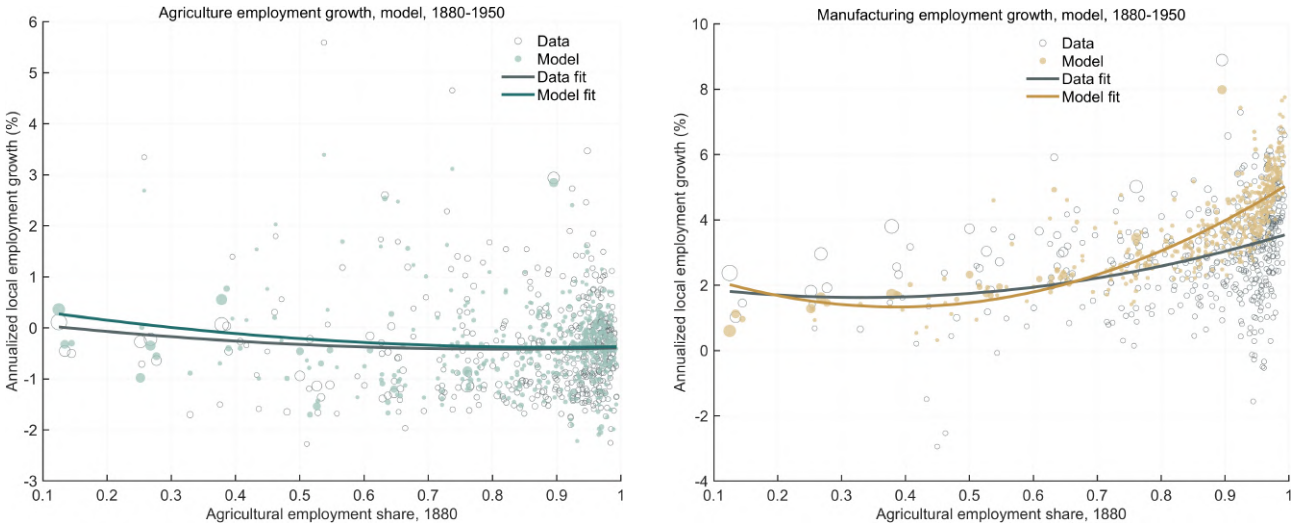
Figure A4.6.3: Recovered productivity versus independent data



(a) Agricultural productivity, 1880 (b) Manufacturing productivity, 1880

Notes: Model-recovered agricultural and manufacturing productivity against agricultural and manufacturing value per worker in the data, by commuting zone.

Figure A4.6.4: Untargeted validation: employment growth, 1880–1950



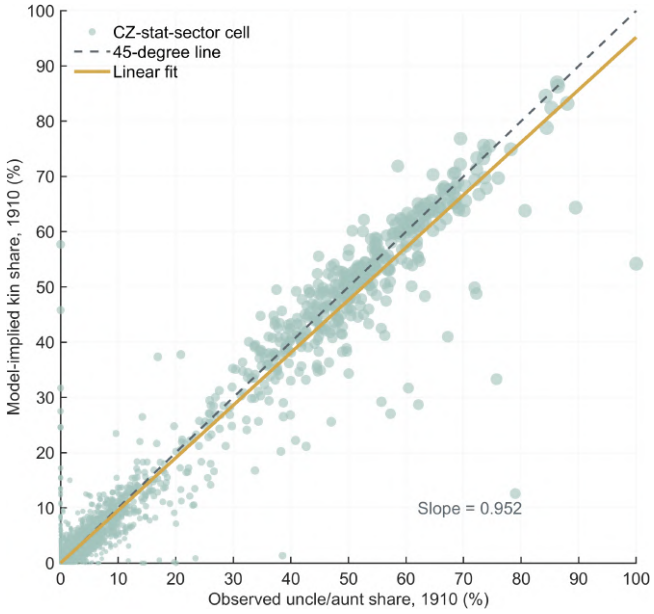
(a) Agricultural employment growth (b) Manufacturing employment growth

Notes: Annualized local employment growth from 1880 to 1950 against the 1880 agricultural employment share, by commuting zone, in the model and in the data.

A4.7 Representative-kin approximation

The quantitative model does not follow every family tree separately after the initial period. Instead, it groups workers by ancestral origin, sector and location and represents their horizontal kin by the average kin distribution within that group. This greatly reduces the number of family

Figure A4.6.5: Untargeted validation: distribution of uncles, 1910



Notes: The figure reports the model-implied versus observed commuting-zone share of uncles and aunts in 1910.

histories the model must track. This appendix uses the linked genealogical data to assess what is lost through this averaging.

The ancestral type g is not an individual family. It is defined by the commuting zone of the linked 1850 ancestor, so many distinct genealogical families can share the same g. I use f below to index these individual families.

The approximation. Consider adults of ancestral type g who live in m and work in j. The model summarizes their horizontal kin by

$$\Phi_{g,t}(m, j; n, k) = r_g \lambda_{g,t}(n, k \mid m, j), \qquad \sum_{n,k} \lambda_{g,t}(n, k \mid m, j) = 1, \tag{A4.7.1}$$

where r_g is total horizontal-kin mass for ancestral type g and $\lambda_{g,t}(n, k \mid m, j)$ is the share of that kin living in location n and working in sector k. Thus, $\Phi_{g,t}$ is the average kin distribution among workers with the same ancestral origin, sector and location.

Descendants from parental state (m, j) choose sectors and locations according to $P_t^O(k \mid m, j, g)$ and $P_{t+1}^N(n \mid m, k, g)$. Adults reaching the same new state (n, k) can come from different

parental states. Their parental composition is

$$\Pr_t(m, j \mid n, k, g) = \frac{L^j_{gmt} P^O_t(k \mid m, j, g) P^N_{t+1}(n \mid m, k, g)}{\sum_{\tilde{m}, \tilde{j}} L^{\tilde{j}}_{g\tilde{m}t} P^O_t(k \mid \tilde{m}, \tilde{j}, g) P^N_{t+1}(n \mid \tilde{m}, k, g)},$$

which gives the law of motion

$$\lambda_{g,t+1}(n', k' \mid n, k) = \sum_{m,j} \Pr_t(m, j \mid n, k, g) P^O_t(k' \mid m, j, g) P^N_{t+1}(n' \mid m, k', g). \tag{A4.7.2}$$

The model therefore represents workers with the same ancestral origin, sector and location by their average kin distribution. This averaging can affect the model in two ways. First, choice probabilities are nonlinear in kin connections, so choices evaluated at the average kin distribution need not equal average choices across individual families. Second, the law of motion follows kin links, so averaging can matter if families that contribute more kin also have systematically different transition rates. I examine both margins below.

Family-level heterogeneity. I first measure how much transition behavior differs across families within the groups used by the model. Let $c = (g, m, j, t)$ denote such a group and let $\widehat{p}_{fc}$ be the fraction of observed descendants from family f who enter manufacturing. Because family-level transition rates are noisy when only a few descendants are observed, I subtract the variation expected from finite family size:

$$V^{\text{ex}}_c = \text{Var}_f(\widehat{p}_{fc}) - \mathbb{E}_f \left[\frac{\bar{p}_c(1 - \bar{p}_c)}{N_{fc}} \right],$$

where N_{fc} is the number of observed descendants and $\bar{p}_c$ is the group-level transition rate. I also report

$$\text{RMSE}^{\text{net}}_c = \left[\max \left\{ \mathbb{E}_f \left[(\widehat{p}_{fc} - \bar{p}_c)^2 \right] - \mathbb{E}_f \left[\frac{\bar{p}_c(1 - \bar{p}_c)}{N_{fc}} \right], 0 \right\} \right]^{1/2},$$

which measures the typical family-level deviation from the group average after removing variation due to finite family size.

Choices at average versus individual kin distributions. I next test directly whether replacing individual kin distributions by their group average changes sector and location choices. I hold wages, prices, switching costs, migration costs, parental connections and continuation values fixed at their 1880 equilibrium values. For each worker i, let Φ_i denote the observed

horizontal-kin distribution and let $\bar{\Phi}_c$ denote its mean within $c = (g, m, j)$. I compare

$$J_c^O = \frac{1}{N_c} \sum_{i \in c} P^O(M \mid \Phi_i) - P^O(M \mid \bar{\Phi}_c), \qquad J_{cn}^N = \frac{1}{N_c} \sum_{i \in c} P^N(n \mid \Phi_i) - P^N(n \mid \bar{\Phi}_c). \quad \text{(A4.7.3)}$$

The first term averages choice probabilities evaluated separately at each worker's observed kin distribution. The second evaluates the same choice problem once at the group-average kin distribution. Their difference therefore measures the effect of averaging kin connections while holding all other equilibrium objects fixed.

Individual kin distributions are constructed from recorded aunts and uncles working in agriculture or manufacturing in modeled CZs. Within each ancestral group, I rescale observed kin by a common factor so that average total horizontal-kin mass equals the calibrated r_g, preserving differences in kin composition and size across individuals. Workers with no recorded eligible kin remain in the sample with zero kin connections whenever their ancestral group contains positive observed kin. I retain groups with at least ten linked children and five parent households. The resulting sample contains 21,808 children in 412 parental-state groups.

For sector choice, I recover the individual-specific index from the equilibrium probability p_c^0:

$$\operatorname{logit} P_i^O(M) = \operatorname{logit} p_c^0 + \frac{b_M(\Phi_i^M - \Phi_c^{0,M}) - b_A(\Phi_i^A - \Phi_c^{0,A})}{\sigma_O},$$

where Φ_i^k is horizontal kin in sector k at the child's birth CZ, $b_A = 0.00375213$, $b_M = 0.004826542$ and $\sigma_O = 0.6$. For location choice, letting $H_i(n) = \sum_k \Phi_i^k(n)$, I compute

$$P_i^N(n \mid m, k, g) \propto P_c^{N,0}(n \mid k) \exp\left\{ \frac{b_H[H_i(n) - H_c^0(n)]}{\sigma_N} \right\}, \qquad b_H = 0.020976, \quad \sigma_N = 0.51.$$

Evaluating the same expressions at $\bar{\Phi}_c$ gives the group-average comparison. Migration probabilities are averaged across the two chosen sectors using the stored sector-choice probabilities, so this exercise isolates averaging in the location decision.

Table A4.7.1: Choice probabilities using individual and average kin distributions

Choice	Average kin (%)	Individual kin (%)	Signed gap (p.p.)	Mean absolute gap (p.p.)
Manufacturing	25.39	25.43	0.046	0.048
Stay in birth CZ	83.73	83.14	-0.586	0.586

Notes: "Individual kin" reports the average probability obtained by evaluating the choice separately for each observed kin distribution. "Average kin" evaluates the same choice at the group-average kin distribution. Signed gap is individual kin minus average kin. Mean absolute gap is the child-weighted mean absolute difference across groups. Migration probabilities are additionally weighted by stored sector-choice probabilities. All other choice inputs are held fixed. The sample contains 21,808 children in 412 parental-state groups for both decisions.

Kin-weighted transitions. The second question is whether families that contribute more kin also have systematically different transition rates. Let K_{fc} denote the horizontal-kin mass contributed by family f and p_{fc} its transition rate. The transition relevant for the law of motion is $\mathbb{E}_f[K_{fc}p_{fc} \mid c]$. Replacing family-specific kin mass and transition rates by their group averages instead gives $\mathbb{E}_f[K_{fc} \mid c]\,\mathbb{E}_f[p_{fc} \mid c]$. Their difference is $\mathrm{Cov}_c(K_f, p_f) = \mathbb{E}_f[K_{fc}p_{fc} \mid c] - \mathbb{E}_f[K_{fc} \mid c]\mathbb{E}_f[p_{fc} \mid c]$. I report this difference relative to the exact kin-weighted transition,

$$C_c = \frac{\mathrm{Cov}_c(K_f, p_f)}{\mathbb{E}_f[K_{fc}p_{fc} \mid c]}.$$

A value close to zero means that averaging across families has little effect on the kin flow carried into the next generation. I construct the same statistic for the joint sector-location transition, which is closest to the object propagated in equation (A4.7.2).

Table A4.7.2: Diagnostics for the Representative-Family Approximation

Year	Cells	Share pos. excess	Median excess	Median RMSE	W. median RMSE	W. median rel. cov.	W. median joint resid.	Median occ.–mig. cov.
1880	434	0.923	0.018	0.122	0.141	0.039	0.020	0.000
1910	438	0.961	0.020	0.138	0.141	0.047	0.016	0.000
1930	407	0.926	0.012	0.104	0.107	0.031	0.017	0.000

Notes: The table reports diagnostics for the representative-family approximation using all (g, n, k, t) cells. "Share pos. excess" is the share of cells in which family-level sector transition rates exhibit excess variance relative to binomial sampling noise. RMSE is the de-noised root mean squared deviation between family-level transition rates and the cell-level representative-family transition rate. Weighted statistics use family-capital cell mass as weights. "W. median rel. cov." is the mass-weighted median of the relative covariance between family-specific sector probabilities and family-capital stocks. "W. median joint resid." compares the joint transition object $E[P_i^O P_i^N K_i]$ to the representative-family counterpart. "Median occ.–mig. cov." is the within-cell covariance between sector and migration probabilities.

Results. The data show substantial family-level heterogeneity: more than 90% of groups have positive excess variance and the adjusted RMSE is roughly 10–14 percentage points. Despite this heterogeneity, evaluating choices at the group-average kin distribution produces only small aggregate differences. The manufacturing probability changes by 0.046 percentage points and the probability of remaining in the birth CZ by 0.586 points.

The kin-weighted transition check gives a similar conclusion. The median discrepancy is approximately 3–5% for sector transitions and about 2% for joint sector-location transitions. The model therefore averages over meaningful differences across individual families, but the two margins through which this averaging affects aggregate choices and kin flows are considerably smaller.

A5 Quantitative analysis

A5.1 Solution algorithm

I solve the model by separating the within-period temporary equilibrium from the intertemporal sequential equilibrium, following the logic of Caliendo et al. (2019) and Eckert and Kleineberg (2024). Conditional on a guess for labor, family connections and continuation values, wages and prices can be solved independently at each date. These market outcomes determine sector and location choices. The resulting choices then update the labor distribution, the distribution of relatives and the value functions. I iterate on the full path until the market equilibrium, choices, laws of motion and value functions are mutually consistent.

Roots and branches. All counterfactuals use the same recovered non-family fundamentals and structural parameters and start from the same observed initial labor allocation. For the baseline roots-and-branches exercises, parental connections remain active throughout. I consider three treatments of connections to horizontal relatives. In the economy without connections to horizontal relatives, I set the horizontal terms in equations (8) and (12) to zero. In the constant-1880 economy, I hold the observed 1880 distribution of horizontal relatives constant at all subsequent dates. In the endogenous economy, the distribution of horizontal relatives evolves according to equations (15)–(17). Within each family-connection specification, I compare the policy economy with the same economy without the policy. A complementary exercise below also examines changes in parental connections.
For any equilibrium outcome Y_t, the effect of roots and the additional effect of branches can be decomposed as

$$\mathcal{I}_t = Y_t^{\text{constant}} - Y_t^{\text{none}}, \tag{A5.1.1}$$

$$\mathcal{P}_t = Y_t^{\text{endo}} - Y_t^{\text{constant}}. \tag{A5.1.2}$$

Temporary equilibrium. Fix a date t, a trial labor distribution L_t and the date-t fundamentals. First aggregate employment over ancestral types,

$$L_{nt}^k = \sum_g L_{gnt}^k, \qquad L_{nt} = \sum_{k \in \{A,M\}} L_{nt}^k.$$

Conditional on these quantities, the temporary equilibrium consists of wages, prices, trade shares and expenditures $\{w_t, p_t, \pi_t, X_t\}$. For a trial wage vector, equations (18) and (19) determine trade shares and sectoral prices. Equations (5) and (22) then determine sectoral expenditures. Wages are updated until the production-market conditions in equation (21) hold in every sector and location. I normalize one nominal wage because only relative nominal prices

are identified.

Algorithm 1 Temporary equilibrium at date t

Require: Trial labor distribution L_t, date-t fundamentals and structural parameters.

Aggregate $L_{nt}^k = \sum_g L_{gnt}^k$ and $L_{nt} = \sum_k L_{nt}^k$.
Initialize wages $w_t^{(0)}$.
Set the market-clearing residual above tolerance.
while the market-clearing residual is above tolerance **do**
    Compute trade shares π_t from equation (18).
    Compute sectoral price indices p_t from equation (19).
    Compute PIGL expenditure shares $\delta_k(w_{nt}^j, p_{nt})$ from equation (5).
    Aggregate sectoral expenditures X_t from equation (22).
    Evaluate excess demand using equation (21).
    Update wages.
end while
return $\{w_t, p_t, \pi_t, X_t\}$.

This block is static conditional on L_t. Horizontal relatives do not enter equations (18)–(22) directly. They affect the temporary equilibrium only through the labor distribution generated by earlier sector and location choices.

Sequential equilibrium. The dynamic solution requires the labor and continuation-value paths used to make choices to agree with the paths generated by those choices. Let $\mathcal{T} = \{1880, 1910, 1930, 1950, 1970\}$ denote the historical solution dates and let $\mathcal{T}^E$ additionally include the terminal extension used to evaluate continuation values. I iterate on $\{L_t, V_t\}_{t \in \mathcal{T}^E}$, holding the observed initial state fixed. The trial labor path determines wages and prices at each date, while the distribution of relatives Φ_t is generated forward from the initial distribution. The terminal extension uses the same value function as in 1950.

Each iteration has four steps. First, I solve wages and prices at every date given the trial labor path. Second, I move forward through time: children choose sectors, then locations as adults and these choices determine the next period's labor distribution and the distribution of relatives. Third, I move backward through time and recompute opportunity and continuation values using the paths generated in the forward step. Finally, I update the trial labor and continuation-value paths and repeat until they agree with the paths implied by the model. Sector and location choices can be solved sequentially because the family connections relevant for each transition are determined by the previous generation.

Algorithm 2 Sequential equilibrium and changing family connections

Require: Initial state (L_{t_0}, Φ_{t_0}), path of fundamentals, terminal extension and a family-connection specification.

Initialize trial paths $\{L_t^{(0)}, V_t^{(0)}\}_{t \in \mathcal{T}^E}$.
while the outer fixed-point residual is above tolerance **do**
A. Temporary equilibria
  for each historical date $t \in \mathcal{T}$ **do**
    Solve $\{w_t^{(s)}, p_t^{(s)}, \pi_t^{(s)}, X_t^{(s)}\}$ given $L_t^{(s)}$ with Algorithm 1.
  end for
B. Forward pass
  Set $(\widetilde{L}_{t_0}, \Phi_{t_0})$ to the observed initial state.
  for each historical transition $t \to t+1$ **do**
    Construct sector-specific family connections from Φ_t, form sector values using $V_{t+1}^{(s)}$ and compute P_t^O from equation (9).
    Compute children's opportunity values O_t using the distribution of relatives and the updated continuation value at $t+1$, with the terminal extension used after the final date.
    Construct destination-specific family amenities and form destination values using the trial market outcomes at $t+1$. Compute P_{t+1}^N from equation (13).
    Update $\widetilde{L}_{t+1}$ from equation (20).
    if the distribution of horizontal relatives changes endogenously **then**
      Update Φ_{t+1} using equations (15)–(17).
    else if the 1880 distribution of horizontal relatives is constant **then**
      Set $\Phi_{t+1} = \Phi_{1880}$.
    else
      Set the horizontal component of Φ_{t+1} to zero.
    end if
  end for

Algorithm 2 Sequential equilibrium and changing family connections (continued)

C. Backward pass

for each historical date $t \in \mathcal{T}$, from the last to the first **do**

Compute updated continuation values $\widetilde{V}_t$ from equation (14), using the wages and prices from block A and the distributions of relatives from block B.

end for

D. Update

Compute the maximum change between $\{\widetilde{L}_t, \widetilde{V}_t\}$ and the current trial paths together with the maximum market-clearing residual.

Update the labor path by damping: $L_t^{(s+1)} = (1-\omega)L_t^{(s)} + \omega\widetilde{L}_t$.

Update continuation values analogously: $V_t^{(s+1)} = (1-\omega)V_t^{(s)} + \omega\widetilde{V}_t, \qquad 0 < \omega \leq 1,$ leaving the initial state and terminal extension fixed.

end while

return the converged paths $\{L_t, \Phi_t, V_t, O_t, P_t^O, P_t^N, w_t, p_t, \pi_t, X_t\}$.

At convergence, wages and prices clear markets, workers' sector and location choices are optimal and the labor and continuation-value paths generated by those choices coincide with the paths used to solve the model.

Moving voucher construction. The grant in equation (A5.5.3) enters only the payoff from eligible origin–destination moves and the common lump-sum tax is included in household income. The voucher affects equilibrium wages and prices indirectly because the it changes location choices and therefore subsequent labor allocations.

Calibrating the grant is an outer root problem around Algorithm 2. For a candidate G, I solve the full sequential equilibrium, compute realized take-up $F(G)$ and expenditure $GF(G)$ and update the grant until expenditure equals the target in equation (A5.5.4). Because take-up is endogenous, the equilibrium is re-solved after every change in G. When the benchmark policy is compared across family-connection specifications, the grant is recalibrated separately within each regime so that realized fiscal expenditure is held constant.

Implementation of the additional counterfactuals. The additional exercises use the same solution algorithm and change one element at a time. To measure the effect of family connections created by the relocation policy, I keep the policy-induced 1880 sector and location allocation unchanged but, from 1910 onward, give each cohort the parental connections and connections to horizontal relatives it would have had without the policy. Wages, prices and choices are then allowed to adjust normally. The difference from the full policy economy therefore captures the effects of the new family connections created by the initial relocation and avoids holding these connections fixed across generations.

The destination-targeting exercises instead change which origin–destination moves are eligible for the grant. When comparing destinations with more or fewer existing relatives at the same initial migration response, I choose the grant in each case to generate the same increase in 1880 movers rather than the same expenditure. Finally, the channel decompositions change one component of family connections at a time and re-solve the model.

A5.2 Family roots

This appendix reports additional results underlying Section 6.1. The main text focuses on how the initial distribution of horizontal relatives affects the manufacturing share. I first report corresponding effects on migration, the wage bill and geographic dispersion. I then show how these effects evolve over time and separate the role of the number of relatives from the role of family geography.

Additional outcomes. Table A5.2.1 reports the 1950 effects of the initial distribution of horizontal relatives on manufacturing, migration, the wage bill and cross-CZ dispersion in manufacturing shares. As in the main text, I separate agricultural, manufacturing and local connections to horizontal relatives by introducing the three channels in every possible order and averaging their marginal contributions.

Table A5.2.1: Channels of the initial distribution of horizontal relatives

	Mfg. share (p.p.)	Migration (p.p.)	Wage bill (%)	Mfg. dispersion (p.p.)
Agricultural connections	-1.200	-0.344	-1.250	0.549
Manufacturing connections	0.142	0.024	0.156	0.054
Local connections	-0.253	-1.704	-0.502	0.154
Total initial-distribution effect	-1.311	-2.024	-1.596	0.757

Notes: Exact three-factor Shapley decomposition of the 1950 effect of holding the observed 1880 distribution of horizontal relatives constant relative to an economy without horizontal relatives. Parental connections and all non-family fundamentals remain unchanged. Components add to totals before rounding.

The results show a clear distinction between the sectoral and spatial channels. Agricultural connections account for most of the decline in manufacturing and the wage bill, while local connections account for most of the decline in migration. The initial distribution of relatives also increases cross-CZ dispersion in manufacturing shares, consistent with greater persistence in the sectoral and spatial allocation observed in 1880.

The effect of roots over time. Table A5.2.2 traces the same decomposition across model dates for the manufacturing share and the wage bill. Parents evolve in both economies, so the comparison continues to isolate the effect of the 1880 distribution of horizontal relatives.

Table A5.2.2: Initial distribution of horizontal relatives: channels

	1910	1930	1950	1970
Panel A. Manufacturing share (p.p.)				
Agricultural connections	−0.69	−1.19	−1.20	−1.20
Manufacturing connections	0.08	0.10	0.14	0.08
Local connections	0.37	0.32	−0.25	−1.16
Total	−0.25	−0.77	−1.31	−2.28
Panel B. Wage bill (%)				
Agricultural connections	−0.98	−1.53	−1.25	−1.15
Manufacturing connections	0.12	0.16	0.16	0.09
Local connections	0.34	0.02	−0.50	−1.13
Total	−0.52	−1.35	−1.60	−2.19

Notes: Difference between an economy with the distribution of horizontal relatives held at its observed 1880 pattern and an economy without horizontal relatives. Parents evolve in both economies. The three rows give the Shapley contributions of agricultural, manufacturing and local connections. The wage bill is measured relative to the economy without horizontal relatives.

The initial effect becomes more negative over time. Agricultural connections account for most of the early decline in manufacturing and the wage bill. The contribution of local connections is initially positive but becomes increasingly negative, accounting for a large share of the additional decline by 1970. Manufacturing connections partly offset these effects throughout.

Number of relatives and family geography. The comparison with an economy without horizontal relatives combines two features of the observed 1880 distribution: workers have horizontal relatives and those relatives are unevenly distributed across space. To separate these margins, I construct an intermediate economy that preserves the total mass of horizontal relatives within each ancestral type but distributes relatives across CZs in proportion to population. Table A5.2.3 reports the results.

The geography of relatives accounts for nearly the entire manufacturing effect. When the same mass of horizontal relatives is instead distributed across CZs in proportion to population, the manufacturing share falls by only 0.030 percentage points. The additional effect of the observed geography is −1.281 points. The pattern is even stronger for migration.

Sectoral change within CZs and employment reallocation across CZs. Finally, I ask whether the lower national manufacturing share comes from lower manufacturing shares within CZs or from a shift of employment toward CZs with lower manufacturing shares. For

Table A5.2.3: Quantity and geography of the initial distribution of horizontal relatives, 1950

	Mfg. share (p.p.)	Migration (p.p.)	Wage bill (%)	Mfg. dispersion (p.p.)
Observed 1880 family connections	-1.311	-2.024	-1.596	0.757
Family connections, population-proportional geography	-0.030	0.064	0.010	-0.238
Additional effect of observed family geography	-1.281	-2.089	-1.606	0.995

Notes: The first row compares the observed 1880 distribution of horizontal relatives with an economy without horizontal relatives. The second preserves the same total mass of horizontal relatives but distributes relatives across CZs in proportion to population. The final row is the difference between the observed distribution and this population-proportional benchmark. Parental connections and all non-family fundamentals remain unchanged.

counterfactual c,

$$\Delta S_c^M = \underbrace{\sum_n e_{n0}\Delta s_{nc}^M}_{\text{within CZs}} + \underbrace{\sum_n \Delta e_{nc} s_{n0}^M}_{\text{across CZs}} + \underbrace{\sum_n \Delta e_{nc}\Delta s_{nc}^M}_{\text{interaction}}, \tag{A5.2.1}$$

where e_{n0} denotes the benchmark employment share of CZ n and s_{n0}^M its manufacturing share.

Table A5.2.4: Within- and across-CZ effects of initial horizontal relatives, 1950

	Within CZs	Across CZs	Interaction	Total
Initial horizontal relatives	-0.688	-0.603	-0.020	-1.311

Notes: Decomposition of the 1950 manufacturing-share effect of the initial distribution of horizontal relatives. "Within CZs" captures changes in local manufacturing shares holding the spatial distribution of employment at its benchmark value. "Across CZs" captures changes in the spatial distribution of employment holding local manufacturing shares at their benchmark values. The interaction captures the joint movement of the two margins. Entries are percentage points and add to the total before rounding.

The within- and across-CZ components are similar in magnitude. The initial distribution of horizontal relatives therefore lowers the national manufacturing share through both lower manufacturing shares within places and a spatial distribution of employment tilted toward less manufacturing-intensive places.

A5.3 Family branches

This appendix reports additional results underlying Section 6.3. To measure the effect of branches, I compare the full model, in which the distribution of horizontal relatives changes with workers' sector and location choices, with an otherwise identical economy in which the distribution of horizontal relatives remains fixed at its observed 1880 pattern. Parental connections and all non-family fundamentals evolve identically in both economies. I then report a complementary exercise in which parental sector connections are also held at their 1880 distribution.

Construction of the main-text statistics. Let Y_t^{full} denote an outcome in the full model and Y_t^{constant} the same outcome when the distribution of horizontal relatives remains fixed at its observed 1880 pattern. Because the two economies start from the same 1880 allocation, the share of the 1880–1970 change in outcome Y explained by changing family connections, reported in Panel C of Table 4, is

$$\mathcal{S}_Y = 100\,\frac{Y_{1970}^{\text{full}} - Y_{1970}^{\text{constant}}}{Y_{1970}^{\text{full}} - Y_{1880}^{\text{full}}}. \tag{A5.3.1}$$

The same calculation is applied to the national manufacturing share, the manufacturing share of the initially least industrialized third of CZs, the top–bottom manufacturing-share gap and the dispersion of CZ manufacturing shares. A positive share therefore means that allowing family connections to change moves the outcome in the same direction as its full-model change between 1880 and 1970.

The bottom and top thirds are defined once using CZ manufacturing shares in 1880 and are held constant thereafter. A group's manufacturing share at date t is total manufacturing employment in that group divided by its total employment. The South is the Census South region. The population share living in manufacturing-majority CZs is

$$\mathcal{M}_t = \sum_n e_{nt}\,\mathbf{1}\left\{s_{nt}^M > \frac{1}{2}\right\}, \tag{A5.3.2}$$

where e_{nt} is CZ n's share of national employment and s_{nt}^M its manufacturing share. I measure cross-CZ dispersion by the employment-weighted standard deviation

$$\text{GeoSD}_t = \left[\sum_n e_{nt}\left(s_{nt}^M - S_t^M\right)^2\right]^{1/2}, \tag{A5.3.3}$$

where S_t^M is the national manufacturing share. The top–bottom gap is the manufacturing share of CZs in the top 1880 third minus that of CZs in the bottom third. For the timing statistic, I linearly interpolate the full-model manufacturing share between model dates and record the first year in which it reaches the constant-1880 economy's 1970 manufacturing share.

Additional aggregate outcomes. The main text focuses on the manufacturing share and its spatial incidence. Table A5.3.1 reports corresponding effects on the wage bill, agricultural employment, migration and the dispersion of manufacturing shares across CZs.

Detailed channels of family branches. Table A5.3.2 reports the detailed decomposition underlying Panel 11a. The spatial contribution is separated into persistence at the previous generation's origin and connections to destinations. The sectoral contribution is separated

Table A5.3.1: Additional consequences of changing connections to horizontal relatives

	Evolving vs. held at 1880
Wage bill in 1970 (%)	+1.78
Cumulative wage bill, 1910–1970 (%)	+0.86
Agricultural employment in 1970 (%)	−9.2
Migration rate in 1970 (p.p.)	+1.13
SD of CZ manufacturing shares (p.p.)	−1.23
CZs with a manufacturing majority	+13

Notes: Effects compare the full model with an otherwise identical economy in which the distribution of horizontal relatives remains fixed at its observed 1880 pattern. The wage bill, cumulative wage bill and agricultural employment are percentage changes relative to the fixed-1880 economy. Migration and manufacturing dispersion are percentage-point changes.

analogously into same-sector persistence and connections to the alternative sector.
The spatial contribution is close to zero in 1950 because origin persistence and destination pull offset one another, while the sectoral contribution is already positive. By 1970, the two channels contribute 0.911 and 1.142 percentage points to the manufacturing share.

Within-CZ adjustment and effects across generations. Table A5.3.3 reports two complementary accounting exercises. Panel A separates the manufacturing effect into changes in sectoral composition within CZs and changes in the spatial distribution of employment. Panel B varies how long the distribution of horizontal relatives is allowed to change before being held constant.
By 1970, approximately three quarters of the manufacturing effect operates within CZs. Most of the long-run gain emerges only after additional generations of sector switching and migration.

Changing composition of horizontal relatives. Table A5.3.4 reports the changes in the composition of horizontal relatives that underlie the effects of branches.
Manufacturing persistence roughly quadruples between 1880 and 1970 while agriculture persistence halves. Most new destination connections are supplied by manufacturing relatives, so sectoral and spatial changes in family connections increasingly reinforce one another.

Robustness to a smaller stock of relatives. Table A5.3.5 repeats the decompositions after reducing each dynasty's stock of non-parental relatives, r_g, by 30 percent. The effects operating through these relatives fall roughly proportionally. The 1880 distribution of horizontal relatives lowers the 1970 manufacturing share by 1.44 percentage points rather than 2.27, while allowing these connections to evolve raises it by 1.28 points rather than 2.03. Both effects are therefore about two thirds as large as in the main specification. The ranking of the channels is unchanged: agricultural and local connections account for most of the initial drag, while sector

Table A5.3.2: Family branches: channels

	1910	1930	1950	1970
A. Manufacturing share (pp)				
Spatial channels				
Origin persistence	-0.459	-0.439	-0.161	0.608
Destination pull	0.074	0.049	0.172	0.305
Spatial total	-0.384	-0.392	0.009	0.911
Sectoral channels				
Sector persistence	0.060	0.237	0.502	0.738
Cross-sector pull	-0.022	0.136	0.214	0.402
Sectoral total	0.037	0.375	0.718	1.142
Total effect of branches	**-0.347**	**-0.017**	**0.727**	**2.053**
B. Wage bill (%)				
Spatial channels				
Origin persistence	-0.614	-0.470	-0.211	0.411
Destination pull	0.109	0.082	0.200	0.271
Spatial total	-0.504	-0.393	-0.014	0.680
Sectoral channels				
Sector persistence	0.087	0.286	0.519	0.713
Cross-sector pull	-0.030	0.170	0.226	0.384
Sectoral total	0.056	0.461	0.748	1.100
Total effect of branches	**-0.448**	**0.069**	**0.734**	**1.780**

Notes: Contributions of family branches, obtained from the hierarchical Shapley decomposition underlying Figure 11a. Panel A reports contributions to the national manufacturing employment share, in percentage points. Panel B reports contributions to the aggregate wage bill, in percent of the wage bill in the economy with ties fixed at their 1880 values.

persistence and origin persistence account for most of the subsequent gain. The contribution of parents is almost unchanged because the exercise varies only the stock of non-parental relatives. Combining both margins, changing family connections raise the 1970 manufacturing share by 8.18 percentage points, compared with 8.99 points in the main specification.

A5.4 Effects beyond 1970

Roots and branches beyond 1970. Figure A5.4.1 extends the aggregate decomposition in Figure 9 beyond the main 1970 horizon by assuming constant fundamentals after 1970 and imposing the steady state in 2010. The extended path illustrates that branches may eventually outweigh roots.

Table A5.3.3: Spatial decomposition and timing of family branches

Panel A. Spatial decomposition of manufacturing effects

	1910	1930	1950	1970
Within-CZ manufacturing shares	-0.3	-0.2	0.5	1.5
Across-CZ employment reallocation	0.0	0.1	0.2	0.5

Panel B. Manufacturing effects by period of endogenous adjustment

Family connections change through	1970 effect (pp)
1910	-0.132
1930	1.522
1950	1.829
1970 (full adjustment)	2.053

Notes: Effects on the national manufacturing employment share, in percentage points. Panel A decomposes the contribution of changes in connections to relatives into changes in manufacturing shares within CZs and changes in the geographic distribution of employment. Panel B reports the 1970 effect when relatives are allowed to evolve through the indicated year and then held fixed. Both panels use an economy with relatives fixed at their 1880 distribution as the benchmark.

Table A5.3.4: Changes in the sectoral composition of horizontal family connections

Panel A. Sectoral orientation of connections to relatives

	1880	1910	1930	1950	1970
Agricultural persistence	0.452	0.528	0.326	0.276	0.223
Agriculture to manufacturing	0.044	0.107	0.159	0.213	0.207
Manufacturing to agriculture	0.261	0.132	0.110	0.076	0.053
Manufacturing persistence	0.116	0.234	0.321	0.423	0.463

Panel B. Sectoral sources of new destination connections

	1910	1930	1950	1970
Agriculture	16.4	18.4	13.9	13.4
Manufacturing	83.6	81.6	86.1	86.6

Notes: Panel A reports mass-weighted connections to relatives classified by the sectors of successive cohorts. Persistence refers to connections within the same sector, while cross-sector connections link agriculture and manufacturing. Panel B reports the percentage of newly formed destination connections supplied by relatives in each sector.

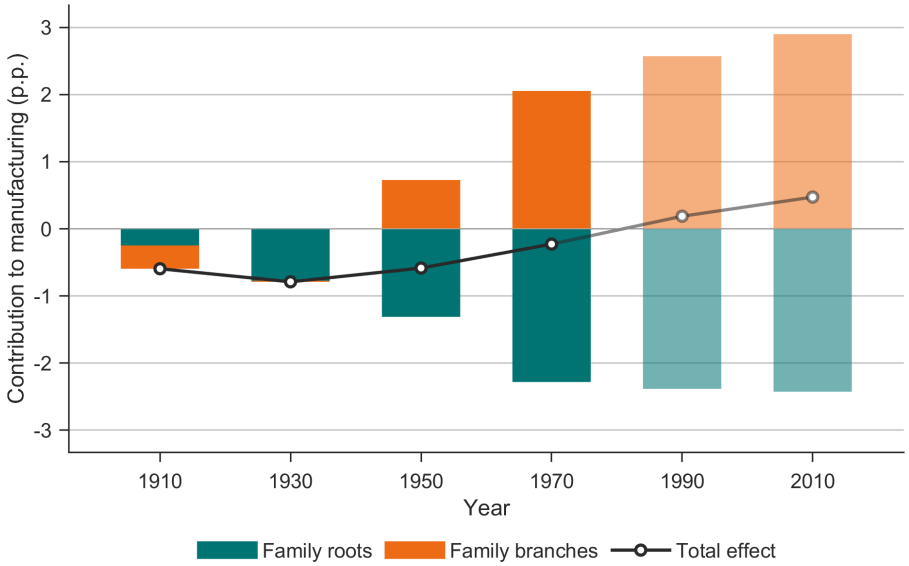
Extended-horizon effects of family roots and branches. Figure A5.4.2 extends the decompositions of family roots and branches through 2010. Panel A5.4.2a shows that roots remain a drag on structural transformation, although their composition changes. While agri-

Table A5.3.5: Robustness to a smaller stock of non-parental relatives, 1970

	Manufacturing share (p.p.)		Output (%)	
	Main	$0.7r_g$	Main	$0.7r_g$
Panel A. Initial 1880 distribution of relatives				
Agricultural connections	−1.19	−0.84	−1.14	−0.80
Manufacturing connections	0.08	0.06	0.10	0.07
Local connections	−1.16	−0.66	−1.13	−0.67
Total	**-2.27**	**-1.44**	**-2.18**	**-1.40**
Panel B. Evolving family connections				
Origin persistence	0.57	0.33	0.50	0.28
Destination pull	0.33	0.20	0.25	0.14
Sector persistence	0.77	0.52	0.81	0.54
Cross-sector pull	0.36	0.24	0.37	0.24
Parents	6.96	6.89	7.51	7.38
Total	**8.99**	**8.18**	**9.44**	**8.58**

Notes: The $0.7r_g$ specification reduces each dynasty's stock of non-parental relatives by 30 percent relative to the main specification. Panel A reports the effect of the 1880 distribution of non-parental relatives relative to an economy without non-parental relatives. Panel B reports the effect of allowing family connections to evolve rather than keeping their 1880 distribution constant and includes the contribution from evolving parental connections. Manufacturing effects are percentage-point changes in the national manufacturing share. Output effects are percent changes.

Figure A5.4.1: Extended-horizon contribution of family roots and branches

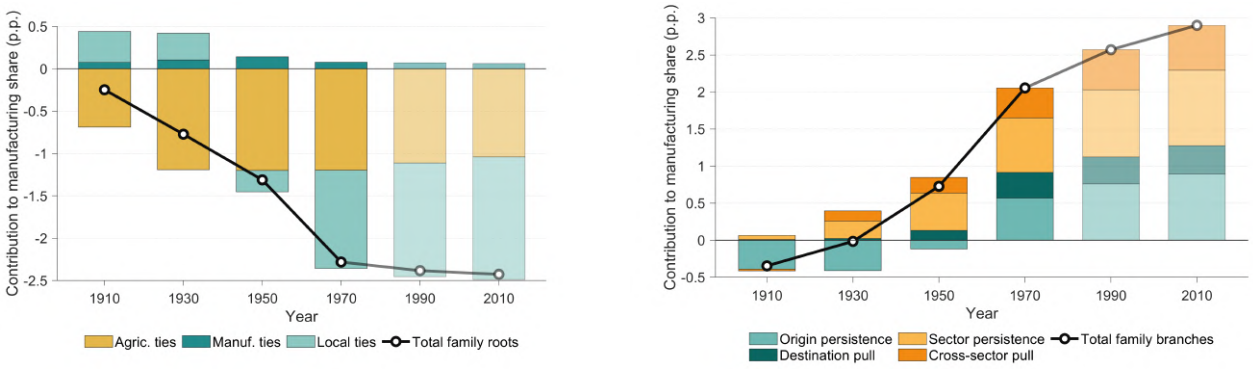


Notes: Contributions to the national manufacturing share in percentage points. Roots compare the fixed-1880 distribution of relatives with no relative support; branches compare the endogenous distribution with the fixed-1880 distribution. The line is the total effect.

cultural connections initially account for most of the negative effect, local family ties become increasingly important as the economy develops. Panel A5.4.2b shows that branches continue to

strengthen, with both sectoral and spatial connections contributing to the manufacturing gains. Beyond 1970, branches eventually more than offset the persistent drag from roots, turning the overall contribution of family ties positive.

Figure A5.4.2: Extended-horizon effects of family roots and branches



(a) Family roots

(b) Family branches

Notes: The figure reports Shapley decompositions of the contributions of family roots and branches to the national manufacturing share, in percentage points. Panel (a) separates roots into agricultural, manufacturing and local connections, comparing the fixed-1880 family distribution with an economy without family support. Panel (b) separates branches into origin persistence, destination pull, sector persistence and cross-sector pull, comparing the endogenous family distribution with the fixed-1880 distribution. Lines report the total effect of each component. Beyond 1970, economic fundamentals are held constant at their 1970 levels, with a steady state imposed in 2010.

A5.5 Moving voucher

Policy eligibility, financing and calibration. The voucher is offered only in 1880. I rank commuting zones by their manufacturing share of employment and designate the bottom 30 percent as eligible origins, $\mathcal{L}$. The resulting 137 of 455 CZs contain 20.9 percent of the 1880 population but only 2.25 percent of national manufacturing employment. The targeted origins are the same in every exercise. Figure A5.5.1 maps the 137 targeted commuting zones.
The policy pays a common grant G to workers who leave an eligible CZ for a destination outside $\mathcal{L}$. Unless otherwise stated, every destination outside $\mathcal{L}$ is eligible. The budget equals one percent of the 1880 national wage bill,

$$B^* = 0.01\, WB_{1880} = \$119.4 \text{ million}. \tag{A5.5.1}$$

It is financed by a uniform lump-sum tax

$$T = \frac{B^*}{\sum_n L_{n,1880}} = \$34.84 \tag{A5.5.2}$$

per person. For a worker in sector j, with origin o and destination d, the consumption component of destination utility is

$$\widetilde{v}^{j}_{do,1880} = v\big(w^{j}_{d,1880} - T + G\,\mathbf{1}\{o \in \mathcal{L},\ d \notin \mathcal{L}\}, p_{d,1880}\big)\,. \tag{A5.5.3}$$

The voucher therefore changes household income at eligible destinations.
The number of recipients depends on the grant. Let $F(G)$ be the number of workers who receive a grant of G. I choose the grant separately for each treatment of family connections so that total payments equal the budget:

$$B_{\text{relocation}}(G) = G\,F(G) = B^{*}. \tag{A5.5.4}$$

I solve for the grant alongside the 1880 allocation, using the procedure in Appendix A5.1. When the voucher induces more moves, take-up changes and so does the grant that exhausts the budget. All comparisons therefore use the same total expenditure.

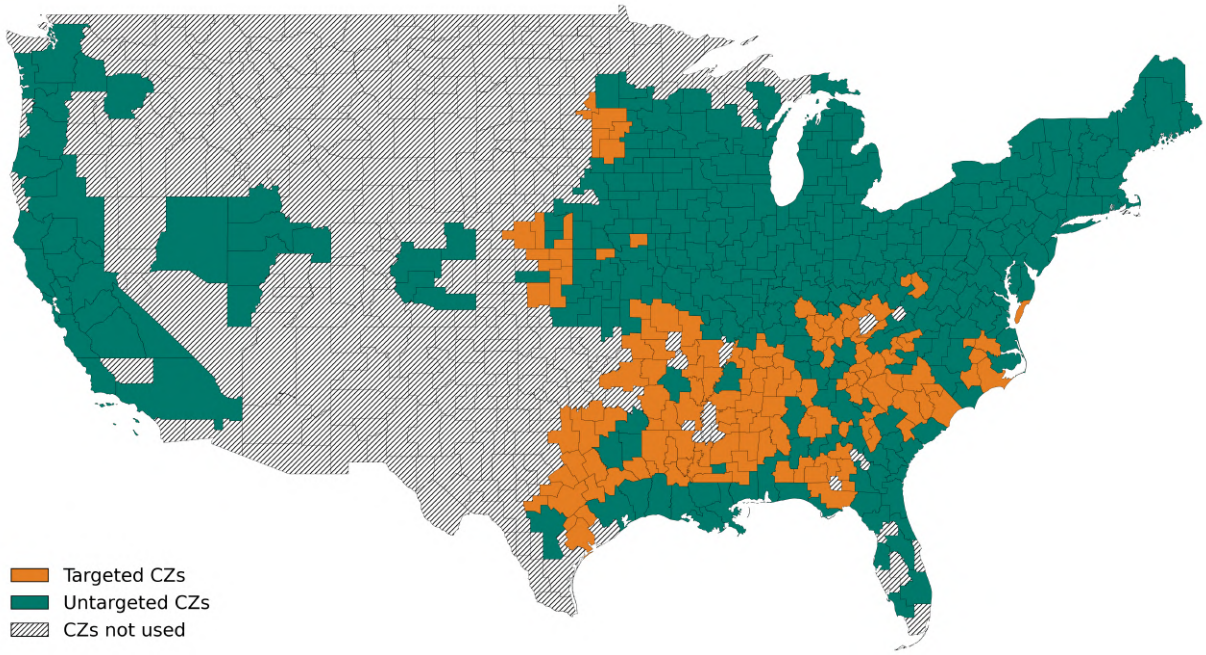
Table A5.5.1: Relocation-voucher calibration and take-up

	No horizontal family ties	Fixed 1880 family ties	Changing family ties
Relocation grant ($)	707	724	723
Grant recipients (000s)	168.8	164.8	165.1
Recipients / targeted population (%)	23.56	23.01	23.04

Notes: Each counterfactual spends one percent of the 1880 national wage bill. The relocation grant is paid to workers who move from a targeted CZ to a destination outside $\mathcal{L}$. The grant is chosen separately in each family-connection specification so that realized expenditure equals the common policy budget.

Counterfactual construction and outcome measures. To isolate the effect of voucher-created connections to horizontal relatives, I compare the full model with an economy in which generations from 1910 onward receive the connections they would have had without the voucher. Parental connections continue to evolve as in the full model. Workers still value family connections and respond to wages and prices when choosing sectors and locations. The comparison preserves the initial response to the voucher but excludes the connections created by subsequent changes in migration and sector choices.
Targeted families include everyone living in $\mathcal{L}$ in 1880 and their descendants, wherever those descendants later live. I measure manufacturing and earnings relative to the economy without the voucher. Let N^{q}_{t} be the number of CZ moves in year t under policy q, and let N^{0}_{t} be the number without the policy. The voucher initially induces $\Delta N_{1880} = N^{\text{policy}}_{1880} - N^{0}_{1880}$ additional

Figure A5.5.1: Commuting zones targeted by the relocation voucher



Notes: The figure shows the commuting zones used in the policy counterfactuals. Targeted CZs, shown in orange, are the 137 commuting zones in the bottom 30 percent of the 1880 manufacturing-employment share. Teal CZs are included in the quantitative analysis but are not targeted. Hatched CZs are not used in the quantitative analysis.

moves. To express later migration relative to that initial response, I calculate

$$\mathcal{R}_T^q = 100 \frac{\sum_{t \in \{1910,1930,1950,1970\}, t \leq T} \left(N_t^q - N_t^0\right)}{\Delta N_{1880}}. \tag{A5.5.5}$$

The total number of additional moves per 100 initially induced movers, including the 1880 moves, is $100+\mathcal{R}_T^q$. The additional migration resulting from changes in connections to horizontal relatives is the difference in $\mathcal{R}_T$ between the full model and the comparison. Earnings are calculated using the wages that result in each case.

Tracking policy groups across economies. To evaluate how the voucher affects different workers and their descendants, I divide the 1880 population into four groups. The first consists of workers from targeted origins who move to a destination outside the targeted set $\mathcal{L}$ under the voucher. I refer to these workers as movers or voucher recipients. The second consists of the remaining workers from targeted origins, whom I refer to as those left behind. The third consists of workers already living outside $\mathcal{L}$ whose family roots are in a targeted CZ and the fourth includes all remaining workers. Group membership is defined in the voucher economy and held fixed across counterfactuals.

I track these groups by assigning each worker a group tag in 1880. Within each commuting

zone, ancestral type and sector, tagged workers follow the same location- and sector-choice probabilities as other workers. Their descendants receive the tag, allowing me to follow each group's population mass across generations.

For movers and those left behind, I define group membership using the 1880 migration decisions in the voucher economy. To identify the corresponding groups in the no-voucher and fixed-family-connections economies, consider a cell $c = (g, n, j)$ with $n \in \mathcal{L}$. Let ℓ_c^q denote the probability of moving outside $\mathcal{L}$ in economy q. In each counterfactual, I assign all workers who move to the movers group, along with a fraction

$$\alpha_c^q = \frac{\ell_c^{\text{voucher}} - \ell_c^q}{1 - \ell_c^q} \tag{A5.5.6}$$

of those who stay. This ensures that the movers group has the same initial share as in the voucher economy. All remaining workers belong to the left-behind group. This matching requires $\ell_c^{\text{voucher}} \geq \ell_c^q$, so additional voucher movers are drawn from counterfactual stayers. Workers retain their counterfactual locations and descendants receive their group assignments. The other two groups are defined by predetermined characteristics and require no adjustment. Group sizes differ by less than 0.04 percent across economies.

Consumption-equivalent welfare. Following the consumption-and-amenity components of Yang (2024), I measure welfare using the flow utility experienced by workers and their descendants. I apply the measure to the groups defined above. Let a denote the group whose welfare is evaluated, which can also include the entire population. For a worker of type (g, n, j) choosing destination m in generation s, flow utility in economy $q \in \{0, \text{voucher}\}$ is

$$\begin{aligned} u_{gnjm,s}^q &= \log\left(C_{gnjm,s}^q \mathcal{B}_{gnjm,s}^q\right), \\ C_{gnjm,s}^q &= v\left(e_{gnjm,s}^q, p_{ms}^q\right), \\ \mathcal{B}_{gnjm,s}^q &= \exp\left[\mathcal{A}_{gms}^j - \rho \log L_{ms}^q + \theta_H H_{gnjs}^q(m)\right]. \end{aligned} \tag{A5.5.7}$$

Here, $e_{gnjm,s}^q$ denotes consumption expenditure, equal to disposable labor earnings in the model. In 1880, expenditure includes the lump-sum tax and, for eligible movers, the relocation grant, as specified in equation (A5.5.3). In subsequent generations, expenditure equals labor earnings. The amenity component captures non-family amenities, congestion and the utility from proximity to relatives. This welfare measure focuses on consumption and amenities and therefore does not directly include moving costs, sector-entry costs or the option value of alternative choices.

For a group a and starting generation t, let $\omega_{gnjm,s|t,a}^q$ denote the share of workers in generation s belonging to that group or its descendants who have type (g, n, j) and choose destination m in economy q. These shares sum to one within each generation and are calculated separately in

the two economies using the group assignments described above. Average flow utility for the group and its descendants is

$$\bar{u}^{q}_{s|t,a} = \sum_{g,n,j,m} \omega^{q}_{gnjm,s|t,a} \log\left(C^{q}_{gnjm,s}\mathcal{B}^{q}_{gnjm,s}\right). \tag{A5.5.8}$$

I measure the welfare effect of the voucher as the discounted change in average flow utility,

$$\Delta\mathcal{W}_{t,a} = (1-\beta)\sum_{\ell=0}^{\infty}\beta^{\ell}\left(\bar{u}^{\text{voucher}}_{s_\ell|t,a} - \bar{u}^{0}_{s_\ell|t,a}\right), \tag{A5.5.9}$$

where s_ℓ denotes the ℓ-th generation, starting at $s_0 = t$, and β is the parental altruism factor. The factor $1-\beta$ normalizes the discounted sum to a permanent per-generation equivalent. The measure captures changes in consumption utility and amenities, including those resulting from the reallocation of workers and their descendants across sectors and places. Family ties enter directly through amenities, while migration and sectoral choices determine where workers experience these benefits. Because the distribution of workers can differ between economies, I take the difference in their respective average flow utilities rather than holding population shares fixed.

To express this welfare change in consumption-equivalent terms, consider a permanent proportional increase of $1+\lambda_{t,a}/100$ in the indirect consumption-utility index of every generation in the no-voucher economy, holding the allocation of workers fixed. Because flow utility is logarithmic in this index, the increase adds $\log(1+\lambda_{t,a}/100)$ to average flow utility in every generation. The corresponding normalized discounted welfare gain is

$$\begin{aligned}\Delta\mathcal{W}^{\text{comp}}_{t,a} &= (1-\beta)\sum_{\ell=0}^{\infty}\beta^{\ell}\log\left(1+\frac{\lambda_{t,a}}{100}\right)\\ &= \log\left(1+\frac{\lambda_{t,a}}{100}\right).\end{aligned} \tag{A5.5.10}$$

Equating this expression to the welfare change induced by the voucher, $\Delta\mathcal{W}^{\text{comp}}_{t,a} = \Delta\mathcal{W}_{t,a}$, and solving for $\lambda_{t,a}$ gives

$$\lambda_{t,a} = 100\left[\exp\left(\Delta\mathcal{W}_{t,a}\right) - 1\right], \tag{A5.5.11}$$

as reported in equation (42). The measure is the permanent percentage increase in the indirect consumption-utility index that would generate the same change in discounted consumption-and-amenity welfare as the voucher. Because consumption preferences are nonhomothetic, this does not generally correspond to an equivalent percentage change in income. I apply the same calculation to different groups and starting generations. For example, the welfare of movers and those left behind is evaluated by following their respective group tags and descendants from

1910 onward.

The value of proximity to relatives. To obtain the compensating transfers in Figure 12, I calculate how much income would make up for the loss of support from horizontal relatives when a worker moves. For each origin–destination pair and sector k, I find G^{kin} such that

$$\log\left(v\left(w_d^k + G^{\text{kin}}, p_d\right)\right) - \log\left(v\left(w_d^k, p_d\right)\right) = \theta_H \psi_R^H \left[H^{\text{orig}}(o) - H^{\text{dest}}(o,d)\right]. \tag{A5.5.12}$$

The transfer is negative if family support is greater at the destination. Across candidate pairs, the median is \$157, the 75th percentile is \$249 and the 90th percentile is \$353. About 6 percent of pairs have a negative value. Median transfers are similar for destinations with high and low initial family support, at \$151 and \$162. The largest source of family-related moving costs is therefore the support workers give up at the origin.

Robustness across policy sizes. Table A5.5.2 reports results for budgets between 0.25 and 1 percent of the 1880 wage bill. I also examine budgets up to 10 percent. These larger programs use a proportional tax because they cannot be financed by the same uniform lump-sum tax. The qualitative results remain similar. Over the smaller budgets shown in the table, the share of the manufacturing effect arising through family connections declines gradually as the program grows.

Policy-induced changes in connections to horizontal relatives account for 33–41 percent of the 1970 manufacturing effect and 22–32 percent of the earnings effect across these budgets.

Family-connection channels. Changes in connections to horizontal relatives can matter because relatives live in different places or because they work in different sectors. I separate these two channels with a Shapley decomposition. Let F be the outcome when all family connections respond to the policy and E the outcome when connections to horizontal relatives from 1910 onward remain at their no-policy values. In E^{loc}, only the location component is held at its no-policy value; in E^{sec}, only the sector component is held fixed. The contributions are

$$\Delta^{\text{loc}} = \frac{1}{2}\left[(F - E^{\text{loc}}) + (E^{\text{sec}} - E)\right], \tag{A5.5.13}$$

$$\Delta^{\text{sec}} = \frac{1}{2}\left[(F - E^{\text{sec}}) + (E^{\text{loc}} - E)\right], \tag{A5.5.14}$$

and add up to the contribution of horizontal relatives $F - E$. Parental connections evolve as in the full model in every arm. Table A5.5.3 reports the decomposition across budgets.

At the benchmark budget, horizontal relatives contribute 0.124 points in these solutions, with about 72 percent of this contribution operating through location connections.

Table A5.5.2: Relocation-voucher effects across budgets

	Budget (% of 1880 wage bill)			
	0.25	0.50	0.75	1.00
Induced movers in 1880 (000s)	4.7	9.2	13.7	18.1
Panel A. Manufacturing employment of targeted families, 1970 (%)				
Full model	0.108	0.217	0.327	0.439
Without family-connection changes	0.064	0.139	0.214	0.294
Share from family-connection changes (%)	41.0	35.8	34.7	33.1
Panel B. Earnings of targeted families, 1970 (%)				
Full model	0.047	0.093	0.139	0.186
Without family-connection changes	0.032	0.069	0.105	0.145
Share from family-connection changes (%)	32.1	25.2	24.3	22.1

Notes: Each column reports the effects of a relocation voucher with the indicated budget. "Without family-connection changes" assigns cohorts from 1910 onward the connections to horizontal relatives of the corresponding no-policy economy, while parental connections evolve as in the full model. Manufacturing and earnings are percent changes relative to the corresponding no-policy economy. The final row of each panel reports the share of the full-model effect attributable to policy-induced changes in connections to horizontal relatives.

Table A5.5.3: Family-connection channels of the voucher effect on manufacturing in 1970

Budget (% of 1880 wage bill)	0.25	0.50	0.75	1.00
Horizontal relatives	0.045	0.080	0.115	0.124
Location	0.028	0.054	0.068	0.089
Sector	0.017	0.026	0.046	0.035

Notes: Contributions to the percent effect of the relocation voucher on manufacturing employment of targeted families in 1970. The contribution of horizontal relatives is the difference between the full policy and the economy that assigns connections to horizontal relatives their no-policy values from 1910 onward, with parental connections evolving as in the full model. It is decomposed into location and sector connections with the Shapley decomposition in equation (A5.5.14).

Relatives at destinations and targeting. This exercise changes which destinations qualify for the voucher, keeping the targeted origins and budget fixed. I consider each origin–destination pair (o, d), with $o \in \mathcal{L}$ and $d \notin \mathcal{L}$. When the relevant values differ across sectors or ancestral groups within an origin, I average them using that origin's 1880 population weights.

I first rank destinations by how attractive they would be without connections to horizontal relatives. For an origin o, let $P^0(o, d)$ be the predicted probability of moving to destination d in 1880 when connections to horizontal relatives are excluded. It depends on migration costs, wages, amenities and prices, but not on the observed distribution of horizontal relatives. I keep the quarter of destinations outside $\mathcal{L}$ with the highest $P^0(o, d)$, denoted $C(o)$. Within this set,

I rank destinations by existing family connections at destinations, $H^{\text{dest}}(o, d)$, the horizontal-relative component of the destination amenity in equation (12). High-access targeting subsidizes moves to the upper half of this ranking and low-access targeting to the lower half. Both rules therefore select destinations with relatively high predicted migration without these family connections, though they do not match the two halves on that probability within the selected quarter.

Table A5.5.4: Relocation vouchers at destinations with high and low family support

	High access	Low access
Mean existing destination family support, H^{dest}	0.01195	0.00109
Grant, common budget ($)	913	1,922
Induced movers (000s)	20.4	6.7
Fiscal cost per induced mover ($)	5,852	17,929
Later moves per 100 induced movers, through 1950	10.1	2.9
Share of 1950 manufacturing effect from policy-induced family connections (%)	45.9	5.5
Equal number of first-generation movers		
Grant ($)	355	1,917
Budget ($ million)	35.5	118.8
Induced movers (000s)	6.66	6.63

Notes: Both rules begin from destinations in the top quartile of predicted migration without connections to horizontal relatives. High- and low-access destinations are the upper and lower halves of this set according to existing family connections at destinations. The common-budget comparison spends one percent of the 1880 national wage bill and recalibrates the grant separately under each rule. Induced movers are additional 1880 movers relative to the no-policy endogenous economy. Later moves report the additional post-1880 migration generated through policy-induced changes in family connections, per 100 initial induced movers.

With the same budget, high-access targeting induces about three times as many movers and costs roughly one third as much per induced mover. It also generates more than three times as many later moves per initial mover. Policy-induced family connections explain 46 percent of the manufacturing effect in 1950 under high-access targeting, compared with 6 percent under low-access targeting. If I instead require both rules to induce the same number of initial movers, low-access targeting needs 3.35 times the budget.

The value to relatives by type of move. I divide moves out of targeted CZs into six types using information known before the voucher. Origins are classified as having stronger or weaker local family roots according to the share of residents whose ancestral CZ is the origin itself. The cutoff is the population-weighted median across targeted origins. Among the candidate destinations $C(o)$, I distinguish those with more or fewer existing relatives at destinations, $H^{\text{dest}}(o, d)$. Combining the two origin groups and two destination groups gives four types. All other destinations are split by whether the origin has many or few local relatives.

For each type c, I consider the same \$723 voucher offered only for moves of that type. I compare the full economy, in which these moves change the connections to horizontal relatives available to later generations (F_c), with one in which the initial moves occur but subsequent connections to horizontal relatives remain at their no-policy values (N_c). Their welfare difference captures the effects of the connections created or lost by the subsidized moves, rather than the direct benefit of receiving the voucher. To compare those effects with grants expressed in dollars, I convert each group's change in welfare into the equivalent one-time change in its 1880 earnings and divide the sum by the number of induced movers:

$$\hat{e}_c = \frac{\sum_g L_g \, \bar{e}_g \left[\exp\left(\left(V_g^{F_c} - V_g^{N_c}\right)/\mu_{g,0}\right) - 1\right]}{\Delta N_{1880}^c}, \tag{A5.5.15}$$

where L_g is the number of workers in 1880 group g, $\bar{e}_g$ is their average 1880 earnings, $\mu_{g,0}$ is the marginal value of their log income and ΔN_{1880}^c is the number of additional initial moves induced by this restricted voucher. The expression in square brackets converts the welfare effect for a member of group g into a proportional change in 1880 earnings; multiplying by $L_g \bar{e}_g$ gives its dollar equivalent. Summing across groups and dividing by the induced moves gives the average spillover value per additional mover. A negative value means that the losses imposed through family connections exceed the gains. This calculation is made using the original uniform-grant amount, before grants are adjusted.

For the uniform voucher, this calculation gives −\$115 per induced move. Of that amount, −\$77 comes from changes in origin connections, −\$35 from destination connections and −\$4 from their interaction, up to rounding. Table A5.5.5 reports the values by type of move.

Table A5.5.5: Types of moves, value to relatives and adjusted grants

	Value to relatives (\$)	Share of spending (%) Uniform	Family-adjusted	Adjusted grant (\$)
Dense origin → many relatives	-167	27.4	21.1	598
Dense origin → few relatives	144	3.8	5.7	909
Thin origin → many relatives	-46	39.8	38.6	719
Thin origin → few relatives	212	12.3	19.3	977
Other moves, dense origin	-199	3.1	2.1	566
Other moves, thin origin	-32	13.6	13.2	733
All moves (uniform voucher)	-115	100.0	100.0	

Notes: Values to relatives are defined in equation (A5.5.15) and calculated using the \$723 grant restricted to each type of move. Spending shares include payments to all grant recipients in 1880, including workers who would have moved without the voucher.

Movers and stayers. For each group I report earnings at the wages of each economy, the manufacturing share and own-generation welfare expressed as the percentage change in that

generation's consumption-utility index. The dynasty column instead reports the 1910 lifetime consumption equivalent in equation (42), which includes descendants.

Table A5.5.6: Movers, stayers and their descendants: effects of the voucher

	1910	1930	1950	1970	Dynasty
Panel A. Movers and their descendants					
Earnings (%)	1.54	2.15	0.00	0.08	
Manufacturing share (pp)	-0.17	1.91	0.96	0.76	
Welfare (%)	6.09	-0.78	-1.90	-0.73	4.41
of which through other relatives	0.09	0.09	-0.01	-0.06	0.09
Panel B. Stayers and their descendants					
Earnings (%)	0.90	0.15	0.04	0.22	
Manufacturing share (pp)	0.09	0.19	-0.09	0.17	
Welfare (%)	-0.06	-0.50	-0.01	0.00	-0.14
of which through other relatives	-1.02	-0.55	-0.36	-0.29	-0.90

Notes: One-percent moving voucher; per person, relative to the same people in the no-voucher economy. Movers are the workers who leave the targeted CZs in the voucher economy and stayers the ones who do not. Welfare by generation is own-generation welfare, expressed as a percentage change in the consumption-utility index, while "Dynasty" is the 1910 lifetime measure in equation (42), which includes descendants. "Through other relatives" is the full model minus the economy with connections to other relatives held at their no-policy values from 1910 onward.

The voucher raises the earnings of movers by 1.5 percent and those of their children by 2.1 percent, and it raises the manufacturing share of their descendants by about one percentage point through 1970. The earnings gain fades after 1930, while the manufacturing gain persists. Over the family line, movers gain 4.4 percent in welfare, and changes in extended-family connections contribute almost nothing to this gain. Stayers earn slightly more, because labor becomes scarcer in the targeted CZs and their manufacturing share barely changes. Yet extended-family connections lower their welfare in every generation, by 1.02 percent in 1910 and 0.90 percent over the family line, as relatives who leave take with them the support that families at home rely on. Other effects, mainly higher local earnings, offset most of this loss, so that stayers end up roughly as well off as without the voucher (-0.14 percent). The cost of the extended-family externality therefore falls on the families who stay, not on those who move.

Destination family connections and moving vouchers. Existing relatives at destinations make the initial move easier, while relocated workers create connections that can facilitate migration and sector entry for other family members in later generations. To distinguish these effects, I compare vouchers offered to destinations with different levels of existing family connections. For each origin, I first select the quarter of destinations with the highest predicted migration when connections to aunts and uncles are excluded. Within this set, the high-access policy subsidizes moves to the half with more initial connections to relatives at destinations,

while the low-access policy subsidizes moves to the half with fewer. Both policies have the same budget.

Figure A5.5.2 shows that high-access targeting induces about three times as many initial moves per dollar as low-access targeting. The cost per induced mover is $5,852 rather than $17,929. Through 1950, high-access targeting also generates 10.1 additional later moves per 100 initial movers, compared with 2.9 under low-access targeting. Changes in family connections account for 46 percent of the high-access policy's manufacturing effect in 1950, compared with 6 percent under low-access targeting. Existing destination connections therefore both increase the immediate response to relocation assistance and strengthen its effects on other relatives and later generations.

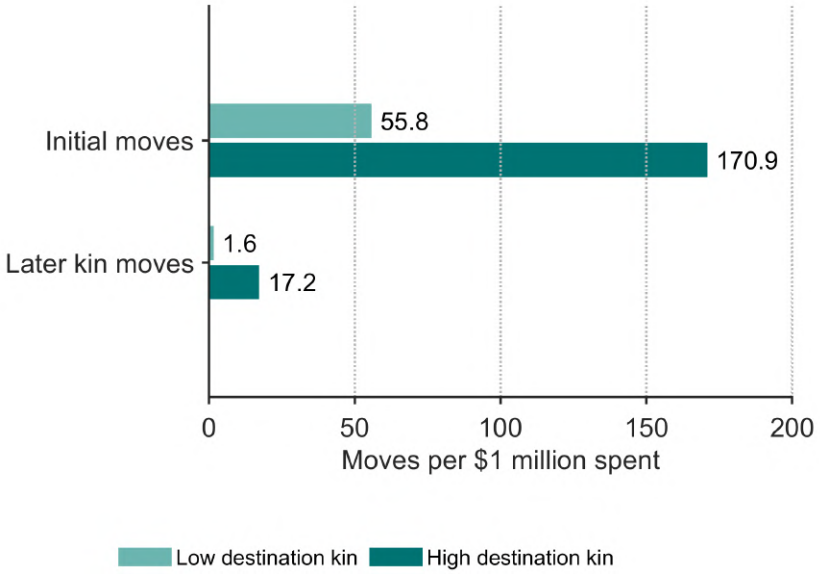


Figure A5.5.2: Destination family connections and the effectiveness of moving vouchers

Notes: The figure compares initial and subsequent moves per $1 million of policy expenditure when vouchers are offered to destinations with high or low initial connections to aunts and uncles. Eligible destinations are drawn from the quarter with the highest predicted migration after excluding these family connections, then divided into two halves by the number of relatives already at the destination. Subsequent moves are measured through 1950. Both policies have the same budget.

Accounting for family ties. Table A5.5.7 compares uniform relocation vouchers with grants adjusted for their welfare effects on other family members, holding the total budget fixed. Accounting for these effects reduces welfare losses among stayers and increases their long-run manufacturing employment and earnings, while modestly improving aggregate welfare.

Table A5.5.7: Uniform and family-adjusted relocation vouchers

	Uniform grant	Family-adjusted grant
Welfare in 1880 (%)		
All workers	-0.259	-0.248
Welfare by group in 1910 (%)		
Targeted families	0.90	0.92
Movers	4.41	4.26
Stayers	-0.14	-0.06
Manufacturing employment in 1970 (%)		
Targeted families	0.44	0.52
Movers	0.94	0.94
Stayers	0.26	0.36
Average earnings in 1970 (%)		
Targeted families	0.19	0.23
Movers	0.08	0.12
Stayers	0.22	0.26
Spending redirected (%)	–	8.8

Notes: Both policies spend one percent of the 1880 national wage bill. The uniform grant pays $723 per eligible move, while the family-adjusted grant accounts for the estimated welfare effects on other family members, holding spending fixed. All outcomes are relative to the no-policy economy. Targeted families originate in eligible commuting zones in 1880; movers leave these zones, while stayers remain. These groups are held fixed across counterfactuals and followed together with their descendants. Welfare is measured in consumption equivalents. Total welfare in 1880 includes descendants, grants and financing taxes, while group welfare in 1910 measures changes from 1910 onward. Manufacturing employment and earnings are measured in 1970. Spending redirected is the share of the budget reallocated relative to the uniform grant.